MECH230 Dynamics Notes

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Preface

These notes are prepared for the MECH230 Dynamics course at the American University of Beirut (AUB), Lebanon. The content is adapted primarily from O'Reilly's Dynamics Primer, with selected examples and exercises drawn from Meriam, Kraige & Bolton's (MKB) and Hibbeler's classic textbooks. Some segments of these notes are adapted from the course notes prepared by Professor Kem Kamrin (UC Berkeley) for his ME104 course.

The following notes begin with simple examples to build a strong foundation, enabling students to confidently tackle more complex problems. They are not comprehensive; students should refer to the course textbooks as needed.

I am also thankful to Professor Oliver O'Reilly for his excellent book on dynamics, which has been an invaluable resource for both me and my students, as well as for his many helpful discussions and advice on the subject.

I am also grateful for the perspectives on dynamics of Professor Noel Perkins (University of Michigan–Ann Arbor), Professor James Casey (UC Berkeley), and Dr. Evan Hemingway, which I gained while TAing Berkeley's ME104.

If you find any mistakes in these notes, please let me know by filling out this form.

Theresa E. Honein

Part I

Preliminaries

1 Active Learning

Check out these two excellent videos of Professor Noel Perkins from the University of Michigan at Ann Arbor explaining the active learning techniques he uses to teach dynamics. We will be using these techniques ourselves. The techniques include:

- **Having an agenda for the day.** Students can add topics they want to discuss too.
- **Frequently checking student understanding:**
  - "Raise your hand if you understand 90% of this or above."
- **Encouragement:**
  - "Ask me a question."
  - "Persevere. Someone else is going to have this question."
  - "I had a great question from one of your classmates."
- **Think → Share:**
  - "I would like you to do the first step on your own. Take a min."
  - "Show of hands: I did it, or I know how to do it."
- **Think → Pair → Share:** * "I would like you to take 5 min to talk to your neighbor and come up with a strategy that will solve this problem."
  - "Show of hands: my neighbor convinced me this will work."
  - "Show of hands: I am not convinced yet."

https://www.youtube.com/watch?v=wHEys-JHeb8

https://www.youtube.com/watch?v=96j69u4v-wE

2 A Note on Units

Quantity	Dimensional Symbol	SI Units		U.S. Customary Units	
		Unit	Symbol	Unit	Symbol
Mass	M	Base units: kilogram	kg	slug	—
Length	L	Base units: meter*	m	Base units: foot	ft
Time	T	Base units: second	s	Base units: second	sec
Force	F	newton	N	Base units: pound	lb

*Also spelled *metre*.

Figure 2.1: Source: MKB

In this class, we will use both SI and US customary units.

In the US customary units, the pound is used both as a unit of force (lbf) and as a unit of mass (lbm). The lbm is the amount of mass which weighs 1 lbf under standard conditions (at latitude 45° and sea level). In MKB, almost exclusively the unit of slug is used for mass. In addition, they use the symbol lb to mean pound force.

Notice that seconds is abbreviated as s in SI units and sec in US customary units.

For the gravitational acceleration use the values $g = 9.81$ m s^{-2} in SI units and $g = 32.2$ ft sec^{-2} in US customary units.

> **!** Note!
>
> When solving a problem, keep all variables in the unit system they are given. For example, you may convert g to kg, but don't convert g to slugs!

2.1 Summary

In U.S. customary units, the **slug** is the unit of mass: 1 slug = 1 lb·sec^2/ft. Use $g = 32.2$ ft/sec^2 in U.S. customary units and $g = 9.81$ m/s^2 in SI. Keep all variables in the unit system they are given.

2.2 Exercises

The following problems are from Set 02 – Units.

1. [MKB 1/2] Familiarise yourself with the U.S. unit of mass (slugs). Note that 1 kg ≈ 0.0685 slugs. *(ans.* $W = 14\,720$ *N or* 3310 *lb;* $m = 102.8$ *slugs)*

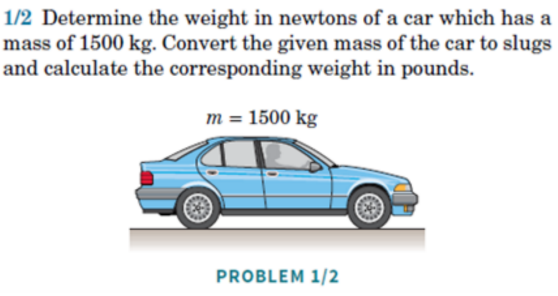


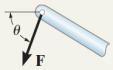
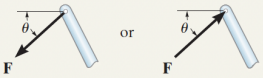
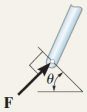
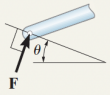
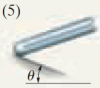
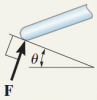
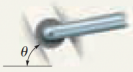
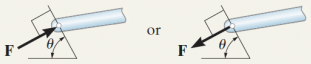
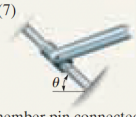
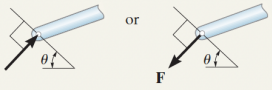
Figure 2.2: MKB 1/2.

2. Read this article about the Mars Climate Orbiter loss and reflect on the importance of consistent units.

3 Free Body Diagrams

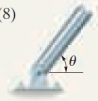
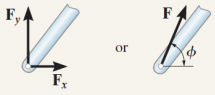
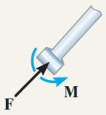
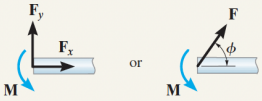
Recall the following table taken from the Hibbeler Statics textbook. It summarizes the constraint resultant forces and moments due to various types of supports.

TABLE 5–1 Supports for Rigid Bodies Subjected to Two-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(1) cable		One unknown. The reaction is a tension force which acts away from the member in the direction of the cable.
(2) weightless link		One unknown. The reaction is a force which acts along the axis of the link.
(3) roller		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(4) rocker		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(5)  smooth contacting surface		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(6)  roller or pin in confined smooth slot		One unknown. The reaction is a force which acts perpendicular to the slot.
(7)  member pin connected to collar on smooth rod		One unknown. The reaction is a force which acts perpendicular to the rod.

continued

TABLE 5–1 Continued

Types of Connection	Reaction	Number of Unknowns
(8)  smooth pin or hinge		Two unknowns. The reactions are two components of force, or the magnitude and direction ϕ of the resultant force. Note that ϕ and θ are not necessarily equal [usually not, unless the rod shown is a link as in (2)].
(9) member fixed connected to collar on smooth rod		Two unknowns. The reactions are the couple moment and the force which acts perpendicular to the rod.
(10) fixed support		Three unknowns. The reactions are the couple moment and the two force components, or the couple moment and the magnitude and direction ϕ of the resultant force.

Think!

Question: What is the general rule which encompasses all the cases in the above table and more?

Answer

Whenever translation in a certain direction is constrained, a constraint force in that direction enforces the constraint. Whenever rotation along a certain axis is constrained, a constraint moment along the same axis results.

Note! Notation

Good notation is important: in print, vectors are usually denoted by boldface, e.g. $\mathbf{v}$ or $\mathbf{F}$. In writing, vectors are either denoted by an overhead arrow, e.g. $\vec{v}$, or by a line or squiggle underneath, e.g. $\underline{v}$. We opt for the latter notation since we reserve the top of a vector for other accents.

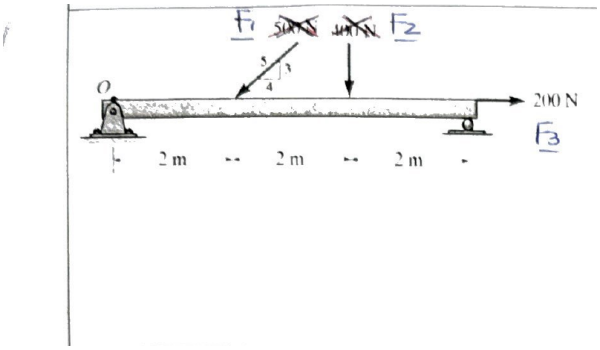
For a massless inextensible string, the tension is the same at all points along the string.

For a frictionless massless pulley, the tension in the string on either side of the pulley is the same.

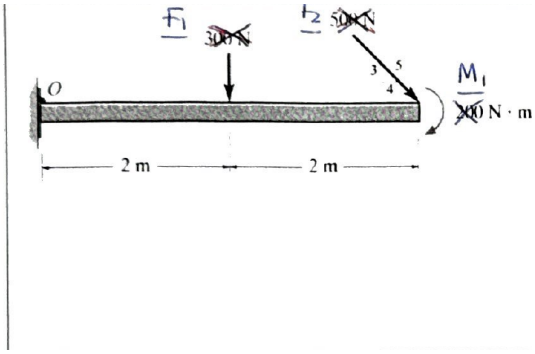
3.1 Exercises

The following problems are from Set 00 – Free Body Diagrams. Gravity acts vertically down; all problems are set up in the vertical plane.

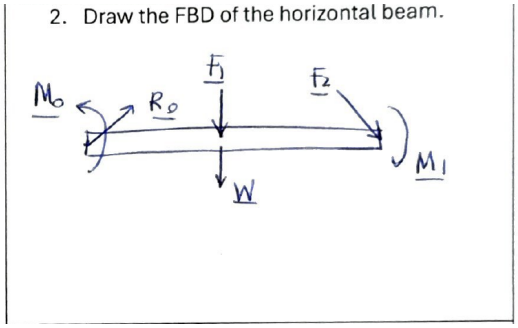
1. Draw the FBD of the horizontal beam.



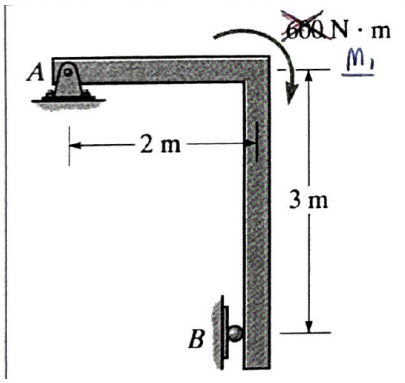
2. Draw the FBD of the horizontal beam.



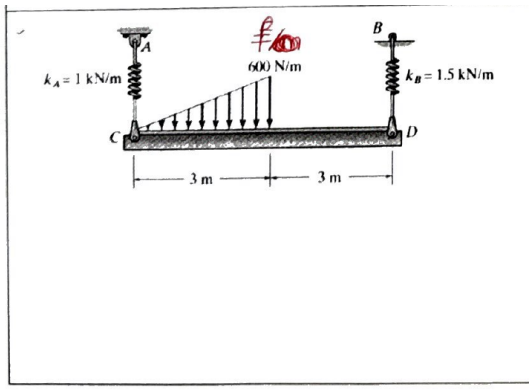
Answer



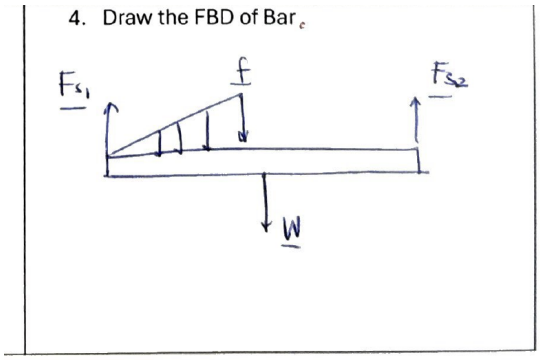
3. Draw the FBD of angled beam AB.



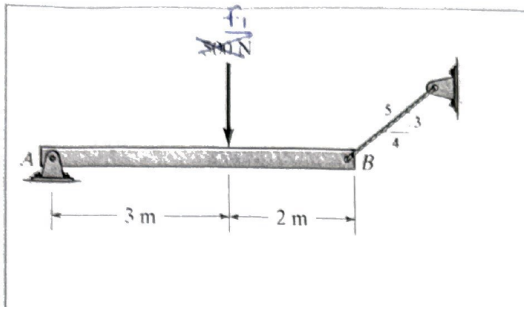
4. Draw the FBD of the bar.



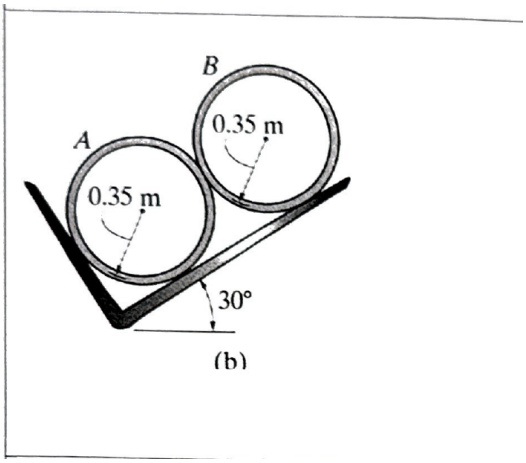
Answer



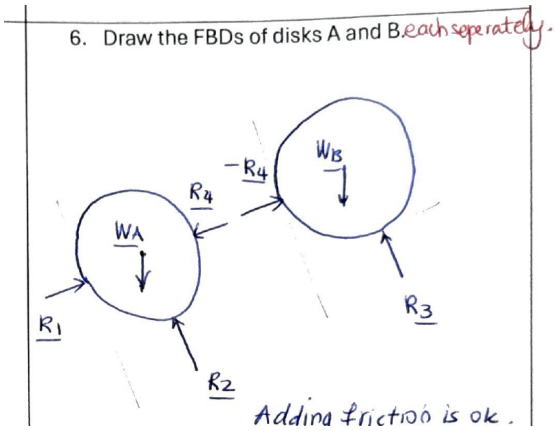
5. Draw the FBD of beam AB.



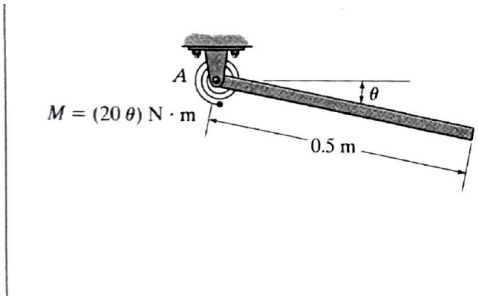
6. Draw the FBDs of disks A and B each separately.



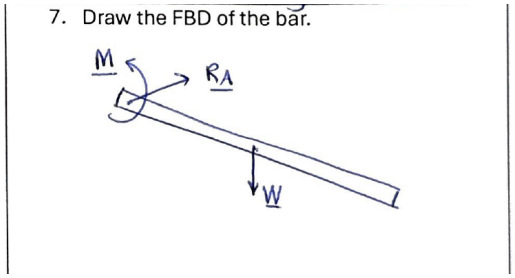
Answer



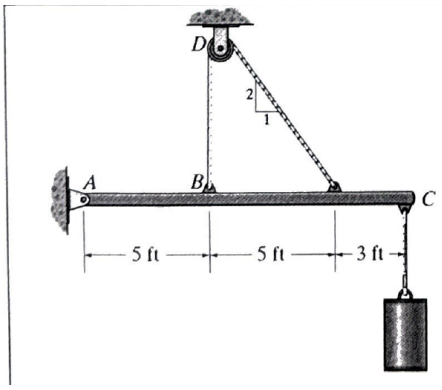
7. Draw the FBD of the bar.



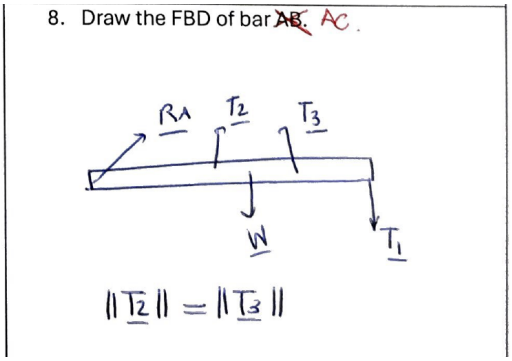
Answer



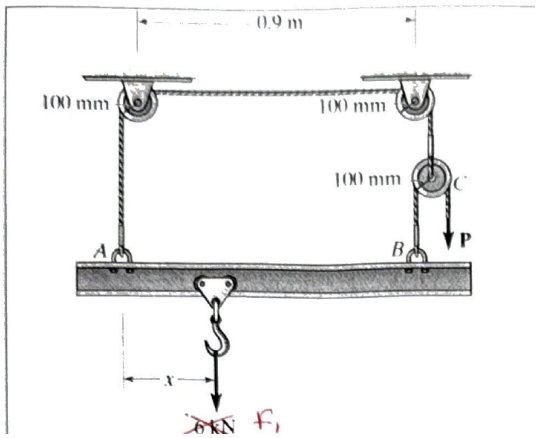
8. Draw the FBD of bar AB.



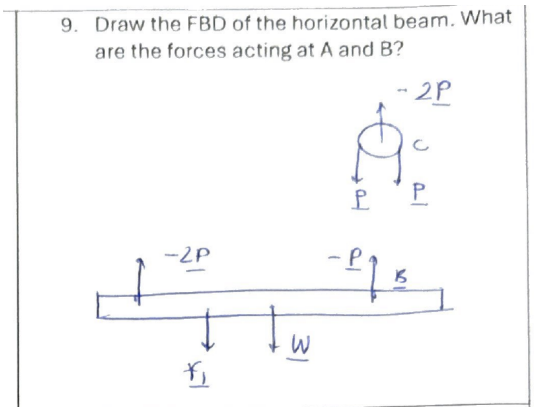
Answer



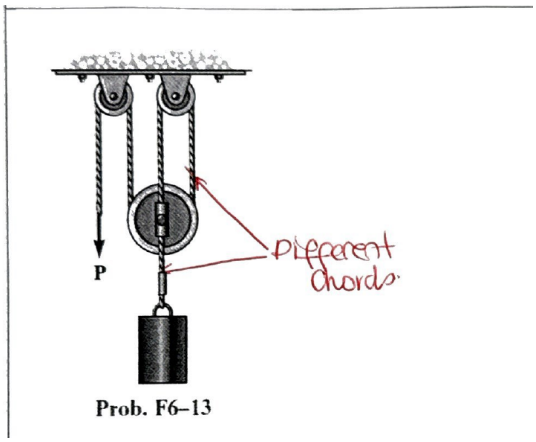
9. Draw the FBD of the horizontal beam. What are the forces acting at A and B?



Answer



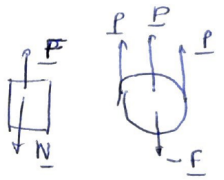
10. The cylinder of mass m is in equilibrium. What forces act on it? Gravity acts vertically downwards (same direction as the drawn force **P**).



Prob. F6–13

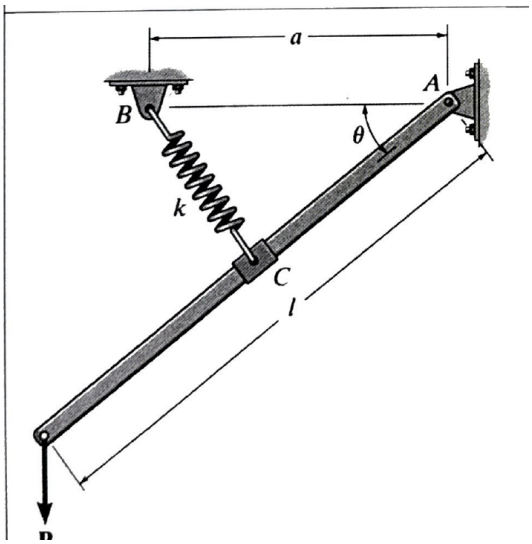
Answer

10. The cylinder of mass m is in equilibrium. What the force acting on it? Gravity acts vertically downwards (same direction as the drawn force **P**).

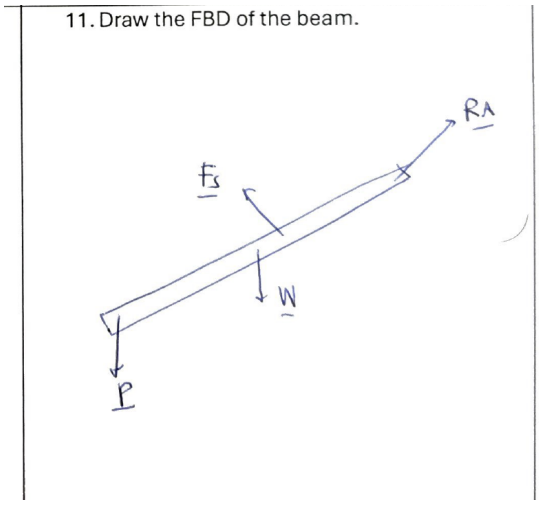


$F = 3P$

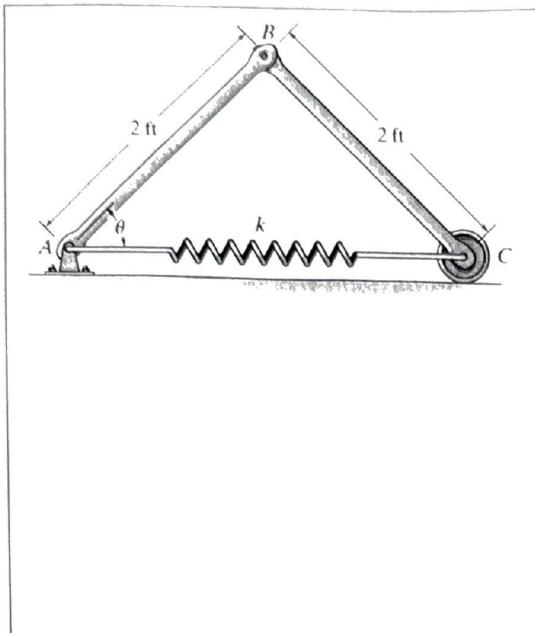
11. Draw the FBD of the beam.



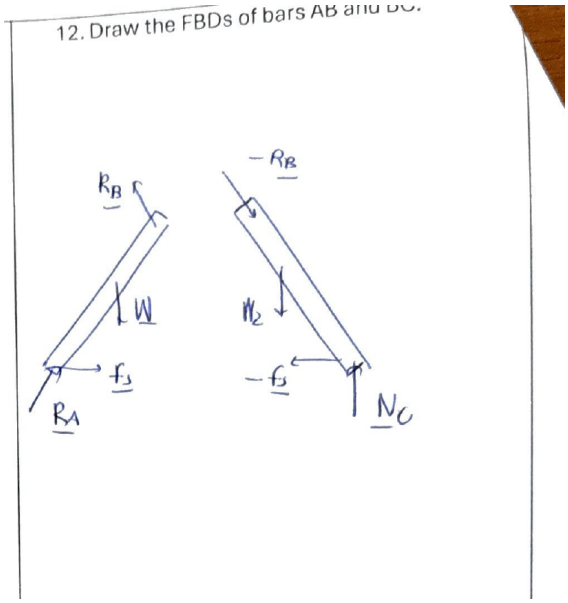
Answer



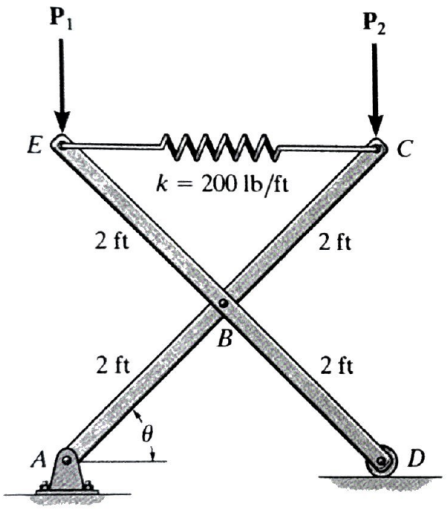
12. Draw the FBDs of bars AB and BC.



Answer



13. Draw the FBDs of bars AC and ED.


P1
P2
E
C
k = 200 lb/ft
2 ft
2 ft
B
2 ft
2 ft
θ
A
D

4 Vector Calculus

4.1 Scaling Vectors

We can scale the vector $\mathbf{a}$.

Think!

Question: Consider a vector $\mathbf{a}$. Plot the vectors $2\mathbf{a}$, $-\mathbf{a}$, $\frac{1}{2}\mathbf{a}$

Answer

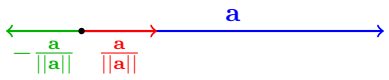
...

Think!

Question: Give the expression for a unit vector in the direction of $\mathbf{a}$? Opposite to $\mathbf{a}$.

Answer

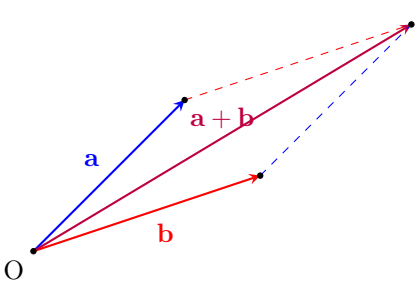
- $\frac{\mathbf{a}}{\|\mathbf{a}\|}$ is a unit vector in the direction of $\mathbf{a}$.
- $-\frac{\mathbf{a}}{\|\mathbf{a}\|}$ is a unit vector in the direction opposite to that of $\mathbf{a}$.



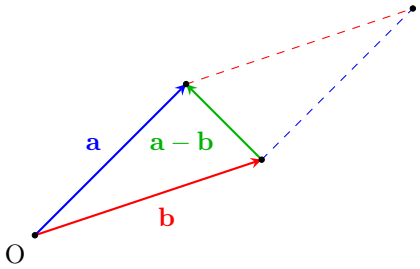
4.2 Vector summation and subtraction

Vectors can be added using the parallelogram law or by placing vectors in a head to tail sequence.

The vector difference $\mathbf{a} - \mathbf{b}$ points from the tip of $\mathbf{b}$ to the tip of $\mathbf{a}$.



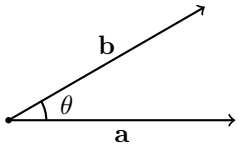
(a) Vector addition via parallelogram law and tip to tail.



(b) Vector subtraction as $\mathbf{a} - \mathbf{b}$ using the parallelogram law. This can be also done using head to tail.

4.3 The dot (scalar) product

Consider the three dimensional Euclidean space $\mathbb{E}^3 = (\mathbb{R}^3, \cdot)$ which is the three-dimensional space $\mathbb{R}^3$ and a dot product defined as follows.



Let $\mathbf{a}, \mathbf{b} \in \mathbb{E}^3$.

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos(\theta). \tag{4.1}$$

Think!

Question: What is the result of $\mathbf{a} \cdot \mathbf{a}$?

Answer

The magnitude of a vector $\mathbf{a}$ is given by

$$\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \implies \|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}. \tag{4.2}$$

Think!

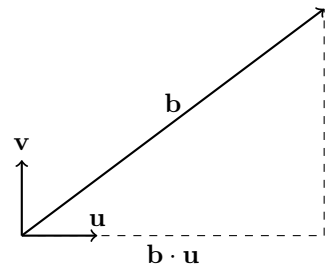
Question: Is $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$? Why?

Answer

Yes, the dot product is commutative. This can be shown using the definition of the dot product.

4.4 Vector Projections

Suppose $\mathbf{u}$ is a unit vector: $\mathbf{u} \cdot \mathbf{u} = 1$.



Think!

Question: Calculate $\mathbf{b} \cdot \mathbf{u}$

Answer

$$\mathbf{b} \cdot \mathbf{u} = \|\mathbf{b}\| \underbrace{\|\mathbf{u}\|}_{1} \cos(\theta) = \|\mathbf{b}\| \cos(\theta). \tag{4.3}$$

Notice the right triangle in the above figure. Verify that $\mathbf{b} \cdot \mathbf{u}$ is the orthogonal projection of $\mathbf{b}$ in the direction of $\mathbf{u}$.

Suppose $\mathbf{v} \cdot \mathbf{v} = 1$ ($\mathbf{v}$ is a unit vector), and $\mathbf{v} \cdot \mathbf{u} = 0$ ($\mathbf{v}$ and $\mathbf{u}$ are orthogonal/perpendicular).

$$\mathbf{b} \cdot \mathbf{v} = \|\mathbf{b}\| \cos\left(\frac{\pi}{2} - \theta\right) = \|\mathbf{b}\| \sin(\theta). \tag{4.4}$$

This is the projection of $\mathbf{b}$ in the direction perpendicular to $\mathbf{u}$. Hence, we can write

$$\mathbf{b} = (\mathbf{b} \cdot \mathbf{u})\,\mathbf{u} + (\mathbf{b} \cdot \mathbf{v})\,\mathbf{v} \tag{4.5}$$

$$= \underbrace{\|\mathbf{b}\|}_{\text{magnitude}} \left(\underbrace{\cos(\theta)\mathbf{u} + \sin(\theta)\mathbf{v}}_{\text{direction}} \right). \tag{4.6}$$

Remember that $\cos^2(\theta) + \sin^2(\theta) = 1$.

4.5 Cross Product

We also define the cross product

$$\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \, |\sin(\theta)| \mathbf{n} \tag{4.7}$$

where $\mathbf{n}$ is a unit vector normal to the plane formed by $\mathbf{a}$ and $\mathbf{b}$. The direction of $\mathbf{n}$ is given by the right hand rule.

Think!

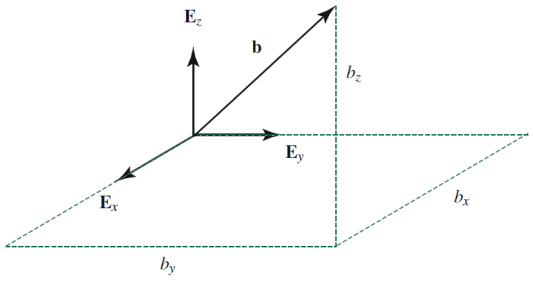
Question: Is $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}$?

Answer

No, the cross product is not commutative. These vectors are opposite.

4.6 Vector Representation Using Bases

We can assign a fixed (right handed) Cartesian basis for $\mathbb{E}^3$ denoted by $\{\mathbf{E}_x, \mathbf{E}_y, \mathbf{E}_z\}$. These three vectors are orthonormal (i.e. they each have a unit magnitude and are mutually perpendicular). Other common notations for orthonormal bases are $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ and $\{\mathbf{I}, \mathbf{J}, \mathbf{K}\}$. We will reserve upper case letters for fixed bases and lower case letters for moving bases.



Think!

Question: Calculate $\mathbf{E}_x \times \mathbf{E}_x$, $\mathbf{E}_x \times \mathbf{E}_y$, etc.

Answer

...

Think!

Question: Calculate $\mathbf{E}_x \cdot \mathbf{E}_x$, $\mathbf{E}_x \cdot \mathbf{E}_y$, etc.

Answer

...

We can represent any vector $\mathbf{b}$ as

$$\mathbf{a} = a_x\mathbf{E}_x + a_y\mathbf{E}_y + a_z\mathbf{E}_z, \tag{4.8}$$
$$\mathbf{b} = b_x\mathbf{E}_x + b_y\mathbf{E}_y + b_z\mathbf{E}_z. \tag{4.9}$$

We can write the scalar product and cross product of these vectors in terms of their Cartesian components.

$$\begin{aligned}
\mathbf{a}\cdot\mathbf{b} &= (a_x\mathbf{E}_x + a_y\mathbf{E}_y + a_z\mathbf{E}_z)\cdot(b_x\mathbf{E}_x + b_y\mathbf{E}_y + b_z\mathbf{E}_z) && (4.10)\\
&= a_xb_x + a_yb_y + a_zb_z && (4.11)\\
\mathbf{a}\times\mathbf{b} &= (a_x\mathbf{E}_x + a_y\mathbf{E}_y + a_z\mathbf{E}_z)\times(b_x\mathbf{E}_x + b_y\mathbf{E}_y + b_z\mathbf{E}_z) && (4.12)\\
&= (a_yb_z - a_zb_y)\mathbf{E}_x + (a_zb_x - a_xb_z)\mathbf{E}_y + (a_xb_y - a_yb_x)\mathbf{E}_z && (4.13)
\end{aligned}$$

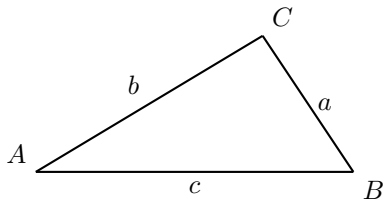
4.7 Differentiation of Vectors

$$\begin{aligned}
\frac{d\mathbf{u}}{dt} &= \frac{du_x}{dt}\mathbf{E}_x + \frac{du_y}{dt}\mathbf{E}_y + \frac{du_z}{dt}\mathbf{E}_z + u_x\frac{d\mathbf{E}_x}{dt} + u_y\frac{d\mathbf{E}_y}{dt} + u_z\frac{d\mathbf{E}_z}{dt} && (4.14)\\
&= \frac{du_x}{dt}\mathbf{E}_x + \frac{du_y}{dt}\mathbf{E}_y + \frac{du_z}{dt}\mathbf{E}_z && (4.15)
\end{aligned}$$

$$\frac{d}{dt}(\mathbf{u}\cdot\mathbf{v}) = \frac{d\mathbf{u}}{dt}\cdot\mathbf{v} + \mathbf{u}\cdot\frac{d\mathbf{v}}{dt} \tag{4.16}$$

$$\frac{d}{dt}(\mathbf{u}\times\mathbf{v}) = \frac{d\mathbf{u}}{dt}\times\mathbf{v} + \mathbf{u}\times\frac{d\mathbf{v}}{dt} \tag{4.17}$$

4.8 Sine Law and Cosine Law



Referring to the above triangle with interior angles A, B, and C and with side lengths a, b, and c, the sine law and cosine law are respectively

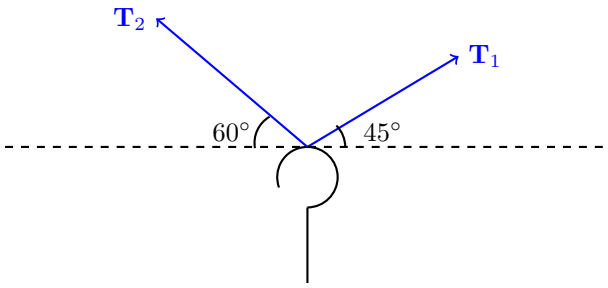
$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}, \tag{4.18}$$

$$c^2 = a^2 + b^2 - 2ab\cos(C). \tag{4.19}$$

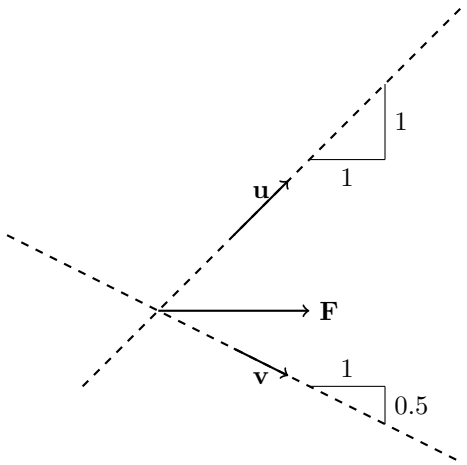
4.9 Exercises

The following problems are from Set 01 – Vector Calculus.

1. Two chords are pulling on a hook with tensions $T_1 = 100$ N and $T_2 = 50$ N as shown. Using the parallelogram rule, determine the magnitude of the resultant force and the angle it makes with the horizontal. Do not introduce a Cartesian basis.



2. Resolve the horizontal force $\mathbf{F}$ into components along the $\mathbf{u}$ and $\mathbf{v}$ basis and determine the magnitudes of the components, noting that $\mathbf{u} \cdot \mathbf{u} = 1$ and $\mathbf{v} \cdot \mathbf{v} = 1$



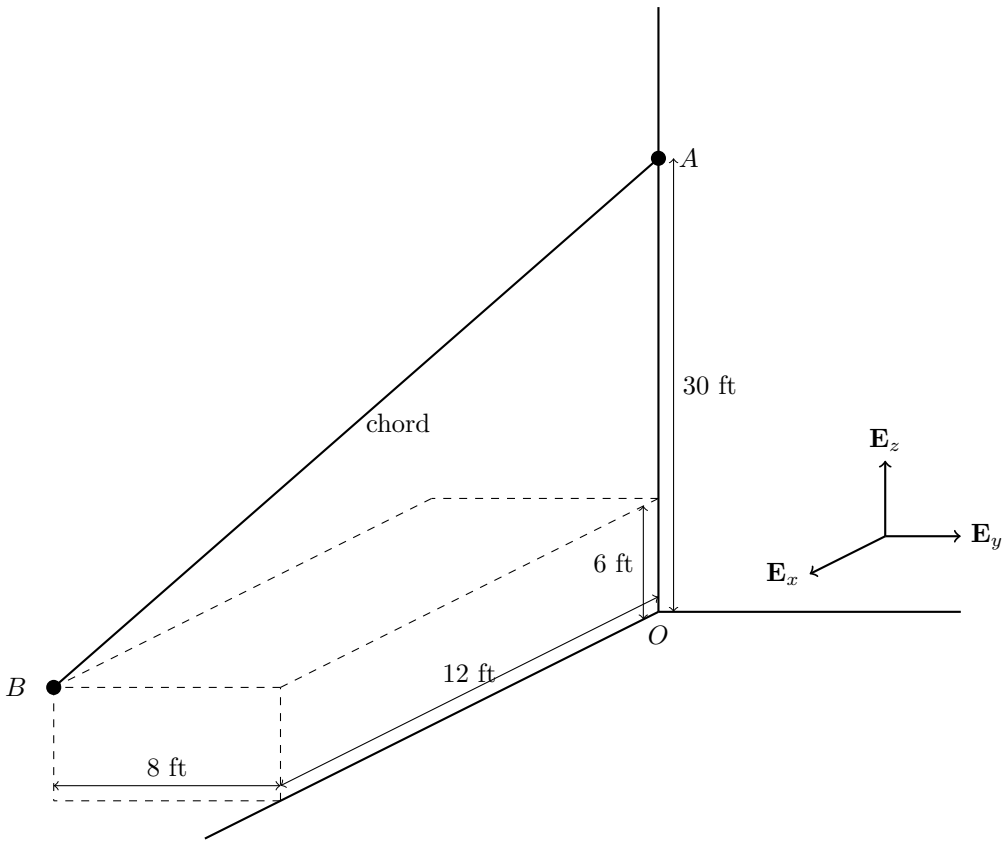
3. In Exercise 1, take unit vectors **I** and **J** to point horizontally leftwards and vertically upwards. The unit vector **K** completes the right-handed triad. Repeat Exercise 1 by resolving all forces in the Cartesian $\{\mathbf{I}, \mathbf{J}, \mathbf{K}\}$ basis.

4. A particle moves in space with velocity

$$\mathbf{v}(t) = 2t^2\mathbf{E}_1 - \mathbf{E}_2 + (-5t + 3)\mathbf{E}_3 \quad \text{m/s}. \tag{4.20}$$

(a) Calculate $\mathbf{v}$ at $t = 1$ s. (b) Calculate $\mathbf{a}(t) \equiv \dot{\mathbf{v}}(t)$. (c) Given $\mathbf{r}(0) = 2\mathbf{E}_3$, find $\mathbf{r}(t)$.

5. A child at B pulls a chord attached to the wall at A with a force of 70 N.



(a) Express the position vector $\mathbf{r}_A$ in Cartesian coordinates.
(b) Express the position vector $\mathbf{r}_B$ in Cartesian coordinates.
(c) Draw $\mathbf{r}_{B/A} = \mathbf{r}_B - \mathbf{r}_A$. Does it point from B to A or from A to B?
(d) Draw $\mathbf{r}_{A/B}$.
(e) What is the tension force $\mathbf{T}$ in the cable?

6. Referring to Exercises 1 and 3:

(a) Calculate $\mathbf{T}_1 \times \mathbf{T}_2$ without using a basis.
(b) Express $\mathbf{T}_1$ and $\mathbf{T}_2$ in the Cartesian basis from Exercise 3 and verify part (a).
(c) What is $\mathbf{T}_2 \times \mathbf{T}_1$?

5 Fundamental Concepts of Solid Mechanics

This section is adapted from chapter 1 of Introduction to Solid Mechanics - An Integrated Approach by Lubliner and Papadopoulos (Lubliner and Papadopoulos 2017).

- **Mechanics**
  - Mechanics, as a scientific discipline, is the study of the behavior of bodies subject to forces and displacements (and, to some extent, heating and cooling).
- **Solid Mechanics**
  - Solid mechanics is the branch of mechanics dealing specifically with solid bodies.
- **Body**
  - A body, for the purpose of mechanics, is a portion of matter that, at a given moment in time, occupies a certain region in space.
  - **Particle**
    - If the region occupied by the body can be idealized as being of negligible extent (and thus reducible to a point), the body is called a particle.
  - **Finite Body**
    - A body that is not reducible to a point is called a finite body.
- **Continuum**
  - A finite body occupying a connected region every portion of which contains some matter is called a continuous body or, simply, a continuum.
- **Material Points**
  - At any point in a region occupied by a continuum, the matter in its immediate (infinitesimal) neighborhood can also be thought of as constituting a particle, and we may thus speak of the particles (also called material points) of a continuum.
- **Displacement**
  - A body is said to undergo a displacement when some or all of its particles are moved to occupy different positions in space.
- **Configuration**

  - The correspondence between the particles and the positions occupied by them at a given time is known as the body's configuration.

- **Deformation**
  - The displacement of a body is said to be rigid if the distances between all pairs of particles are the same in any two configurations. Otherwise, the body is said to undergo deformation.

- **Kinematics**
  - The branch of mechanics dealing with the displacement and motion of bodies is called kinematics.

- **Force**
  - Forces are interactions between bodies that cause them to move, and more specifically to accelerate, relative to each other (unless prevented from doing so by other forces).

5.1 Galileo, Kepler, Newton

5.1.1 Gravitational force

All bodies on or near the earth are subject to the earth's gravitational force and, as shown by Galileo in his famous experiments, will fall toward the center of the earth (unless prevented from doing so by other forces) with the same acceleration; the magnitude of this acceleration is denoted by g.

Historical Perspective: Galileo Galilei (1564–1642)

Figure 5.1: Portrait by Justus Sustermans, 1636 (Wikimedia Commons, public domain).

Italian astronomer, physicist, and engineer, often called the father of observational astronomy and of the modern scientific method. His experiments on falling bodies and inclined planes were the first systematic study of kinematics, and his advocacy of a heliocentric solar system brought him into famous conflict with the Church.

It was later shown by Newton, on the basis of Kepler's laws of planetary motion, that the magnitude of the acceleration (with respect to a frame of reference based on the "fixed" stars) due to gravitational force between any two bodies (idealized as particles) is inversely proportional to the square of the distance between them, and for each body it is proportional to the quantity of matter (or mass) of the other body. The product of mass and acceleration is therefore equal and opposite for the two bodies and may be identified with the force exerted on each body by the other.

Historical Perspective: Isaac Newton (1642–1727)

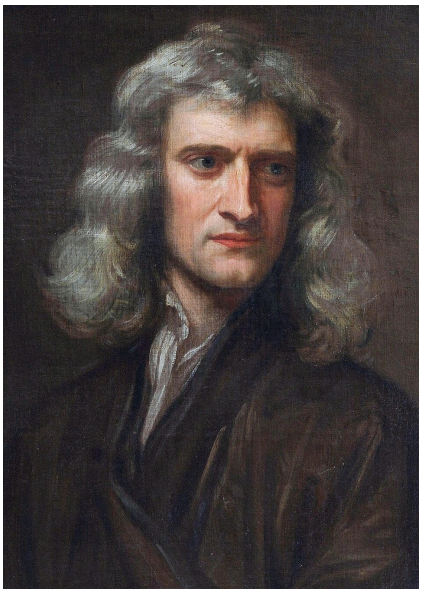


Figure 5.2: Portrait by Godfrey Kneller, 1689 (Wikimedia Commons, public domain).

English mathematician and physicist. Building on Galileo's and Kepler's work, Newton's 1687 *Philosophiæ Naturalis Principia Mathematica* established the three laws of motion and the law of universal gravitation, unifying the mechanics of falling bodies on Earth with the motion of the planets for the first time.

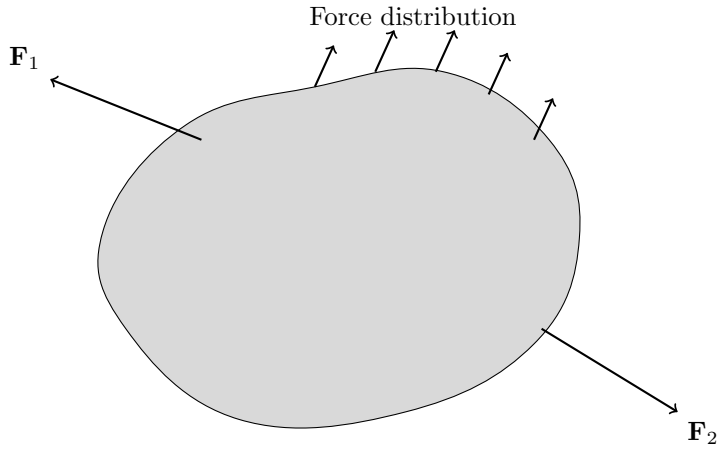
https://youtu.be/icrTzMozNgo

5.1.2 Equilibrium

If a body's acceleration is zero, then it is at rest or in a uniform motion. In this case, the forces is said to be in equilibrium. The study of equilibrium is known as statics and constitutes one of the main subjects of this book. Traditionally, statics deals only (or at least primarily) with rigid bodies, while deformable bodies are studied in courses historically called strength of materials (or, more recently, mechanics of materials). In this book (Lubliner and Papadopoulos 2017), the statics of rigid and deformable solid bodies will be studied in an integrated manner.

6 A Continuum Mechanics Perspective

Consider a body $\mathcal{B}$ in three-dimensional Euclidean space $\mathbb{E}^3$.



In general, bodies are deformable continua. We depict continua as in the above figure and refer to them as *continuum potatoes*.

Point or distributed forces or moments can be applied to a body's surface or throughout its volume. For example, a body resting on a table has contact forces on the surface touching the table, while weight acts on all points in the body.

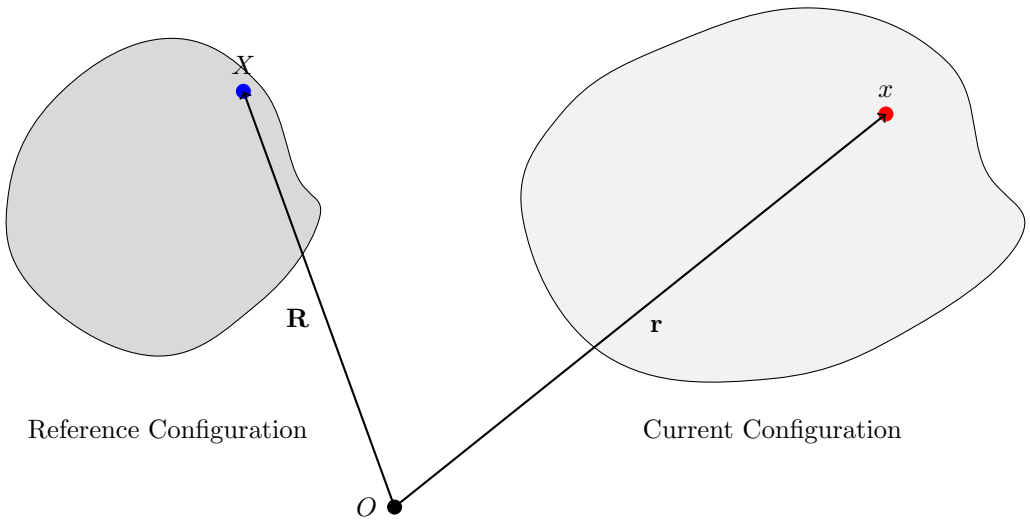
Think!

Question: How do we know if a body has moved or deformed?

Answer

...

Consider a reference configuration for this body. A reference configuration is any configuration that can be physically occupied by the body — it could be the initial configuration, but need not be.



We denote the position vector from a chosen fixed origin to a material point in the reference configuration as $\mathbf{R}$. The position vector to the same material point in the current configuration is $\mathbf{r}$.

The motion of the body is described by

$$\mathbf{r} = \chi(\mathbf{R}, t). \tag{6.1}$$

Think!

Question: What makes a body rigid?

Answer

In this class, we consider only rigid bodies. If we choose two points on a rigid body, the distance between them is always constant. Also, angles between material lines on a rigid body are constant.

Think!

Question: Under what conditions do we consider a rigid body to be a particle?

Answer

In the beginning of the class, we consider particles. Particles are idealized rigid bodies whose rotational inertia is ignored.

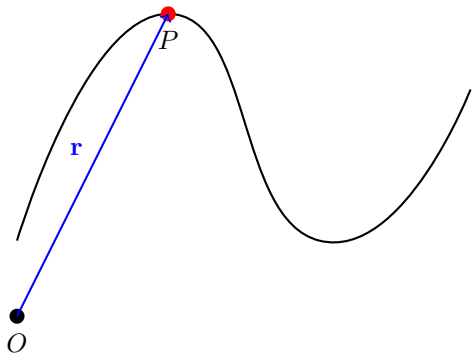
Part II

Particle Dynamics

7 Particle Kinematics

7.1 Position, Velocity, and Acceleration

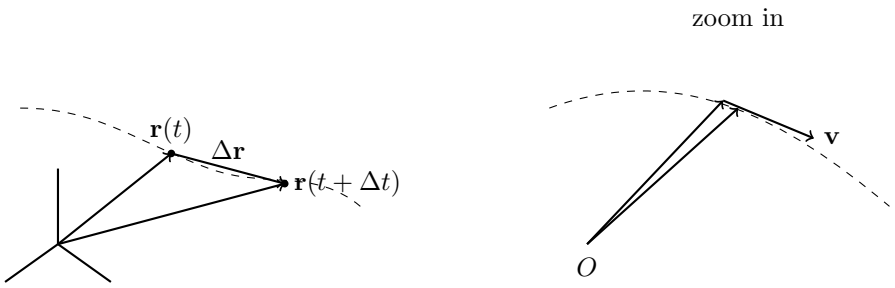
Consider a particle moving in $\mathbb{E}^3$. Its position vector relative to a fixed origin O is denoted by $\mathbf{r} \in \mathbb{E}^3$. As time t evolves, so does $\mathbf{r}(t)$.



Let Δt be a finite time elapsed:

$$\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t). \tag{7.1}$$

As $\Delta t \to 0$, Δt becomes dt and $\Delta \mathbf{r}$ becomes $d\mathbf{r}$. The 'd' means differential. We denote the time derivative of $\mathbf{r}$ by $\dot{\mathbf{r}}$ or $\frac{d\mathbf{r}}{dt}$.



The (absolute) velocity vector $\mathbf{v}$ of the particle is the time rate of change of the position vector:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}. \tag{7.2}$$

Think!

Question: What is the direction of the velocity vector?

Answer

$\mathbf{v}$ is along $d\mathbf{r}$ which is always tangent to the path.

The speed of the particle is the magnitude of the velocity vector $v = \|\mathbf{v}\|$.

The (absolute) acceleration is

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \dot{\mathbf{v}} = \ddot{\mathbf{r}} = \frac{d^2\mathbf{r}}{dt^2}. \tag{7.3}$$

7.2 Arc-Length and the Unit Tangent Vector

Define the arc-length parameter s such that

$$\frac{ds}{dt} = \|\mathbf{v}\|. \tag{7.4}$$

$\frac{ds}{dt}$ is the speed of the particle. Recall that $\mathbf{v} = \frac{d\mathbf{r}}{dt}$, so

$$\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\|. \tag{7.5}$$

So ds is the length of the differential $d\mathbf{r}$ along the curve. Integrating:

$$s(t) - s_0 = \int_{t_0}^{t} \frac{ds}{dt}(\tau) d\tau = \int_{t_0}^{t} \sqrt{\mathbf{v}(\tau) \cdot \mathbf{v}(\tau)}\, d\tau. \tag{7.6}$$

Notes:

- $s(t) - s_0$ is the distance traveled by the particle along its path $\mathcal{C}$ during the time interval $[t_0, t]$.
- $s(t_0) = s_0$ where t_0 and s_0 are initial conditions.
- The dummy variable τ is used (as opposed to t) when evaluating this integral because we are integrating the speed as it varies between t_0 and t. The variable of integration is different than the upper limit of integration.

Think!

Question: Under what conditions is $s(t)$ invertible so that $t(s)$ is well defined?

Answer

If the function $s(t)$ is one-to-one, it can be inverted, and time can be written in terms of arc-length $t = t(s)$.

Think!

Question: Why is

$$s(t) - s_0 \neq \|\mathbf{r}(t) - \mathbf{r}_0\| \,? \tag{7.7}$$

Under what conditions does the equality hold?

Answer

The equality holds under rectilinear motion or over an infinitesimal time interval.

We define the unit tangent vector $\mathbf{e}_t$:

$$\frac{\mathbf{v}}{\|\mathbf{v}\|} = \mathbf{e}_t, \qquad \mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt} = \dot{s}\mathbf{e}_t. \tag{7.8}$$

Think!

Question: Verify that $\mathbf{e}_t$ is a unit vector.

Answer

...

Note the relationships

$$\mathbf{v} = \frac{d\mathbf{r}(t)}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt},$$
$$\mathbf{a} = \frac{d\mathbf{v}(t)}{dt} = \frac{d^2\mathbf{r}}{ds^2}\left(\frac{ds}{dt}\right)^2 + \frac{d\mathbf{r}}{ds}\frac{d^2s}{dt^2}.$$

Notice: the acceleration vector is due to changes in both the magnitude and direction of the velocity vector.

Think!

Question: Is the unit vector $\mathbf{e}_t$ constant in general? Under what conditions is it constant?

Answer

No. It is constant only during rectilinear motion in a fixed direction.

7.2.1 Example

Example

Question: Consider a particle moving in space with position vector

$$\mathbf{r}(t) = R_0 \left(\cos(\omega t)\mathbf{E}_x + \sin(\omega t)\mathbf{E}_y\right). \tag{7.9}$$

Calculate $\mathbf{v}$, $\mathbf{a}$, $\mathbf{v} \times \mathbf{a}$, $\|\mathbf{r}\|$. Describe the motion of the particle. Show that the motion satisfies:

$$\ddot{x} + \omega^2 x = 0, \qquad \ddot{y} + \omega^2 y = 0. \tag{7.10}$$

Answer

We calculate the velocity and acceleration vectors of the particle as follows:

$$\begin{aligned}
\mathbf{v} &= \frac{d\mathbf{r}}{dt} = R_0\omega\left(-\sin(\omega t)\,\mathbf{E}_x + \cos(\omega t)\,\mathbf{E}_y\right), \\
\mathbf{a} &= \frac{d\mathbf{v}}{dt} = -R_0\omega^2\left(\cos(\omega t)\,\mathbf{E}_x + \sin(\omega t)\,\mathbf{E}_y\right).
\end{aligned}$$

As you take these derivatives, remember that:

- $\dot{\mathbf{E}}_x = \mathbf{0}$ and $\dot{\mathbf{E}}_y = \mathbf{0}$ since the basis $\{\mathbf{E}_x, \mathbf{E}_y\}$ is fixed.
- The chain rule: $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$.

Then,

$$\begin{aligned}
\mathbf{r} \times \mathbf{a} &= R_0\left(\cos(\omega t)\mathbf{E}_x + \sin(\omega t)\mathbf{E}_y\right) \times \left(-R_0\omega^2\left(\cos(\omega t)\mathbf{E}_x + \sin(\omega t)\mathbf{E}_y\right)\right) \\
&= -R_0^2\omega^2\Big(\cos^2(\omega t)\underbrace{\mathbf{E}_x \times \mathbf{E}_x}_{\mathbf{0}} + \cos(\omega t)\sin(\omega t)\left(\underbrace{\mathbf{E}_x \times \mathbf{E}_y}_{\mathbf{E}_z} + \underbrace{\mathbf{E}_y \times \mathbf{E}_x}_{-\mathbf{E}_z}\right) \\
&\qquad + \sin^2(\omega t)\underbrace{\mathbf{E}_y \times \mathbf{E}_y}_{\mathbf{0}}\Big) \\
&= \mathbf{0}.
\end{aligned}$$

Alternatively, notice that

$$\mathbf{a} = -\omega^2 \mathbf{r},$$

so

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times (-\omega^2 \mathbf{r}) = -\omega^2 (\mathbf{r} \times \mathbf{r}) = \mathbf{0}.$$

For practice, let us also calculate $\mathbf{r} \cdot \mathbf{a}$:

$$\begin{aligned}
\mathbf{r} \cdot \mathbf{a} &= R_0 \left(\cos(\omega t)\mathbf{E}_x + \sin(\omega t)\mathbf{E}_y\right) \cdot \left(-R_0\omega^2 \left(\cos(\omega t)\mathbf{E}_x + \sin(\omega t)\mathbf{E}_y\right)\right) \\
&= -R_0^2\omega^2 \Bigg(\cos^2(\omega t) \underbrace{\mathbf{E}_x \cdot \mathbf{E}_x}_{1} + \cos(\omega t)\sin(\omega t) \left(\underbrace{\mathbf{E}_x \cdot \mathbf{E}_y}_{0} + \underbrace{\mathbf{E}_y \cdot \mathbf{E}_x}_{0} \right) + \sin^2(\omega t) \underbrace{\mathbf{E}_y \cdot \mathbf{E}_y}_{1} \Bigg) \\
&= -R_0^2\omega^2 \left(\cos^2(\omega t) + \sin^2(\omega t)\right) \\
&= -R_0^2\omega^2.
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\mathbf{r} \cdot \mathbf{a} &= \mathbf{r} \cdot (-\omega^2 \mathbf{r}) \\
&= -\omega^2 (\mathbf{r} \cdot \mathbf{r}) \\
&= -\omega^2 \|\mathbf{r}\|^2 \\
&= -\omega^2 R_0^2.
\end{aligned}$$

Noting that $\mathbf{a} = -\omega^2 \mathbf{r}$, we can show that the given $\mathbf{r}$ satisfies the given differential equations:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times (-\omega^2 \mathbf{r}) = -\omega^2 \mathbf{r} \times \mathbf{r} = \mathbf{0}.$$

For $\mathbf{r} \cdot \mathbf{a}$:

$$\mathbf{r} \cdot \mathbf{a} = -R_0^2\omega^2 \left(\cos^2(\omega t) + \sin^2(\omega t)\right) = -R_0^2\omega^2.$$

7.3 Cartesian Coordinates

$\{\mathbf{E}_x, \mathbf{E}_y, \mathbf{E}_z\}$ is a basis for $\mathbb{E}^3$. Since this basis is fixed, we have:

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z, \tag{7.11}$$

$$\mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y + \dot{z}\mathbf{E}_z, \tag{7.12}$$

$$\mathbf{a} = \ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y + \ddot{z}\mathbf{E}_z. \tag{7.13}$$

Later we will also use the cylindrical polar basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{E}_z\}$ and the Serret-Frenet triad $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$.

7.4 Rectilinear Motion

- *rectus* = straight, *linea* = line

Due to constraints or initial conditions, the body moves on a straight line, e.g. a car on a straight road or a ball tossed vertically.

We take $\mathbf{E}_x$ parallel to the line and $\mathbf{c}$ to be a constant vector. Then:

$$\mathbf{r}(t) = x(t)\mathbf{E}_x + \mathbf{c}, \tag{7.14}$$

$$\mathbf{v}(t) = \dot{x}\mathbf{E}_x = v(t)\mathbf{E}_x, \tag{7.15}$$

$$\mathbf{a}(t) = \ddot{x}\mathbf{E}_x = a(t)\mathbf{E}_x. \tag{7.16}$$

$\frac{ds}{dt} = \left|\frac{dx}{dt}\right|$, so unless $\dot{x} > 0$ or $\dot{x} < 0$ throughout, x and s cannot be easily interchanged as this convention flips $\mathbf{e}_t$ based on the motion.

7.4.1 Identities for Rectilinear Motion

Recall the definitions $\mathbf{v} = \frac{d\mathbf{r}}{dt}$ and $\mathbf{a} = \frac{d\mathbf{v}}{dt}$. For the case of rectilinear motion along $\mathbf{E}_x$, projecting these equations along $\mathbf{E}_x$ yields respectively $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$. Applying the chain rule to the latter equation, we obtain

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}. \tag{7.17}$$

where we assumed that $v(t)$ can be written in terms of x.

We will now examine three useful identities for rectilinear motion that relate s, v, and a.

$$ds = v\,dt, \qquad dv = a\,dt, \qquad v\,dv = a\,ds. \tag{7.18}$$

7.4.2 Given Acceleration as a Function of Time

From $dv = a\,dt$, we can integrate this equation to get the signed speed as a function of time.

$$v(t) - v(t_0) = \int_{t_0}^{t} a(\tilde{t})\,d\tilde{t}. \tag{7.19}$$

From $ds = v\,dt$:

$$s(t_1) - s(t_0) = \int_{s(t_0)}^{s(t_1)} d(\tilde{s}) = \int_{t_0}^{t_1} v(\tilde{t})\,d\tilde{t}. \tag{7.20}$$

Note the use of the dummy variables $\tilde{s}$ and $\tilde{t}$ in the above integrals.

7.4.3 Given Acceleration as a Function of Speed

Given $a = a(v)$, we can use again

$$dv = adt \tag{7.21}$$

to get

$$\begin{aligned} dt &= \frac{dv}{a(v)} \\ t(v) - t(v_0) &= \int_{v_0}^{v} \frac{d\tilde{v}}{a(\tilde{v})} \end{aligned} \tag{7.22}$$

Also, using the chain rule

$$a = \frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds} \tag{7.23}$$

If $a = a(v)$, then

$$\begin{aligned} ds &= \frac{vdv}{a} \\ s(t) - s_0 &= \int_{v_0}^{v} \frac{\tilde{v}d\tilde{v}}{a(\tilde{v})}. \end{aligned} \tag{7.24}$$

7.4.4 Given Acceleration as a Function of Displacement

Starting with $a = v\,dv/ds$, if $a = a(s)$, we can rearrange this equation to get

$$\int_{v(s_0)}^{v(s)} v\,dv = \int_{s_0}^{s} a(\tilde{s})\,d\tilde{s}.$$

Evaluating the left-hand side gives

$$\frac{1}{2}\left(v^2(s) - v^2(s_0)\right) = \int_{s_0}^{s} a(\tilde{s})\,d\tilde{s}.$$

Equivalently,

$$v^2(s) = v^2(s_0) + 2\int_{s_0}^{s} a(\tilde{s})\,d\tilde{s}.$$

7.4.5 Constant Acceleration

7.4.5.1 Example: Gravity

Throw a ball upward and take your hand to be the origin.

Here,

$$\mathbf{a} = -g\,\mathbf{e}_t, \qquad g = 32.2\ \frac{\text{ft/s}}{\text{s}}.$$

Note that $\mathbf{e}_t$ always points in the direction of increasing s (upward). Therefore,

$$a = -g.$$

Using

$$dv = a\,dt,$$

we obtain

$$\int_{v(t_0)}^{v(t)} dv = -g \int_{t_0}^{t} d\tilde{t},$$
$$v(t) - v(t_0) = -g(t - t_0).$$

Letting $v(t_0) = v_0$ and choosing the time origin so that $t_0 = 0$, we find

$$v(t) = v_0 - gt.$$

To determine the position as a function of time, use

$$ds = v\, dt.$$

Integrating,

$$\int_{s_0}^{s(t)} ds = \int_0^t \left(v_0 - g\tilde{t}\right) d\tilde{t},$$
$$s(t) - s_0 = v_0 t - \frac{g}{2} t^2.$$

Hence,

$$s(t) = s_0 + v_0 t - \frac{g}{2} t^2.$$

In this example, $s_0 = 0$.

7.4.5.2 Example: Vehicle Braking

For constant acceleration, the relation

$$v\,dv = a\,ds$$

is particularly useful.

Integrating,

$$\int_{v(s_0)}^{v(s)} v\,dv = a\int_{s_0}^{s} d\tilde{s},$$
$$\frac{1}{2}\left(v^2(s) - v^2(s_0)\right) = a(s - s_0).$$

Therefore,

$$v^2(s) - v^2(s_0) = 2a(s - s_0),$$

or, in increment notation,

$$\Delta(v^2) = 2a\,\Delta s.$$

This equation is commonly used to determine braking distance when the deceleration is approximately constant.

7.5 Summary

Kinematics in Cartesian coordinates:

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z, \tag{7.25}$$
$$\mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y + \dot{z}\mathbf{E}_z, \tag{7.26}$$
$$\mathbf{a} = \ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y + \ddot{z}\mathbf{E}_z. \tag{7.27}$$

Rectilinear motion (along $\mathbf{E}_x$): $\mathbf{r} = x\mathbf{E}_x$, $\mathbf{v} = \dot{x}\mathbf{E}_x$, $\mathbf{a} = \ddot{x}\mathbf{E}_x$, and $a = v\,dv/dx$.

7.6 Exercises

The following problems are from Set 03 – Rectilinear Motion.

1. [MKB 2/24] Solve $a(x)$ as a piecewise function. *(ans. $v = 8$ ft/sec)*

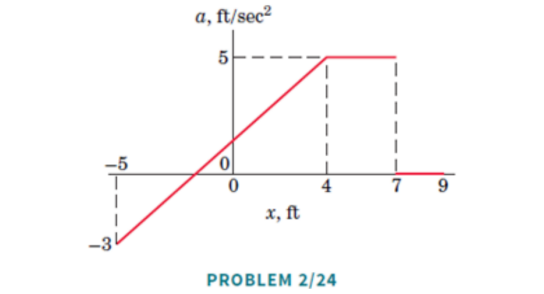


Figure 7.1: MKB 2/24.

2. [MKB 2/28] Take $\mathbf{E}_x$ along the horizontal; origin at the location of the plane when the parachute deploys ($v = 200$ mi/hr). Convert mi to ft and hr to sec. *(ans. $s = 5810$ ft)*

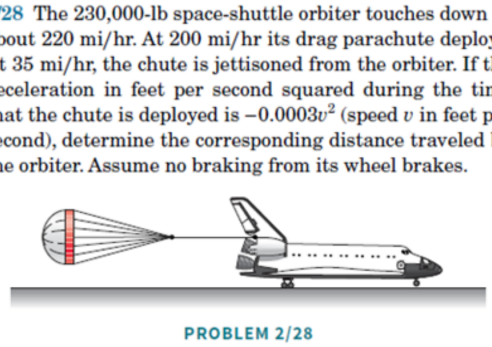


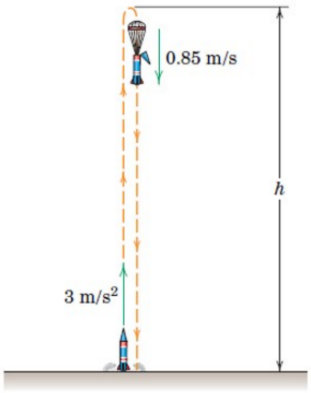
Figure 7.2: MKB 2/28.

3. [MKB 2/25] Take $\mathbf{E}_y$ vertically upwards; origin at the initial position of the rocket. The acceleration is

$$a = \begin{cases} 3 \text{ m/s}^2 & 0 \le t < 8\,\text{s} \\ -9.81 \text{ m/s}^2 & 8\,\text{s} \le t < t_{\text{top}} \\ 0 & t_{\text{top}} < t \le t_{\text{end}} \end{cases} \tag{7.28}$$

(ans. $h = 125.4$ m, total time $= 157.9$ s)

2/25 A model rocket is launched from rest with a constant upward acceleration of 3 m/s^2 under the action of a small thruster. The thruster shuts off after 8 seconds, and the rocket continues upward until it reaches its apex. At apex, a small chute opens which ensures that the rocket falls at a constant speed of 0.85 m/s until it impacts the ground. Determine the maximum height h attained by the rocket and the total flight time. Neglect aerodynamic drag during ascent, and assume that the mass of the rocket and the acceleration of gravity are both constant.

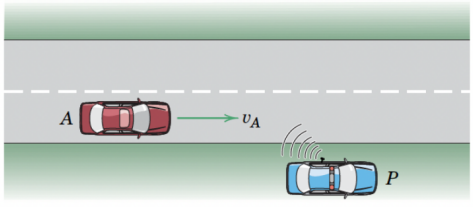


PROBLEM 2/25

Figure 7.3: MKB 2/25.

4. [MKB 2/22]

2/22 Car A is traveling at a constant speed $v_A = 130$ km/h at a location where the speed limit is 100 km/h. The police officer in car P observes this speed via radar. At the moment when A passes P, the initially stationary police car begins to accelerate at the constant rate of 6 m/s^2 until a speed of 160 km/h is achieved, and that speed is then maintained. Determine the distance required for the police officer to overtake car A. Neglect any nonrectilinear motion of P.



PROBLEM 2/22

Figure 7.4: MKB 2/22.

8 Particle Kinetics

8.1 The Balance of Linear Momentum

A particle is endowed with an inertial measure m, called its mass.

The linear momentum of a particle is defined to be

$$\mathbf{G} = m\mathbf{v}. \tag{8.1}$$

Taking the time derivative:

$$\dot{\mathbf{G}} = \dot{m}\mathbf{v} + m\dot{\mathbf{v}}. \tag{8.2}$$

Think!

Question: Can you think of systems whose mass is not conserved?

Answer

In this class, we consider cases where mass is conserved, i.e. $\dot{m} = 0$. This is not the case, for example, for a launching rocket that is losing mass. The $\dot{m}\mathbf{v}$ term is useful in control volume analysis.

Then, in our case of interest:

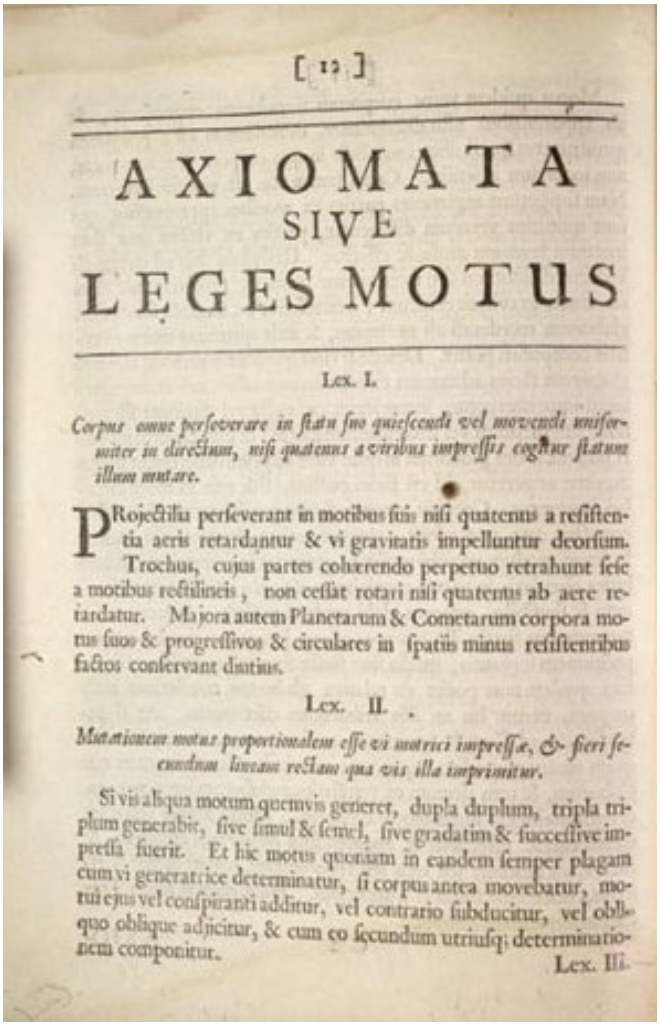
$$\dot{\mathbf{G}} = m\dot{\mathbf{v}} = m\mathbf{a}. \tag{8.3}$$

We postulate the following balance law/axiom/postulate to be true:

$$\mathbf{F} = \dot{\mathbf{G}}, \tag{8.4}$$

where $\mathbf{F}$ is the resultant external force acting on the particle. This is known as the **Balance of Linear Momentum**, **Newton's Second Law**, or **Euler's First Law**.

Historical Perspective: Newton's *Principia* (1687)



[12]

AXIOMATA
SIVE
LEGES MOTUS

Lex. I.

Corpus omne perseverare in statu suo quiescendi vel movendi uniformiter in directum, nisi quatenus a viribus impressis cogitur statum illum mutare.

Projectilia perseverant in motibus suis nisi quatenus a resistentia aeris retardantur & vi gravitatis impelluntur deorsum. Trochus, cujus partes cohærendo perpetuo retrahunt sese a motibus rectilineis, non cessat rotari nisi quatenus ab aere retardatur. Majora autem Planetarum & Cometarum corpora motus suos & progressivos & circulares in spatiis minus resistentibus factos conservant diutius.

Lex. II.

Mutationem motus proportionalem esse vi motrici impressæ, & fieri secundum lineam rectam qua vis illa imprimitur.

Si vis aliqua motum quemvis generet, dupla duplum, tripla triplum generabit, sive simul & semel, sive gradatim & successive impressa fuerit. Et hic motus quoniam in eandem semper plagam cum vi generatrice determinatur, si corpus antea movebatur, motui ejus vel conspiranti additur, vel contrario subducitur, vel obliquo obliquè adjicitur, & cum eo secundum utriusq; determinationem componitur.

Lex. III.

Figure 8.1: *Axiomata sive Leges Motus* ("Axioms, or Laws of Motion"), Book I of the first edition of Newton's *Philosophiæ Naturalis Principia Mathematica*, 1687 (Wikimedia Commons, public domain).

This is the page on which Newton first stated, in Latin, the three laws of motion used throughout this chapter – including the balance of linear momentum above, and Newton's Third Law, stated later in this chapter.

Important: **a** here is the absolute acceleration vector of the particle.

Think!

Question: Under what conditions is the linear momentum $\mathbf{G}$ conserved?

Answer

An unbalanced net force $\mathbf{F} \neq \mathbf{0}$ results in a change of $\mathbf{G}$. If $\mathbf{F} = \mathbf{0}$, then $\mathbf{G} = \mathbf{c}$ where $\mathbf{c}$ is a constant vector. This is Newton's First Law.

If $\mathbf{c} = \mathbf{0}$, we recover statics.

8.1.1 Representations and Projections

Vectors are represented on a basis. For example:

$$\mathbf{F} = F_x\mathbf{E}_x + F_y\mathbf{E}_y + F_z\mathbf{E}_z, \tag{8.5}$$

$$\mathbf{a} = a_x\mathbf{E}_x + a_y\mathbf{E}_y + a_z\mathbf{E}_z. \tag{8.6}$$

Think!

Question: How many independent scalar equations can we obtain from $\mathbf{F} = m\mathbf{a}$? How do we obtain them?

Answer

We get useful scalar equations by projecting $\mathbf{F} = \dot{\mathbf{G}}$ along unit directions:

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_x \implies F_x = ma_x. \tag{8.7}$$

8.2 Newton's Third Law

When dealing with the forces of interaction between particles, a particle and a rigid body, or rigid bodies, we also invoke Newton's third law: "For every action there is an equal and opposite reaction." For example, the force exerted by a surface on a particle is equal in magnitude and opposite in direction to the force exerted by the particle on the surface.

8.3 Four Steps to Problem Solving Using the BoLM

1. Specify the system you are considering. Pick a frame (choose an origin and a coordinate system), and express $\mathbf{r}$, $\mathbf{v}$ and $\mathbf{a}$ using that coordinate system.

2. Draw a Free Body Diagram (FBD), i.e. model of the forces.
3. Write the balance of linear momentum for the system: $\mathbf{F} = \dot{\mathbf{G}}$.
4. Do the analysis: project $\mathbf{F} = \dot{\mathbf{G}}$ to get useful equations.

8.3.1 Example: Projectile Motion

Projectile motion is a curvilinear motion in the plane $\mathbb{E}^2$.

1. Choose O at the launch point, $\mathbf{E}_x$ along the ground, $\mathbf{E}_y$ against gravity.

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y, \quad \mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y, \quad \mathbf{a} = \ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y. \tag{8.8}$$

2. Force model:

Figure 8.2: FBD

$$\mathbf{W} = mg\left(-\mathbf{E}_y\right). \tag{8.9}$$

3. Balance of linear momentum:

$$mg(-\mathbf{E}_y) = m\left(\ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y\right). \tag{8.10}$$

Projecting:

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_x \implies 0 = m\ddot{x} \implies \ddot{x} = 0, \tag{8.11}$$

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_y \implies -mg = m\ddot{y} \implies \ddot{y} = -g. \tag{8.12}$$

4. Integrating:

$$\dot{x}(t) = \dot{x}_0, \quad x(t) = \dot{x}_0 t, \tag{8.13}$$

$$\dot{y}(t) = -gt + \dot{y}_0, \quad y(t) = -\frac{g}{2}t^2 + \dot{y}_0 t + y_0. \tag{8.14}$$

Remark: You can also project $\mathbf{F} = m\mathbf{a}$ in the $\mathbf{E}_z$ direction, but since there are no forces in that direction and the initial velocity in that direction is 0, you conclude that the motion is planar.

8.3.2 Example: Projectile Motion with Viscous Drag

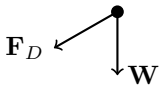
(Primer Section 1.5.3)

Going through the four steps, we get

1. Again,

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y, \quad \mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y, \quad \mathbf{a} = \ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y. \tag{8.15}$$

2. Force model — weight and Stokes drag:



$$\mathbf{W} = mg(-\mathbf{E}_y), \tag{8.16}$$

$$\mathbf{F}_D = \underbrace{c_s \|\mathbf{v}\|}_{\text{magnitude}} \underbrace{\left(-\frac{\mathbf{v}}{\|\mathbf{v}\|}\right)}_{\text{unit direction}} = -c_s\mathbf{v}. \tag{8.17}$$

The Stokes drag (1851) applies at low speeds in viscous fluids.

3. Balance of linear momentum:

$$\mathbf{F} = \dot{\mathbf{G}}, \tag{8.18}$$

$$-mg\mathbf{E}_y - c_s\left(\dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y\right) = m\left(\ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y\right). \tag{8.19}$$

Projecting:

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_x \implies -c_s\dot{x} = m\ddot{x} \tag{8.20}$$

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_y \implies -mg - c_s\dot{y} = m\ddot{y} \tag{8.21}$$

Given initial conditions $x_0, \dot{x}_0, y_0, \dot{y}_0$, these are an initial value problem (IVP).

4. Solving the x-equation (let $v_x = \dot{x}$):

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_x \implies -c_s\dot{x} = m\ddot{x}, \tag{8.22}$$

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_y \implies -mg - c_s\dot{y} = m\ddot{y}. \tag{8.23}$$

These equations give ODEs for $x(t)$ and $y(t)$. Given IC's $x_0, \dot{x}_0, y_0, \dot{y}_0$, we have an initial value problem (IVP).

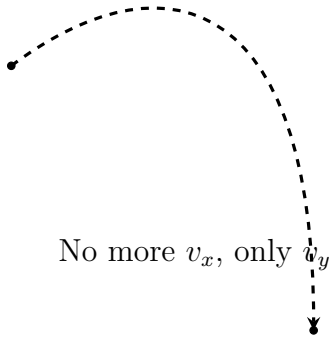
The x-equation is a first order ODE for $v_x = \dot{x}$:

$$\begin{aligned} -c_s v_x &= m\dot{v}_x \\ -\frac{c_s}{m} dt &= \frac{dv_x}{v_x} \\ -\frac{c_s}{m}\int_0^t d\tilde{t} &= \int_{v_{x_0}}^{v_c(t)} \frac{d\tilde{v}_x}{\tilde{v}_x} \\ -\frac{c_s}{m} t &= \ln\left(\frac{v_x}{v_{x_0}}\right) \\ v_x &= v_{x_0} e^{-\frac{c_s t}{m}} \end{aligned} \tag{8.24}$$

Note that

$$\lim_{t\to\infty} v_x = 0. \tag{8.25}$$

Thus, the terminal velocity of the particle is only along $\mathbf{E}_y$.



Solving the y-equation:

$$dt = \frac{m dv_y}{-mg - c_s v_y} \tag{8.26}$$

$$\int_0^t d\tilde{t} = m\int_{v_{y_0}}^{v_y(t)} \frac{d\tilde{v}_y}{-mg - c_s \tilde{v}_y} \tag{8.27}$$

$$v_y = -\frac{mg}{c_s} + \frac{1}{c_s}\left(mg + c_s v_{y_0}\right) e^{-c_s t/m} \tag{8.28}$$

As $t \to \infty$, $v_y \to -\frac{mg}{c_s}$. Thus, the terminal velocity

$$\mathbf{v}_{\text{term}} = \frac{mg}{c_s}\left(-\mathbf{E}_y\right). \tag{8.29}$$

$$v_y = -\frac{mg}{c_s} + \frac{1}{c_s}\left(mg + c_s v_{y_0}\right) e^{-c_s t/m}. \tag{8.30}$$

8.3.3 Example: Projectile Motion with Bluff Body Pressure Drag

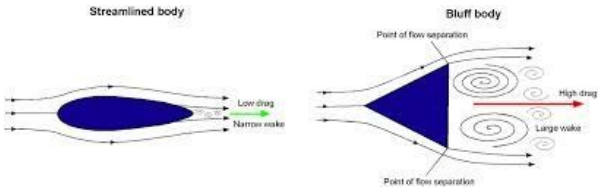


Figure 8.3: Bluff body pressure drag

$$\mathbf{F}_D = \frac{1}{2} m C_D v^2 \left(-\frac{\mathbf{v}}{\|\mathbf{v}\|} \right), \tag{8.31}$$

where C_D is the drag coefficient, ρ is the density of the fluid, and A is the projected area in the direction of motion.

The equations of motion in this case (see primer section 1.6) need to be solved numerically. One way to do this is described in the video below.

https://youtu.be/RMkMK32vqyM

8.4 Summary

Balance of linear momentum (BoLM / Newton's 2nd law / Euler's 1st law):

$$\mathbf{F} = \dot{\mathbf{G}} = m\mathbf{a}. \tag{8.32}$$

The four steps:

1. Choose an origin; draw basis vectors; write and differentiate the position vector $\mathbf{r}$.
2. Draw the free body diagram; write expressions for all forces.
3. Write the vector BoLM equation $\sum \mathbf{F} = m\mathbf{a}$.
4. Project along chosen directions and analyse to answer the question.

8.5 Exercises

The following problems are from Set 04 – Balance of Linear Momentum.

1. [MKB 2/75] Take $\mathbf{E}_x$ along the incline and $\mathbf{E}_y$ perpendicular to it (upwards); origin at A. Follow the 4 steps. *(ans. $\theta = (90^\circ + \alpha)/2$)*

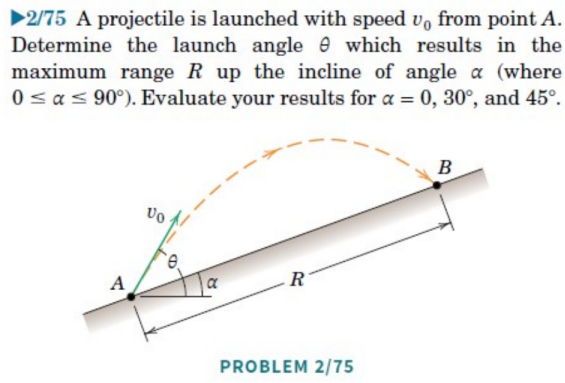


Figure 8.4: MKB 2/75.

2. [MKB 3/004] Consider the whole truck system; follow the 4 steps to find the truck's acceleration, then isolate a trailer to find the drawbar tension. *(ans.* $T = 13.33$ *kN,* $a = 0.667$ m/s^2 *)*

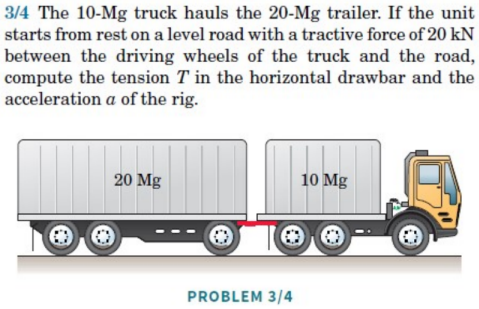
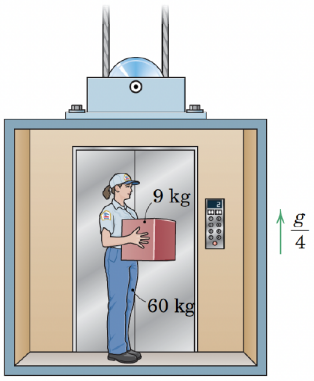


Figure 8.5: MKB 3/004.

https://youtu.be/LAl3ZUiUpus

3. [MKB 3/005] For each part, determine your system and follow the 4 steps. *(ans.* $R = 846$ *N,* $L = 110.4$ *N)*

3/5 A 60-kg woman holds a 9-kg package as she stands within an elevator which briefly accelerates upward at a rate of $g/4$. Determine the force R which the elevator floor exerts on her feet and the lifting force L which she exerts on the package during the acceleration interval. If the elevator support cables suddenly and completely fail, what values would R and L acquire?



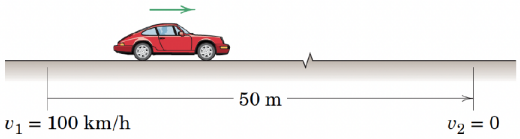
PROBLEM 3/5

/006] Follow the four steps for solving this problem also. (ans. $F = 2$

Figure 8.6: MKB 3/005.

4. [MKB 3/006] Follow the four steps. *(ans.* $F = 2890$ *N)*

3/6 During a brake test, the rear-engine car is stopped from an initial speed of 100 km/h in a distance of 50 m. If it is known that all four wheels contribute equally to the braking force, determine the braking force F at each wheel. Assume a constant deceleration for the 1500-kg car.



PROBLEM 3/6

Figure 8.7: MKB 3/006.

5. [OOR Exercise 1.3]

5. [OOR Exercise 1.3]

Exercise 1.3: Consider a particle of mass m that lies at rest on a horizontal surface. A vertical gravitational force $-mg\mathbf{E}_y$ acts on the particle. Draw a free-body diagram of the particle. If one follows the four steps, then why is it a mistake to write $mg\mathbf{E}_y$ for the normal force instead of $N\mathbf{E}_y$, where N is unknown, in the free-body diagram?

Figure 8.8: O'Reilly Primer, Exercise 1.3.

6. [OOR Exercise 1.5]

6. [OOR Exercise 1.5]

Exercise 1.5: The motion of a particle is such that its position vector $\mathbf{r}(t) = 3t\mathbf{E}_x + 4t\mathbf{E}_y + 10\mathbf{E}_z$ (meters). Show that the path of the particle is a straight line and that the particle moves along this line at a constant speed of 5 meters per second. Using this information, show that the arc-length parameter s is given by $s(t) = 5(t - t_0) + s_0$. Finally, show that the particle moves 50 meters along its path every 10 seconds.

Figure 8.9: O'Reilly Primer, Exercise 1.5.

7. [OOR Exercise 1.8]

7. [OOR Exercise 1.8]

Exercise 1.8: A projectile is launched at time $t_0 = 0$ seconds from a location $\mathbf{r}(t_0) = \mathbf{0}$. The initial velocity of the projectile is $\mathbf{v}(t_0) = v_0\cos(\alpha)\mathbf{E}_x + v_0\sin(\alpha)\mathbf{E}_y$. Here, v_0 and α are constants. During its flight, a vertical gravitational force $-mg\mathbf{E}_y$ acts on the projectile. Modeling the projectile as a particle of mass m, show that its path is a parabola:

$$y(x) = -\left(\frac{g}{2v_0^2\cos^2(\alpha)}\right)x^2 + \tan(\alpha)x.$$

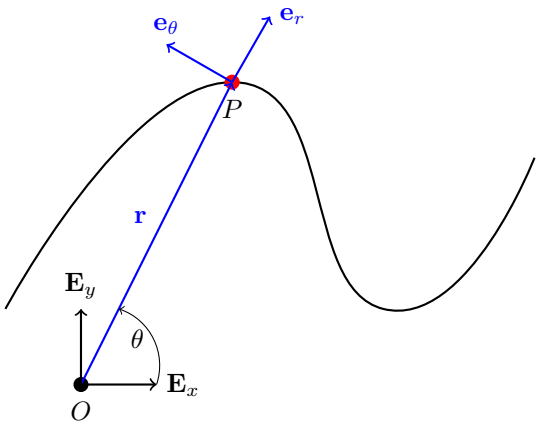
Why is this result not valid when $\alpha = \pm 90°$?

Figure 8.10: O'Reilly Primer, Exercise 1.8.

9 Cylindrical Polar Coordinate System

The cylindrical polar coordinate system is three-dimensional. We start by introducing the two-dimensional polar basis, then extend it to three dimensions.

9.1 Polar Basis



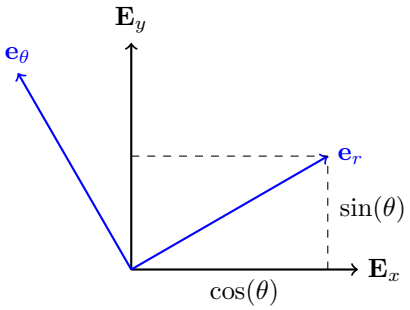
Consider a particle in motion in the two dimensional Euclidean space $\mathbb{E}^2$. Define:

$$r = \sqrt{x^2 + y^2} \geq 0, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right). \tag{9.1}$$

So

$$x = r\cos(\theta), \qquad y = r\sin(\theta) \tag{9.2}$$

Define the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ as a rotation of $\{\mathbf{E}_x, \mathbf{E}_y\}$ by angle θ (positive CCW about $\mathbf{E}_z$):



Think!

Question: Express $\mathbf{e}_r$ and $\mathbf{e}_\theta$ in the Cartesian basis.

Answer

$$\mathbf{e}_r = \cos(\theta)\mathbf{E}_x + \sin(\theta)\mathbf{E}_y, \tag{9.3}$$
$$\mathbf{e}_\theta = -\sin(\theta)\mathbf{E}_x + \cos(\theta)\mathbf{E}_y. \tag{9.4}$$

$\{\mathbf{e}_r, \mathbf{e}_\theta\}$ is an orthonormal basis: $\mathbf{e}_r \cdot \mathbf{e}_r = 1$, $\mathbf{e}_\theta \cdot \mathbf{e}_\theta = 1$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$.

- $\mathbf{e}_r$ points in the direction of increasing r.
- $\mathbf{e}_\theta$ points in the direction of increasing θ.

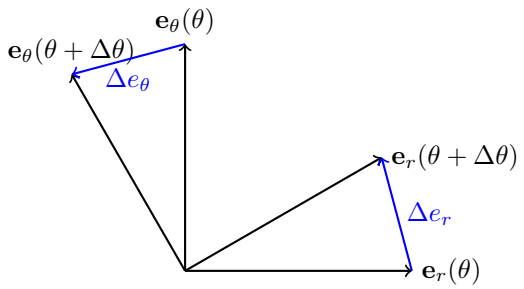
Think!

Question: Calculate $\dot{\mathbf{e}}_r (= \dot{\theta}\mathbf{e}_\theta)$ and $\dot{\mathbf{e}}_\theta (= \dot{\theta}\mathbf{e}_r)$.

Answer

$$\begin{aligned} \mathbf{e}_r = \mathbf{e}_r(\theta) &\implies \frac{d\mathbf{e}_r}{d\theta} = -\sin(\theta)\mathbf{E}_x + \cos(\theta)\mathbf{E}_y = \mathbf{e}_\theta \\ \mathbf{e}_\theta = \mathbf{e}_\theta(\theta) &\implies \frac{d\mathbf{e}_\theta}{d\theta} = -\cos(\theta)\mathbf{E}_x - \sin(\theta)\mathbf{E}_y = -\mathbf{e}_r \end{aligned} \tag{9.5}$$

You can also see this result geometrically,



Thus, for small $d\theta > 0$,

$$\begin{aligned} &\mathbf{e}_r(\theta+d\theta)-\mathbf{e}_r(\theta) \text{ points in the direction of } \mathbf{e}_\theta(\theta) \\ &\mathbf{e}_\theta(\theta+d\theta)-\mathbf{e}_\theta(\theta) \text{ points in the direction of } -\mathbf{e}_r(\theta) \end{aligned} \tag{9.6}$$

https://www.youtube.com/watch?v=r3lLIkWfMVk

9.2 Kinematics in Polar Coordinates

We can write the position vector to the particle as $\mathbf{r} = r\mathbf{e}_r$ (with $r \geq 0$). Note that $\mathbf{e}_r$ is a function of θ. Both r and θ are functions of time t.

Think!

Question: Show that $\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$ and $\mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta$.

Answer

$$\begin{aligned} \mathbf{v} &= \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r \\ &= \dot{r}\mathbf{e}_r + r\frac{d\mathbf{e}_r}{d\theta}\dot{\theta} \\ &= \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta. \\ \mathbf{a} &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta \\ &= \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta. \end{aligned}$$

Note the components of acceleration vector:

- $r\ddot{\theta}$: tangential
- $\ddot{r}$: radial
- $r\dot{\theta}^2$: centripetal

- $2\dot{r}\dot{\theta}$: Coriolis

Historical Perspective: Gaspard-Gustave de Coriolis (1792–1843)

Coriolis was a French mathematician and mechanical engineer at the École Polytechnique. His 1835 memoir *Sur les équations du mouvement relatif des systèmes de corps* worked out, for the first time in full generality, the extra terms that appear when Newton's laws are written for an observer in a rotating frame – motivated not by weather or ocean currents, but by the mechanics of rotating machinery such as waterwheels and turbines. The term $2\dot{r}\dot{\theta}$ above is the one that now carries his name.

The same underlying fact – that differentiating a vector expressed in a basis which is itself rotating produces terms beyond the plain derivatives of its components – had already appeared, without a name, in the Serret-Frenet formulae for a particle tracing a fixed space curve. It took much longer, well into the following century, for "the Coriolis effect" to enter popular use, once physicists and meteorologists adopted it to explain the deflection of winds and ocean currents on the rotating Earth.

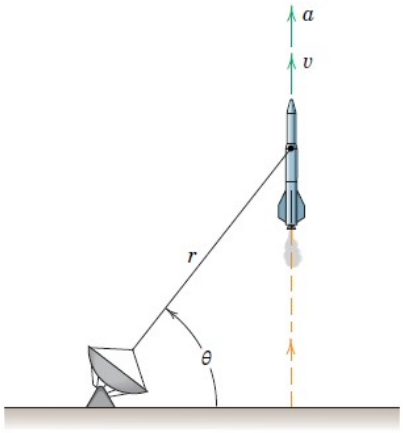
We may also introduce the notation

$$\begin{aligned}\mathbf{r} &= r\mathbf{e}_r + 0\mathbf{e}_\theta \\ \mathbf{v} &= v_r\mathbf{e}_r + v_\theta\mathbf{e}_\theta \\ \mathbf{a} &= a_r\mathbf{e}_r + a_\theta\mathbf{e}_\theta\end{aligned}$$

Remark: $a_r \neq \dot{v}_r$**.** Be careful: $v_r = \dot{r}$ so $\dot{v}_r = \ddot{r}$, but $a_r = \ddot{r} - r\dot{\theta}^2 \neq \dot{v}_r$ in general.

Remark: When to use polar vs Cartesiancoordinates? For rectilinear motions, Cartesian coordinates are usually easiest. For problems where r is directly measured (e.g. radar tracking as in MKB 2/121.), polar coordinates are natural.

2/121 SS The rocket is fired vertically and tracked by the radar station shown. When θ reaches 60°, other corresponding measurements give the values $r = 9$ km, $\ddot{r} = 21$ m/s^2, and $\dot{\theta} = 0.02$ rad/s. Calculate the magnitudes of the velocity and acceleration of the rocket at this position.

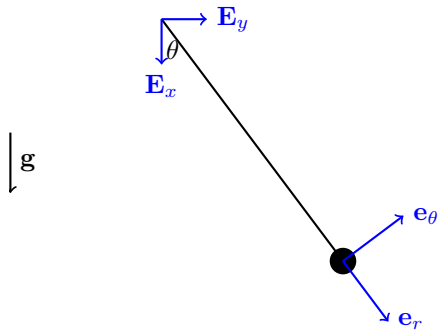


PROBLEM 2/121

Figure 9.1: MKB 2/121.

9.3 Example: Nonlinear Pendulum

Consider a particle of mass m suspended by a massless inextensible string of length ℓ. Find the equation of motion of the particle and the tension in the string.



The planar motion of the particle is constrained by the string. This constraint can be expressed mathematically in Cartesian and polar coordinates respectively as

$$\sqrt{x^2 + y^2} = \ell \tag{9.7}$$

$$r = \ell = \text{const.} \implies \dot{r} = 0,\ \ddot{r} = 0. \tag{9.8}$$

Note that the constraint is simpler in polar coordinates. This is a common theme: the right choice of coordinates can simplify the problem significantly.

Think!

Question: How many degrees of freedom does this system have?

Answer

One. A particle in the plane has 2 DOFs, but the constraint $r = \ell$ removes one. Knowing $\theta(t)$ completely describes the motion.

Following the four steps:

1. Our system is the particke. Choosing polar coordinates, we write its position, velocity, and acceleration as

$$\mathbf{r} = \ell \mathbf{e}_r, \quad \mathbf{v} = \ell\dot{\theta}\mathbf{e}_\theta, \quad \mathbf{a} = \ell\ddot{\theta}\mathbf{e}_\theta - \ell\dot{\theta}^2\mathbf{e}_r.$$

Orienting $\mathbf{e}_\theta$

$\mathbf{e}_\theta$ points in the direction of increasing θ, determined by the right-hand rule: point the thumb along $\mathbf{E}_z$; your fingers curl in the direction of increasing θ.

2. Forces:

$$\mathbf{T} = T\left(-\mathbf{e}_r\right), \quad \mathbf{W} = mg\left(\cos(\theta)\mathbf{e}_r - \sin(\theta)\mathbf{e}_\theta\right).$$

Remark: Tension

$\mathbf{T}$ is a constraint force keeping the particle on a circular path. If $T > 0$, the wire is in tension. The tension is the same throughout because the wire is massless.

3. Balance of linear momentum:

$$mg\mathbf{E}_x - T\mathbf{e}_r = m\ell\left(\ddot{\theta}\mathbf{e}_\theta - \dot{\theta}^2\mathbf{e}_r\right).$$

4. Projecting along $\mathbf{e}_\theta$ (hides unknown tension) and results in the equation of motion for θ:

$$-mg\sin(\theta) = m\ell\ddot{\theta} \implies \ddot{\theta} + \frac{g}{\ell}\sin(\theta) = 0$$

which can be solved numerically for $\theta(t)$ given IC's $\theta(0), \dot{\theta}(0)$. This is a nonlinear ODE. For small θ, we can linearise $\sin(\theta) \approx \theta$ and solve the resulting simple harmonic motion equation by hand

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0$$

Projecting along $\mathbf{e}_r$ gives the tension:

$$T = m\left(\ell\dot{\theta}^2 + g\cos(\theta)\right)$$

9.4 Types of Accelerations Explained

The goal of this section is to give physical intuition for the different components of acceleration in polar coordinates.

Recall:

$$\mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta.$$

Think!

Question: Can you think of physical systems where the acceleration of the particle is purely tangential, purely radial, purely centripetal, or purely Coriolis?

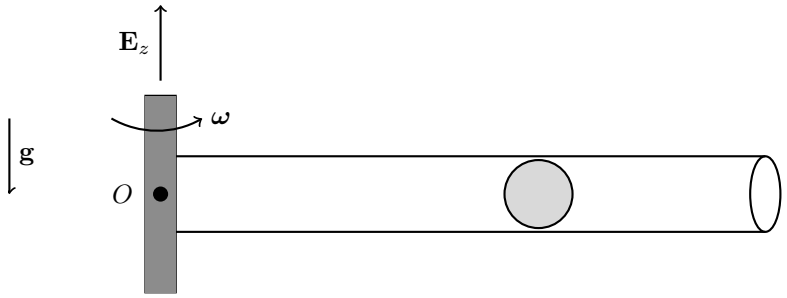
Answer

...

- **Radial** $\ddot{r}$: particle on a spring in rectilinear motion.
- **Centripetal** $-r\dot{\theta}^2$: particle on a circle at constant speed — the normal force steers the particle.
- **Tangential** $r\ddot{\theta}$: It is not possible to have purely tangential acceleration without a nonzero centripetal acceleration. A particle pushed along a circle with varying speed constitutes a minimal example that has both tangential and centripetal acceleration.
- **Coriolis** $2\dot{r}\dot{\theta}$: It is not possible to have purely Coriolis acceleration without a nonzero centripetal acceleration. A minimal example of nonzero Coriolis acceleration is a particle in a tube rotating at constant angular velocity. The Coriolis acceleration is due to both the rotation of the tube and the radial motion of the particle inside the tube.

9.4.1 Example: Particle in a Rotating Tube

Consider a particle of mass m inside a tube rotating at constant angular velocity ω about the vertical axis. The tube is horizontal and frictionless. Find the equation of motion for $r(t)$ and the normal force on the particle.



Following the four steps:

1. Kinematics: Here $\dot{\theta} = \omega = \text{const.}$, $\ddot{\theta} = 0$, so

$$\mathbf{r} = r\mathbf{e}_r, \quad \mathbf{v} = \dot{r}\mathbf{e}_r + r\omega\mathbf{e}_\theta, \quad \mathbf{a} = \left(\ddot{r} - r\omega^2\right)\mathbf{e}_r + 2\dot{r}\omega\mathbf{e}_\theta.$$

2. Assume no friction: $\mathbf{N} = N_\theta\mathbf{e}_\theta + N_z\mathbf{E}_z$, $\mathbf{W} = -mg\mathbf{E}_z$.

3. The balance of linear momentum is

$$\mathbf{F} = \dot{\mathbf{G}},$$
$$N_\theta\mathbf{e}_\theta + N_z\mathbf{E}_z - mg\mathbf{E}_z = m\left(\ddot{r} - r\omega^2\right)\mathbf{e}_r + 2m\dot{r}\omega\mathbf{e}_\theta.$$

4. Projecting this equation along $\mathbf{e}_r$ and $\mathbf{e}_\theta$ gives,

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right)\cdot\mathbf{e}_r \implies \ddot{r} - \omega^2 r = 0 \quad \text{(EOM for } r(t)\text{)},$$
$$\left(\mathbf{F} = \dot{\mathbf{G}}\right)\cdot\mathbf{e}_\theta \implies N_\theta = 2m\dot{r}\omega \quad \text{(Coriolis reaction)}.$$

> **Coriolis acceleration**
>
> As the particle moves outward, $r\omega$ increases. The tube must supply the $\mathbf{e}_\theta$ acceleration $2\dot{r}\omega$ so the particle stays inside.

9.4.2 Example: Particle Pushed in a Circle

Consider a particle of mass m on a horizontal circular guide. The particle is pushed by a force $\mathbf{P}$ in the $\mathbf{e}_\theta$ around the circle of radius R at constant angular velocity ω. Find the normal force on the particle.

Following the four steps:

1.

$$\begin{aligned}\mathbf{r} &= R\mathbf{e}_r + H\mathbf{E}_z \\ \mathbf{v} &= R\dot{\theta}\mathbf{e}_\theta \\ \mathbf{a} &= R\ddot{\theta}\mathbf{e}_\theta - R\dot{\theta}^2\mathbf{e}_r\end{aligned}$$

2. The normal force $\mathbf{N}$ from the track has components in the $\mathbf{e}_r$ and $\mathbf{E}_z$ directions.

$$\begin{aligned}\mathbf{P} &= P\mathbf{e}_\theta \quad (\mathbf{P} = P\mathbf{E}_y \text{ only at initial instant.}) \\ \mathbf{W} &= mg\left(-\mathbf{E}_z\right) = -mg\mathbf{E}_z \\ \mathbf{N} &= N_r\left(-\mathbf{e}_r\right) + N_z\mathbf{E}_z\end{aligned}$$

3. The balance of linear momentum is

$$-N_r\mathbf{e}_r + \left(N_z - mg\right)\mathbf{E}_z + P\mathbf{e}_\theta = mR\left(\ddot{\theta}\mathbf{e}_\theta - \dot{\theta}^2\mathbf{e}_r\right)$$

4. Projecting along $\mathbf{e}_\theta$, $\mathbf{e}_r$, and $\mathbf{E}_z$ gives

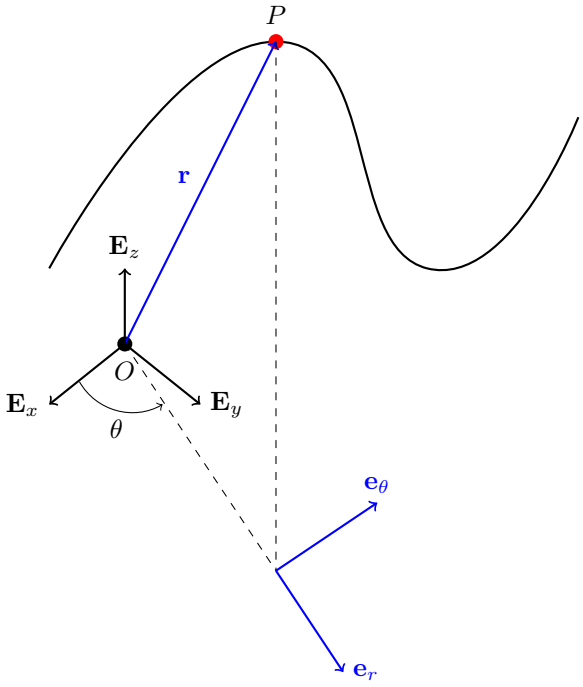
$$\begin{aligned}\left(\mathbf{F} = \dot{G}\right)\cdot\mathbf{e}_\theta &\quad P = mR\ddot{\theta} \\ \left(\mathbf{F} = \dot{G}\right)\cdot\mathbf{e}_r &\quad N_r = mR\dot{\theta}^2 \\ \left(\mathbf{F} = \dot{G}\right)\cdot\mathbf{E}_z &\quad N_z = mg\end{aligned}$$

   The first of these equations hides $\mathbf{N}$ and provides an EOM for $\theta(t)$ that can be solved given the ICs $\theta_0, \dot{\theta}_0$.

The second and third of these equations provide the constraint force components. N_r supplies the centripetal acceleration which steers the particle. N_z holds up the particle and ensure $\ddot{z} = 0$.

9.5 Cylindrical-Polar Coordinates

Consider a particle in motion in three-dimensional Euclidean space $\mathbb{E}^3$. We can extend the polar coordinate system by adding a vertical coordinate z and a vertical basis vector $\mathbf{E}_z$. The radial unit vector $\mathbf{e}_r$ lies along the projection of $\mathbf{r}$ in the $\{\mathbf{E}_x, \mathbf{E}_y\}$ plane.



In summary,

$$\begin{bmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{E}_z \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \\ \mathbf{E}_z \end{bmatrix}.$$

The position, velocity, acceleration of the particle are

$$\begin{aligned} \mathbf{r} &= r\mathbf{e}_r + z\mathbf{E}_z, \\ \mathbf{v} &= \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{E}_z, \\ \mathbf{a} &= \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta + \ddot{z}\mathbf{E}_z. \end{aligned}$$

https://youtu.be/FMEV8mmVqEE

9.6 Summary

Cylindrical polar basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{E}_z\}$:

$$\mathbf{e}_r = \cos\theta\,\mathbf{E}_x + \sin\theta\,\mathbf{E}_y, \qquad \mathbf{e}_\theta = -\sin\theta\,\mathbf{E}_x + \cos\theta\,\mathbf{E}_y, \tag{9.9}$$

$$\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r. \tag{9.10}$$

Position, velocity, acceleration:

$$\mathbf{r} = r\mathbf{e}_r + z\mathbf{E}_z, \tag{9.11}$$

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta + \dot z\,\mathbf{E}_z, \tag{9.12}$$

$$\mathbf{a} = (\ddot r - r\dot\theta^2)\mathbf{e}_r + (r\ddot\theta + 2\dot r\dot\theta)\mathbf{e}_\theta + \ddot z\,\mathbf{E}_z. \tag{9.13}$$

9.7 Lecture Videos

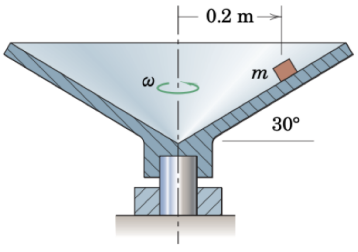
https://youtu.be/pj7kitk0VcI

9.8 Exercises

In both the following problems, set up a cylindrical polar coordinate system whose origin is taken to be at the vertex of the cone.

a.

3/67 The small object is placed on the inner surface of the conical dish at the radius shown. If the coefficient of static friction between the object and the conical surface is 0.30, for what range of angular velocities ω about the vertical axis will the block remain on the dish without slipping? Assume that speed changes are made slowly so that any angular acceleration may be neglected.

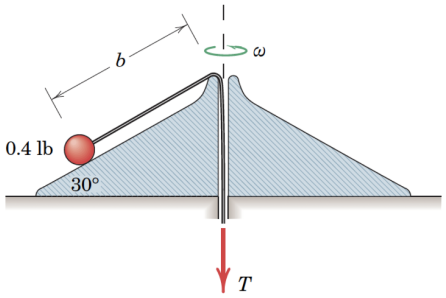


PROBLEM 3/67

See an animation of the cylindrical-polar basis for a particle moving aroubd a cone here.

b.

3/193 **SS** The 0.4-lb ball and its supporting cord are revolving about the vertical axis on the fixed smooth conical surface with an angular velocity of 4 rad/sec. The ball is held in the position $b = 14$ in. by the tension T in the cord. If the distance b is reduced to the constant value of 9 in. by increasing the tension T in the cord, compute the new angular velocity ω and the work $U'_{1\text{-}2}$ done on the system by T.

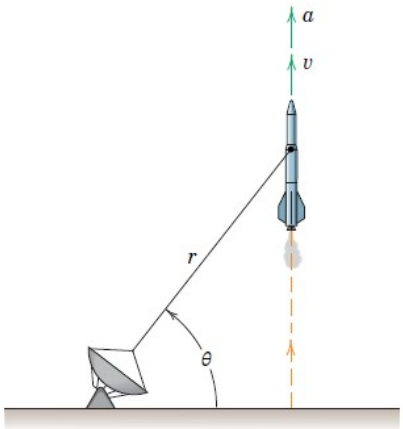


PROBLEM 3/193

The following problems are from Set 05 – Cylindrical Polar Coordinates.

1. [MKB 2/121] Take the origin at the satellite; $\mathbf{E}_x$ rightwards, $\mathbf{E}_y$ upwards. Write the position in both cylindrical and Cartesian coordinates. *(ans. $v = 360$ m/s, $a = 20.1$ m/s^2)*

2/121 **SS** The rocket is fired vertically and tracked by the radar station shown. When θ reaches 60°, other corresponding measurements give the values $r = 9$ km, $\ddot{r} = 21$ m/s^2, and $\dot{\theta} = 0.02$ rad/s. Calculate the magnitudes of the velocity and acceleration of the rocket at this position.



PROBLEM 2/121

Figure 9.2: MKB 2/121.

https://youtu.be/wbUBexp8xfc

2. [MKB 02-105] Take $\mathbf{E}_x$ rightwards and $\mathbf{E}_y$ upwards; write the car's position in both bases. *(ans.* $\dot{r} = 47.7$ *ft/sec,* $\dot{\theta} = -41.0$ *deg/sec)*

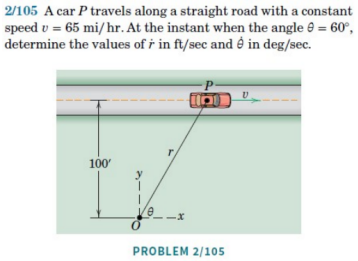


Figure 9.3: MKB 02-105.

3. [MKB 02-126] Write $\mathbf{r}_P = (0.75 + \ell)\mathbf{e}_r$ m and differentiate. If the setup is in the vertical plane, find the force applied by the robot arm. *(ans.* $v = 0.296$ *m/s,* $a = 0.345$ *m/s*^{2}*)*

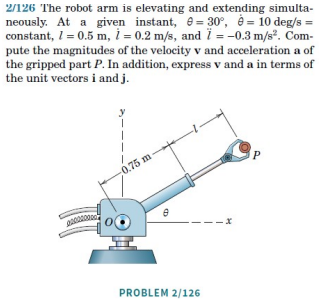
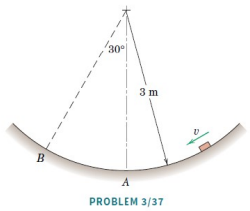


Figure 9.4: MKB 02-126.

4. [MKB 03-037] Apply the 4 steps at points A and B independently; origin at the centre of curvature. *(ans.* $N_A = 10.89$ *N,* $N_B = 8.30$ *N)*

3/37 The small 0.6-kg block slides with a small amount of friction on the circular path of radius 3 m in the vertical plane. If the speed of the block is 5 m/s as it passes point A and 4 m/s as it passes point B, determine the normal force exerted on the block by the surface at each of these two locations.



PROBLEM 3/37

Figure 9.5: MKB 03-037.

5. [Primer Exercise 2.6]

5. [Primer Exercise 2.6]

Exercise 2.6: Consider the projectile problem discussed in Section 1.6 of Chapter 1. Using a cylindrical polar coordinate system, show that the equations governing the motion of the particle are

$$m\ddot{r} - mr\dot{\theta}^2 = -mg\sin(\theta), \quad mr\ddot{\theta} + 2m\dot{r}\dot{\theta} = -mg\cos(\theta), \quad m\ddot{z} = 0.$$

Notice that, in contrast to using Cartesian coordinates to determine the governing equations, solving these differential equations is nontrivial.

Figure 9.6: O'Reilly Primer, Exercise 2.6.

6. [Primer Exercise 2.7]

6. [Primer Exercise 2.7]

Exercise 2.7: Consider a spherical bead of mass m and radius R that is placed inside a long cylindrical tube. The inner radius of the tube is R, and the tube is pivoted so that it rotates in a horizontal plane. Furthermore, the contact between the tube and the bead is smooth. Here, the bead is modeled as a particle of mass m. Now suppose that the tube is whirled at a constant angular speed Ω (radians per second). The whirling motion of the tube is such that the velocity vector of the bead is $\mathbf{v} = \dot{r}\mathbf{e}_r + \Omega r\mathbf{e}_\theta$. Show that the equation governing the motion of the bead is

$$\ddot{r} - \Omega^2 r = 0,$$

and the force exerted by the tube on the particle is $mg\mathbf{E}_z + 2m\dot{r}\Omega\mathbf{e}_\theta$.

Figure 9.7: O'Reilly Primer, Exercise 2.7.

7. [Primer Exercise 2.8]

7. [Primer Exercise 2.8]

Exercise 2.8: Consider the case where the bead is initially at rest relative to the center of the whirling tube at a location $r = r_0$. Using the solution to the differential equation $\ddot{r} - \Omega^2 r = 0$ recorded in Section A.5.3 of Appendix A, show that, unless $r_0 = 0$, the bead discussed in the previous exercise will eventually exit the whirling tube.

Figure 9.8: O'Reilly Primer, Exercise 2.8.

10 Constrained Motion

The **degree of freedom (DOF)** of a system is the number of independent coordinates needed to describe a configuration of the system.

Examples:

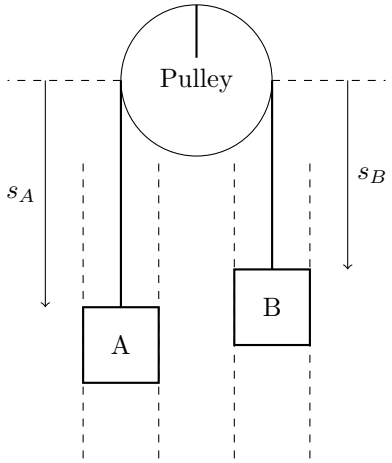
System	DOFs	Constraints	Constraint equation(s)
A particle free to move in $\mathbb{E}^2$	2	0	–
A particle constrained to move on a curve in $\mathbb{E}^2$	1	1	(curve equation, implicit)
Two particles free to move in $\mathbb{E}^2$	4	0	–
Two particles in $\mathbb{E}^2$ constrained to move on the same curve	2	2	(curve equation for each particle, implicit)
Two particles in $\mathbb{E}^2$ connected by a rigid rod	3	1	$\left\|\mathbf{r}_{B/A}\right\| = l \,\forall t$
Two particles in $\mathbb{E}^2$ connected by a rigid rod *and* constrained to move on a curve	1	3	$\left\|\mathbf{r}_{B/A}\right\| = l \,\forall t$; curve equation for each particle (implicit)

The **number of constraints** is the number of DOFs the unconstrained system would have (2 per free particle in $\mathbb{E}^2$) minus the actual number of DOFs; each independent constraint equation removes exactly one DOF.

Example: Consider two particles in smooth vertical slots. If A and B move independently, the system has 2 DOFs. If connected by an inextensible cable, the system has 1 DOF.

The inextensibility constraint $\ell = \text{const.}$ can be written as:

$$\ell = s_A + c + s_B \tag{10.1}$$
$$\Delta\ell = 0 = \Delta_{t_1 \to t_2} s_A + \Delta_{t_1 \to t_2} s_B \tag{10.2}$$
$$\implies \Delta s_B = -\Delta s_A \tag{10.3}$$
$$\dot{\ell} = 0 = \dot{s}_A + \dot{s}_B \tag{10.4}$$
$$\implies \dot{s}_B = -\dot{s}_A. \tag{10.5}$$



10.1 Summary

For **pulley–chord systems**: (1) choose a datum, (2) define distances from the datum, (3) write a chord equation per chord. Each chord equation and its derivatives give additional constraints.

10.2 Exercises

The following problems are from Set 06 – Constrained Motion.

1. [MKB 2/099] Determine the $\mathbf{e}_r$ and $\mathbf{e}_\theta$ components of the acceleration of pin P; origin at the centre of the circular slot. *(ans.* $a_n = 66.0$ *mm/s*^{2}*,* $a_t = 29.7$ *mm/s*^{2}*)*

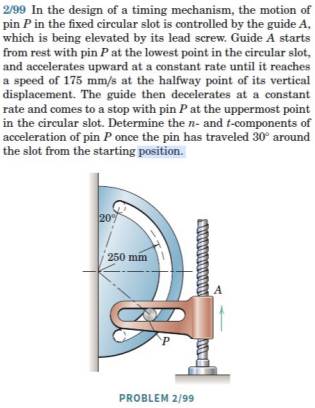


Figure 10.1: MKB 2/099.

https://youtu.be/Bt5neJj4Zhc

2. [03-056] Set up two polar coordinate systems with origins O and O'. Write position vectors of P in each basis. *(ans.* $N = 2.89$ *N towards* O', $R = 1.599$ *N)*

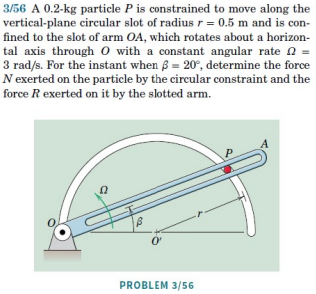
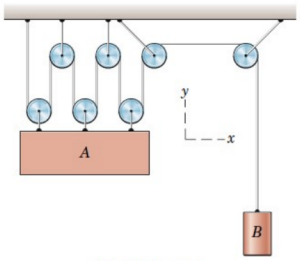


Figure 10.2: MKB 03-056.

3. [MKB 02-172] Use $\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = 3.5\mathbf{E}_y$ m/s. *(ans.* $\mathbf{v}_A = -0.5\mathbf{E}_y$ *m/s,* $\mathbf{v}_B = 3\mathbf{E}_y$ *m/s)*

2/172 At the instant represented, $\mathbf{v}_{B/A} = 3.5\mathbf{j}$ m/s. Determine the velocity of each body at this instant. Assume that the upper surface of A remains horizontal.



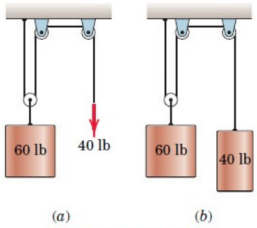
PROBLEM 2/172

Figure 10.3: MKB 02-172.

4. [MKB 03-021] Draw FBDs of the massless pulleys to find forces on the 60-lb cylinders. *(ans. (a) $a = 10.73$ ft/sec^2 up; (b) $a = 2.93$ ft/sec^2 up)*

. (a) $a = 10.73$ ft/sec^2 up, (b) $a = 2.93$ ft/sec^2 up)

3/21 SS Determine the vertical acceleration of the 60-lb cylinder for each of the two cases. Neglect friction and the mass of the pulleys.

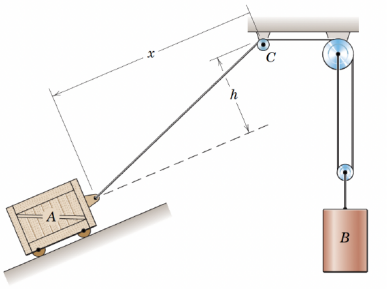


PROBLEM 3/21

Figure 10.4: MKB 03-021.

5. [MKB 2-177]

2/177 **SS** Determine an expression for the velocity v_A of the cart A down the incline in terms of the upward velocity v_B of cylinder B.



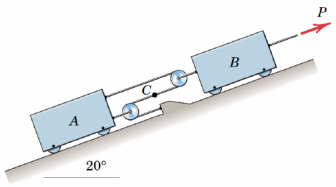
PROBLEM 2/177

Figure 10.5: MKB 2/177.

6. [MKB 2-179]

ans. $v_{B/A} = 1$ ft/sec, $a_{B/A} = 2$ ft/sec^2, $v_C = 4$ ft/sec (all directed up incline))

2/179 Under the action of force P, the constant acceleration of block B is 6 ft/sec^2 up the incline. For the instant when the velocity of B is 3 ft/sec up the incline, determine the velocity of B relative to A, the acceleration of B relative to A, and the absolute velocity of point C of the cable.



PROBLEM 2/179

Figure 10.6: MKB 2/179.

11 The Serret-Frenet Basis

In general, the path of a particle is a curve in space — we refer to such curves as *space curves*.

Consider a fixed curve $\mathcal{C}$ embedded in $\mathbb{E}^3$. Let the position vector of a point $P \in \mathcal{C}$ be $\mathbf{r}$. In Cartesian coordinates, $\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z$.

Recall the arc-length parameter s associated with $\mathcal{C}$:

$$\left(\frac{ds}{dt}\right)^2 = \frac{d\mathbf{r}}{dt} \cdot \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\frac{dx}{dt} + \frac{dy}{dt}\frac{dy}{dt} + \frac{dz}{dt}\frac{dz}{dt}.$$

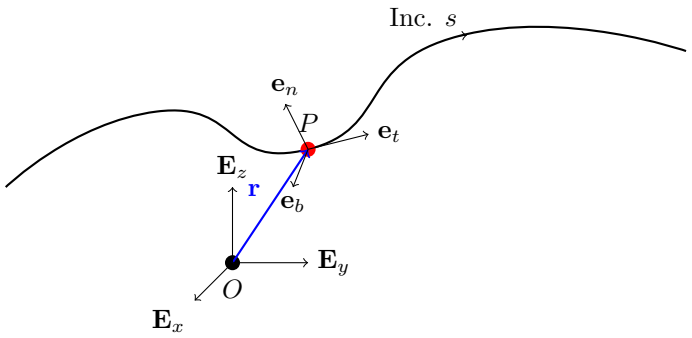
This parameter uniquely identifies a point P on $\mathcal{C}$, giving the representation $\mathbf{r} = \hat{\mathbf{r}}(s)$.

Historical Perspective: Frenet, Serret, and the Moving Basis

The unit vectors $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ carry the names of two French mathematicians who never worked together. Jean Frédéric Frenet first derived the formulas describing how this triad turns along a curve in his 1847 doctoral thesis at Toulouse, publishing only a summary in 1852. Joseph Alfred Serret arrived at the same relations independently and published them in full in 1851, and it is largely through his more widely circulated version that both the formulas and the triad came to carry both names.

The real subtlety of this basis is that $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ rotates as the particle moves along $\mathcal{C}$, even though the curve itself is fixed in space. Differentiating a vector expressed in a rotating basis introduces extra terms beyond the plain derivatives of its components. We meet this same fact again, under its own name, as the *Coriolis acceleration*, when we turn to rotating frames in cylindrical polar coordinates.

11.1 The Serret-Frenet Triad



The Serret-Frenet basis $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ is defined at the point P; it depends on the curve and the particle's location on it.

Think!

Question: How do we define the unit tangent vector?

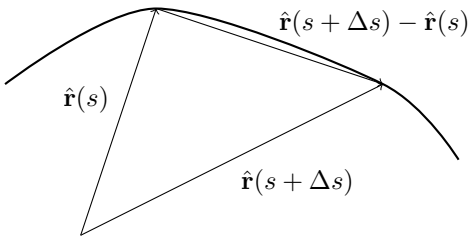
Answer

Recall the unit tangent vector:

$$\mathbf{e}_t = \hat{\mathbf{e}}_t(s) = \frac{d\mathbf{r}}{ds} = \lim_{\Delta s \to 0} \frac{\hat{\mathbf{r}}(s + \Delta s) - \hat{\mathbf{r}}(s)}{\Delta s}. \tag{11.1}$$

The hat $(\hat{\cdot})$ indicates that this function is now expressed in terms of another variable, s. $\mathbf{e}_t$ is known as the unit tangent vector to $\mathcal{C}$ at the point P.

For general motion, $||\Delta\mathbf{r}|| \neq ds$ except in the infinitesimal sense: $\hat{\mathbf{r}}(s + \Delta s) - \hat{\mathbf{r}}(s) \to \Delta s \mathbf{e}_t$ as $\Delta s \to 0$ and $d\mathbf{r}$ is tangent to the curve.



We need to determine the expression for the acceleration in the Serret-Frenet basis.

$$\mathbf{a} = \dot{\mathbf{v}} = \dot{v}\mathbf{e}_t + v\dot{\mathbf{e}}_t \tag{11.2}$$

where

$$\dot{\mathbf{e}}_t = \frac{d\hat{\mathbf{e}}_t}{ds}\frac{ds}{dt} = v\frac{d\mathbf{e}_t}{ds}. \tag{11.3}$$

The derivative $\frac{d\hat{\mathbf{e}}_t}{ds}$ only depends on the curve that the particle is tracing.

Think!

Question: What do we know about the direction of $\frac{d\hat{\mathbf{e}}_t}{ds}$?

Answer

We can show that $\mathbf{e}_t$ and $\frac{d\hat{\mathbf{e}}_t}{ds}$ are perpendicular by differentiating $\mathbf{e}_t \cdot \mathbf{e}_t = 1$:

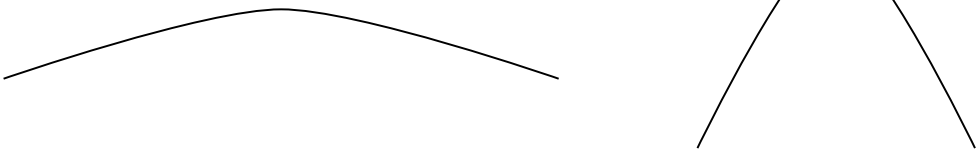
$$\frac{d}{ds}(\mathbf{e}_t \cdot \mathbf{e}_t) = \mathbf{e}_t \cdot \frac{d\mathbf{e}_t}{ds} = 0. \tag{11.4}$$

Think!

Question: What do we know about the magnitude of $\frac{d\hat{\mathbf{e}}_t}{ds}$?

Answer

Consider two curves. Draw the unit tangent vectors at the same distance along the curve, ds, apart and draw their difference, $\Delta\mathbf{e}_t$.



For which of these two curves is the magnitude of $\frac{\Delta\hat{\mathbf{e}}_t}{\Delta s}$ higher?

As the curve begins to resemble a line, the magnitude of $\frac{d\hat{\mathbf{e}}_t}{ds}$ becomes smaller. As the *curvature* of a curve increases, so does the magnitude of $\frac{d\hat{\mathbf{e}}_t}{ds}$.

Note! Derivatives of a unit vector

It is a common mistake to assume that the derivative of a unit vector is necessarily a unit vector. Recall: $\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta$ is not generally a unit vector.

Hence, we define:

$$\kappa\mathbf{e}_n = \frac{d\mathbf{e}_t}{ds}, \tag{11.5}$$

where $\kappa = \left\| \frac{d\mathbf{e}_t}{ds} \right\| \geq 0$ and $\|\mathbf{e}_n\| = 1$.

- $\mathbf{e}_n$ is the unit **principal normal** vector.
- κ is the **curvature** of $\mathcal{C}$ at P.
- The **radius of curvature** is $\rho = 1/\kappa$.

For any point on a curve, three nearby non-collinear points define the **osculating circle** (radius ρ, center of curvature).

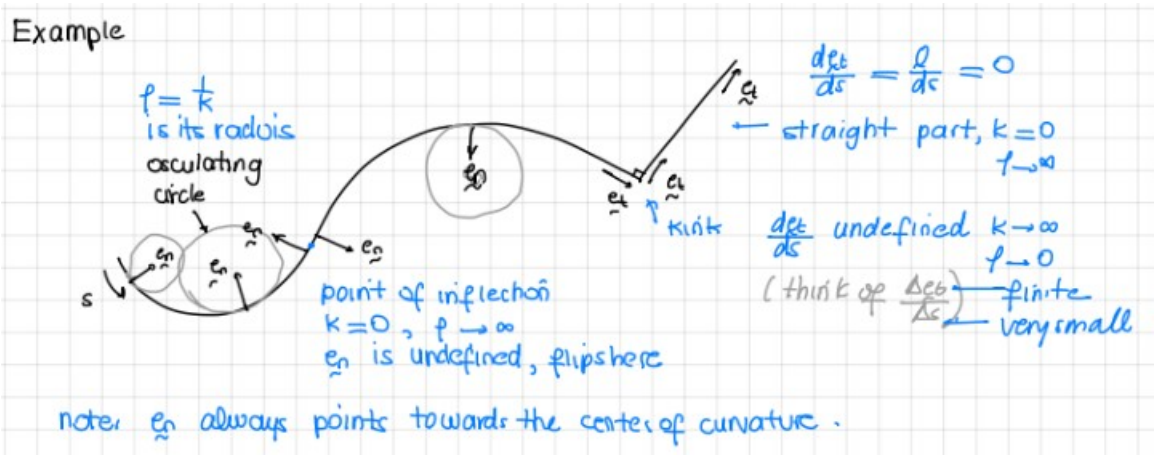


Figure 11.1: Serret-Frenet triad example

Think!

Question: When is $\mathbf{e}_n$ not uniquely defined?

Answer

When $d\mathbf{e}_t/ds = \mathbf{0}$ (i.e. $\kappa = 0$), such as on a straight line, at an inflection point, or at a corner. In this case, $\mathbf{e}_n$ is defined as any unit vector perpendicular to $\mathbf{e}_t$.

Example: In problem 03/040, we choose the Serret-Frenet basis on the straight portion of the curve to be such that it transitions smoothly to the curved portion of the curve.

The unit **binormal** vector:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n.$$

Concluding remarks:

- $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ is orthonormal and right-handed.

- The **osculating plane** is spanned by $\mathbf{e}_t$ and $\mathbf{e}_n$; it contains the osculating circle.
- The **rectifying plane** is spanned by $\mathbf{e}_t$ and $\mathbf{e}_b$.
- The **normal plane** is spanned by $\mathbf{e}_n$ and $\mathbf{e}_b$.

A vector can be expressed in the Serret-Frenet basis as in any other basis. The Serret-Frenet basis is not a fixed basis.

$$\mathbf{b} = b_t\mathbf{e}_t + b_n\mathbf{e}_n = b_x\mathbf{E}_x + b_y\mathbf{E}_y + b_z\mathbf{E}_z. \tag{11.6}$$

Check out the following animation showing the osculating circle and video placing the Serret-Frenet basis on a bobsled.

https://www.youtube.com/watch?v=q2I1oHxPgaQ

https://www.youtube.com/shorts/NyphZOJggFk?feature=share

https://www.youtube.com/watch?v=0ACqRREH180

https://www.youtube.com/watch?v=j2PnkdtIdQ8

11.2 Kinematics

Position, velocity, and acceleration in the Serret-Frenet basis:

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z = \mathbf{r}(t) = \mathbf{r}(\widehat{s(t)}).$$
$$\mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y + \dot{z}\mathbf{E}_z = \frac{d\mathbf{r}}{ds}\frac{ds}{dt} = \frac{ds}{dt}\mathbf{e}_t = v\mathbf{e}_t.$$
$$\mathbf{a} = \dot{\mathbf{v}} = \frac{d}{dt}\left(\frac{ds}{dt}\mathbf{e}_t\right) = \frac{d^2}{dt^2}\mathbf{e}_t + \frac{ds}{dt}\frac{d\mathbf{e}_s}{ds} = \frac{d^2}{dt^2}\mathbf{e}_t + \frac{ds}{dt}\frac{d\mathbf{e}_t}{ds}\frac{ds}{dt} = \dot{v}\mathbf{e}_t + \kappa v^2\mathbf{e}_n.$$

11.3 Kinetics

$$\mathbf{F} = m\mathbf{a} \tag{11.7}$$
$$F_t\mathbf{e}_t + F_n\mathbf{e}_n + F_b\mathbf{e}_b = m\left(\dot{v}\mathbf{e}_t + \kappa v^2\mathbf{e}_n\right). \tag{11.8}$$

Note that forces in the binormal direction should always balance.

11.3.1 Example: Problem 03/040 and Roller Coaster Design

Think!

Question: Consider the design of a roller coaster that is a circular ring. Comment on its safety.

Answer

At the transition between the circular and straight portions, there is a jump in curvature, hence a jump in acceleration, hence a jump in the normal force — producing a jerk. For a circular loop, the normal force goes from 0 to a finite value instantaneously. This could snap the rider's neck. Instead, we use two connected **clothoids** (Euler spirals), whose curvature is proportional to arc length.

11.4 The Serret-Frenet Formulae

We want the derivatives of the Serret-Frenet basis vectors with respect to s.

We already have:

$$\frac{d\mathbf{e}_t}{ds} = \kappa \mathbf{e}_n. \tag{11.9}$$

Since $\mathbf{e}_b \cdot \mathbf{e}_b = 1$, we have $\frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_b = 0$.

Also from $\mathbf{e}_t \cdot \mathbf{e}_b = 0$, we get $\frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_t = 0$. So $\frac{d\mathbf{e}_b}{ds}$ is parallel to $\mathbf{e}_n$:

$$\mathbf{e}_t \cdot \mathbf{e}_b = 0 \tag{11.10}$$

$$\frac{d\mathbf{e}_t}{ds} \cdot \mathbf{e}_b + \frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_t = 0 \tag{11.11}$$

$$\kappa \mathbf{e}_n \cdot \mathbf{e}_b + \frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_t = 0 \tag{11.12}$$

$$\frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_t = 0. \tag{11.13}$$

Consequently, we define

$$\frac{d\mathbf{e}_b}{ds} = -\tau \mathbf{e}_n, \tag{11.14}$$

where τ is the **torsion** of $\mathcal{C}$ at P. The negative sign in the above formula is a convention.

Then:

$$\frac{d\mathbf{e}_n}{ds} = \frac{d}{ds}\left(\mathbf{e}_b \times \mathbf{e}_t\right) \tag{11.15}$$

$$= \frac{d\mathbf{e}_b}{ds} \times \mathbf{e}_t + \mathbf{e}_b \times \frac{d\mathbf{e}_t}{ds} \tag{11.16}$$

$$= (-\tau \mathbf{e}_n) \times \mathbf{e}_t + \mathbf{e}_b \times \kappa \mathbf{e}_n \tag{11.17}$$

$$= -\kappa \mathbf{e}_t + \tau \mathbf{e}_b. \tag{11.18}$$

In summary (the Serret-Frenet formulae):

$$\begin{bmatrix} \frac{d\mathbf{e}_t}{ds} \\ \frac{d\mathbf{e}_n}{ds} \\ \frac{d\mathbf{e}_b}{ds} \end{bmatrix} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} \mathbf{e}_t \\ \mathbf{e}_n \\ \mathbf{e}_b \end{bmatrix}. \tag{11.19}$$

Define the **Darboux vector** $\omega_{SF} = \kappa \mathbf{e}_b + \tau \mathbf{e}_t$ so that:

$$\frac{d\mathbf{e}_i}{ds} = \omega_{SF} \times \mathbf{e}_i, \quad i = t, n, b. \tag{11.20}$$

https://youtu.be/OB13c0WlUSg

11.4.1 Example: A Particle on a Helix

For a helix $x = R\cos(\theta)$, $y = R\sin(\theta)$, $z = \alpha R\theta$:

$$\mathbf{r} = R\mathbf{e}_r + \alpha R\theta \mathbf{E}_z = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z. \tag{11.21}$$

$$\kappa = \frac{1}{R(1+\alpha^2)}, \qquad \tau = \frac{\alpha}{(1+\alpha^2)R}. \tag{11.22}$$

11.5 The Curvature Formula for a Plane Curve

Consider a point P tracing a plane curve with $y = f(x)$, $z = z_0$, so that

$$\mathbf{r} = x\mathbf{E}_x + f(x)\mathbf{E}_y + z_0\mathbf{E}_z. \tag{11.23}$$

Determining the arc-length parameter s:

$$\begin{aligned}
&\frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{E}_x + \frac{df}{dx}\frac{dx}{dt}\mathbf{E}_y, \\
&\left(\frac{ds}{dt}\right)^2 = \left(1 + \left(\frac{df}{dx}\right)^2\right)\left(\frac{dx}{dt}\right)^2, \\
&\frac{ds}{dt} = \frac{dx}{dt}\sqrt{1 + \left(\frac{df}{dx}\right)^2}, \\
&s = s(x) = \int_{x_0}^{x} \sqrt{1 + \left(\frac{df}{du}\right)^2}\, du + s(x_0).
\end{aligned} \tag{11.24}$$

The tangent vector:

$$\mathbf{e}_t = \frac{d\mathbf{r}}{ds} = \frac{d\mathbf{r}}{dx}\frac{dx}{ds} = \frac{1}{\sqrt{1 + \left(\frac{df}{dx}\right)^2}}\left(\mathbf{E}_x + \frac{df}{dx}\mathbf{E}_y\right). \tag{11.25}$$

The $\kappa\mathbf{e}_n$ expression:

$$\kappa\mathbf{e}_n = \frac{d\mathbf{e}_t}{ds} = \frac{d\mathbf{e}_t}{dx}\frac{dx}{ds} = \frac{\frac{d^2f}{dx^2}}{\left(1 + \left(\frac{df}{dx}\right)^2\right)^2}\left(\mathbf{E}_y - \frac{df}{dx}\mathbf{E}_x\right). \tag{11.26}$$

Recalling that $\mathbf{e}_n$ is a unit vector and $\kappa \geq 0$:

$$\begin{aligned}
&\kappa = \kappa(x) = \frac{\left|\frac{d^2f}{dx^2}\right|}{\left(\sqrt{1 + \left(\frac{df}{dx}\right)^2}\right)^3}, \\
&\mathbf{e}_n = \frac{\operatorname{sgn}\left(\frac{d^2f}{dx^2}\right)}{\sqrt{1 + \left(\frac{df}{dx}\right)^2}}\left(\mathbf{E}_y - \frac{df}{dx}\mathbf{E}_x\right),
\end{aligned} \tag{11.27}$$

where $\operatorname{sgn}(a) = 1$ if $a > 0$ and -1 if $a < 0$.

The binormal vector and torsion:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \operatorname{sgn}\left(\frac{d^2f}{dx^2}\right)\mathbf{E}_z, \tag{11.28}$$

and because this vector is piecewise constant, the torsion of the plane curve is $\tau = 0$.

11.6 Alternative Notation for Plane Curves

This section is not usually covered.

For plane curves, some texts define the angle $\beta = \beta(s)$ such that:

$$\mathbf{e}_t = \cos(\beta)\mathbf{E}_x + \sin(\beta)\mathbf{E}_y, \qquad \bar{\mathbf{e}}_n = \cos(\beta)\mathbf{E}_y - \sin(\beta)\mathbf{E}_x. \tag{11.29}$$

Then $\kappa = \left|\frac{d\beta}{ds}\right|$ and $\frac{d\beta}{ds}$ is the rate of rotation of the triad about $\mathbf{E}_z$.

11.7 Summary

Serret–Frenet basis $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$:

$$\mathbf{v} = v\,\mathbf{e}_t, \qquad \mathbf{a} = \dot{v}\,\mathbf{e}_t + \kappa v^2\,\mathbf{e}_n, \qquad \mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n. \tag{11.30}$$

The curvature is $\kappa = 1/\rho$ (radius of curvature ρ). The BoLM gives $F_t = m\dot{v}$ and $F_n = m\kappa v^2$.

11.8 Exercises

The following problems are from Set 07 – The Serret–Frenet Basis.

1. [MKB 2/079] Analyse the acceleration vector in the Serret–Frenet basis.

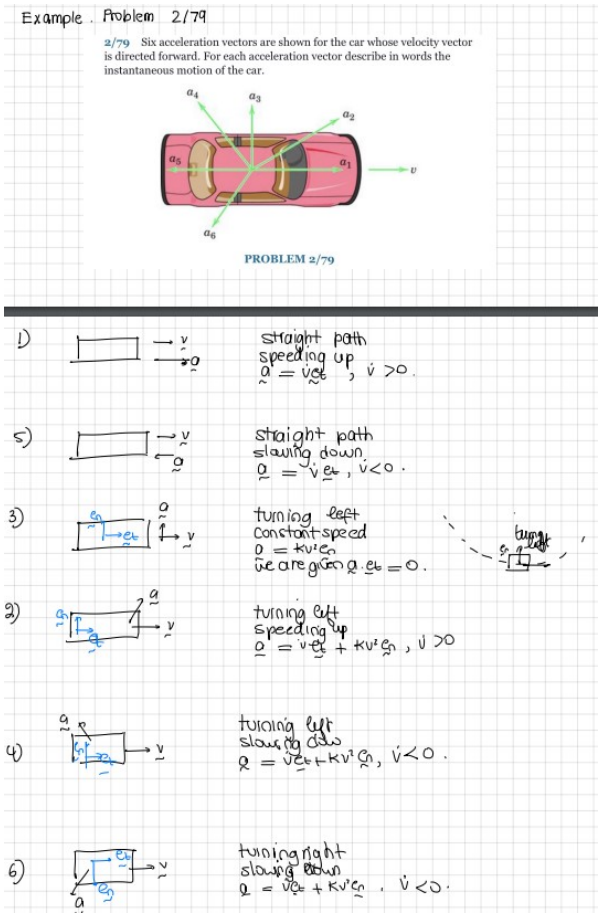
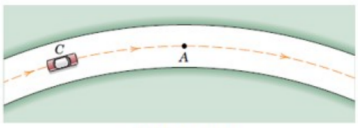


Figure 11.2: MKB 2/079.

2. [MKB 02-090] Note that $\dot{v} \neq \|\mathbf{a}\|$; acceleration has both tangential and normal components. *(ans. $v = 20$ m/s)*

he velocity vector. (ans. $v = 20$ m/s or 72 km/h)

2/90 The car C increases its speed at the constant rate of 1.5 m/s^2 as it rounds the curve shown. If the magnitude of the total acceleration of the car is 2.5 m/s^2 at point A where the radius of curvature is 200 m, compute the speed v of the car at this point.

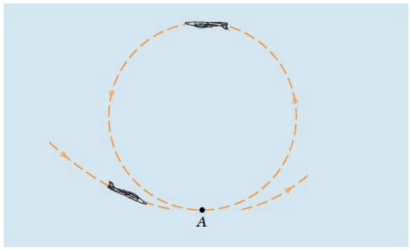


PROBLEM 2/90

Figure 11.3: MKB 02-090.

3. [MKB 02-091] *(ans. $\rho = 1709$ m)*

2/91 At the bottom A of the vertical inside loop, the magnitude of the total acceleration of the airplane is $3g$. If the airspeed is 800 km/h and is increasing at the rate of 20 km/h per second, calculate the radius of curvature ρ of the path at A.



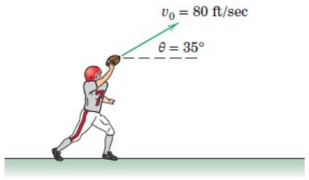
PROBLEM 2/91

Figure 11.4: MKB 02-091.

4. [02-097] Follow the 4 steps; express $\mathbf{e}_t$ and $\mathbf{e}_n$ in the Cartesian basis. *(ans. at $t = 1$ s: $\dot{v} = -6.58$ ft/s^2, $\rho = 142.2$ ft; at $t = 2$ s: $\dot{v} = 8.75$ ft/s^2, $\rho = 149.7$ ft)*

time in the Cartesian basis. Calculate $\mathbf{e}_t$ from its definition and calculating
t throughout the motion of the projectile $\mathbf{e}_t \times \mathbf{e}_n = -\mathbf{E}_z$. (ans. for $t = 1$ s:
^{2}, $\rho = 142.2$ ft, for $t = 2$ s: $\dot{v} = 8.75$ ft/sec^2, $\rho = 149.7$ ft)

2/97 A football player releases a ball with the initial conditions shown in the figure. Determine the radius of curvature ρ of the path and the time rate of change $\dot{v}$ of the speed at times $t = 1$ sec and $t = 2$ sec, where $t = 0$ is the time of release from the quarterback's hand.



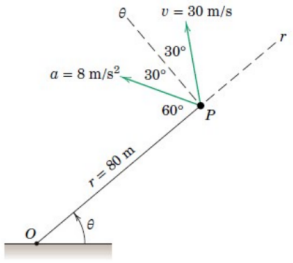
PROBLEM 2/97

Figure 11.5: MKB 02-097.

5. [02-103] *(ans.* $(x_C, y_C) = (22.5, -22.9)$ *m)*

6. [02-199] Draw $\mathbf{e}_t$ and $\mathbf{e}_n$ first. *(ans.* $\dot{r} = 15$ *m/s,* $\dot{\theta} = 0.325$ *rad/s,* $\rho = 129.9$ *m)*

2/199 At the instant depicted, assume that the particle P, which moves on a curved path, is 80 m from the pole O and has the velocity v and acceleration a as indicated. Determine the instantaneous values of $\dot{r}$, $\ddot{r}$, $\dot{\theta}$, $\ddot{\theta}$, the n- and t-components of acceleration, and the radius of curvature ρ.

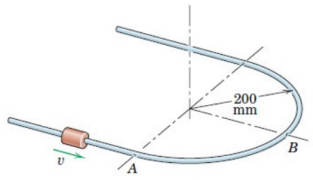


PROBLEM 2/199

Figure 11.6: MKB 02-199.

7. [03-040] Note the jump in normal force at the transition from straight to curved path. *(ans. (a)* $R = 1.177$ *N; (b)* $R = 1.664$ *N)*

3/40 The 120-g slider has a speed $v = 1.4$ m/s as it passes point A of the smooth guide, which lies in a horizontal plane. Determine the magnitude R of the force which the guide exerts on the slider (a) just before it passes point A of the guide and (b) as it passes point B.



PROBLEM 3/40

Figure 11.7: MKB 03-040.

8. [03-041] What does "weightless" mean? *(ans.* $\rho = 24\,000$ *ft)*

the four steps to solve this problem. Think: what does weightless mean? (ans.

3/41 A jet transport plane flies in the trajectory shown in order to allow astronauts to experience the "weightless" condition similar to that aboard orbiting spacecraft. If the speed at the highest point is 600 mi/hr, what is the radius of curvature ρ necessary to exactly simulate the orbital "free-fall" environment?

PROBLEM 3/41

Figure 11.8: MKB 03-041.

9. [03-043] Draw the Serret–Frenet basis carefully; the figure hints at the direction of $\mathbf{e}_n$. *(ans.* $v = 29.1$ *m/s,* $N = 12.36$ *kN)*

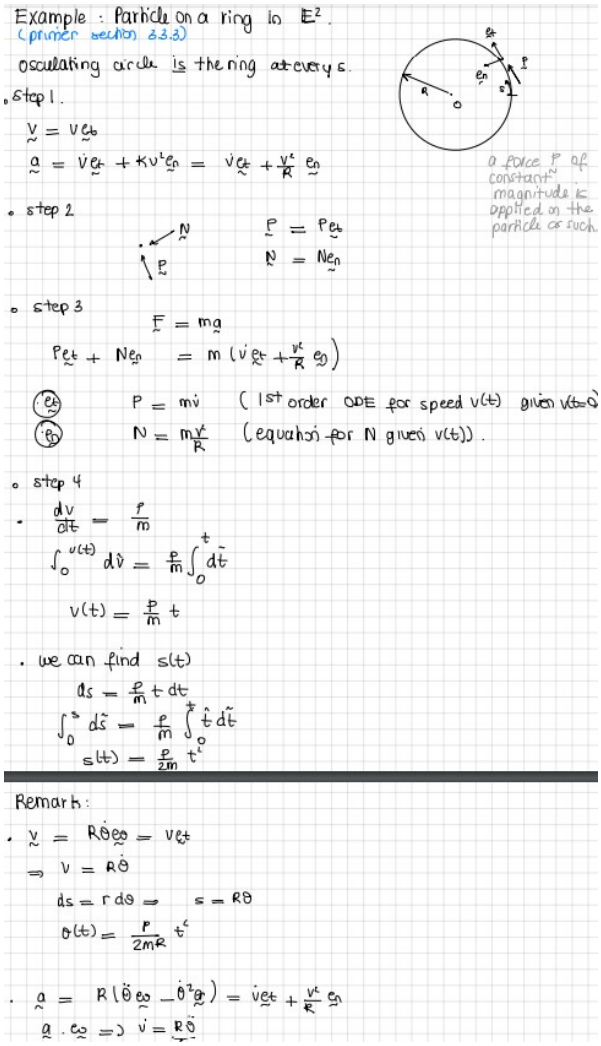


Figure 11.9: Problem 03/043.

10. [OOR 3.1]

10. [OOR 3.1]

Exercise 3.1: For the space curve $\mathbf{r} = x\mathbf{E}_x + ax\mathbf{E}_y$, show that

$$\mathbf{e}_t = \frac{1}{\sqrt{1+a^2}}(\mathbf{E}_x + a\mathbf{E}_y), \qquad \kappa = 0.$$

In addition, show that $\mathbf{e}_n$ is any vector perpendicular to $\mathbf{e}_t$, for example,

$$\mathbf{e}_n = \frac{1}{\sqrt{1+a^2}}(-a\mathbf{E}_x + \mathbf{E}_y).$$

Why is the torsion τ of this curve 0?

Figure 11.10: O'Reilly Primer, Exercise 3.1.

11. [OOR 3.2]

11. [OOR 3.2]

Exercise 3.2: Calculate the Serret-Frenet basis vectors for the space curve

$$\mathbf{r} = x\mathbf{E}_x + \frac{x^3}{3}\mathbf{E}_y.$$

It is convenient to describe these vectors as functions of x. In addition, show that the arc-length parameter s as a function of x is given by

$$s = s(x) = \int_{x_0}^{x} \sqrt{1+u^4}\,du + s(x_0).$$

Figure 11.11: O'Reilly Primer, Exercise 3.2.

12 Spring Force

12.1 Introduction

Think!

Question: What does it mean for a spring to be linear?

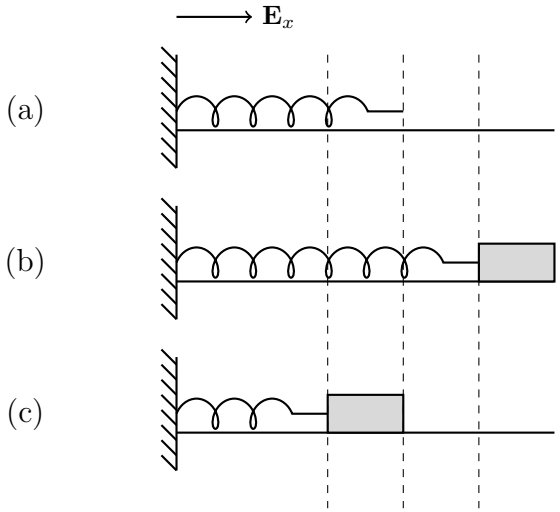
Answer

Consider a mass m suspended by a spring causing an extension δ. Suspending a mass $2m$ causes double the extension, 2δ.

Hooke's Law: The force generated by a spring is assumed to be linearly proportional to its extension/compression.

12.2 The Simple Harmonic Oscillator

Consider a block of mass m attached to a linear spring of stiffness k and unstretched length ℓ_0. Denote the current length by ℓ and the stretch by $\varepsilon = \ell - \ell_0$.



Think!

Question: What is the stretch in each case? What is the spring force?

Answer

(a) Unstretched: Its current length is ℓ_0 and its stretch is $\varepsilon = 0$.

(b) Extended: Its current length is ℓ and its stretch is $\varepsilon = \ell - \ell_0 > 0$. The spring force is $\mathbf{F}_s = -k\varepsilon\mathbf{E}_x$ pointing to the left.

(c) Compressed: Its current length is ℓ and its stretch is $\varepsilon = \ell - \ell_0 < 0$. The spring force is $\mathbf{F}_s = -k\varepsilon\mathbf{E}_x$ pointing to the right.

Note! Prescribing the spring force

The spring force always seeks to return the particle to the unstretched state. The same expression applies whether the spring is extended or compressed.

https://youtu.be/WtTDHW2JUVY

12.3 Formalism

In general:

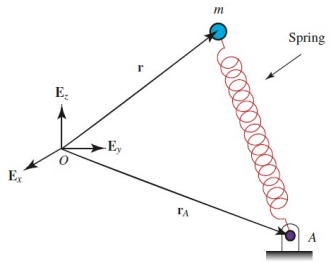


Figure 12.1: General spring configuration

Think!

Question: What is the prescription of the spring force in this general case?

Answer

...

The stretch is:

$$\varepsilon = \|\mathbf{r} - \mathbf{r}_A\| - \ell_0, \tag{12.1}$$

where ℓ_0 is the unstretched length. The magnitude the friction force $\mathbf{F}_s$ is a statement of Hooke's law.

$$\|\mathbf{F}_s\| = |K\left(\|\mathbf{r} - \mathbf{r}_A\| - \ell_0\right)|. \tag{12.2}$$

The spring force is:

$$\mathbf{F}_s = -K\left(\|\mathbf{r} - \mathbf{r}_A\| - \ell_0\right)\frac{\mathbf{r} - \mathbf{r}_A}{\|\mathbf{r} - \mathbf{r}_A\|}. \tag{12.3}$$

This direction is correct in both tension and compression.

https://youtu.be/YiOZregJx9w

12.4 Choosing the Origin

12.4.1 Horizontal Simple Harmonic Oscillator

Think!

Question: Derive the equations of motion of the harmonic oscillator with the origin at the free end of the spring when it is unstretched.

Answer

The equation of motion is

$$\ddot{x} + \frac{k}{m}x = 0, \tag{12.4}$$

where x is measured from the undeformed position.

Think!

Question: Derive the equations of motion with the origin at the fixed end of the spring.

Answer

...

12.4.2 Vertical Simple Harmonic Oscillator

By analogy, the equation of motion of a vertical harmonic oscillator is:

$$\ddot{x} + \frac{k}{m}x = 0, \tag{12.5}$$

where x is measured from the **static equilibrium** position.

https://www.youtube.com/watch?v=YE9N2f2qTqg

12.5 Summary

The spring force on a mass m at position $\mathbf{r}$, attached to a spring (stiffness K, unstretched length ℓ_0) with base at $\mathbf{r}_A$:

$$\mathbf{F}_s = -K(\|\mathbf{r} - \mathbf{r}_A\| - \ell_0)\,\frac{\mathbf{r} - \mathbf{r}_A}{\|\mathbf{r} - \mathbf{r}_A\|}. \tag{12.6}$$

12.6 Exercises

Complete this introduction to vibrations.

The following problems are from Set 09 – Spring Force (Chapter 8 of MKB). In each problem, find the equation of motion of the block.

1. [08-002] *(ans. $\omega_n = 12$ rad/s, $f_n = 1.910$ Hz)*

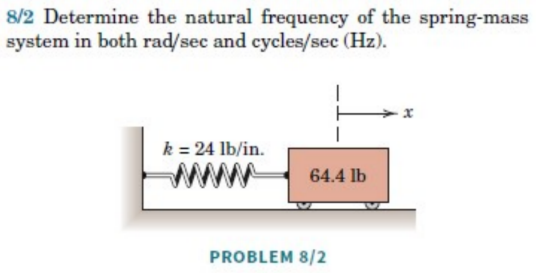
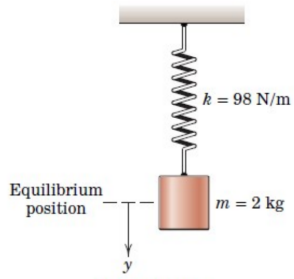


Figure 12.2: MKB 08-002.

2. [08-004] First determine the static deflection of the spring; take the origin at the statically deflected position. *(ans. $\delta_{st} = 0.200$ m, $\tau = 0.898$ s, $v_{\max} = 0.7$ m/s)*

8/4 For the spring-mass system shown, determine the static deflection δ_{st}, the system period τ, and the maximum velocity v_{max} which result if the cylinder is displaced 100 mm downward from its equilibrium position and released from rest.



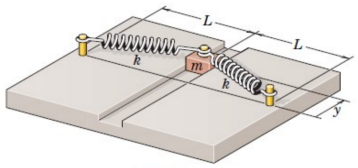
PROBLEM 8/4

Figure 12.3: MKB 08-004.

https://youtu.be/YE9N2f2qTqg

3. [08-019] *(ans.* $\ddot{y} + \frac{3k}{mL^2}y = 0$*)*

▶8/19 The slider of mass m is confined to the horizontal slot shown. The two springs each of constant k are linear. Derive the nonlinear equation of motion for small values of y, retaining terms of order y^3 and larger. Both springs are unstretched when $y = 0$. Neglect friction.



PROBLEM 8/19

Figure 12.4: MKB 08-019.

13 Friction Force

Most theories of friction arise from studies by the French scientist Charles Augustin de Coulomb (1736–1806).

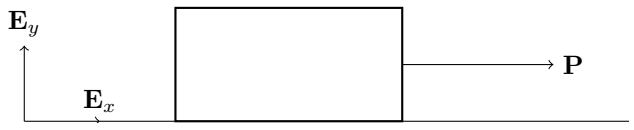
Historical Perspective: Charles-Augustin de Coulomb (1736–1806)

Figure 13.1: Portrait of Charles-Augustin de Coulomb (Wikimedia Commons, public domain).

French physicist and military engineer. Beyond his eponymous law of electrostatic force, Coulomb's 1785 prize-winning memoir *Théorie des machines simples* established the empirical laws of dry friction – that friction is roughly proportional to normal force and largely independent of contact area and sliding speed – still used in engineering today.

13.1 Introduction

Consider a block on a rough horizontal surface pushed with force $\mathbf{P} = P\mathbf{E}_x$.

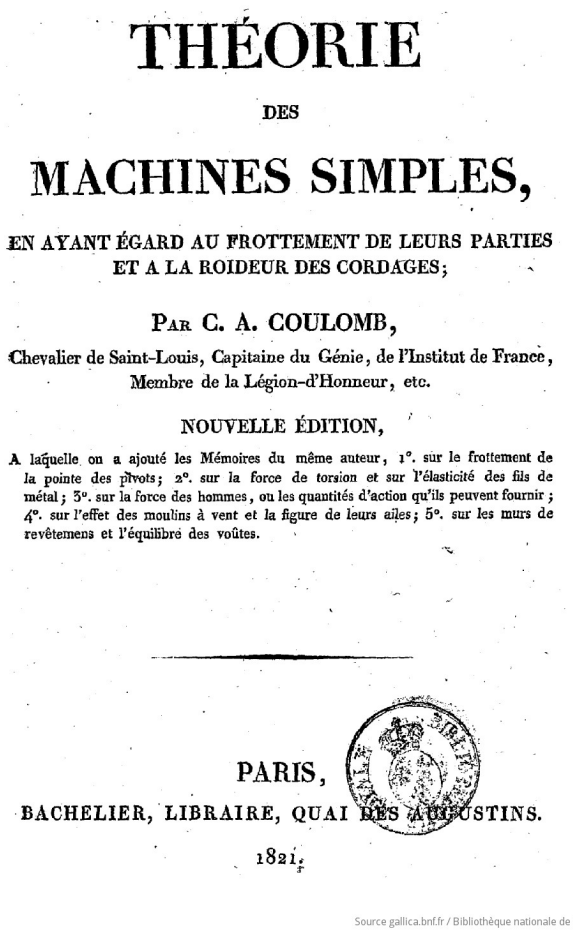


Upon increasing P, we observe:

- For small P, the block remains at rest.
- Beyond a critical value $P = P^*$, the block starts to move.
- Once in motion, a constant value $P = P^{**}$ is required to maintain constant speed.
- Both P^* and P^{**} are proportional to the normal force $\|\mathbf{N}\|$.

13.2 Formalism: Coulomb's Theory of Friction

Historical Perspective: Coulomb's Original Memoir (1785)



THÉORIE

DES

MACHINES SIMPLES,

EN AYANT ÉGARD AU FROTTEMENT DE LEURS PARTIES ET A LA ROIDEUR DES CORDAGES;

PAR C. A. COULOMB,

Chevalier de Saint-Louis, Capitaine du Génie, de l'Institut de France, Membre de la Légion-d'Honneur, etc.

NOUVELLE ÉDITION,

A laquelle on a ajouté les Mémoires du même auteur, 1°. sur le frottement de la pointe des pivots; 2°. sur la force de torsion et sur l'élasticité des fils de métal; 3°. sur la force des hommes, ou les quantités d'action qu'ils peuvent fournir; 4°. sur l'effet des moulins à vent et la figure de leurs ailes; 5°. sur les murs de revêtemens et l'équilibre des voûtes.

PARIS,

BACHELIER, LIBRAIRE, QUAI DES AUGUSTINS.

1821.

Source gallica.bnf.fr / Bibliothèque nationale de France

Figure 13.2: Page from Coulomb's *Théorie des machines simples, en ayant égard au frottement de leurs parties et à la roideur des cordages*, 1785 (Gallica / Bibliothèque nationale de France, public domain).

This prize memoir, submitted to the French Academy of Sciences, contains Coulomb's systematic experiments on sliding and rolling friction that gave rise to the formalism presented below.

We distinguish two types of friction: static and kinetic.

13.2.1 Static Friction

When the relative velocity between two surfacesis zero ($\mathbf{v}_{rel} = \mathbf{0}$), the friction force $\mathbf{F}_f$ is **unknown** and lies along the contact surface. We must solve for both its magnitude and direction. We say the static friction force points in the direction that **impedes motion**.

Think!

Question: Consider a block on a rough inclined surface that is not moving. Draw the FBD and write the equation of motion. Do you have as many equations as unknowns?

Answer

Yes. The friction and normal forces have unknown magnitudes, but the block's acceleration is known (it matches the surface acceleration). There are as many equations as unknowns.

The magnitude of the static friction force satisfies the **static friction criterion**:

$$\|\mathbf{F}_f\| \leq \mu_s \|\mathbf{N}\|, \tag{13.1}$$

where μ_s is the static friction coefficient.

13.2.2 Kinetic Friction

When two bodies are in relative motion ($\mathbf{v}_{rel} \neq \mathbf{0}$), the friction force is **known** (both in magnitude and direction):

$$\mathbf{F}_f = \mu_k \|\mathbf{N}\| \left(-\frac{\mathbf{v}_{rel}}{\|\mathbf{v}_{rel}\|} \right), \tag{13.2}$$

where μ_k is the kinetic friction coefficient. This prescription is used to solve for the motion of the system.

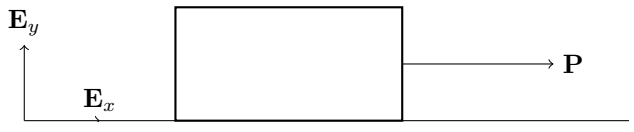
Remarks:

- μ_k is also written μ_d (dynamic friction coefficient).
- Kinetic friction is a prescription, not a constraint force.
- The magnitude of kinetic friction does not depend on speed.

https://youtu.be/_fvCFtpL3c8

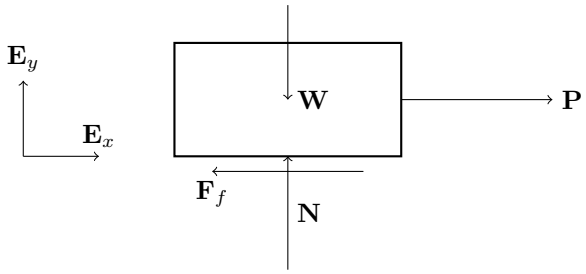
13.2.3 Example: Crate Pushed with Incremental Force

Consider a block on a rough horizontal surface pushed with force $\mathbf{P} = P\mathbf{E}_x$. At $t = 0$, the block is initially at rest and $P = 0$, then P is increased linearly.



1. Kinematics: $\mathbf{r} = x\mathbf{E}_x + y_0\mathbf{E}_y$, $\mathbf{v} = \dot{x}\mathbf{E}_x$, $\mathbf{a} = \ddot{x}\mathbf{E}_x$.
2. FBD with forces $\mathbf{P} = P\mathbf{E}_x$, $\mathbf{W} = -mg\mathbf{E}_y$, $\mathbf{N} = N\mathbf{E}_y$, $\mathbf{F}_f = F_f(-\mathbf{E}_x)$.

   When kinetic: $\mathbf{F}_f = -\mu_k \|\mathbf{N}\| \operatorname{sgn}(\dot{x})\mathbf{E}_x$.



3. BoLM:

$$P\mathbf{E}_x - mg\mathbf{E}_y + N\mathbf{E}_y - F_f\mathbf{E}_x = m\ddot{x}\mathbf{E}_x. \tag{13.3}$$

4. Let $P > 0$.

Initially, when the block is at rest, static friction is acting on it. Setting $\ddot{x} = 0$:

$$P = F_f, \quad N = mg. \tag{13.4}$$

The block stays at rest while $P \leq \mu_s mg$. The maximum static value is $P^* = \mu_s mg$.

If P exeeds this value P^*, then sliding begins, and the friction force becomes kinetic $\mathbf{F}_f = -\mu_k mg\mathbf{E}_x$ and:

$$\ddot{x} = \frac{P}{m} - \mu_k g. \tag{13.5}$$

Here, we assumed that $\operatorname{sgn}(\dot{x}) \geq 0$ because $\mathbf{P}$ always points to the right. The acceleration vector of the box is:

$$\mathbf{a} = \ddot{x}\mathbf{E}_x = \left(\frac{P}{m} - \mu_k g\right)\mathbf{E}_x. \tag{13.6}$$

If the block moves at constant speed, then $\ddot{x} = 0$ and $P^{**} = \mu_k mg$.

13.2.4 Example: A Box on a Ramp

(MKB problem 03/002.)

When it is unknown whether the block is in motion:

1. Assume sticking: $\mathbf{v}_{rel} = \mathbf{0}$.
2. Calculate the static friction force $\mathbf{F}_{trial}$.
3. Check the static friction criterion.
    - If satisfied: the block sticks. Analysis complete.
    - If not satisfied: the block slides. Apply kinetic friction and resolve.

Think!

If the block is initially placed with zero velocity on the ramp, and the static friction criterion is found to be violated, then $\mathbf{v}_{rel} = 0$ and we have an issue prescribing the kinetic friction force. At this transitional case, we take the kinetic friction force to point in the same direction as the static friction force had the static friction criterion not been violated.
In summary, if we

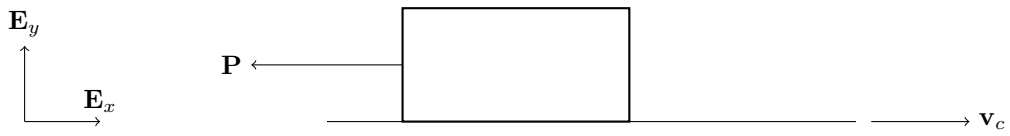
$$\mathbf{F}_f = \begin{cases} \mathbf{F}_{trial}, \text{ solved by BoLM and satisfying } ||\mathbf{F}_{trial}|| \leq \mu_s \mathbf{N} & \mathbf{v}_{rel} = 0 \text{ (friction is static)}, \\ -\mu_k ||\mathbf{N}|| \frac{\mathbf{F}_{trial}}{||\mathbf{F}_{trial}||} & \mathbf{v}_{rel} = 0 \text{ but } ||\mathbf{F}_{trial}|| \geq \mu_s \mathbf{N} \text{ (friction is kin} \\ -\mu_k ||\mathbf{N}|| \frac{\mathbf{v}_{rel}}{||\mathbf{v}_{rel}||} & \mathbf{v}_{rel} \neq 0 \text{ (friction is kinetic)} \end{cases} \tag{13.7}$$

13.3 Relative Velocity

A common misconception: if an object is moving, the friction acting on it must be kinetic. Counter-example: a crate on a moving truck. If the crate moves with the truck (same velocity), the friction between them is **static**. Only if the crate slides on the truck bed is the friction kinetic.

Other examples:

- **Box held fixed on a moving conveyor belt:** The belt moves, so $\mathbf{v}_{rel} = -\mathbf{v}_c \neq \mathbf{0}$ — friction is kinetic. The box is stationary in space but moving relative to the belt.



- **Person standing in an accelerating bus:** Friction is static. The impending motion of the person (relative to the bus) is toward the rear, so the static friction force points toward the front.
- **Car rolling without slipping:** Friction is static and points in the direction of motion, opposing the tendency of the tire to slip backward.

13.4 Normal and Friction Force Prescription

- For a particle moving on a **space curve**: $\mathbf{N} = N_n\mathbf{e}_n + N_b\mathbf{e}_b$ and $\mathbf{F}_f = F_f\mathbf{e}_t$.
- For a particle moving on a **surface** with normal $\mathbf{n}$ and tangent directions $\mathbf{t}_1$, $\mathbf{t}_2$: $\mathbf{N} = N\mathbf{n}$ and $\mathbf{F}_f = F_{f1}\mathbf{t}_1 + F_{f2}\mathbf{t}_2$.

https://www.youtube.com/watch?v=XNdP7Nk850s

13.5 Summary

- **Static friction**: $\|\mathbf{F}_f\| \le \mu_s\|\mathbf{N}\|$.
- **Kinetic (Coulomb) friction**: $\mathbf{F}_f = -\mu_k\|\mathbf{N}\|\dfrac{\mathbf{v}_{\text{rel}}}{\|\mathbf{v}_{\text{rel}}\|}$.

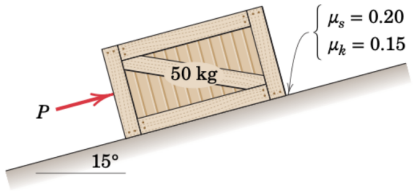
Strategy: assume sticking; check the static criterion. If violated, resolve with kinetic friction.

13.6 Exercises

The following problems are from Set 10 – Friction.

1. [MKB 03-002] Consider taking $\mathbf{E}_x$ along and $\mathbf{E}_y$ perpendicular to the incline. Assume sticking first; check the static criterion. *(ans. (a) $a = 1.118$ m/s^2 down incline; (b) $a = 0$; (c) $a = 2.04$ m/s^2 up incline)*

3/2 The 50-kg crate is stationary when the force P is applied. Determine the resulting acceleration of the crate if (*a*) $P = 0$, (*b*) $P = 150$ N, and (*c*) $P = 300$ N.

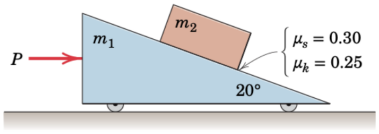


PROBLEM 3/2

Figure 13.3: MKB 03-002.

2. [MKB 03-030] Apply the 4 steps twice: first for the combined system, then for mass m_2 alone. *(ans.* $0.0577(m_1 + m_2)g \le P \le 0.745(m_1 + m_2)g$*)*

3/30 Determine the range of applied force P over which the block of mass m_2 will not slip on the wedge-shaped block of mass m_1. Neglect friction associated with the wheels of the tapered block.



PROBLEM 3/30

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Figure 13.4: MKB 03-030.

3. [MKB 03-047] Compare the Serret–Frenet basis here with Problem 03-043 from Set 07. *(ans.* $N = 156.2$ *lb,* $a = 27.7$ *ft/sec*^{2}*)*

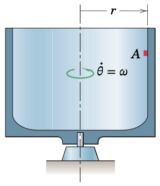
3/47 A 180-lb snowboarder has speed $v = 15$ ft/sec when in the position shown on the halfpipe. Determine the normal force on his snowboard and the magnitude of his total acceleration at the instant depicted. Use a value $\mu_k = 0.10$ for the coefficient of kinetic friction between the snowboard and the surface. Neglect the weight of the snowboard and assume that the mass center G of the snowboarder is 3 feet from the surface of the snow.

PROBLEM 3/47

Figure 13.5: MKB 03-047.

4. [03-049] Set up a cylindrical polar coordinate system with origin at the base of the cylinder axis; apply the 4 steps to the small object A. *(ans.* $\omega = \sqrt{\mu_s g/r}$*)*

3/49 A small object A is held against the vertical side of the rotating cylindrical container of radius r by centrifugal action. If the coefficient of static friction between the object and the container is μ_s, determine the expression for the minimum rotational rate $\dot{\theta} = \omega$ of the container which will keep the object from slipping down the vertical side.

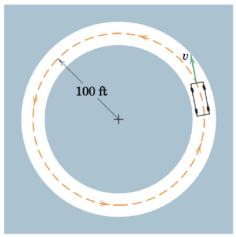


PROBLEM 3/49

Figure 13.6: MKB 03-049.

5. [03-050] This determines the maximum lateral acceleration before slipping. *(ans.* $a_n = 0.818g$, $F = 2460$ *lb)*

3/50 The standard test to determine the maximum lateral acceleration of a car is to drive it around a 200-ft-diameter circle painted on a level asphalt surface. The driver slowly increases the vehicle speed until he is no longer able to keep both wheel pairs straddling the line. If this maximum speed is 35 mi/hr for a 3000-lb car, determine its lateral acceleration capability a_n in g's and compute the magnitude F of the total friction force exerted by the pavement on the car tires.

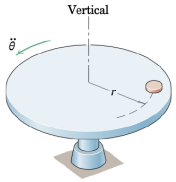


PROBLEM 3/50

Figure 13.7: MKB 03-050.

6. [03-071] Polar coordinate system with origin at the centre of the disk. *(ans.* $N = \frac{\mu_s g}{4\pi^2}\left(\frac{1}{r\alpha} - 1\right)$*)*

3/71 A small coin is placed on the horizontal surface of the rotating disk. If the disk starts from rest and is given a constant angular acceleration $\ddot{\theta} = \alpha$, determine an expression for the number of revolutions N through which the disk turns before the coin slips. The coefficient of static friction between the coin and the disk is μ_s.

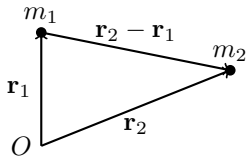


PROBLEM 3/71

Figure 13.8: MKB 03-071.

https://youtu.be/RDFfE0tFTnk

14 Gravitational Force Model

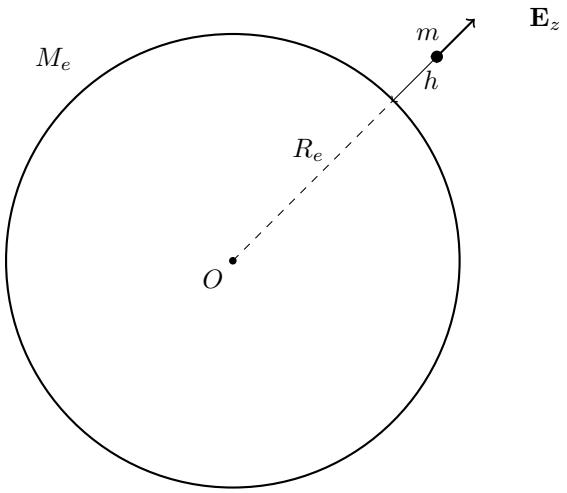


Call $\mathbf{r}_2 - \mathbf{r}_1 = \mathbf{r}_{2/1}$. According to Newton's law of gravitation, the force on m_1 due to m_2 is:

$$\mathbf{F}_1 = G\frac{m_1 m_2}{\left\|\mathbf{r}_{2/1}\right\|^2}\frac{\mathbf{r}_{2/1}}{\left\|\mathbf{r}_{2/1}\right\|} = G\frac{m_1 m_2}{\left\|\mathbf{r}_{2/1}\right\|^3}\mathbf{r}_{2/1}. \tag{14.1}$$

where $G = 6.67408 \times 10^{-11}$ m^3 kg^{-1} s^{-2} is the gravitational constant.

14.1 Example: Attraction Between Earth and an Object of Mass m



$$\mathbf{F} = G\frac{M_e m}{(R_e + h)^2}\left(-\mathbf{E}_z\right). \tag{14.2}$$

Since $h \ll R_e$, we have $(R_e + h)^2 \approx R_e^2$, giving:

$$\mathbf{F} = G\frac{M_e}{R_e^2} m \left(-\mathbf{E}_z\right) = mg\left(-\mathbf{E}_z\right), \tag{14.3}$$

where $g = GM_e/R_e^2 = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$.

Earth's parameters:

- $M_e = 5.972 \times 10^{24}$ kg
- $R_e = 6.371 \times 10^6$ m

https://youtu.be/nHnDuCWtacY

15 Power, Work, and Energy

15.1 Power and Work

The **mechanical power** of a force $\mathbf{F}$ acting on a particle with absolute velocity $\mathbf{v}$ is:

$$\mathcal{P} = \mathbf{F} \cdot \mathbf{v}. \tag{15.1}$$

The **work** done by $\mathbf{F}$ over the interval $[t_1, t_2]$ is:

$$W_{\mathbf{F},12} = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{v}\, dt = \int_{t_1}^{t_2} \mathbf{F} \cdot d\mathbf{r}, \tag{15.2}$$

since $d\mathbf{r} = \mathbf{v}\, dt$. So power is the derivative of work: $\mathcal{P} = \dot{W}$.

In SI units: work is in Newton meters (Joules) and power in Watts.

Notes:

- There are many representations of this integral depending on the coordinate system:
  - Cartesian: $d\mathbf{r} = dx\mathbf{E}_x + dy\mathbf{E}_y + dz\mathbf{E}_z$
  - Polar: $d\mathbf{r} = dr\mathbf{e}_r + r\, d\theta\mathbf{e}_\theta$
  - Serret-Frenet: $d\mathbf{r} = ds\, \mathbf{e}_t$
- This integral is **path dependent** — we need to know the particle's path to compute it.

Think!

Question: In what case would the work of a force be zero?

Answer

The work is zero if the force is perpendicular to the displacement/velocity, or if the force is applied at a fixed point.

Think!

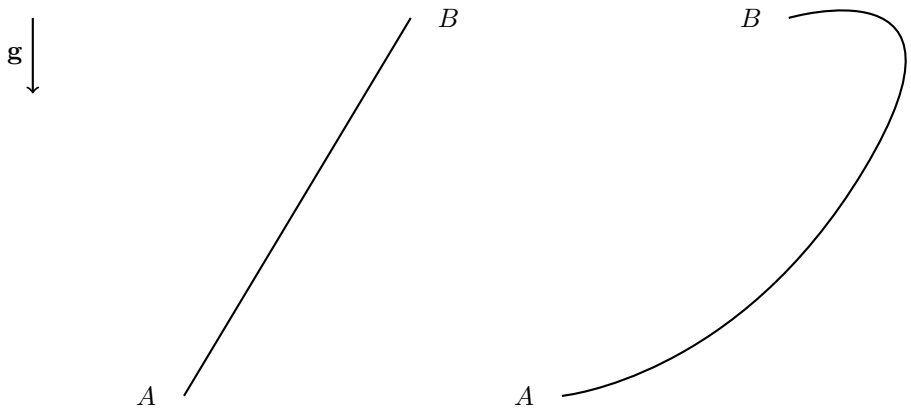
Question: When is the work of a force positive? When is it negative?

Answer

...

15.1.1 Example: Particle on a Rail

Consider two points A and B in the vertical plane connected by a smooth rail. A particle slides along the rail from B to A.



Think!

Question: What is the work done by forces on an object traveling from A to B on (a) a straight rail or (b) a curved path?

Answer

In both cases $\mathbf{F} = \mathbf{W} + \mathbf{N}$. Since $\mathbf{N} \perp d\mathbf{r}$, the normal force does no work. The work of weight is:

$$W_{\mathbf{W},1-2} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} mg\mathbf{E}_y \cdot d\mathbf{r} = mg\mathbf{E}_y \cdot \int_{\mathbf{r}_1}^{\mathbf{r}_2} d\mathbf{r} = mg\mathbf{E}_y \cdot \Delta\mathbf{r}_{1-2}.$$

$\Delta\mathbf{r}_{1-2}$ is the same for both rails, so weight does the same work regardless of path.

Think!

Question: What changes if the rail is rough?

Answer

With friction:

$$W_{1-2} = W_{\mathbf{W},1-2} + \int_{s_1}^{s_2} \mu_k ||\mathbf{N}||(-\mathbf{e}_t) \cdot ds\, \mathbf{e}_t = W_{\mathbf{W},1-2} - \mu_k \int_{s_1}^{s_2} ||\mathbf{N}|| ds.$$

Friction work is path dependent and dissipative ($W_{\mathbf{F}_f,1-2} < 0$). On the other hand, the work of the weight is positive.

Think!

Question: Can you think of a situation where the normal force is not workless?

Answer

A particle in an elevator is a common example. In this case, the normal force is not perpendicular to the motion, and it does work.

15.2 Kinetic Energy

The **kinetic energy** of a particle:

$$T = \frac{m}{2} \mathbf{v} \cdot \mathbf{v}. \tag{15.3}$$

In different bases:

$$T = \frac{1}{2} m(v_x^2 + v_y^2 + v_z^2) = \frac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2) = \frac{1}{2} m v^2.$$

15.3 Energy

Energy is defined as the ability to perform work.

15.4 The Work-Energy Theorem

The rate of change of kinetic energy equals the mechanical power of the resultant force $\mathbf{F}$ acting on the particle:

$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v} = \mathcal{P}. \tag{15.4}$$

Proof:

$$\frac{d}{dt}T = \frac{d}{dt}\left(\frac{1}{2}m\mathbf{v} \cdot \mathbf{v}\right) = m\mathbf{a} \cdot \mathbf{v} = \mathbf{F} \cdot \mathbf{v}.$$

Integral form (with respect to time):

$$T_2 - T_1 = W_{\mathbf{F},12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r}. \tag{15.5}$$

https://www.youtube.com/watch?v=bjpTKehJV7s

15.5 Implications of $\dot{T} = \mathbf{F} \cdot \mathbf{v}$ through examples

- Consider a free falling particle. The weight is the only force acting on it. If the particle is going up (opposite to the weight), $\dot{T} < 0$. If the particle is going down (same direction of the weight), $\dot{T} > 0$.
- Consider a person standing on a train while the train is accelerating:

  $$\mathbf{F} = \mathbf{W} + \mathbf{F}_f + \mathbf{N} \tag{15.6}$$
  $$\mathbf{F} \cdot \mathbf{v} = \mathbf{F}_f \cdot \mathbf{v} \geq 0 \implies \dot{T} > 0. \tag{15.7}$$

  The kinetic energy of the person is increasing.
- Consider a box sliding on a rough surface:

  $$\mathbf{F} = \mathbf{F}_f + \mathbf{W} + \mathbf{N} \tag{15.8}$$
  $$\mathbf{F} \cdot \mathbf{v} = \mathbf{F}_f \cdot \mathbf{v} < 0 \implies \dot{T} < 0. \tag{15.9}$$

  The kinetic energy of the box is decreasing until it stops.
- Consider a particle on a smooth rail in the horizontal plane:

  $$\mathbf{F} = \mathbf{W} + \mathbf{N} \tag{15.10}$$
  $$\mathbf{F} \cdot \mathbf{v} = 0 \implies \dot{T} = 0. \tag{15.11}$$

  The kinetic energy of the particle is conserved.

The same analysis/conclusions can be achieved using the work-energy theorem.

15.6 Another way to derive the Work-Energy theorem

Starting with the balance of linear momentum we can derive the work-energy theorem:

$$\mathbf{F} = m\mathbf{a} \tag{15.12}$$

$$\mathbf{F} \cdot d\mathbf{r} = m\mathbf{a} \cdot d\mathbf{r} \tag{15.13}$$

$$\int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r} = \int_{r_1}^{r_2} m\mathbf{a} \cdot d\mathbf{r} \tag{15.14}$$

$$W_{1-2} = \int_{t_1}^{t_2} m\mathbf{a} \cdot \mathbf{v}\, dt = \int_{t_1}^{t_2} \dot{T}\, dt = T_2 - T_1, \tag{15.15}$$

where the work of a force is

$$W_{1-2} = \int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r}.$$

15.7 Conservative Forces

Forces whose work depends only on the endpoints of a path are **conservative**. From the previous examples, weight (constant force) is conservative; friction is not.

A force $\mathbf{F}_c$ is conservative if there exists a scalar **potential energy** function $U = U(\mathbf{r})$ such that:

$$\mathbf{F}_c = -\frac{\partial U}{\partial \mathbf{r}} = -\text{grad}_{\mathbf{r}} U. \tag{15.16}$$

Then:

$$W_{1-2} = \int_{\mathbf{r}(t_1)}^{\mathbf{r}(t_2)} \mathbf{F}_c \cdot d\mathbf{r} = -\int_{\mathbf{r}(t_1)}^{\mathbf{r}(t_2)} \frac{\partial U}{\partial \mathbf{r}} \cdot d\mathbf{r} = -U_2 + U_1 = -\Delta_{1-2} U. \tag{15.17}$$

If potential energy decreases ($U_2 \leq U_1$), the work is positive and kinetic energy increases.

Think!

Question: Consider a particle thrown from the top of a building. Describe the changes in kinetic and potential energy of the particle during its motion.

Answer

When the particle is still high, its potential energy is high and its kinetic energy is low. When the particle has traveled downward, its potential energy is low and its kinetic energy is high.

15.7.1 Constant Forces

Constant force $\mathbf{C}$ is conservative with potential $U = -\mathbf{C} \cdot \mathbf{r}$.

Proof: Propose $U = -\mathbf{C} \cdot \mathbf{r} = U(\mathbf{r}) = -C_x x - C_y y - C_z z$.

$$-\text{grad}_r U = C_x \mathbf{E}_x + C_y \mathbf{E}_y + C_z \mathbf{E}_z = \mathbf{C}. \tag{15.18}$$

We found a U from which $\mathbf{C}$ is derivable.

The weight near Earth's surface: $U_{\mathbf{W}} = mgy$.

15.7.2 Spring Force

The spring force is conservative with potential:

$$U_s = \frac{1}{2} K \varepsilon^2.$$

$U_s > 0$ in both compression and tension — it represents the spring's capacity to do work.

Proof:

$$-\frac{\partial U_s}{\partial \mathbf{r}} = K\varepsilon \frac{\mathbf{r}_A - \mathbf{r}}{||\mathbf{r}_A - \mathbf{r}||} = \mathbf{F}_s.$$

15.7.3 Gravitational Force

The gravitational force $\mathbf{F}_G = G\frac{M_e m}{(R_e+h)^2}(-\mathbf{e}_r)$ is conservative with:

$$U = -\frac{GM_e m}{r}.$$

Drag, friction, tension in inextensible cables, and normal forces are nonconservative.

15.8 Energy and Its Conservation

If only conservative forces act, define total mechanical energy $E = T + U$. Then:

$$\dot{T} = \mathbf{F}_c \cdot \mathbf{V} = -\frac{\partial U}{\partial \mathbf{r}} \cdot \dot{\mathbf{r}} = -\dot{U}$$
$$\dot{T} = -\dot{U} \implies \dot{E} = \dot{T} + \dot{U} = 0.$$

E is conserved when all work-doing forces are conservative, ie. $\mathbf{F} \cdot \mathbf{v} = \mathbf{F}_c \cdot \mathbf{v}$.

If nonconservative forces also act:

$$E_B - E_A = W_{\mathbf{F}_{nc}, AB}. \tag{15.19}$$

15.8.1 Example: Smooth vs. Rough Rail

Smooth rail: $\mathbf{F} \cdot \mathbf{v} = \mathbf{W} \cdot \mathbf{v} + \mathbf{N} \cdot \mathbf{v} = \mathbf{W} \cdot \mathbf{v} = -\dot{U}$, so $\dot{E} = 0$.

Rough rail: $\dot{E} = \mathbf{F}_f \cdot \mathbf{v} < 0$, so energy bleeds off over time. Integrating:

$$E(t_2) = E(t_1) + W_{\mathbf{F}_f, 1-2} < E(t_1).$$

15.9 Summary

Power of force $\mathbf{F}$ on particle with velocity $\mathbf{v}$: $P = \mathbf{F} \cdot \mathbf{v}$.

Work–energy theorem: $T_B - T_A = W_{\mathbf{F},AB}$, where $T = \frac{1}{2} m \mathbf{v} \cdot \mathbf{v}$.

Conservative forces satisfy $W_{F_c,AB} = -(U_B - U_A)$. Including non-conservative work:

$$(T_B + U_B) - (T_A + U_A) = W_{\mathbf{F}_{nc},AB}. \tag{15.20}$$

If all work-doing forces are conservative, **energy is conserved**: $E = T + U = \text{const}$.

15.10 Exercises

The following problems are from Set 11 – Power, Work and Energy.

1. [MKB 03-128] Only the gravitational force acts on the spacecraft, so energy is conserved. Use conservation of energy. *(ans.* $v_B = 26\,300$ *km/h)*

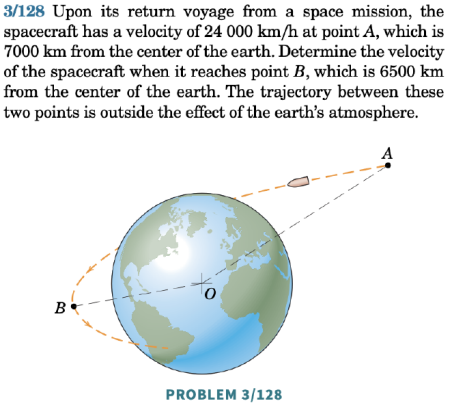
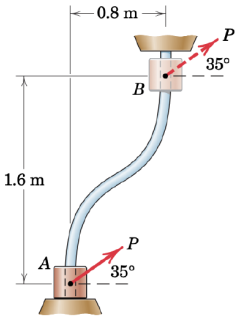


Figure 15.1: MKB 03-128.

2. [MKB 03-080] *(ans. (a) no motion; (b)* $v_B = 5.62$ *m/s)*

3/80 The 2-kg collar is at rest in position A when the constant force P is applied as shown. Determine the speed of the collar as it passes position B if (a) $P = 25$ N and (b) $P = 40$ N. The curved rod lies in a vertical plane, and friction is negligible.

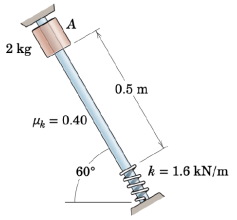


PROBLEM 3/80

Figure 15.2: MKB 03-080.

3. [MKB 03-084] *(ans. (a) $v = 2.56$ m/s; (b) $x = 98.9$ mm)*

3/84 The 2-kg collar is released from rest at A and slides down the inclined fixed rod in the vertical plane. The coefficient of kinetic friction is 0.40. Calculate (a) the velocity v of the collar as it strikes the spring and (b) the maximum deflection x of the spring.

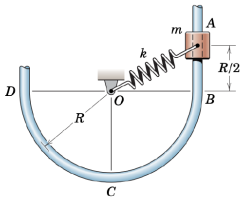


PROBLEM 3/84

Figure 15.3: MKB 03-084.

4. [MKB 03-094] *(ans. (a) $N_B = 48$ N right; (b) $N'_B = 29.4$ N right; (c) $N_C = 17.63$ N down; (d) $N_D = 29.4$ N left)*

3/94 The collar of mass m is released from rest while in position A and subsequently travels with negligible friction along the vertical-plane circular guide. Determine the normal force (magnitude and direction) exerted by the guide on the collar (a) just before the collar passes point B, (b) just after the collar passes point B (i.e., the collar is now on the curved portion of the guide), (c) as the collar passes point C, and (d) just before the collar passes point D. Use the values $m = 0.4$ kg, $R = 1.2$ m, and $k = 200$ N/m. The unstretched length of the spring is $0.8R$.

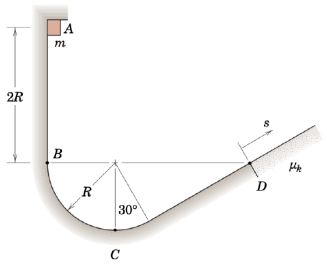


PROBLEM 3/94

Figure 15.4: MKB 03-094.

5. [MKB 03-101] *(ans. (a) $N_B = 4mg$; (b) $N_C = 7mg$; (c) $s = \frac{4R}{\sqrt{3}(1+\mu_k)}$)*

3/101 The small slider of mass m is released from rest while in position A and then slides along the vertical-plane track. The track is smooth from A to D and rough (coefficient of kinetic friction μ_k) from point D on. Determine (a) the normal force N_B exerted by the track on the slider just after it passes point B, (b) the normal force N_C exerted by the track on the slider as it passes the bottom point C, and (c) the distance s traveled along the incline past point D before the slider stops.



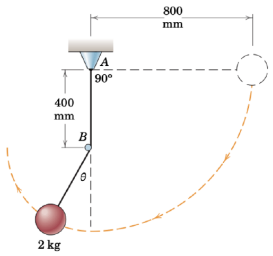
PROBLEM 3/101

Figure 15.5: MKB 03-101.

6. [MKB 03-263] *(ans. $R = 46.7$ N)*

6. [MKB 03-263] (ans. $R = 46.7$ N)

3/263 The simple 2-kg pendulum is released from rest in the horizontal position. As it reaches the bottom position, the cord wraps around the smooth fixed pin at B and continues in the smaller arc in the vertical plane. Calculate the magnitude of the force R supported by the pin at B when the pendulum passes the position $\theta = 30°$.



PROBLEM 3/263

Figure 15.6: MKB 03-263.

7. [OOR Exercise 5-1]

7. [OOR Exercise 5-1]

Exercise 5.1: Give examples to illustrate the following statement: "Every constant force is conservative, but not all conservative forces are constant."

Figure 15.7: O'Reilly Primer, Exercise 5.1.

8. [OOR Exercise 5-4]

8. [OOR Exercise 5-4]

Exercise 5.4: A particle is moving on a smooth horizontal plane. A gravitational force $-mg\mathbf{E}_z$ acts on the particle. If the plane is given a vertical motion, then why does the normal force acting on the particle perform work? Using this example, show that the normal force is not a conservative force.

Figure 15.8: O'Reilly Primer, Exercise 5.4.

9. [OOR Exercises 5-6 & 5-7]

9. [OOR Exercise 5-6 & 5-7]

Exercise 5.6: A particle is free to move on a smooth plane $z = 0$. It is attached to a fixed point O by a linear spring of stiffness K and unstretched length ℓ_0. A gravitational force $-mg\mathbf{E}_z$ acts on the particle. Starting from the work-energy theorem, prove that

$$E = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{K}{2}\left(r - \ell_0\right)^2$$

is conserved.

Exercise 5.7: Consider the same system discussed in Exercise 5.6 but in this case assume that the surface is rough. Show that

$$\dot{E} = -\mu_d mg\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}.$$

Figure 15.9: O'Reilly Primer, Exercises 5.6 and 5.7.

10. [OOR Exercises 5-8 & 5-9]

19. [OOR Exercise 5-8 & 5-9]

Exercise 5.8: For any vector $\mathbf{b}$ show that

$$\frac{d}{dt}\|\mathbf{b}\| = \frac{\mathbf{b}\cdot\dot{\mathbf{b}}}{\|\mathbf{b}\|}.$$

Using this result, show that if the magnitude of $\mathbf{b}$ is constant, then any change in $\mathbf{b}$ must be normal to $\mathbf{b}$ or $\dot{\mathbf{b}} = \mathbf{0}$. It is illuminating to apply the identity to the vectors $\mathbf{e}_r$ and $\mathbf{E}_x$.

Exercise 5.9: Newton's gravitational force on a particle of mass m due to a body of mass M is

$$\mathbf{F}_N = -\frac{GMm}{\|\mathbf{r}\|^2}\frac{\mathbf{r}}{\|\mathbf{r}\|},$$

where G is a constant. With the assistance of the identity established in Exercise 5.8, show that $\mathbf{F}_N$ is conservative with a potential energy

$$U_N = -\frac{GMm}{\|\mathbf{r}\|}.$$

Figure 15.10: O'Reilly Primer, Exercises 5.8 and 5.9.

16 Momenta and Impulses of Particles

16.1 Linear Momentum and Its Conservation

Recall the linear momentum $\mathbf{G} = m\mathbf{v}$.

16.1.1 Linear Impulse and Linear Momentum

The integral form of the balance of linear momentum:

$$\mathbf{G}(t_1) - \mathbf{G}(t_0) = \int_{t_0}^{t_1} \mathbf{F}\, dt. \tag{16.1}$$

The time integral of a force is its **linear impulse**. This form is more general than $\mathbf{F} = m\mathbf{a}$ because it does not require $\mathbf{v}$ to be differentiable.

https://youtu.be/-AHQOYy3VBk

16.1.2 Conservation of Linear Momentum

Think!

Question: Under what conditions is the linear momentum of a system conserved?

Answer

...

Think!

Question: Under what conditions is the linear momentum of a system conserved in a direction $\mathbf{c}$?

Answer

Suppose that the component of $\mathbf{G}$ in the direction of a given vector $\mathbf{c}$ is

conserved:

$$\frac{d}{dt}(\mathbf{G}\cdot\mathbf{c}) = 0. \tag{16.2}$$

This means that

$$\dot{\mathbf{G}}\cdot\mathbf{c} + \mathbf{G}\cdot\dot{\mathbf{c}} = \mathbf{F}\cdot\mathbf{c} + \mathbf{G}\cdot\dot{\mathbf{c}} = 0. \tag{16.3}$$

Thus, given a vector $\mathbf{c}$,

$$\mathbf{G}\cdot\mathbf{c} \text{ is conserved if, and only if, } \mathbf{F}\cdot\mathbf{c} + \mathbf{G}\cdot\dot{\mathbf{c}} = 0.$$

If $\mathbf{c}$ is a constant vector, $\mathbf{G}\cdot\mathbf{c}$ is conserved if, and only if, $\mathbf{F}\cdot\mathbf{c} = 0$. From the BoLM, this means that there is no force in this constant direction.

https://youtu.be/_3a0CbiLJ18

16.1.3 Example

Example

Question: Is the linear momentum of a projectile conserved? In a certain direction?

Answer

With $\mathbf{W} = -mg\mathbf{E}_y$, linear momentum is conserved in the $\mathbf{E}_x$ direction (no force there) but not in $\mathbf{E}_y$.

16.2 The Moment of a Force

Think!

Question: How do you calculate the moment of a force about a point?

Answer

The moment of force $\mathbf{F}$ applied at A about point P is:

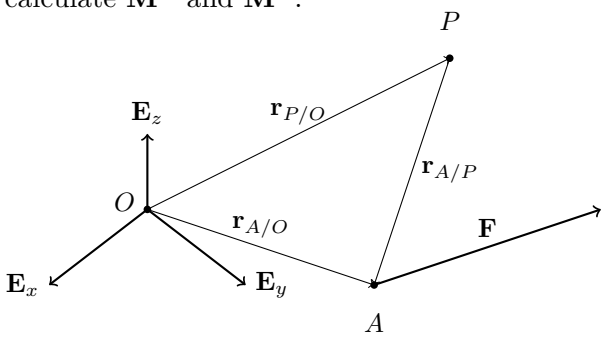
$$\mathbf{M}^P = (\mathbf{r}_A - \mathbf{r}_P)\times\mathbf{F}. \tag{16.4}$$

Example

Question: Given

$$\begin{aligned}\mathbf{F} &= F_x\mathbf{E}_x + F_y\mathbf{E}_y + F_z\mathbf{E}_z,\\ \mathbf{r}_P &= -2\mathbf{E}_x + 3\mathbf{E}_y + \mathbf{E}_z,\\ \mathbf{r}_A &= -\mathbf{E}_x + 2\mathbf{E}_y - \mathbf{E}_z,\end{aligned}$$

calculate $\mathbf{M}^O$ and $\mathbf{M}^P$.



Answer

...

https://youtu.be/qiFdls4aFsw

16.3 Angular Momentum and Its Conservation

Let $\mathbf{r}$ be the position vector of a particle relative to a fixed point O, and $\mathbf{v}$ its absolute velocity. The **angular momentum** relative to O:

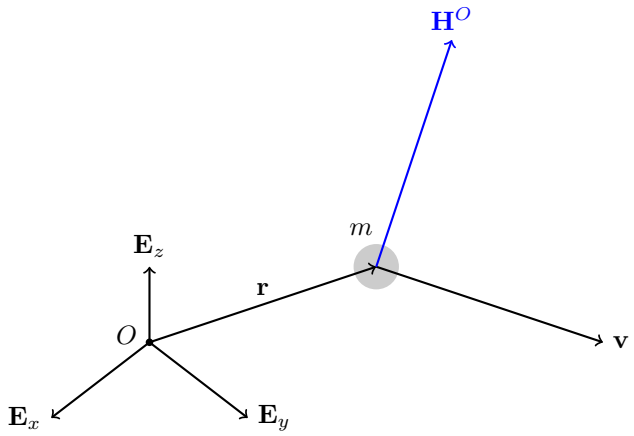
$$\mathbf{H}^O = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{G}. \tag{16.5}$$

In Cartesian coordinates:

$$\mathbf{H}^O = m\left(y\dot{z} - z\dot{y}\right)\mathbf{E}_x + m\left(z\dot{x} - x\dot{z}\right)\mathbf{E}_y + m\left(x\dot{y} - y\dot{x}\right)\mathbf{E}_z. \tag{16.6}$$

In cylindrical-polar coordinates:

$$\mathbf{H}^O = -mzr\dot{\theta}\mathbf{e}_r + m\left(z\dot{r} - r\dot{z}\right)\mathbf{e}_\theta + mr^2\dot{\theta}\mathbf{E}_z. \tag{16.7}$$



https://youtu.be/RohK7Sk-pCg

16.3.1 Angular Momentum Theorem

From the BoLM:

$$\dot{\mathbf{H}}^O = \frac{d}{dt}(\mathbf{r} \times m\mathbf{v}) = \underbrace{\mathbf{v} \times m\mathbf{v}}_{\mathbf{0}} + \mathbf{r} \times m\dot{\mathbf{v}} = \mathbf{r} \times \mathbf{F}. \tag{16.8}$$

So $\dot{\mathbf{H}}^O = \mathbf{M}^O$ (the angular momentum theorem).

https://youtu.be/5oHdJJfxGvk

16.3.2 Conservation of Angular Momentum

Think!

Question: Under what conditions is the angular momentum of a system conserved?

Answer

...

Think!

Question: Under what conditions is $\mathbf{H}^O \cdot \mathbf{c}$ conserved?

Answer

Suppose that the component of $\mathbf{H}^O$ in the direction of a given vector $\mathbf{c}$ is

conserved:

$$\frac{d}{dt}\left(\mathbf{H}^O \cdot \mathbf{c}\right) = 0. \tag{16.9}$$

Then,

$$\frac{d}{dt}\left(\mathbf{H}^O \cdot \mathbf{c}\right) = \dot{\mathbf{H}}^O \cdot \mathbf{c} + \mathbf{H}^O \cdot \dot{\mathbf{c}} = (\mathbf{r} \times \mathbf{F}) \cdot \mathbf{c} + \mathbf{H}^O \cdot \dot{\mathbf{c}}. \tag{16.10}$$

Consequently, for a given vector $\mathbf{c}$,

$$\mathbf{H}^O \cdot \mathbf{c} \text{ is conserved if, and only if, } (\mathbf{r} \times \mathbf{F}) \cdot \mathbf{c} + \mathbf{H}^O \cdot \dot{\mathbf{c}} = 0.$$

If $\mathbf{c}$ is a constant vector, then $\mathbf{H}^O \cdot \mathbf{c}$ is conserved if, and only if, $(\mathbf{r} \times \mathbf{F}) \cdot \mathbf{c} = 0$.

https://youtu.be/9TFLI5Tjiv8

16.3.3 Central Force Problems

Example

Question: Consider the motion of the Earth and the Moon. What are the forces acting on the Moon? Are any quantities conserved?

Answer

Yes. Energy and angular momentum are conserved.

A **central force** problem is one where $\mathbf{F}$ is parallel to $\mathbf{r}$. The angular momentum theorem implies $\dot{\mathbf{H}}^O = \mathbf{r} \times \mathbf{F} = \mathbf{0}$, so

$$\mathbf{H}^O = h\mathbf{h} = \text{constant} = \mathbf{r} \times m\mathbf{v}, \tag{16.11}$$

where h and $\mathbf{h}$ are constant.

The vectors $\mathbf{r}$ and $\mathbf{v}$ form a plane with constant unit normal $\mathbf{h}$. This plane passes through O and is fixed. Given initial conditions $\mathbf{r}(t_0)$ and $\mathbf{v}(t_0)$, we can choose a cylindrical polar coordinate system such that $\mathbf{E}_z = \mathbf{h}$, $\mathbf{r} = r\mathbf{e}_r$, and $\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$. To do this, it suffices to choose $\mathbf{E}_z$ so that

$$\mathbf{H}^O = h\mathbf{E}_z = \mathbf{r}(t_0) \times m\mathbf{v}(t_0). \tag{16.12}$$

16.3.4 Kepler's Problem

The gravitational force on a planet of mass m by the sun of mass M is conservative:

$$\begin{aligned} \mathbf{F} &= -\frac{GmM}{\|\mathbf{r}\|^2}\frac{\mathbf{r}}{\|\mathbf{r}\|} = -\frac{\partial U}{\partial \mathbf{r}}, \\ U &= -\frac{GmM}{\|\mathbf{r}\|}. \end{aligned} \tag{16.13}$$

The BoLM in cylindrical polar coordinates yields:

$$\begin{aligned} m\ddot{r} - mr\dot{\theta}^2 &= -\frac{GMm}{r^2}, \\ mr\ddot{\theta} + 2m\dot{r}\dot{\theta} &= 0. \end{aligned} \tag{16.14}$$

Two conserved quantities:

$$E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{GMm}{r}, \tag{16.15}$$

$$h = \mathbf{H}^O \cdot \mathbf{E}_z = mr^2\dot{\theta}. \tag{16.16}$$

Watch this video on Kepler's laws.

16.3.5 Particle on a Smooth Cone

Show that $\mathbf{H}^O \cdot \mathbf{E}_z$ is conserved.

16.4 Summary

Linear impulse–momentum: $\int_{t_A}^{t_B} \mathbf{F}\, dt = \mathbf{G}_B - \mathbf{G}_A$.

Angular momentum about fixed O: $\mathbf{H}^O = \mathbf{r} \times m\mathbf{v}$.

BoAM: $\mathbf{M}^O = \dot{\mathbf{H}}^O$.

Conservation of $\mathbf{G}$ if $\mathbf{F} = \mathbf{0}$; conservation of $\mathbf{H}^O$ if $\mathbf{M}^O = \mathbf{0}$.

https://youtu.be/MEwpRoLB7KM

16.5 Exercises

The following problems are from Set 12 – Impulse and Momentum.

1. [MKB 03-149] Straightforward application of the linear impulse–momentum equation. *(ans.* $v = 1.218$ *m/s down)*

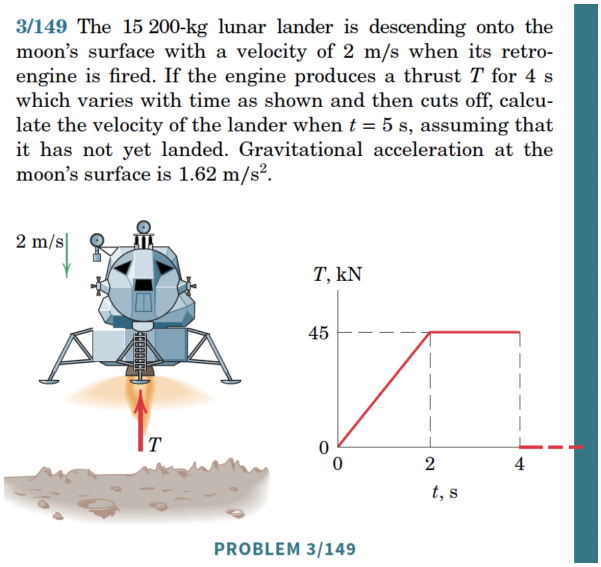


Figure 16.1: MKB 03-149.

2. [MKB 03-159] The block is subjected to a time-varying force $P(t)$ shown in the plot; $P = 0$ for $t > 3$ s. *(ans.* $t_s = 3.69$ *s)*

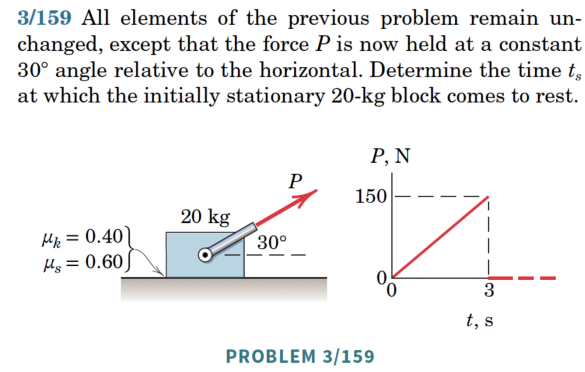
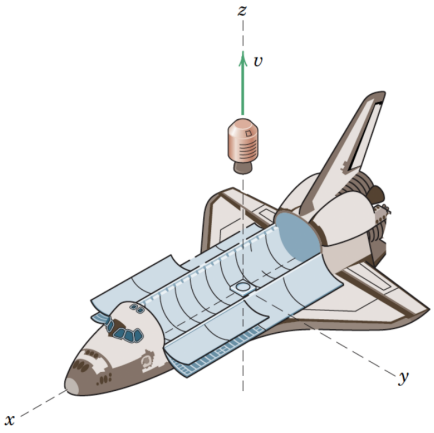


Figure 16.2: MKB 03-159.

https://youtu.be/ugj4fefUN4c

3. [MKB 03-161] Use Newton's third law. Recall $\int_{x_A}^{x_B} f(x)\,dx = F_{\text{avg}}(x_B - x_A)$. *(ans.* $v_f = 0.00264$ *m/s,* $F_{\text{avg}} = 59.5$ *N)*

3/161 The space shuttle launches an 800-kg satellite by ejecting it from the cargo bay as shown. The ejection mechanism is activated and is in contact with the satellite for 4 s to give it a velocity of 0.3 m/s in the z-direction relative to the shuttle. The mass of the shuttle is 90 Mg. Determine the component of velocity v_f of the shuttle in the minus z-direction resulting from the ejection. Also find the time average F_{av} of the ejection force.



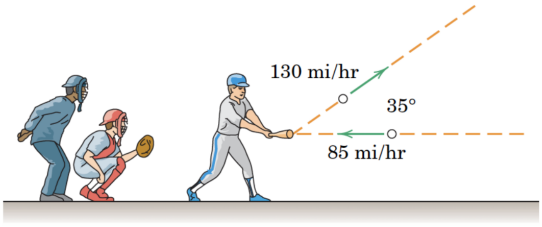
PROBLEM 3/161

Figure 16.3: MKB 03-161.

https://youtu.be/Q-F7ew3jFuQ

4. [03-167] *(ans.* $R_x = 559$ *lb,* $R_y = 218$ *lb)*

3/167 The baseball is traveling with a horizontal velocity of 85 mi/hr just before impact with the bat. Just after the impact, the velocity of the $5\frac{1}{8}$-oz ball is 130 mi/hr directed at 35° to the horizontal as shown. Determine the x- and y-components of the average force **R** exerted by the bat on the baseball during the 0.005-sec impact. Comment on the treatment of the weight of the baseball (*a*) during the impact and (*b*) over the first few seconds after impact.

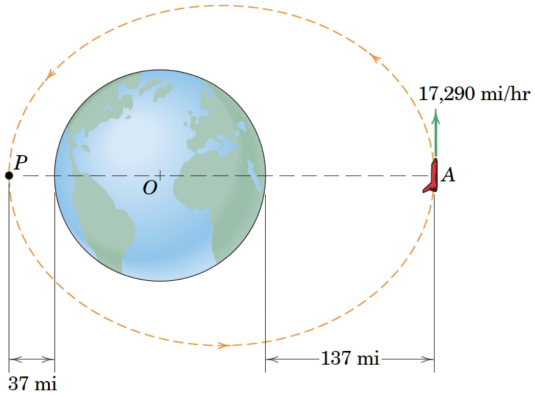


PROBLEM 3/167

Figure 16.4: MKB 03-167.

5. [03-177] Central force problem – what quantities are conserved? *(ans.* $v_P = 17\,723$ *mi/hr)*

3/177 Just after launch from the earth, the space-shuttle orbiter is in the 37 × 137–mi orbit shown. At the apogee point A, its speed is 17,290 mi/hr. If nothing were done to modify the orbit, what would be its speed at the perigee P? Neglect aerodynamic drag. (Note that the normal practice is to add speed at A, which raises the perigee altitude to a value that is well above the bulk of the atmosphere.)

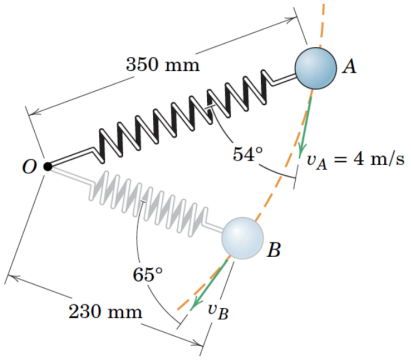


PROBLEM 3/177

Figure 16.5: MKB 03-177.

6. [03-185] *(ans.* $v_B = 5.43$ *m/s)*

3/185 A particle of mass m moves with negligible friction on a horizontal surface and is connected to a light spring fastened at O. At position A the particle has the velocity $v_A = 4$ m/s. Determine the velocity v_B of the particle as it passes position B.



PROBLEM 3/185

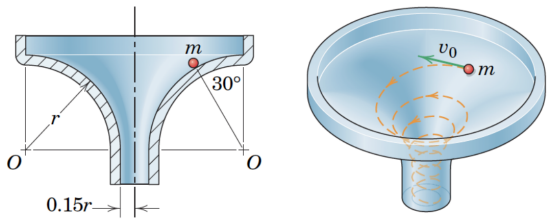
Figure 16.6: MKB 03-185.

https://youtu.be/V3ussTTlrs8

7. [03-181] *(ans.* $|H| = 389\ N \cdot m \cdot s$, $|M| = 260\ N \cdot m$*)*

8. [03-192] Label r as R and the requested angle θ as β. The particle moves on a surface of revolution $z^2 + (r - 1.15R)^2 = R^2$. Use conservation of total energy and $\mathbf{E}_z$-component of $\mathbf{H}^O$. *(ans.* $\theta = 52.9^\circ$*)*

3/192 A particle is launched with a horizontal velocity $v_0 = 0.55$ m/s from the 30° position shown and then slides without friction along the funnel-like surface. Determine the angle θ which its velocity vector makes with the horizontal as the particle passes level O-O. The value of r is 0.9 m.



PROBLEM 3/192

Figure 16.7: MKB 03-192.

https://youtu.be/jjHTcoIAQmY

17 Collisions

17.1 Collision of Two Bodies

Think!

Question: What happens in reality when two cars collide?

Answer

In reality, colliding bodies deform — possibly permanently — and their rotational inertia may change.

Ignoring rotational motion, the simplest model uses a mass particle for each body. Since particles do not deform, we introduce the **coefficient of restitution** e.

17.2 Frictionless, Oblique, Central Impact of Two Bodies

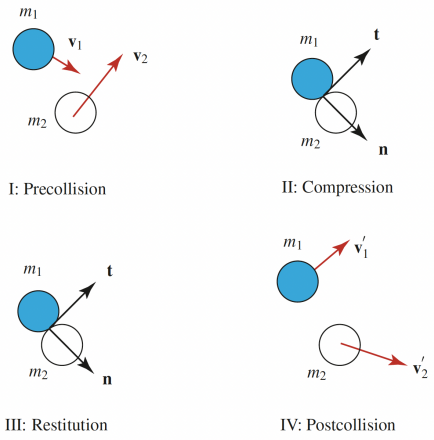


Fig. 6.4 *Four stages of a collision.*

Figure 17.1: Figure from O. M. O'Reilly's Dynamics Primer

17.2.1 Assumptions

- Model each body as a mass particle.
- We ignore the rotational inertias of bodies (purely translational motion).
- As the bodies are in a state of purely translational motion, all the points on the body have the same velocity.
- The position vector of the mass particle is defined to coincide with the center of mass of the body it models.
- No friction between particles during collision — all impact forces are along $\mathbf{n}$.
- The duration of compression $(t_1 - t_0)$ and restitution $(t_2 - t_1)$ are very small.
- At time t_1, both particles have the same velocity along $\mathbf{n}$: v_{II}.
- $\mathbf{F}_{1C} = -\mathbf{F}_{2C}$, $\mathbf{F}_{1R} = -\mathbf{F}_{2R}$.

17.2.2 Impact Stages

Let $\mathbf{n}$ be the common unit normal between the bodies when they collide. This vector is normal to the lateral surfaces of both bodies at the unique point of contact. Define a set of orthonormal vectors $\{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$ by selecting $\mathbf{t}_1$ and $\mathbf{t}_2$ to be unit tangent vectors to the lateral surface of both bodies at the point of contact. The impact is assumed to be such that these vectors are constant for the duration of the impact. At the end of stage II, the velocities of the bodies in the direction of $\mathbf{n}$ are assumed to be equal $(= v_{II})$.

17.2.3 Linear Impulses During Impact

Think!

Question: Draw the FBDs of m_1 and m_2 separately during compression and restitution.

Answer

- $\mathbf{F}_{1_d} = F_{1_d}\mathbf{n}$: force exerted by body 2 on body 1 during compression.
- $\mathbf{F}_{1_r} = F_{1_r}\mathbf{n}$: force exerted by body 2 on body 1 during restitution.
- $\mathbf{F}_{2_d}$, $\mathbf{F}_{2_r}$: reaction forces by body 1 on body 2.
- All other forces have resultants $\mathbf{R}_1$ and $\mathbf{R}_2$.

Since the collision duration is very small, the impulses of $\mathbf{R}_1$ and $\mathbf{R}_2$ vanish. Define the time instants:

- t_0: time just before collision
- t_1: time at the end of compression
- t_2: time at the end of restitution

Approximating:

$$\begin{aligned}
\int_{t_0}^{t_1} \Big(\mathbf{F}_{1_d}(\tau) + \mathbf{R}_1(\tau)\Big)\, d\tau &\approx \int_{t_0}^{t_1} \mathbf{F}_{1_d}(\tau)\, d\tau, \\
\int_{t_1}^{t_2} \Big(\mathbf{F}_{1_r}(\tau) + \mathbf{R}_1(\tau)\Big)\, d\tau &\approx \int_{t_1}^{t_2} \mathbf{F}_{1_r}(\tau)\, d\tau, \\
\int_{t_0}^{t_1} \Big(\mathbf{F}_{2_d}(\tau) + \mathbf{R}_2(\tau)\Big)\, d\tau &\approx \int_{t_0}^{t_1} \mathbf{F}_{2_d}(\tau)\, d\tau, \\
\int_{t_1}^{t_2} \Big(\mathbf{F}_{2_r}(\tau) + \mathbf{R}_2(\tau)\Big)\, d\tau &\approx \int_{t_1}^{t_2} \mathbf{F}_{2_r}(\tau)\, d\tau.
\end{aligned} \tag{17.1}$$

We assume equal and opposite collisional forces:

$$\begin{aligned}
\mathbf{F}_{1_r} &= -\mathbf{F}_{2_r}, \\
\mathbf{F}_{1_d} &= -\mathbf{F}_{2_d}.
\end{aligned} \tag{17.2}$$

We now define the **coefficient of restitution**:

$$e = \frac{\int_{t_1}^{t_2} \mathbf{F}_{1_r}(\tau) \cdot \mathbf{n}\, d\tau}{\int_{t_0}^{t_1} \mathbf{F}_{1_d}(\tau) \cdot \mathbf{n}\, d\tau} = \frac{\int_{t_1}^{t_2} \mathbf{F}_{2_r}(\tau) \cdot \mathbf{n}\, d\tau}{\int_{t_0}^{t_1} \mathbf{F}_{2_d}(\tau) \cdot \mathbf{n}\, d\tau}. \tag{17.3}$$

Think!

Question: What does $e = 0$ and $e = 1$ mean?

Answer

- $e = 1$: perfectly **elastic** collision — compression impulse equals restitution impulse.
- $e = 0$: perfectly **plastic** collision — no restitution impulse.
- In general $0 \leq e \leq 1$, determined experimentally.

17.2.4 The Coefficient of Restitution in Terms of Velocities

To write the coefficient of restitution in a more convenient form using velocities, we first write the linear impulse-linear momentum equations.

Think!

Question: For the above impulses, complete the linear impulse-linear momentum equations.

Answer

$$
\begin{aligned}
m_1 v_{II} - m_1 \mathbf{v}_1 \cdot \mathbf{n} &= \int_{t_0}^{t_1} \mathbf{F}_{1_d}(\tau) \cdot \mathbf{n}\, d\tau, \\
m_2 v_{II} - m_2 \mathbf{v}_2 \cdot \mathbf{n} &= \int_{t_0}^{t_1} \mathbf{F}_{2_d}(\tau) \cdot \mathbf{n}\, d\tau, \\
m_1 \mathbf{v}_1' \cdot \mathbf{n} - m_1 v_{II} &= \int_{t_1}^{t_2} \mathbf{F}_{1_r}(\tau) \cdot \mathbf{n}\, d\tau = e \int_{t_0}^{t_1} \mathbf{F}_{1_d}(\tau) \cdot \mathbf{n}\, d\tau, \\
m_2 \mathbf{v}_2' \cdot \mathbf{n} - m_2 v_{II} &= \int_{t_1}^{t_2} \mathbf{F}_{2_r}(\tau) \cdot \mathbf{n}\, d\tau = e \int_{t_0}^{t_1} \mathbf{F}_{2_d}(\tau) \cdot \mathbf{n}\, d\tau.
\end{aligned}
\tag{17.4}
$$

From these 4 equations, we can isolate the common velocity v_{II} of the particles along $\mathbf{n}$ at time t_1:

$$
v_{II} = \frac{\mathbf{v}_1' \cdot \mathbf{n} + e\mathbf{v}_1 \cdot \mathbf{n}}{1+e} = \frac{\mathbf{v}_2' \cdot \mathbf{n} + e\mathbf{v}_2 \cdot \mathbf{n}}{1+e}. \tag{17.5}
$$

From this, we get the familiar expression for the coefficient of restitution:

$$
e = \frac{\mathbf{v}_2' \cdot \mathbf{n} - \mathbf{v}_1' \cdot \mathbf{n}}{\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}}. \tag{17.6}
$$

https://youtu.be/0GcIPJgpwN4

Note!

When one object impacts a massive constrained object (e.g. a wall), the velocity of the massive object is unaffected by the collision.

17.3 Example: The Most General Planar Impact Problem

Example

Question: Given particles m_1 and m_2 with pre-impact velocities $\mathbf{v}_1$ and $\mathbf{v}_2$, find the post-impact velocities. The coefficient of restitution e is given.

Answer

There are 4 unknowns (components of $\mathbf{v}_1'$ and $\mathbf{v}_2'$ along $\mathbf{n}$, $\mathbf{t}_1$, $\mathbf{t}_2$). Along the tangent directions: velocities are unchanged. Along $\mathbf{n}$: we have the conservation of linear momentum and the restitution equation.

https://youtu.be/4zI-bUmKLKI

17.4 Energy Loss During Impact

Pre-impact and post-impact kinetic energies:

$$T = \frac{1}{2}m_1\mathbf{v}_1 \cdot \mathbf{v}_1 + \frac{1}{2}m_2\mathbf{v}_2 \cdot \mathbf{v}_2, \tag{17.7}$$

$$T' = \frac{1}{2}m_1\mathbf{v}_1' \cdot \mathbf{v}_1' + \frac{1}{2}m_2\mathbf{v}_2' \cdot \mathbf{v}_2'. \tag{17.8}$$

The energy lost:

$$\begin{aligned} T - T' &= \frac{1}{2}m_1\left((\mathbf{v}_1 \cdot \mathbf{n})^2 - (\mathbf{v}_1' \cdot \mathbf{n})^2\right) + \frac{1}{2}m_2\left((\mathbf{v}_2 \cdot \mathbf{n})^2 - (\mathbf{v}_2' \cdot \mathbf{n})^2\right) \\ &= \frac{m_1 m_2}{2(m_1 + m_2)}\left(\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}\right)^2 (1 - e^2). \end{aligned} \tag{17.9}$$

Hence, for $0 \leq e \leq 1$, kinetic energy cannot increase as a result of impact.

17.5 Negative Values of the Coefficient of Restitution

Think!

Question: What does a negative e mean for a ball impacting a wall?

Answer

When the coefficient of restitution is negative,

$$e = \frac{\mathbf{v}_2' \cdot \mathbf{n} - \mathbf{v}_1' \cdot \mathbf{n}}{\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}} \leq 0, \tag{17.10}$$

and hence $\mathbf{v}_1' \cdot \mathbf{n} - \mathbf{v}_2' \cdot \mathbf{n}$ and $\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}$ have the same sign. As a result, the colliding bodies pass through each other during the course of the impact. To eliminate this behavior, it is generally assumed that e is positive.

17.6 Exercises

The following problems are from Set 13 – Collisions. In all of the following problems, specify the system you are considering. Are there any conservations in that system? Remember that the linear momentum of a system is conserved in a fixed direction if the sum of forces acting on the system along this direction is zero.

1. [MKB 03-142]

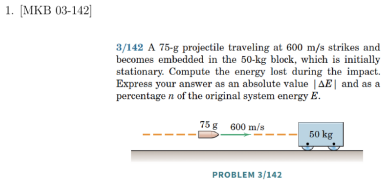


Figure 17.2: MKB 03-142.

2. [MKB 03-196]

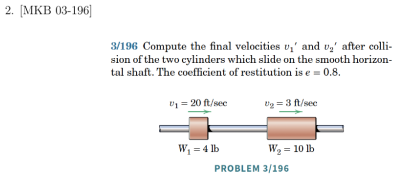


Figure 17.3: MKB 03-196.

3. [MKB 03-197]

3. [MKB 03-197]

3/197 The two bodies have the masses and initial velocities shown in the figure. The coefficient of restitution for the collision is $e = 0.3$, and friction is negligible. If the time duration of the collision is 0.025 s, determine the average impact force which is exerted on the 3-kg body.

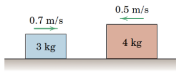


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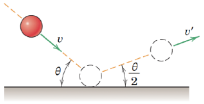
Figure 17.4: MKB 03-197.

4. [03-201]

4. [03-201]

3/201 Determine the value of the coefficient of restitution e for which the outgoing angle is one-half of the incoming angle θ as shown. Evaluate your general expression for $\theta = 40°$.



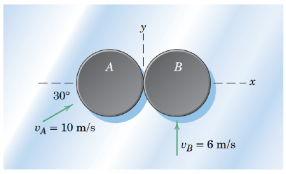
PROBLEM 3/201

Figure 17.5: MKB 03-201.

5. [03-212]

5. [03-212]

3/212 Two identical hockey pucks moving with initial velocities v_A and v_B collide as shown. If the coefficient of restitution is $e = 0.75$, determine the velocity (magnitude and direction θ with respect to the positive x-axis) of each puck just after impact. Also calculate the percentage loss n of system kinetic energy.



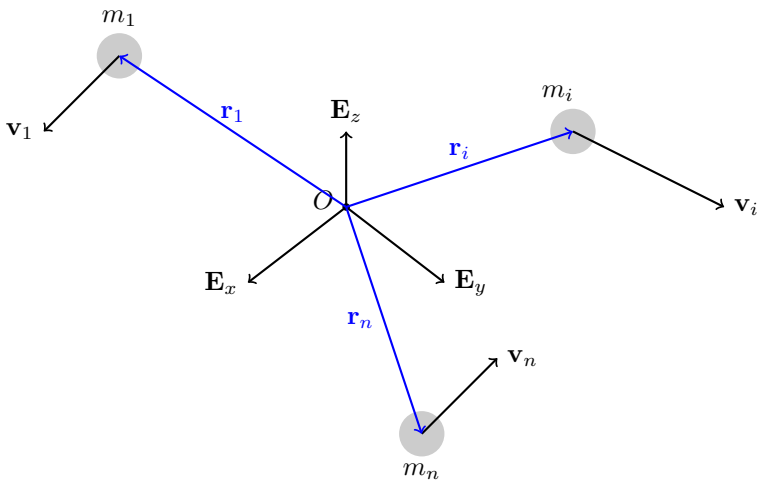
PROBLEM 3/212

Figure 17.6: MKB 03-212.

Part III

System of Particles

18 System of Particles



Consider a system of n particles. A typical particle has mass m_i, position vector $\mathbf{r}_i$ relative to O, velocity $\mathbf{v}_i = \dot{\mathbf{r}}_i$, acceleration $\mathbf{a}_i = \dot{\mathbf{v}}_i$, linear momentum $\mathbf{G}_i = m_i\mathbf{v}_i$, angular momentum relative to some point P $\mathbf{H}_i^P = (\mathbf{r}_i - \mathbf{r}_P) \times \mathbf{G}_i$, and kinetic energy $T_i = \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i$.

In this chapter, we define analogous system-level quantities and derive the balance laws for the system from the balance law for each individual particle.

18.1 Center of Mass

Think!

Question: What is the position vector of the center of mass of a system of particles?

Answer

The center of mass C has position vector

$$\mathbf{r}_C = \frac{1}{m}\sum_{k=1}^{n} m_k \mathbf{r}_k, \tag{18.1}$$

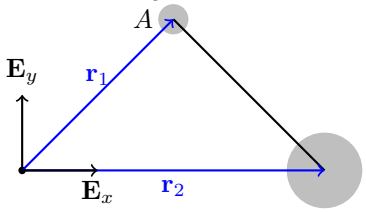
where $m = \sum_{k=1}^{n} m_k$ is the total mass.

Example

Question: Consider an unsymmetrical dumbbell: particle 1 of mass m and particle 2 of mass $3m$ connected by a massless rigid rod, with

$$\mathbf{r}_1 = \mathbf{E}_x + \mathbf{E}_y, \qquad \mathbf{r}_2 = 2\mathbf{E}_x.$$

Calculate $\mathbf{r}_C$. Is C located on the dumbbell?



Answer

$$\mathbf{r}_C = \frac{m\mathbf{r}_1 + 3m\mathbf{r}_2}{m+3m} = \frac{\mathbf{r}_1 + 3\mathbf{r}_2}{4} = \frac{7}{4}\mathbf{E}_x + \frac{1}{4}\mathbf{E}_y.$$

We can check mathematically that C lies on the line connecting m_1 and m_2 by calculating the equation of that line and checking at C belongs to it.

Alternatively, we can define a new basis, $\{\mathbf{u}, \mathbf{v}\}$ and calculate the position of C relative to m_1 on that basis. This should be along $\mathbf{u}$.

Note the useful identity: relative to any reference point A,

$$\mathbf{r}_{C/A} = \frac{\sum_i m_i \mathbf{r}_{i/A}}{\sum_i m_i}. \tag{18.2}$$

Think!

Question: Show that $\sum_{k=1}^{n} m_k(\mathbf{r}_C - \mathbf{r}_k) = \mathbf{0}$ and $\sum_{k=1}^{n} m_k(\mathbf{v}_C - \mathbf{v}_k) = \mathbf{0}$.

Answer

Both follow directly from the definition of the center of mass.

Differentiating the position of the center of mass:

$$\mathbf{v}_C = \frac{1}{m}\sum_{k=1}^{n} m_k \mathbf{v}_k = \frac{1}{m}\sum_{k=1}^{n} \mathbf{G}_k. \tag{18.3}$$

Notice that the velocity of the center of mass is the weighted sum of the velocities of the particles.

Think!

Question: Show that $\sum_{k=1}^{n} m_k(\mathbf{r}_C - \mathbf{r}_k) = \mathbf{0}$ and $\sum_{k=1}^{n} m_k(\mathbf{v}_C - \mathbf{v}_k) = \mathbf{0}$.

Answer

...

18.2 Linear Momentum

The linear momentum of the system is

$$\mathbf{G} = m\frac{d\mathbf{r}}{dt} = \sum_{k=1}^{n} m_k \frac{d\mathbf{r}_k}{dt} = \sum_{k=1}^{n} \mathbf{G}_k. \tag{18.4}$$

The linear momentum of the system equals that of its center of mass.

18.3 Angular Momentum

The angular momentum $\mathbf{H}^P$ of the system of particles relative to a point P, whose position vector relative to O is $\mathbf{r}_P$, is the sum of the individual angular momenta:

$$\mathbf{H}^P = \sum_{k=1}^{n} \mathbf{H}^{P_K} = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times m_k \mathbf{v}_k = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{G}, \tag{18.5}$$

where

$$\mathbf{H}^C = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_C) \times m_k \mathbf{v}_k = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_C) \times m_k (\mathbf{v}_k - \mathbf{v}_C). \tag{18.6}$$

Think!

Question: Show that $\sum_{k=1}^{n}(\mathbf{r}_k - \mathbf{r}_P) \times m_k\mathbf{v}_k = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{G}$.

Answer

Introduce $\mathbf{r}_C - \mathbf{r}_C$ inside the sum and expand.

18.4 Kinetic Energy

The kinetic energy of the system is:

$$T = \sum_{k=1}^{n} \frac{1}{2} m_k \mathbf{v}_k \cdot \mathbf{v}_k = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \sum_{k=1}^{n} m_k (\mathbf{v}_k - \mathbf{v}_C) \cdot (\mathbf{v}_k - \mathbf{v}_C). \tag{18.7}$$

The kinetic energy splits into the kinetic energy of the center of mass plus the kinetic energy relative to the center of mass. In general, $T \neq \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C$.

Think!

Question: Prove $\sum_{k=1}^{n} \frac{1}{2} m_k \mathbf{v}_k \cdot \mathbf{v}_k = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \sum_{k=1}^{n} m_k (\mathbf{v}_k - \mathbf{v}_C) \cdot (\mathbf{v}_k - \mathbf{v}_C)$.

Answer

Add $\mathbf{v}_C - \mathbf{v}_C$ to each $\mathbf{v}_k$, expand, and use $\sum m_k(\mathbf{v}_k - \mathbf{v}_C) = \mathbf{0}$.

18.5 Kinetics: Balance of Linear Momentum

Starting from $\mathbf{F}_i = m_i \mathbf{a}_i$ for each particle and summing:

$$\mathbf{F} = \sum_{k=1}^{n} \mathbf{F}_k = m \mathbf{a}_C. \tag{18.8}$$

The resultant external force equals the total mass times the acceleration of the center of mass.

Note!

The conditions for conservation of linear momentum of a system (completely or along a direction) are analogous to those for a single particle: $\mathbf{F} \cdot \mathbf{c} = 0$ for a constant $\mathbf{c}$.

18.6 Kinetics: Balance of Angular Momentum

Starting from $\mathbf{H}^P = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times m_k \mathbf{v}_k$, the time derivative is:

$$\dot{\mathbf{H}}^P = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times \mathbf{F}_k - \mathbf{v}_P \times \mathbf{G} = \mathbf{M}^P - \mathbf{v}_P \times \mathbf{G}, \tag{18.9}$$

where the resultant moment of the system of forces relative to P is $\mathbf{M}^P = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times \mathbf{F}_k$. The angular momentum theorem becomes

$$\dot{\mathbf{H}}^P = \mathbf{M}^P - \mathbf{v}_P \times \mathbf{G}. \tag{18.10}$$

> **!** Note!
>
> $\dot{\mathbf{H}}^P$ is **not** necessarily equal to $\mathbf{M}^P$. The two agree only in special cases.

Two important special cases:

1. $P = O$ (fixed origin, $\mathbf{v}_O = \mathbf{0}$): $\dot{\mathbf{H}}^O = \mathbf{M}^O = \sum_{k=1}^{n} \mathbf{r}_k \times \mathbf{F}_k$.
2. $P = C$ (center of mass, $\mathbf{v}_C \times \mathbf{G} = \mathbf{0}$): $\dot{\mathbf{H}}^C = \mathbf{M}^C = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_C) \times \mathbf{F}_k$.

> **!** Note!
>
> The conditions for conservation of angular momentum of a system are analogous to those for a single particle.

Example: A pendulum suspended from a cart on smooth horizontal rails — write the BoAM in 3 forms.

18.7 Work-Energy Theorem

The work-energy theorem for a system of particles is:

$$\dot{T} = \sum_{k=1}^{n} \mathbf{F}_k \cdot \mathbf{v}_k, \tag{18.11}$$

and

$$\dot{E} = \sum_{k=1}^{n} \mathbf{F}_{nc_k} \cdot \mathbf{v}_k. \tag{18.12}$$

where $\mathbf{F}_{nc_k}$ is the nonconservative force acting on the kth particle and E is the total energy of the system of particles. Integrating:

$$T(t_2) - T(t_1) = \sum_{k=1}^{n} W_{\mathbf{F}_k,12}, \qquad W_{\mathbf{F}_k,12} = \int_{t_1}^{t_2} \mathbf{F}_k \cdot \mathbf{v}_k \, dt. \tag{18.13}$$

! Note!

The work of a force is the integral of the force dotted with the velocity of the **point of application** of the force.

⚠ Example

Question: Consider the unsymmetrical dumbbell (particle 1 of mass m, particle 2 of mass $3m$, massless rod) moving in the smooth $\{\mathbf{E}_x, \mathbf{E}_y\}$ plane with gravity along $-\mathbf{E}_z$. Is the energy conserved?

Answer

Drawing the FBDs of both masses: ...

18.8 Summary

Kinematics of a system of particles

Consider a system of n particles each with mass m_i and with position vector $\mathbf{r}_i$ from the origin. The total mass of the system of particles is $m = \sum_{i=1}^{n} m_i$. The center of mass of the system of particles C is defined to be located at

$$\mathbf{r}_C = \frac{\sum_{i=1}^{n} m_i \mathbf{r}_i}{m}. \tag{18.14}$$

The velocity and acceleration of the center of mass are then obtained by differentiation $\mathbf{r}_C$.

$$\mathbf{v}_C = \frac{\sum_{i=1}^{n} m_i \mathbf{v}_i}{m}, \quad \text{and} \quad \mathbf{a}_C = \frac{\sum_{i=1}^{n} m_i \mathbf{a}_i}{m}. \tag{18.15}$$

The linear momentum of a system of particles is the sum of the linear momenta of its constituents which is also the linear momentum of the center of mass

$$\mathbf{G} = \sum_{i=1}^{n} \mathbf{G}_i = m\mathbf{v}_C. \tag{18.16}$$

The angular momentum of a system of particles about point P is the sum of the angular momenta of its constituents about P

$$\mathbf{H}^P = \sum_{i=1}^{n} (\mathbf{r}_i - \mathbf{r}_P) \times m_i \mathbf{v}_i. \tag{18.17}$$

This expression can be rewritten as

$$\mathbf{H}^P = \mathbf{H} + (\mathbf{r} - \mathbf{r}_P) \times \mathbf{G}. \tag{18.18}$$

where the angular momentum $\mathbf{H}^C$ of the system about the center of mass can be written equivalently as

$$\mathbf{H} = \sum_{i=1}^{n} (\mathbf{r}_i - \mathbf{r}_C) \times m_k \mathbf{v}_i = \sum_{i=1}^{n} (\mathbf{r}_i - \mathbf{r}_C) \times m_i (\mathbf{v}_i - \mathbf{v}_C). \tag{18.19}$$

The kinetic energy of the system of particles is the sum of the kinetic energies of its constituents. This is not equal to the $\frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C$.

$$T = \sum_{i=1}^{n} T_i = \sum_{i=1}^{n} \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \sum_{i=1}^{n} m_k (\mathbf{v}_i - \mathbf{v}_C) \cdot (\mathbf{v}_i - \mathbf{v}_C). \tag{18.20}$$

Kinetics of a system of particles

The balance of linear momentum for a system of particles is obtained by adding the balance of linear momenta of the individual particles $\mathbf{F}_i = m_i \mathbf{a}_i$

$$\mathbf{F} = \sum \mathbf{F}_i = \sum m_i \mathbf{a}_i = m \mathbf{a}_c. \tag{18.21}$$

Here, $\mathbf{F}$ is the net external force acting on the system. The balance of angular momentum of a system of particles about any point P is

$$\dot{\mathbf{H}}^P = \mathbf{M}^P - \mathbf{v}^P \times \mathbf{G}. \tag{18.22}$$

If P is a fixed point O, then this equation simplifies to $\mathbf{M}^O = \dot{\mathbf{H}}^O$. If P is the center of mass C, then this equation simplifies to $\mathbf{M}^C = \dot{\mathbf{H}}^C$. The work-energy theorem for a system of particles is the sum of the work-energy theorems of the individual particles

$$\sum \dot{T}_i = \sum \mathbf{F}_i \cdot \mathbf{v}_i. \tag{18.23}$$

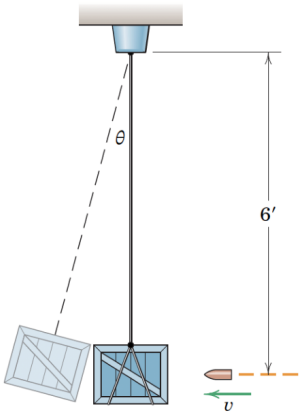
Now that different points in the system have different displacements/velocities, it is important to note that the work of a force i is obtained by dotting $\mathbf{F}_i$ with the velocity/displacement of its point of application.

18.9 Exercises

The following problems are from Set 14 – System of Particles.

1. [MKB 03-168] Show that during the (very brief) collision, the linear momentum of the bullet–pendulum system is conserved. *(ans.* $\theta = 20.7^\circ$, $n = 99.8\%$*)*

3/168 The ballistic pendulum is a simple device to measure projectile velocity v by observing the maximum angle θ to which the box of sand with embedded projectile swings. Calculate the angle θ if the 2-oz projectile is fired horizontally into the suspended 50-lb box of sand with a velocity $v = 2000$ ft/sec. Also find the percentage of energy lost during impact.

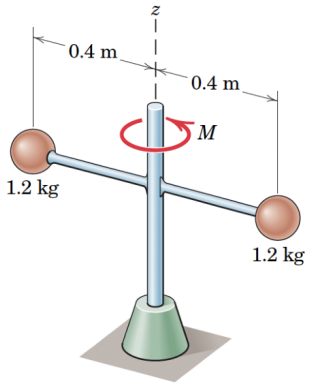


PROBLEM 3/168

Figure 18.1: MKB 03-168.

2. [MKB 03-178] Use the integral form of the BoAM; choose the origin at the green base. *(ans.* $\dot{\theta} = 26.0$ *rad/s)*

3/178 The rigid assembly which consists of light rods and two 1.2-kg spheres rotates freely about a vertical axis. The assembly is initially at rest and then a constant couple $M = 2$ N·m is applied for 5 s. Determine the final angular velocity of the assembly. Treat the small spheres as particles.

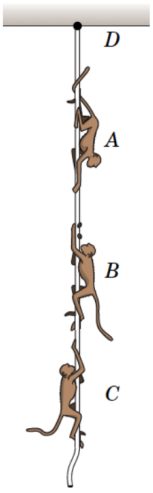


PROBLEM 3/178

Figure 18.2: MKB 03-178.

3. [04-008] Take an appropriate cut in the rope. *(ans.* $T = 58.3$ *lb)*

4/8 Three monkeys A, B, and C weighing 20, 25, and 15 lb, respectively, are climbing up and down the rope suspended from D. At the instant represented, A is descending the rope with an acceleration of 5 ft/sec^2, and C is pulling himself up with an acceleration of 3 ft/sec^2. Monkey B is climbing up with a constant speed of 2 ft/sec. Treat the rope and monkeys as a complete system and calculate the tension T in the rope at D.

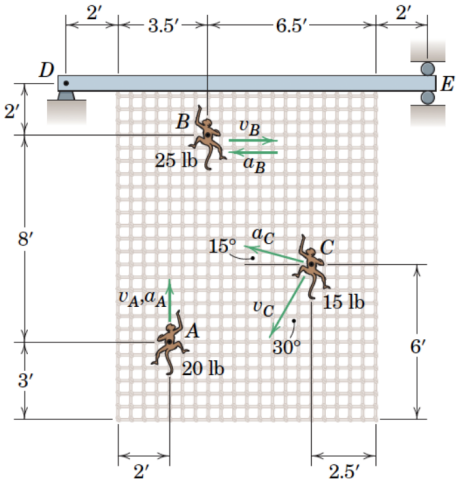


PROBLEM 4/8

Figure 18.3: MKB 04-008.

4. [04-009] *(ans.* $D_x = 1.288$ *lb left,* $D_y = 35.5$ *lb up,* $N_E = 25.6$ *lb up)*

4/9 The monkeys of Prob. 4/8 are now climbing along the heavy rope wall suspended from the uniform beam. If monkeys A, B, and C have velocities of 5, 3, and 2 ft/sec, and accelerations of 1.5, 0.5, and 2 ft/sec^2, respectively, determine the changes in the reactions at D and E caused by the motion and weight of the monkeys. The support at E makes contact with only one side of the beam at a time. Assume for this analysis that the rope wall remains rigid.

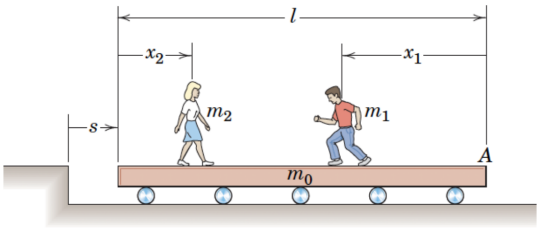


PROBLEM 4/9

Figure 18.4: MKB 04-009.

5. [04-019] Identify a direction of linear momentum conservation; the center of mass remains stationary. *(ans.* $s = \frac{(m_1+m_2)x_{10}-m_2 l}{m_0+m_1+m_2}$*)*

4/19 The man of mass m_1 and the woman of mass m_2 are standing on opposite ends of the platform of mass m_0 which moves with negligible friction and is initially at rest with $s = 0$. The man and woman begin to approach each other. Derive an expression for the displacement s of the platform when the two meet in terms of the displacement x_1 of the man relative to the platform.



PROBLEM 4/19

Figure 18.5: MKB 04-019.

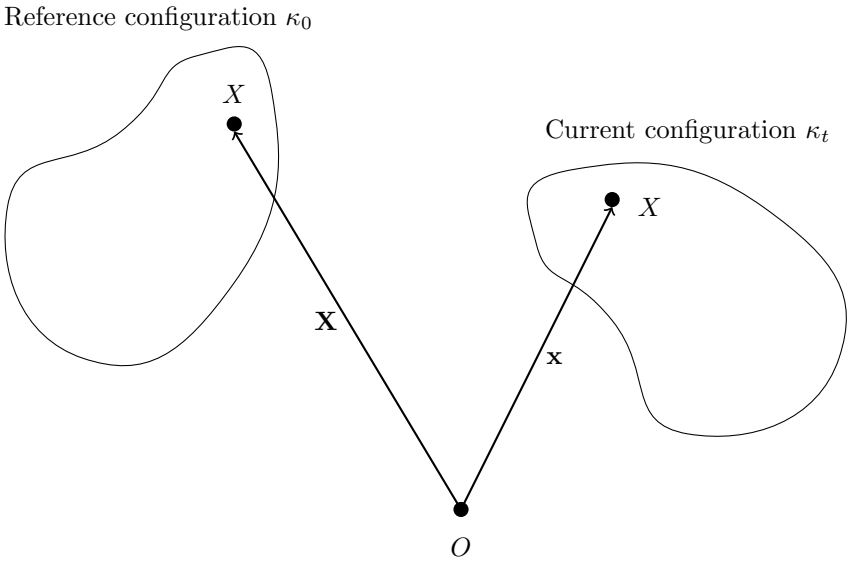
Part IV

Rigid Body Dynamics

19 Kinematics of Rigid Bodies

In this chapter, we will denote the fixed basis by $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$.

Consider a body $\mathcal{B}$ in motion in 3D space.



- A body $\mathcal{B}$ is a collection of material points (mass particles or particles).
- It is convenient to define a fixed reference configuration κ_0 for the body: $\mathbf{X} = \kappa_0$.
- The present (or current) configuration of the body is denoted by κ_t.
- We denote a material point by X. The position of the material point X relative to a fixed origin at the reference configuration is denoted by $\mathbf{X}$ and at time t is denoted by $\mathbf{x}$.
- This motion is invertible; we can use χ to uniquely define the motion as a function of X and t: $\mathbf{x} = \tilde{\chi}(X, t) = \chi(\mathbf{X}, t)$.
- χ (pronounced like "kye") is called the **motion function**. It tells you the current position vector of the material point which, at the reference configuration, had position vector $\mathbf{X}$.

Think!

Question: What is a mathematical statement that means that a body is rigid?

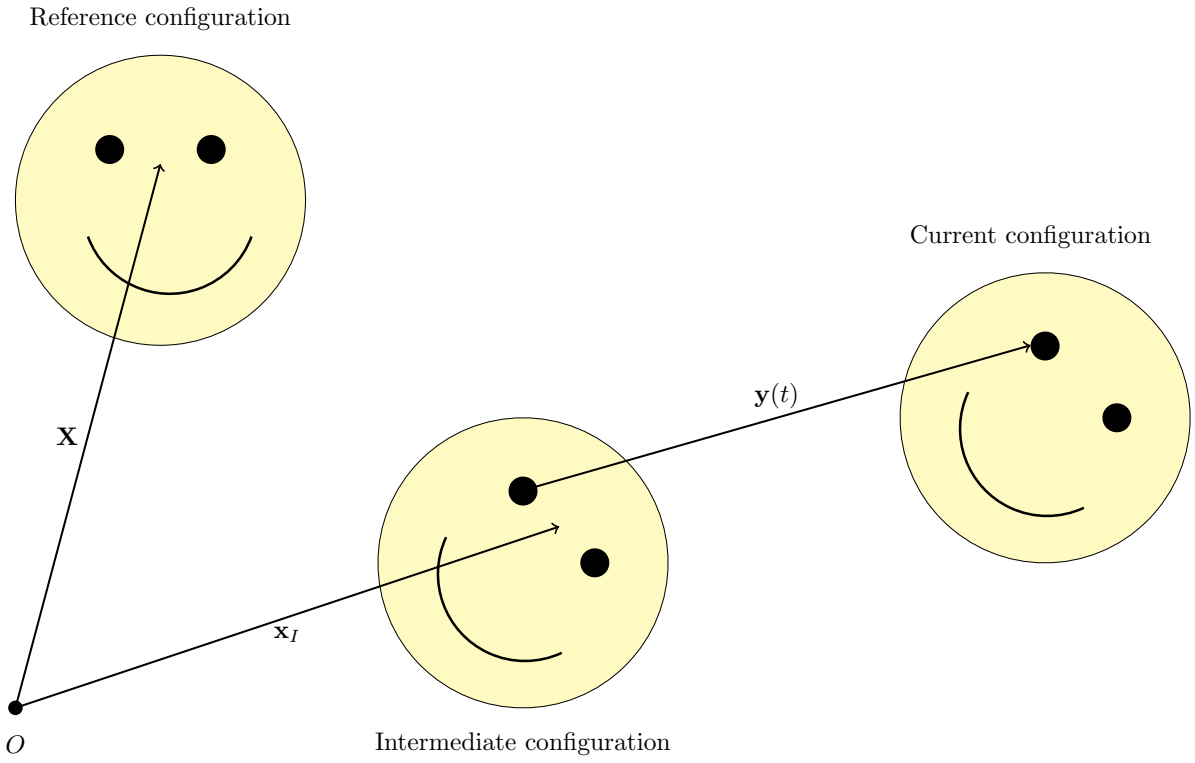
Answer

For rigid bodies the distance between any two mass particles X_1 and X_2 remains constant for all motions:

$$\|\mathbf{x}_1 - \mathbf{x}_2\| = \|\mathbf{X}_1 - \mathbf{X}_2\|. \tag{19.1}$$

The motion preserves angles between lines on the body (unlike elastic bodies or fluids) and preserves orientation (it is not a reflection).

Any rigid body motion can be built in two steps:



1. **Rotate** the object about O until it has the desired orientation (but likely the wrong position). This motion maps every point $\mathbf{X}$ to an intermediate position $\mathbf{x}_I$: $\mathbf{x}_I = \mathbf{Q}(t)\mathbf{X}$ with $\|\mathbf{Q}(t)\mathbf{X}\| = \|\mathbf{X}\|$ for all $\mathbf{X}$.
2. **Translate** via the translation vector $\mathbf{y}(t)$ (without further rotation) from an intermediate configuration to a current configuration.

Then every rigid motion is:

$$\mathbf{x} = \chi(\mathbf{X}, t) = \mathbf{x}_I(\mathbf{X}, t) + \mathbf{y}(t) = \mathbf{Q}(t)\mathbf{X} + \mathbf{y}(t). \tag{19.2}$$

For each rigid motion (and choice of O), $\mathbf{Q}(t)$ and $\mathbf{y}(t)$ are unique.

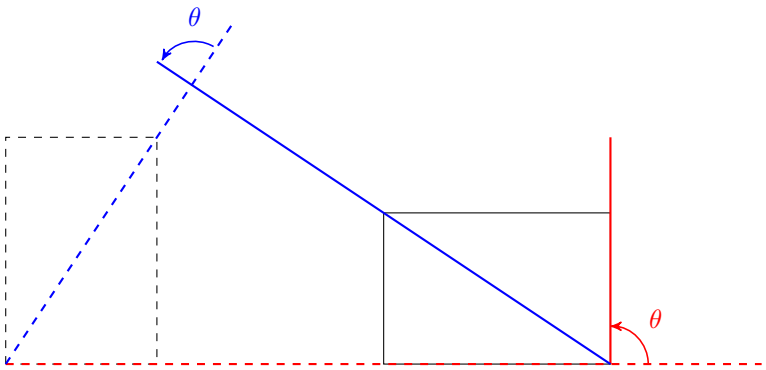
$\mathbf{Q}(t)$ is an object that acts on a vector to produce another vector. If $\mathbf{Q}$ is also linear, then we call it a **tensor**. We will show that $\mathbf{Q}(t)$ is a linear map and is called the **rotation tensor**. Both $\mathbf{Q}$ and $\mathbf{y}$ are the same for all material points of the rigid body(functions of t only) as all material points on the rigid body are rotated and translated by the same amount.

In this class we consider bodies translating and rotatingin the plane, ie. **general plane motion**, where $\omega = \dot{\theta}\mathbf{E}_3$ and $\alpha = \dot{\omega}$. The fixed axis of rotation is $\mathbf{E}_3$.

Think!

Question: How do we know if a body has rotated relative to a reference configuration?

Answer

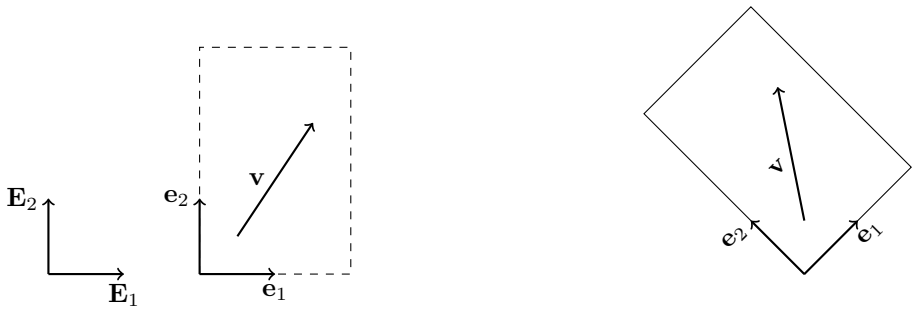


reference configuration current configuration

We need the reference configuration to define rotation. The rotation angle θ is the angular displacement of every material line from reference to current configuration. The axis of rotation is perpendicular to the page.

19.1 Corotational Bases

A *corotational basis* $\{\mathbf{e}_1, \mathbf{e}_2\}$ is a pair of vectors embedded in the material. Initially $\mathbf{e}_i = \mathbf{E}_i$.



reference configuration current configuration

For a rigid body, the components of a material vector $\mathbf{v}$ along the corotational basis are constant throughout the motion:

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2, \tag{19.3}$$

where x_1 and x_2 are constant and $\mathbf{e}_1(t)$, $\mathbf{e}_2(t)$ evolve with the body. Since initially $\mathbf{e}_1 = \mathbf{E}_1$ and $\mathbf{e}_2 = \mathbf{E}_2$, we can write $\mathbf{x}(t=0) = x_1\mathbf{E}_1 + x_2\mathbf{E}_2$.The corotational basis is unaffected by translation.

19.2 Tensors

A tensor $\mathbf{T}$ is a *linear transformation* that transforms a vector into another vector: $\mathbf{Tu} = \mathbf{v}$. A tensor exists independent of any basis and is defined by its action on vectors.

Examples:

- A rotation tensor $\mathbf{Q}$ acting on some vector $\mathbf{v}$ returns a vector $\mathbf{u}$ that is a rotation of $\mathbf{v}$ by some angle θ.
- The identity tensor $\mathbf{I}$ acting on any vector $\mathbf{v}$ returns the same vector $\mathbf{v}$.
- A scaling tensor $\mathbf{S}$ acting on some vector $\mathbf{v}$ returns another the scaled vector $k\mathbf{v}$.

Think!

Question: Is a rotation a linear transformation?

Answer

Yes. To be a linear transformation, a rotation would have to satisfy two rules:

- Rotating $a\mathbf{X}$ is the same as rotating $\mathbf{X}$ and then scaling by a.
- Rotating $\mathbf{X}_1$ and $\mathbf{X}_2$ is the same as rotating $\mathbf{X}_1$, rotating $\mathbf{X}_2$, and then adding the results together.

Since both of these propositions are true for rotations, we conclude that rotations are linear transformations. Thus, rotations are tensors.

Think!

Question: Are tensors and matrices the same?

Answer

We can answer this question using an analogous one: are vectors and arrays the same?

No. A vector is an abstract object independent of any basis; an array is its representation in a specific basis. Similarly, a tensor is not a matrix, but can be expressed as one in a given coordinate system.

We can represent a vector as an array of numbers if we have chosen a basis or coordinate system, but a vector is an abstract object that exists independent of any basis.

Consider for example a position vector $\mathbf{r}$ to a particle P. This vector encodes the position of P relative to the origin (distance and direction) independent of any basis. We could however express this vector in the Cartesian basis as $x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z$. In this basis, we could write the array $[x, y, z]$ to represent this vector.

Analogously, a tensor is not a matrix, but a tensor can be expressed as a matrix in a given coordinate system.

19.3 Tensor (Bun) Product

Consider the mathematical object, termed the **tensor product** of two vectors:

$$\mathbf{a} \otimes \mathbf{b}, \tag{19.4}$$

where $\mathbf{a}$ and $\mathbf{b}$ are vectors. This object is defined to act on vectors to produce a vector as follows:

$$(\mathbf{a} \otimes \mathbf{b})\,\mathbf{u} = (\mathbf{b} \cdot \mathbf{u})\,\mathbf{v}. \tag{19.5}$$

If this object is also linear, then it is a tensor. To verify that this object is linear, we need to show that

$$(\mathbf{a} \otimes \mathbf{b})\,(\mathbf{u} + \mathbf{v}) = (\mathbf{a} \otimes \mathbf{b})\,\mathbf{u} + (\mathbf{a} \otimes \mathbf{b})\,\mathbf{v}, \tag{19.6}$$

$$(\mathbf{a} \otimes \mathbf{b})\,(k\mathbf{u}) = k((\mathbf{a} \otimes \mathbf{b})\,\mathbf{u}). \tag{19.7}$$

For the first,

$$(\mathbf{a} \otimes \mathbf{b})\,(\mathbf{u} + \mathbf{v}) = ((\mathbf{u} + \mathbf{v}) \cdot \mathbf{b})\,\mathbf{a} = \mathbf{u} \cdot \mathbf{b}\mathbf{a} + \mathbf{v} \cdot \mathbf{b}\mathbf{a} = (\mathbf{a} \otimes \mathbf{b})\,\mathbf{u} + (\mathbf{a} \otimes \mathbf{b})\,\mathbf{v}.$$

For the second,

$$(\mathbf{a} \otimes \mathbf{b})\,(k\mathbf{u}) = (k\mathbf{u}) \cdot \mathbf{b}\mathbf{a} = k(\mathbf{u} \cdot \mathbf{b}\mathbf{a}) = k((\mathbf{a} \otimes \mathbf{b})\,\mathbf{u}). \tag{19.8}$$

Think!

Question: What does $2\mathbf{E}_1 \otimes \mathbf{E}_1 + 2\mathbf{E}_2 \otimes \mathbf{E}_2 + 2\mathbf{E}_3 \otimes \mathbf{E}_3$ do to a vector?

Answer

It scales it by a factor of two.

Think!

Question: Show that the identity tensor may be written as $\mathbf{E}_1 \otimes \mathbf{E}_1 + \mathbf{E}_2 \otimes \mathbf{E}_2 + \mathbf{E}_3 \otimes \mathbf{E}_3$.

Answer

Apply the definition of the tensor product term by term on a vector.

Think!

Question: Show that the rotation tensor $\mathbf{Q}$ can be expressed as $\mathbf{Q} = \mathbf{e}_x \otimes \mathbf{E}_x + \mathbf{e}_y \otimes \mathbf{E}_y + \mathbf{e}_z \otimes \mathbf{E}_z$. In other words, show that $\left(\mathbf{e}_x \otimes \mathbf{E}_x + \mathbf{e}_y \otimes \mathbf{E}_y + \mathbf{e}_z \otimes \mathbf{E}_z\right)(\mathbf{X}_A - \mathbf{X}_B) = \mathbf{x}_A - \mathbf{x}_B$ where $\mathbf{X}_A - \mathbf{X}_B = a\mathbf{E}_x + b\mathbf{E}_y$ and $\mathbf{x}_A - \mathbf{x}_B = a\mathbf{e}_x + b\mathbf{e}_y$.

Answer

...

19.4 Matrix Representation of Tensors

Given a basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, any tensor $\mathbf{Q}$ can be written as a sum of dyads (tensor products) on that basis:

$$\mathbf{Q} = \sum_{k=1}^{3} \sum_{\ell=1}^{3} Q_{k\ell}\, \mathbf{e}_k \otimes \mathbf{e}_\ell. \tag{19.9}$$

There are two equivalent perspectives for arriving at the matrix representation of $\mathbf{Q}$, and both lead to the same components Q_{ij}.

19.4.1 Perspective 1: From the Dyadic Expansion

Acting with $\mathbf{Q}$ on a basis vector $\mathbf{e}_j$ and using $\mathbf{e}_\ell \cdot \mathbf{e}_j = 1$ if $\ell = j$ and 0 otherwise:

$$\mathbf{Q}\mathbf{e}_j = \left(\sum_{k=1}^{3} \sum_{\ell=1}^{3} Q_{k\ell}\, \mathbf{e}_k \otimes \mathbf{e}_\ell \right) \mathbf{e}_j = \sum_{k=1}^{3} \sum_{\ell=1}^{3} Q_{k\ell} \left(\mathbf{e}_\ell \cdot \mathbf{e}_j \right) \mathbf{e}_k = \sum_{k=1}^{3} Q_{kj}\, \mathbf{e}_k. \tag{19.10}$$

Taking the dot product with $\mathbf{e}_i$:

$$\left(\mathbf{Q}\mathbf{e}_j \right) \cdot \mathbf{e}_i = \sum_{k=1}^{3} Q_{kj} \left(\mathbf{e}_k \cdot \mathbf{e}_i \right) = Q_{ij}. \tag{19.11}$$

So the coefficient Q_{ij} in the dyadic expansion of $\mathbf{Q}$ is exactly the matrix entry in row i, column j.

19.4.2 Perspective 2: From the Action on a General Vector

Given a basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, every vector $\mathbf{X}$ can be written uniquely as $\mathbf{X} = X_1\mathbf{e}_1 + X_2\mathbf{e}_2 + X_3\mathbf{e}_3$ with components $X_i = \mathbf{X} \cdot \mathbf{e}_i$. By linearity of $\mathbf{Q}$:

$$\mathbf{Q}\mathbf{X} = X_1(\mathbf{Q}\mathbf{e}_1) + X_2(\mathbf{Q}\mathbf{e}_2) + X_3(\mathbf{Q}\mathbf{e}_3). \tag{19.12}$$

Writing the output vector $\mathbf{x} \equiv \mathbf{Q}\mathbf{X}$ in the same basis, $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3$ with $x_i = \mathbf{x} \cdot \mathbf{e}_i = (\mathbf{Q}\mathbf{X}) \cdot \mathbf{e}_i$, and defining

$$Q_{ij} \equiv \left(\mathbf{Q}\mathbf{e}_j \right) \cdot \mathbf{e}_i, \tag{19.13}$$

we get

$$x_1 = Q_{11}X_1 + Q_{12}X_2 + Q_{13}X_3, \tag{19.14}$$
$$x_2 = Q_{21}X_1 + Q_{22}X_2 + Q_{23}X_3, \tag{19.15}$$
$$x_3 = Q_{31}X_1 + Q_{32}X_2 + Q_{33}X_3, \tag{19.16}$$

which is exactly the matrix equation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & Q_{22} & Q_{23} \\ Q_{31} & Q_{32} & Q_{33} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}. \tag{19.17}$$

! Note!

Both perspectives agree on the same definition $Q_{ij} = \left(\mathbf{Q}\mathbf{e}_j\right) \cdot \mathbf{e}_i$. Perspective 1 identifies Q_{ij} as the coefficient of $\mathbf{e}_i \otimes \mathbf{e}_j$ in the dyadic expansion of $\mathbf{Q}$ itself; Perspective 2 identifies it as the entry that makes ordinary matrix-vector multiplication reproduce the action of $\mathbf{Q}$ on components. The matrix representation depends on the chosen basis, but the tensor itself is basis-independent. Transpose, symmetry, and skew-symmetry of a tensor can be defined without reference to any basis, but in practice we identify them through the matrix representation: $\mathbf{T}$ is symmetric if $[\mathbf{T}] = [\mathbf{T}]^T$ and skew if $[\mathbf{T}] = -[\mathbf{T}]^T$.

19.4.3 Rigid Body Motion

In matrix form, rigid body motion is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} Q_{11}(t) & Q_{12}(t) & Q_{13}(t) \\ Q_{21}(t) & Q_{22}(t) & Q_{23}(t) \\ Q_{31}(t) & Q_{32}(t) & Q_{33}(t) \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}. \tag{19.18}$$

For $[\mathbf{Q}(t)]$ to be a rotation matrix: $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$ and $\det([\mathbf{Q}]) = 1$.

Think!

Question: Verify that the matrix representation of the rotation tensor $\mathbf{Q}$ of a rotation about $\mathbf{E}_z$ is

$$\mathtt{Q} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where θ is a function of t. Recall that the rotation tensor can be expressed as

$$\mathbf{Q} = \mathbf{e}_1 \otimes \mathbf{E}_1 + \mathbf{e}_2 \otimes \mathbf{E}_2 + \mathbf{e}_3 \otimes \mathbf{E}_3,$$

and for a rotation about $\mathbf{E}_3$, we have

$$\begin{aligned} \mathbf{e}_1 &= \cos(\theta)\mathbf{E}_1 + \sin(\theta)\mathbf{E}_2, \\ \mathbf{e}_2 &= -\sin(\theta)\mathbf{E}_1 + \cos(\theta)\mathbf{E}_2. \end{aligned}$$

Answer

Compute $Q_{ij} = \left(\mathbf{Q}\mathbf{E}_j\right) \cdot \mathbf{E}_i = \mathbf{e}_j \cdot \mathbf{E}_i$ to recover the matrix.

Think!

Question: A tensor has the following matrix representation on $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$. What does this rotation matrix do to a vector?

$$\mathtt{Q} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Answer

It rotates vector anti-clockwise by $\theta = t$ in the $xy-$plane.

Check this a few examples:

$$\begin{bmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos(t) \\ \sin(t) \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\sin(t) \\ \cos(t) \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

In general, this matrix is a rotation matrix that rotates vectors $\theta(t)$ radians anti-clockwise about $\mathbf{e}_z$.

Note!

This matrix satisfies $\mathtt{Q}\mathtt{Q}^T = \mathtt{I}$ and $\det(\mathtt{Q}) = 1$, so $\mathbf{Q}$ is a **proper orthogonal tensor**. It can be shown that all proper orthogonal tensors are rotations and vice versa.

19.5 The Angular Velocity Vector

From rigid body motion, for any two material points A and B:

$$\mathbf{x}_A = \mathbf{Q}\mathbf{X}_A + \mathbf{q}, \tag{19.19}$$

$$\mathbf{x}_B = \mathbf{Q}\mathbf{X}_B + \mathbf{q}. \tag{19.20}$$

Thus, $(\mathbf{x}_B - \mathbf{x}_A) = \mathbf{Q}(\mathbf{X}_B - \mathbf{X}_A)$, where $\mathbf{Q}$ is a proper orthogonal tensor, meaning that $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$ and $\det(\mathbf{Q}) = 1$. Differentiating:

$$\mathbf{v}_B - \mathbf{v}_A = \dot{\mathbf{Q}}(\mathbf{X}_B - \mathbf{X}_A) = \dot{\mathbf{Q}}\mathbf{Q}^T(\mathbf{x}_B - \mathbf{x}_A). \tag{19.21}$$

The taking the derivative of $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$, we show that $\dot{\mathbf{Q}}\mathbf{Q}^T$ is a skew tensor.

$$\frac{d}{dt}(\mathbf{Q}\mathbf{Q}^T) = \mathbf{0} = \dot{\mathbf{Q}}\mathbf{Q}^T + \mathbf{Q}\dot{\mathbf{Q}}^T, \tag{19.22}$$

$$\dot{\mathbf{Q}}\mathbf{Q}^T = -\mathbf{Q}\dot{\mathbf{Q}}^T = -(\dot{\mathbf{Q}}\mathbf{Q}^T)^T. \tag{19.23}$$

> **!** Note!
>
> Every skew tensor corresponds to a vector cross product. In the Cartesian basis, if $\mathbf{a} = a_1\mathbf{E}_1 + a_2\mathbf{E}_2 + a_3\mathbf{E}_3$ and $\mathbf{b} = b_1\mathbf{E}_1 + b_2\mathbf{E}_2 + b_3\mathbf{E}_3$, then
>
> $$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{E}_1 - (a_1b_3 - a_3)\mathbf{E}_2 + (a_1b_2 - a_2b_1)\mathbf{E}_3, \tag{19.24}$$
>
> which can be re-written in matrix form (using the Cartesian basis) as
>
> $$[\mathbf{a} \times \mathbf{b}] = \underbrace{\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}}_{[\mathbf{a}]_\times} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}. \tag{19.25}$$
>
> The skew matrix $[\mathbf{a}]_\times$ built from the components of $\mathbf{a}$ is the matrix representation of the skew tensor whose axial vector is $\mathbf{a}$. Applying this to $\dot{\mathbf{Q}}\mathbf{Q}^T$ identifies its axial vector as ω, so:

$$\mathbf{v}_B - \mathbf{v}_A = \omega \times (\mathbf{x}_B - \mathbf{x}_A). \tag{19.26}$$

The tensor $\Omega = \dot{\mathbf{Q}}\mathbf{Q}^T$ is the **angular velocity tensor** and ω is the **angular velocity vector**. Taking another derivative:

$$\mathbf{a}_B - \mathbf{a}_A = \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B - \mathbf{x}_A)) \tag{19.27}$$

where α is the angular acceleration.

> **!** Note!
>
> For any corotational basis vector $\mathbf{e}_i$ fixed to the rigid body:
>
> $$\dot{\mathbf{e}}_i = \omega \times \mathbf{e}_i, \quad i = 1, 2, 3. \tag{19.28}$$

19.6 Velocity and Acceleration of Two Material Points

Let A and B be material points on the rigid body. Expressing $\mathbf{x}_{B/A}$ on a corotational basis (constant components):

$$\mathbf{x}_{B/A} = x\mathbf{e}_x + y\mathbf{e}_y, \tag{19.29}$$

$$\mathbf{v}_{B/A} = x\dot{\mathbf{e}}_x + y\dot{\mathbf{e}}_y = \omega \times \mathbf{x}_{B/A}, \tag{19.30}$$

$$\mathbf{a}_{B/A} = \alpha \times \mathbf{x}_{B/A} + \omega \times (\omega \times \mathbf{x}_{B/A}). \tag{19.31}$$

Here, $\omega = \omega\mathbf{E}_z$ and $\alpha = \alpha\mathbf{E}_z$ are the previously defined angular velocity and angular acceleration of the rigid body (in general, ω and α are not necessarily constrained to be along $\mathbf{E}_3$). With $\omega = \omega\mathbf{E}_z$ and $\alpha = \alpha\mathbf{E}_z$:

$$\omega\, d\omega = \alpha\, d\theta, \qquad \omega\, dt = d\theta, \qquad \alpha\, dt = d\omega. \tag{19.32}$$

> **!** Note!
>
> - $\mathbf{v}_{B/A} = \omega \times \mathbf{x}_{B/A}$ is an expression of rigidity: $\frac{d}{dt}(\mathbf{x}_{B/A}) = \mathbf{0}$.
> - The angular velocity and angular acceleration are properties of the rigid body. All material lines on the rigid body rotate with the same angular velocity and angular acceleration.

19.7 Particle in a Noninertial Frame

A noninertial frame is one that accelerates (e.g. an accelerating bus, a merry-go-round).

Consider two points: A is a material point of a rigid body and B moves relative to it, then

$$\mathbf{x}_B - \mathbf{x}_A = x\mathbf{e}_x + y\mathbf{e}_y, \tag{19.33}$$

$$\mathbf{v}_B - \mathbf{v}_A = \omega \times (\mathbf{x}_B - \mathbf{x}_A) + \mathring{\mathbf{x}}_{B/A}, \tag{19.34}$$

$$\mathbf{a}_B - \mathbf{a}_A = \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B - \mathbf{x}_A)) + 2\omega \times \mathring{\mathbf{x}}_{B/A} + \mathring{\mathring{\mathbf{x}}}_{B/A}, \tag{19.35}$$

where $\mathring{\mathbf{x}}_{B/A} = \dot{x}\mathbf{e}_x + \dot{y}\mathbf{e}_y$ and $\mathring{\mathbf{x}}_{B/A} = \ddot{x}\mathbf{e}_x + \ddot{y}\mathbf{e}_y$ are the first and second **corotational rates**. These are the velocity and acceleration of B with respect to A as observed from the noninertial frame (ie. an observer fixed to the rigid body). The term $2\omega \times \mathring{\mathbf{x}}_{B/A}$ is the **Coriolis acceleration** and $\omega \times (\omega \times \mathbf{x}_{B/A})$ is the **centripetal acceleration**.

! Note!

Consider a particle B whose motion is described by a noninertial frame with origin A, angular velocity ω and angular acceleration α. In an inertial frame, the balance of linear momentum for particle B applies:

$$\mathbf{F} = m\mathbf{a}_B, \tag{19.36}$$

$$\mathbf{F} = m\left(\mathbf{a}_A + \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B \tag{19.37}$$

$$\mathbf{F} - m\left(\mathbf{a}_A + \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B - \mathbf{x}_A)) + 2\omega \times \mathring{\mathbf{x}}_{B/A}\right) = m\mathring{\mathring{\mathbf{x}}}_{B/A} \tag{19.38}$$

. The balance of linear momentum for particle B in the noninertial frame is the same as in an inertial frame, except that the fictitious forces $-m\left(\mathbf{a}_A + \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B - \mathbf{x}_A)) + 2\omega \times \mathring{\mathbf{x}}_{B/A}\right)$ must be included.

19.7.1 Merry-go-round

Think!

Question: Watch the below video of people playing catch on a merry-go-round. A ball is thrown on a merry-go-round rotating about fixed center O, with $\mathbf{r}_0 = -R\mathbf{E}_x$ and $\mathbf{v}_0 = v_0\mathbf{E}_x$. The merry-go-round has an angular velocity ω and an angular acceleration α. What are the velocity and acceleration of the ball that is appears to have with respect to an observer on the merry-go-round?
https://www.youtube.com/watch?v=mPsLanVS1Q8

Answer

The ball P is a projectile so its absolute acceleration, velocity and position vectors are respectively

$$\mathbf{a} = -g\mathbf{E}_z, \tag{19.39}$$

$$\mathbf{v} = -gt\mathbf{E}_z + \mathbf{v}_0, \tag{19.40}$$

$$\mathbf{r} = -\frac{g}{2}t^2 + \mathbf{v}_0 t + \mathbf{r}_0. \tag{19.41}$$

Conducting a velocity analysis between the ball and the origin O, we get

$$\mathbf{v} = \omega \times (\mathbf{r} - \mathbf{r}_0) + \mathring{\mathbf{x}}_{P/O}. \tag{19.42}$$

The relative velocity being the only unknown in this equation, we can solve for it:

$$\mathring{\mathbf{x}}_{P/O} = -gt\mathbf{E}_z + v_0\mathbf{E}_x + \omega(v_0 t\mathbf{E}_y - R\mathbf{E}_y). \tag{19.43}$$

Another example of this observed in the Foucault pendulum, where the plane of oscillation of the pendulum appears to rotate with respect to the Earth. The Coriolis effect is responsible for this apparent rotation.

https://www.youtube.com/shorts/kbHYp3H3mDE

19.8 Classifications of Rigid Body Motions

Pure (rectilinear) translation: $\omega = \mathbf{0}$, $\alpha = \mathbf{0}$.

Curvilinear translation: material points follow parallel curved paths; $\omega = \mathbf{0}$, $\alpha = \mathbf{0}$ (e.g. Ferris wheel bucket).

Fixed point rotation: $\omega \neq \mathbf{0}$, fixed point with $\mathbf{v}_O = \mathbf{a}_O = \mathbf{0}$ (e.g. pendulum):

$$\mathbf{a}_P = \alpha \times \mathbf{r}_P + \omega \times (\omega \times \mathbf{r}_P). \tag{19.44}$$

The term $\alpha \times \mathbf{r}_P$ is the tangential acceleration; $\omega \times (\omega \times \mathbf{r}_P)$ is the centripetal acceleration.

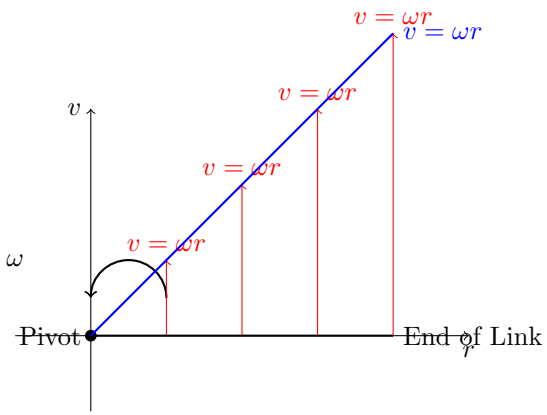
General motion: combined translation and rotation.

19.9 Instantaneous Center of Rotation

Think!

Question: Draw the velocity profile of a rotating link with a fixed end.

Answer



For a body in fixed point rotation, the velocity of each point on the body is perpendicular to the position vector from the axis of rotation to this point.

Think!

Question: Draw the velocity profile of a sphere rotating about its fixed center, or of a random rigid body in fixed point rotation.

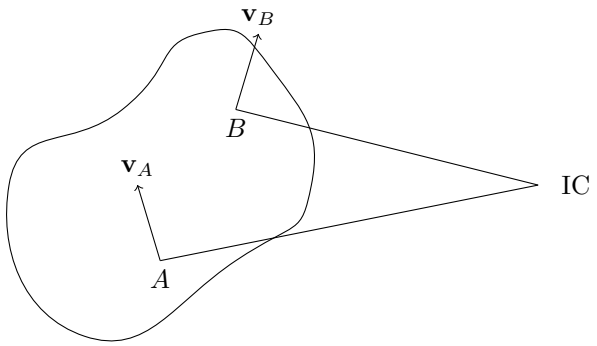
Answer

...

A body in general plane motion is not rotating about a fixed point, but **instantaneously** it rotates about the **instantaneous center of rotation (IC)** — a point with zero velocity at that instant. If $\mathbf{v}_{IC} = \mathbf{0}$, then for any material point A:

$$\mathbf{v}_A = \omega \times (\mathbf{r}_A - \mathbf{r}_{IC}). \tag{19.45}$$

So $\mathbf{r}_{A/IC}$ is perpendicular to $\mathbf{v}_A$. Knowing the directions of $\mathbf{v}_A$ and $\mathbf{v}_B$ for two material points identifies the IC.



For the IC at this instant:

$$\begin{aligned}\mathbf{v}_P &= \omega \times \mathbf{r}_{P/IC},\\ \mathbf{a}_P &= \alpha \times \mathbf{r}_{P/IC} + \omega \times (\omega \times \mathbf{r}_{P/IC}).\end{aligned} \tag{19.46}$$

> **!** Note!
>
> - The IC of a RB does not have to be on the RB itself, however, the IC behaves like a material point of the RB. One could consider a massless extension of the RB which containts the IC.
> - The IC can only be used for velocity analysis since its acceleration is nonzero.
> - The IC generally migrates through space.
> - For a translating rigid body the IC is at infinity.

Procedure:

- Use geometry to find the IC.
- Find $|\omega| = \|\omega\|$ using $\mathbf{v}_P = \omega \times \mathbf{r}_{P/IC}$.
- Get $\mathbf{v}_P$ for any point P.

Examples: Rod falling on corner and Crank slider.

https://youtu.be/E_5JPTllkDU

19.10 Doing Kinematics Geometrically

In some problems, we can use the laws of geometry that are true for every time and are differentiable.

Example: Problem 5/31.

Focus on triangle OAB:

- Let α be the angle OAB.
- Let β be the angle BOA. $\dot{\beta}$ is the angular velocity of bar OA.
- Let θ be the angle OBA. $\dot{\theta}$ is the angular velocity of bar BC.
- The law of sines holds at all times.
- The law of cosines holds at all times.
- The fact that OAB is a triangle holds at all times, so $\beta + \alpha + \theta = \pi$, which we can differentiate.

19.11 Kinematics of Rolling and Sliding

> **Note!**
>
> Terminology: "rolling" = roll without slip; "sliding" = roll with slip.

Consider a rigid body $\mathcal{B}$ in contact with a fixed surface $\mathcal{S}$. Let P be the material point of $\mathcal{B}$ in contact at time t, with unit normal $\mathbf{n}$ to $\mathcal{S}$.

Since P belongs to $\mathcal{B}$:

$$\mathbf{v}_P - \mathbf{v}_C = \omega \times (\mathbf{r}_P - \mathbf{r}_C), \tag{19.47}$$

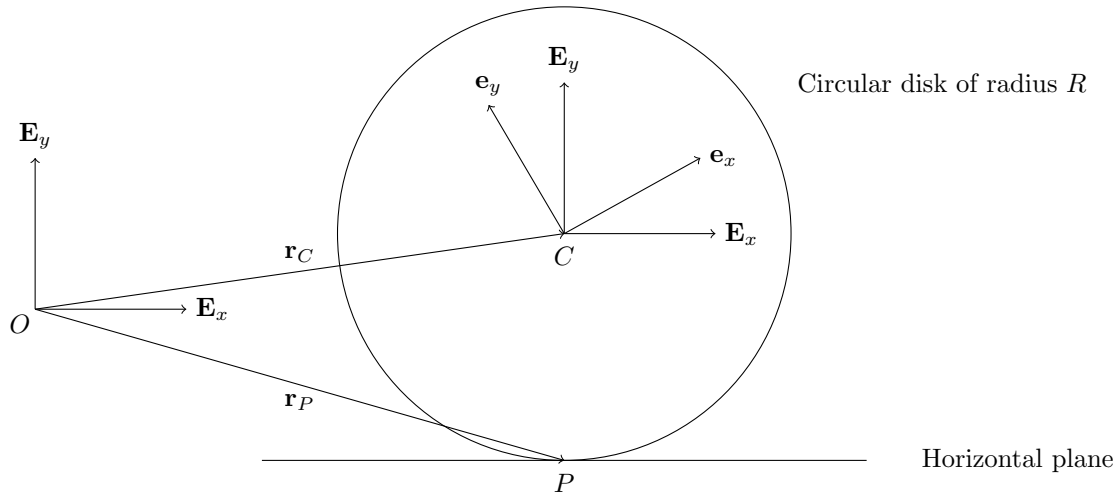
$$\mathbf{a}_P - \mathbf{a}_C = \alpha \times (\mathbf{r}_P - \mathbf{r}_C) + \omega \times (\omega \times (\mathbf{r}_P - \mathbf{r}_C)) \,. \tag{19.48}$$

- **Contact (touching):** $\mathbf{v}_P \cdot \mathbf{n} = 0$. Thus, $\mathbf{v}_C \cdot \mathbf{n} = -\omega \times (\mathbf{r}_P - \mathbf{r}_C) \cdot \mathbf{n}$.
- **Rolling (no slip):** $\mathbf{v}_P = \mathbf{0}$, so $\mathbf{v}_C = -\omega \times (\mathbf{r}_P - \mathbf{r}_C)$. Note: $\mathbf{a}_P \neq \mathbf{0}$ in general.

19.11.1 Example: Rolling Circular Disk

> **Think!**
>
> **Question:** What is the IC of a rolling disk?
>
> **Answer**
>
> The contact point with the surface.

> **Think!**
>
> **Question:** Draw the velocity profile of a wheel that is rolling without slipping on a road.
>
> **Answer**
>
> ...



Consider an upright homogeneous disk of radius R rolling on a plane, with angular velocity $\omega = \omega \mathbf{E}_z$ and angular acceleration $\alpha = \alpha \mathbf{E}_z$. The center of mass C moves in rectilinear motion:

$$\mathbf{r}_C = x\mathbf{E}_x + c\mathbf{E}_y, \qquad \mathbf{v}_C = \dot{x}\mathbf{E}_x, \qquad \mathbf{a}_C = \ddot{x}\mathbf{E}_x. \tag{19.49}$$

Define a corotational basis for the disk:

$$\mathbf{e}_x = \cos(\theta)\mathbf{E}_x + \sin(\theta)\mathbf{E}_y, \qquad \mathbf{e}_y = \cos(\theta)\mathbf{E}_y - \sin(\theta)\mathbf{E}_x, \qquad \mathbf{e}_z = \mathbf{E}_z. \tag{19.50}$$

Since the disk is rolling without slipping, $\mathbf{v}_P = \mathbf{0}$. Then, using the two-point velocity relation between C and P:

$$\mathbf{v}_C - \mathbf{v}_P = \mathbf{v}_C = \dot{x}\mathbf{E}_x = \omega \times (\mathbf{r}_C - \mathbf{r}_P) = \omega \mathbf{E}_z \times R\mathbf{E}_y = -\omega R \mathbf{E}_x, \tag{19.51}$$

so that

$$\dot{x} = -\omega R. \tag{19.52}$$

Differentiating this result with respect to time gives

$$\ddot{x} = -\alpha R. \tag{19.53}$$

Substituting this result into the two-point acceleration relation between C and P,

$$\mathbf{a}_C - \mathbf{a}_P = \alpha \times (\mathbf{r}_C - \mathbf{r}_P) + \omega \times (\mathbf{v}_C - \mathbf{v}_P), \tag{19.54}$$

$$-R\alpha\mathbf{E}_x - \mathbf{a}_P = -\alpha R\mathbf{E}_x - \omega^2 R\mathbf{E}_y, \tag{19.55}$$

$$\mathbf{a}_P = -\omega^2 R\mathbf{E}_y. \tag{19.56}$$

Thus, the acceleration of the contact point P is purely vertical (toward the center).

19.12 Center of Mass

The center of mass C of $\mathcal{B}$ has position vector

$$\mathbf{r}_C = \frac{\int_{\mathcal{B}} \mathbf{r}\, dm}{\int_{\mathcal{B}} dm}, \qquad dm = \rho(x,t)\, dv. \tag{19.57}$$

For rigid bodies, C behaves as a material point:

$$\mathbf{v}_C - \mathbf{v}_A = \omega \times (\mathbf{r}_C - \mathbf{r}_A), \qquad \mathbf{a}_C - \mathbf{a}_A = \alpha \times (\mathbf{r}_C - \mathbf{r}_A) + \omega \times (\omega \times (\mathbf{r}_C - \mathbf{r}_A)). \tag{19.58}$$

Example: Find the center of mass of any body from Set 15.

19.13 Linear Momentum

By definition, the linear momentum $\mathbf{G}$ of a body is the sum of the linear momenta of its constituents:

$$\mathbf{G} = \int_{\mathcal{B}} \mathbf{v}\, dm. \tag{19.59}$$

Equivalently,

$$\mathbf{G} = \int_{\mathcal{B}} \mathbf{v}\, dm = \int_{\mathcal{B}} \frac{d\mathbf{r}}{dt}\, dm = \frac{d}{dt}\left(\int_{\mathcal{B}} \mathbf{r}\, dm\right) = \frac{d}{dt}(m\mathbf{r}_C) = m\mathbf{v}_C. \tag{19.60}$$

https://youtu.be/101EriK4HIQ

19.14 Angular Momenta

The angular momentum of a system relative to any point P is

$$\mathbf{H}^P = \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_P) \times \mathbf{v}\, dm. \tag{19.61}$$

For P being the center of mass C and the fixed origin O, respectively, we have

$$\mathbf{H}^C = \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_C) \times \mathbf{v}\, dm, \qquad \mathbf{H}^O = \int_{\mathcal{B}} \mathbf{r} \times \mathbf{v}\, dm. \tag{19.62}$$

$\mathbf{H}^P$ and $\mathbf{H}^C$ are related through

$$\mathbf{H}^P = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{G}. \tag{19.63}$$

For the special case that P is the origin O, this gives

$$\mathbf{H}^O = \mathbf{H}^C + \mathbf{r}_C \times \mathbf{G}. \tag{19.64}$$

In other words, the angular momentum of a rigid body relative to a fixed point O is the sum of the angular momentum of the rigid body about its center of mass and the angular momentum of its center of mass relative to O.

https://youtu.be/BGa_d4LscWQ

19.14.1 Inertia Tensor

Recall that for any material point on the body, its velocity $\mathbf{v}$ is

$$\mathbf{v} - \mathbf{v}_C = \omega \times (\mathbf{r} - \mathbf{r}_C). \tag{19.65}$$

Define $\pi = \mathbf{r} - \mathbf{r}_C = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z$. Then,

$$\begin{aligned}
\mathbf{H}^C &= \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_C) \times \mathbf{v}\, dm && (19.66)\\
&= \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_C) \times (\mathbf{v}_C + \omega \times (\mathbf{r} - \mathbf{r}_C))\, dm && (19.67)\\
&= \int_{\mathcal{B}} \pi \times (\mathbf{v}_C + \omega \times \pi)\, dm && (19.68)\\
&= \int_{\mathcal{B}} \pi \times \mathbf{v}_C\, dm + \int_{\mathcal{B}} \pi \times (\omega \times \pi)\, dm. && (19.69)
\end{aligned}$$

The first integral vanishes (since $\int_{\mathcal{B}} \pi\, dm = \mathbf{0}$), and applying the BAC-CAB identity to the second gives

$$\mathbf{H}^C = \int_{\mathcal{B}} ((\pi \cdot \pi)\omega - (\pi \cdot \omega)\pi)\, dm = \mathbf{I}^C \omega. \tag{19.70}$$

Substituting $\pi = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z$ and $\omega = \omega_x\mathbf{e}_x + \omega_y\mathbf{e}_y + \omega_z\mathbf{e}_z$ and expanding component by component,

$$\mathbf{H}^C = (I^C_{xx}\omega_x + I^C_{xy}\omega_y + I^C_{xz}\omega_z)\mathbf{e}_x + (I^C_{xy}\omega_x + I^C_{yy}\omega_y + I^C_{yz}\omega_z)\mathbf{e}_y + (I^C_{xz}\omega_x + I^C_{yz}\omega_y + I^C_{zz}\omega_z)\mathbf{e}_z, \tag{19.71}$$

or, in matrix-vector form,

$$\begin{bmatrix} \mathbf{H}^C \cdot \mathbf{e}_x \\ \mathbf{H}^C \cdot \mathbf{e}_y \\ \mathbf{H}^C \cdot \mathbf{e}_z \end{bmatrix} = \begin{bmatrix} I^C_{xx} & I^C_{xy} & I^C_{xz} \\ I^C_{xy} & I^C_{yy} & I^C_{yz} \\ I^C_{xz} & I^C_{yz} & I^C_{zz} \end{bmatrix} \begin{bmatrix} \omega \cdot \mathbf{e}_x \\ \omega \cdot \mathbf{e}_y \\ \omega \cdot \mathbf{e}_z \end{bmatrix}. \tag{19.72}$$

The inertia matrix on the corotational basis:

$$[\mathbf{I}^C] = \begin{bmatrix} I^C_{xx} & I^C_{xy} & I^C_{xz} \\ I^C_{xy} & I^C_{yy} & I^C_{yz} \\ I^C_{xz} & I^C_{yz} & I^C_{zz} \end{bmatrix}, \tag{19.73}$$

with diagonal entries (**moments of inertia**) and off-diagonal entries (**products of inertia**):

$$I^C_{xx} = \int_{\mathcal{B}} (y^2 + z^2)\, dm, \qquad I^C_{yy} = \int_{\mathcal{B}} (x^2 + z^2)\, dm, \qquad I^C_{zz} = \int_{\mathcal{B}} (x^2 + y^2)\, dm, \tag{19.74}$$

$$I^C_{xy} = -\int_{\mathcal{B}} xy\, dm, \qquad I^C_{xz} = -\int_{\mathcal{B}} xz\, dm, \qquad I^C_{yz} = -\int_{\mathcal{B}} yz\, dm. \tag{19.75}$$

When $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ is an eigenbasis of $[\mathbf{I}^C]$, the products of inertia vanish.

> **!** Note!
>
> Some texts (e.g. MKB) instead *define* $I^C_{xy} = \int_{\mathcal{B}} xy\, dm$ (without the minus sign); with that convention the off-diagonal entries of the inertia matrix above must be negated to match.

Using tensors, the same result can be written

$$\mathbf{H}^C = \int_{\mathcal{B}} ((\pi \cdot \pi)\omega - (\pi \cdot \omega)\pi)\, dm \tag{19.76}$$

$$= \int_{\mathcal{B}} (\pi \cdot \pi\, \mathbf{I} - \pi \otimes \pi)\, dm\, \omega \tag{19.77}$$

$$= \mathbf{I}^C \omega, \tag{19.78}$$

where $\mathbf{I}^C$ is the **inertia tensor** of the rigid body about its center of mass C. To find the matrix components of the inertia tensor, first simplify

$$\pi \cdot \pi\, \mathbf{I} = (x^2 + y^2 + z^2)(\mathbf{e}_x \otimes \mathbf{e}_x + \mathbf{e}_y \otimes \mathbf{e}_y + \mathbf{e}_z \otimes \mathbf{e}_z), \tag{19.79}$$

$$\pi \otimes \pi = x^2 \mathbf{e}_x \otimes \mathbf{e}_x + xy\mathbf{e}_x \otimes \mathbf{e}_y + xz\mathbf{e}_x \otimes \mathbf{e}_z + yx\mathbf{e}_y \otimes \mathbf{e}_x + y^2 \mathbf{e}_y \otimes \mathbf{e}_y + yz\mathbf{e}_y \otimes \mathbf{e}_z + zx\mathbf{e}_z \otimes \mathbf{e}_x + zy\mathbf{e}_z \otimes \mathbf{e}_y + z^2 \tag{19.80}$$

Combining these two results and writing the inertia tensor in matrix form recovers

$$[\mathbf{I}^C] = \begin{bmatrix} I^C_{xx} & I^C_{xy} & I^C_{xz} \\ I^C_{xy} & I^C_{yy} & I^C_{yz} \\ I^C_{xz} & I^C_{yz} & I^C_{zz} \end{bmatrix}, \tag{19.81}$$

with the same component formulas given above.

https://youtu.be/BP1N9zFeY98

The moments of inertia are a measure of the mass distribution of a rigid body about the axes of rotation. The greater the moment of inertia, the more difficult it is to change the angular velocity of the rigid body about that axis.

Think!

Question: Is my moment of inertia about an axis through my body greater when my arms are closed or open?

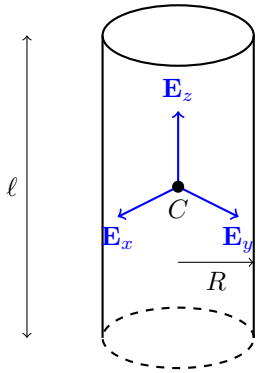
Answer

Greater when arms are open — mass is farther from the axis.

https://www.youtube.com/watch?v=M_YCWDXCwZM&t=152s

Example: Tight rope walker. A tight rope walker holds a pole to increase their moment of inertia against rotating off of the rope. This gives the tight rope walker more time to react.

19.14.2 Example: Solid Cylinder



With $dm = \rho\, r\, dr\, d\theta\, dz$, we have

$$I_{zz}^C = \int_{\mathcal{B}} (x^2 + y^2)\, dm \tag{19.82}$$

$$= \int_{\mathcal{B}} r^2\, dm \tag{19.83}$$

$$= \rho \int_{-\ell/2}^{\ell/2} \int_0^{2\pi} \int_0^R r^3\, dr\, d\theta\, dz \tag{19.84}$$

$$= \rho\, 2\pi\ell \int_0^R r^3\, dr \tag{19.85}$$

$$= \frac{1}{2}\rho\pi\ell R^4. \tag{19.86}$$

Since $V = \pi R^2 \ell$ and $m = \rho\pi\ell R^2$, this simplifies to $I_{zz}^C = \dfrac{mR^2}{2}$.

https://youtu.be/Z5ugvTjmLys

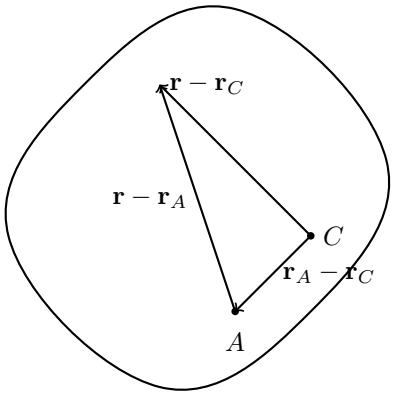
19.14.3 Example: Cylindrical Hoop

Every dm has $r = R$, so:

$$I_{zz}^C = R^2 \int_{\mathcal{B}} dm = mR^2. \tag{19.87}$$

19.15 Parallel Axis Theorem

Suppose we already know $I_{xx}^C, I_{xy}^C, I_{xz}^C$, etc. about the center of mass C. We want $I_{xx}^A, I_{xy}^A, I_{xz}^A$, etc. about a fixed material point A on the rigid body, with axes at A parallel to those at C.



Recall that

$$I_{xx} = \int_{\mathcal{B}} (x^2 + y^2)\, dm, \qquad I_{xy} = -\int_{\mathcal{B}} xy\, dm, \qquad \mathbf{r} - \mathbf{r}_C = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z, \tag{19.88}$$

and let $\mathbf{r}_A - \mathbf{r}_C = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z$. Then,

$$I_{xx}^A = \int_{\mathcal{B}} \left[(y - A_y)^2 + (z - A_z)^2\right]\, dm \tag{19.89}$$

$$= I_{xx}^C + (A_y^2 + A_z^2) \int_{\mathcal{B}} dm - 2A_y \int_{\mathcal{B}} y\, dm - 2A_z \int_{\mathcal{B}} z\, dm. \tag{19.90}$$

Because $\int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_C)\, dm = \mathbf{0}$, we have $\int_{\mathcal{B}} y\, dm = 0$ and $\int_{\mathcal{B}} z\, dm = 0$, so

$$I_{xx}^A = I_{xx}^C + m(A_y^2 + A_z^2). \tag{19.91}$$

The moment of inertia always increases when we move away from the center of mass. Similarly,

$$I_{xy}^A = -\int_{\mathcal{B}} (x - A_x)(y - A_y)\, dm = I_{xy}^C - mA_xA_y. \tag{19.92}$$

https://www.youtube.com/shorts/0CfRKuqjZ7Y

19.15.1 Deriving the Parallel Axis Theorem Using Tensors

Analogous to

$$\mathbf{I}^C = \int_{\mathcal{B}} \left((\mathbf{r} - \mathbf{r}_C) \cdot (\mathbf{r} - \mathbf{r}_C)\, \mathbf{I} - (\mathbf{r} - \mathbf{r}_C) \otimes (\mathbf{r} - \mathbf{r}_C) \right) dm, \tag{19.93}$$

let the inertia tensor of the rigid body $\mathcal{B}$ about any material point P be

$$\mathbf{I}^P = \int_{\mathcal{B}} \left((\mathbf{r} - \mathbf{r}_P) \cdot (\mathbf{r} - \mathbf{r}_P)\, \mathbf{I} - (\mathbf{r} - \mathbf{r}_P) \otimes (\mathbf{r} - \mathbf{r}_P) \right) dm. \tag{19.94}$$

To write $\mathbf{I}^P$ in terms of $\mathbf{I}^C$, introduce $\mathbf{r}_C - \mathbf{r}_C$:

$$\mathbf{I}^P = \int_{\mathcal{B}} \left((\mathbf{r} + \mathbf{r}_C - \mathbf{r}_C - \mathbf{r}_P) \cdot (\mathbf{r} + \mathbf{r}_C - \mathbf{r}_C - \mathbf{r}_P)\, \mathbf{I} - (\mathbf{r} + \mathbf{r}_C - \mathbf{r}_C - \mathbf{r}_P) \otimes (\mathbf{r} + \mathbf{r}_C - \mathbf{r}_C - \mathbf{r}_P) \right) dm \tag{19.95}$$

$$= \ldots \tag{19.96}$$

$$= \|\mathbf{r}_P - \mathbf{r}_C\|^2 \mathbf{I} - m(\mathbf{r}_P - \mathbf{r}_C) \otimes (\mathbf{r}_P - \mathbf{r}_C). \tag{19.97}$$

A second derivation starts from the relationship between the angular momentum of the body about P and about C,

$$\mathbf{H}^P = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{G}. \tag{19.98}$$

Since C and P are both material points on the rigid body, their velocities are related by

$$\mathbf{v}_C = \mathbf{v}_P + \omega \times (\mathbf{r}_C - \mathbf{r}_P). \tag{19.99}$$

Substituting this into the angular momentum relationship,

$$\mathbf{H}^P = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times (\mathbf{v}_P + \omega \times (\mathbf{r}_C - \mathbf{r}_P)) \tag{19.100}$$

$$= \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{v}_P + (\mathbf{r}_C - \mathbf{r}_P) \times (\omega \times (\mathbf{r}_C - \mathbf{r}_P)). \tag{19.101}$$

Using the BAC-CAB identity, $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$, the last term simplifies:

$$(\mathbf{r}_C - \mathbf{r}_P) \times (\omega \times (\mathbf{r}_C - \mathbf{r}_P)) = \omega \left((\mathbf{r}_C - \mathbf{r}_P) \cdot (\mathbf{r}_C - \mathbf{r}_P) \right) - (\mathbf{r}_C - \mathbf{r}_P) \left((\mathbf{r}_C - \mathbf{r}_P) \cdot \omega \right) \tag{19.102}$$

$$= \left(\|\mathbf{r}_C - \mathbf{r}_P\|^2 \mathbf{I} - (\mathbf{r}_C - \mathbf{r}_P) \otimes (\mathbf{r}_C - \mathbf{r}_P) \right) \omega \tag{19.103}$$

$$= \left(\|\mathbf{r}_P - \mathbf{r}_C\|^2 \mathbf{I} - (\mathbf{r}_P - \mathbf{r}_C) \otimes (\mathbf{r}_P - \mathbf{r}_C) \right) \omega. \tag{19.104}$$

Thus, defining

$$\mathbf{I}^P \equiv \mathbf{I}^C + m \|\mathbf{r}_P - \mathbf{r}_C\|^2 \mathbf{I} - m(\mathbf{r}_P - \mathbf{r}_C) \otimes (\mathbf{r}_P - \mathbf{r}_C), \tag{19.105}$$

we obtain

$$\mathbf{H}^P = \mathbf{I}^P \omega + (\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{v}_P. \tag{19.106}$$

In component form, taking the corotational axes at P and C to be parallel with $\mathbf{r}_P - \mathbf{r}_C = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z$,

$$[\mathbf{I}^P] = \begin{bmatrix} I^C_{xx} & I^C_{xy} & I^C_{xz} \\ I^C_{xy} & I^C_{yy} & I^C_{yz} \\ I^C_{xz} & I^C_{yz} & I^C_{zz} \end{bmatrix} + m \begin{bmatrix} A_x^2 + A_y^2 + A_z^2 & 0 & 0 \\ 0 & A_x^2 + A_y^2 + A_z^2 & 0 \\ 0 & 0 & A_x^2 + A_y^2 + A_z^2 \end{bmatrix} - m \begin{bmatrix} A_x^2 & A_xA_y & A_xA_z \\ A_xA_y & A_y^2 & A_yA_z \\ A_xA_z & A_yA_z & A_z^2 \end{bmatrix}, \tag{19.107}$$

recovering the results above.

Combining both derivations, the parallel axis theorem in tensor form is:

$$\mathbf{I}^A = \mathbf{I}^C + m \left\|\mathbf{r}_A - \mathbf{r}_C\right\|^2 \mathbf{I} - m(\mathbf{r}_A - \mathbf{r}_C) \otimes (\mathbf{r}_A - \mathbf{r}_C). \tag{19.108}$$

> **!** Note!
>
> The parallel axis theorem applies only between the center of mass C and another material point — not between any two arbitrary material points. The moment of inertia always increases when moving away from C.

https://youtu.be/8R_cF2e_9nU

19.16 Radius of Gyration

The radius of gyration k_z about the z-axis satisfies $mk_z^2 = I_{zz}$. Similarly, k_x and k_y are defined by $mk_x^2 = I_{xx}$ and $mk_y^2 = I_{yy}$.

https://www.youtube.com/shorts/pRxOpqpKq78

19.17 Angular Momentum About a Moving Point P

For a material point P on the rigid body:

$$\begin{aligned} \mathbf{H}^P &= \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_P) \times \mathbf{v}\, dm && (19.109)\\ &= \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_P) \times [\mathbf{v}_P + \omega \times (\mathbf{r} - \mathbf{r}_P)]\, dm && (19.110)\\ &= \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_P) \times [\omega \times (\mathbf{r} - \mathbf{r}_P)]\, dm + \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_P) \times \mathbf{v}_P\, dm && (19.111)\\ &= \mathbf{I}^P \omega + (\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{v}_P. && (19.112) \end{aligned}$$

This definition of $\mathbf{I}^P$ agrees with the one derived above. If P is chosen to be the center of mass C,

$$\mathbf{H}^C = \mathbf{I}^C\omega + (\mathbf{r}_C - \mathbf{r}_C) \times m\mathbf{v}_P = \mathbf{I}^C\omega. \tag{19.113}$$

If P is chosen to be a fixed point O ($\mathbf{v}_O = \mathbf{0}$),

$$\mathbf{H}^O = \mathbf{I}^O\omega + (\mathbf{r}_O - \mathbf{r}_C) \times m\mathbf{v}_O = \mathbf{I}^O\omega. \tag{19.114}$$

In general:

$$\dot{\mathbf{H}}^P = \mathbf{I}^P\mathring{\omega} + \omega \times (\mathbf{I}^P\omega) + \frac{d}{dt}\left[(\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{v}_P\right]. \tag{19.115}$$

> **!** Note!
>
> As the body moves, the components of $\mathbf{I}^O$, $\mathbf{I}^C$, and $\mathbf{I}^P$ expressed on a corotational basis remain constant.

19.18 Summary

Corotational basis $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{E}_z\}$ fixed to the rigid body:

$$\dot{\mathbf{e}}_x = \omega \times \mathbf{e}_x, \qquad \dot{\mathbf{e}}_y = \omega \times \mathbf{e}_y, \qquad \omega = \dot{\theta}\,\mathbf{E}_z. \tag{19.116}$$

Velocity/acceleration of two points A, B on a rigid body:

$$\mathbf{v}_B - \mathbf{v}_A = \omega \times (\mathbf{r}_B - \mathbf{r}_A), \tag{19.117}$$
$$\mathbf{a}_B - \mathbf{a}_A = \alpha \times (\mathbf{r}_B - \mathbf{r}_A) + \omega \times (\mathbf{v}_B - \mathbf{v}_A). \tag{19.118}$$

Instantaneous center (IC): the point on a body (or its extension) with instantaneous zero velocity. Lies at the intersection of perpendiculars to the velocity vectors of two points.

Particle in a non-inertial frame (point A fixed to body, B moving relative to body):

$$\mathbf{v}_B - \mathbf{v}_A = \omega \times (\mathbf{r}_B - \mathbf{r}_A) + \mathbf{v}_{\text{rel}}, \tag{19.119}$$
$$\mathbf{a}_B - \mathbf{a}_A = \alpha \times (\mathbf{r}_B - \mathbf{r}_A) + \omega \times (\omega \times (\mathbf{r}_B - \mathbf{r}_A)) + 2\omega \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}. \tag{19.120}$$

The Coriolis acceleration is $2\omega \times \mathbf{v}_{\text{rel}}$.

Angular momentum and inertia tensor:

$$H_z^C = I_{zz}^C\,\omega, \qquad I_{zz}^C = \int_{\mathcal{B}} (x^2 + y^2)\,dm. \tag{19.121}$$

Parallel axis theorem: $I_{zz}^A = I_{zz}^C + md^2$ where d is the distance from C to A.

19.19 Exercises

19.19.1 Set 15 – Rigid Body Kinematics

1. [MKB 05-010] *(ans.* $\mathbf{v}_B = -11\mathbf{E}_x$ *m/s,* $\mathbf{a}_B = 22\mathbf{E}_x - 220\mathbf{E}_y$ *m/s*^{2}*)*

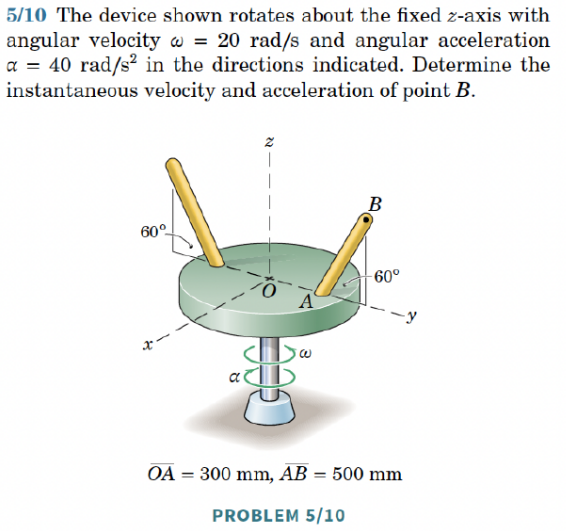
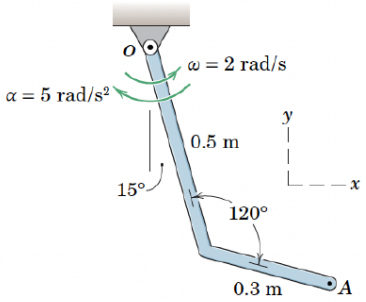


Figure 19.1: MKB 05-010.

2. [MKB 05-013] *(ans.* $\mathbf{v}_A = 1.121\mathbf{E}_x + 0.838\mathbf{E}_y$ *m/s,* $\mathbf{a}_B = -4.48\mathbf{E}_x + 0.1465\mathbf{E}_y$ *m/s*^{2}*)*

5/13 The bent flat bar rotates about a fixed axis through point O with the instantaneous angular properties indicated in the figure. Determine the velocity and acceleration of point A.

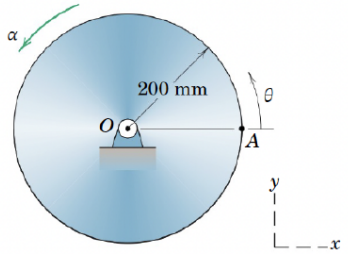


PROBLEM 5/13

Figure 19.2: MKB 05-013.

3. [05-018] *(ans.* $\mathbf{v}_A = -0.374\mathbf{E}_x + 0.1905\mathbf{E}_y$ *m/s,* $\mathbf{a}_A = -0.757\mathbf{E}_x - 0.605\mathbf{E}_y$ *m/s*^{2}*)*

5/18 Point A of the circular disk is at the angular position $\theta = 0$ at time $t = 0$. The disk has angular velocity $\omega_0 = 0.1$ rad/s at $t = 0$ and subsequently experiences a constant angular acceleration $\alpha = 2$ rad/s^2. Determine the velocity and acceleration of point A in terms of fixed $\mathbf{i}$ and $\mathbf{j}$ unit vectors at time $t = 1$ s.

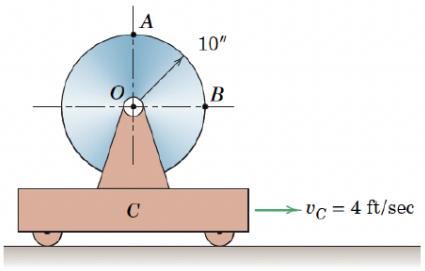


PROBLEM 5/18

Figure 19.3: MKB 05-018.

4. [05-049] *(ans. (a)* $N = 91.7$ *rev/min CCW; (b)* $N = 45.8$ *rev/min CCW; (c)* $N = 45.8$ *rev/min CW)*

5/49 The cart has a velocity of 4 ft/sec to the right. Determine the angular speed N of the wheel so that point A on the top of the rim has a velocity (a) equal to 4 ft/sec to the left, (b) equal to zero, and (c) equal to 8 ft/sec to the right.

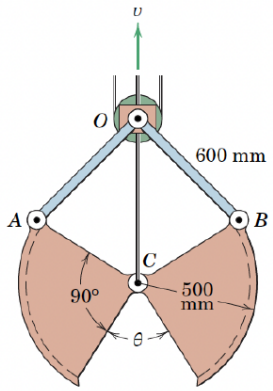


PROBLEM 5/49

Figure 19.4: MKB 05-049.

5. [05-065] *(ans.* $\omega = 0.722$ *rad/s)*

5/65 The elements of a simplified clam-shell bucket for a dredge are shown. The cable which opens and closes the bucket passes through the block at O. With O as a fixed point, determine the angular velocity ω of the bucket jaws when $\theta = 45°$ as they are closing. The upward velocity of the control cable is 0.5 m/s as it passes through the block.

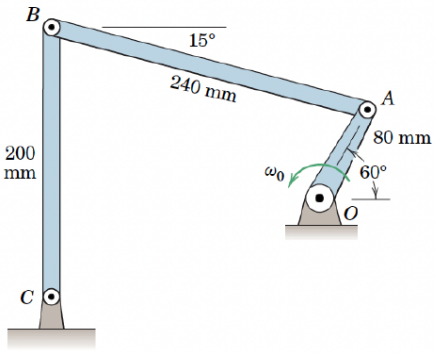


PROBLEM 5/65

Figure 19.5: MKB 05-065.

6. [05-069] *(ans.* $\omega_{AB} = 1.725$ *rad/s CCW,* $\omega_{BC} = 4$ *rad/s CCW)*

5/69 SS A four-bar linkage is shown in the figure (the ground "link" OC is considered the fourth bar). If the drive link OA has a counterclockwise angular velocity $\omega_0 = 10$ rad/s, determine the angular velocities of links AB and BC.



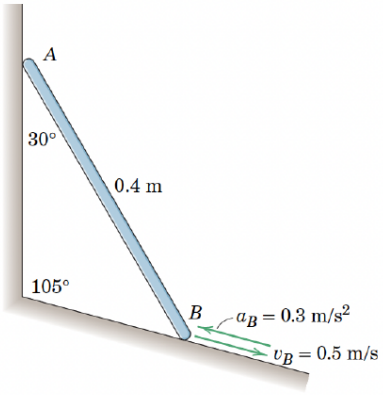
PROBLEM 5/69

Figure 19.6: MKB 05-069.

19.19.2 Set 16 – IC and Particle in Non-inertial Frame

1. [MKB 05-102] *(ans.* $\alpha = 0.286$ *rad/s*^{2}, $a_A = 0.653$ *m/s*^{2} *down)*

5/102 The bar of Prob. 5/66 is repeated here. The ends of the 0.4-m bar remain in contact with their respective support surfaces. End B has a velocity of 0.5 m/s and an acceleration of 0.3 m/s^2 in the directions shown. Determine the angular acceleration of the bar and the acceleration of end A.



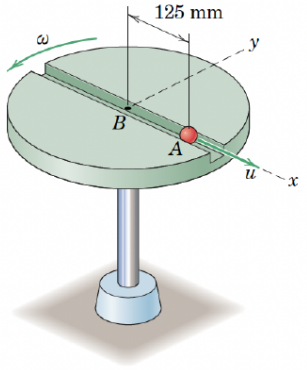
PROBLEM 5/102

Figure 19.7: MKB 05-102.

https://youtu.be/n34Z2jwwnW0

2. [MKB 05-125] *(ans.* $\mathbf{v}_A = 0.1\mathbf{E}_x + 0.25\mathbf{E}_y$ *m/s,* $\beta = 68.2^\circ$*)*

5/125 The disk rotates with angular speed $\omega = 2$ rad/s. The small ball A is moving along the radial slot with speed $u = 100$ mm/s relative to the disk. Determine the absolute velocity of the ball and state the angle β between this velocity vector and the positive x-axis.



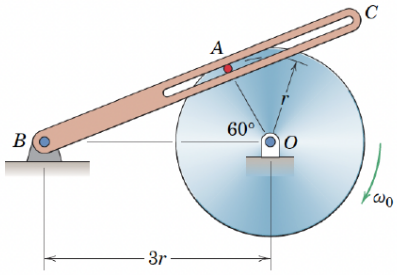
PROBLEM 5/125

Figure 19.8: MKB 05-125.

https://youtu.be/PyM7KmlPirU

3. [05-143] *(ans.* $v_{\text{rel}} = 3.93$ *m/s at* 19.11°, $a_{\text{rel}} = 15.22$ *m/s*^{2} *at* 19.11°, $\omega_{BC} = 1.429$ *rad/s CW,* $\alpha_{BC} = 170.0$ *rad/s*^{2} *CW)*

5/143 The disk rotates about a fixed axis through point O with a clockwise angular velocity $\omega_0 = 20$ rad/s and a counterclockwise angular acceleration $\alpha_0 = 5$ rad/s^2 at the instant under consideration. The value of r is 200 mm. Pin A is fixed to the disk but slides freely within the slotted member BC. Determine the velocity and acceleration of A relative to slotted member BC and the angular velocity and angular acceleration of BC.



PROBLEM 5/143

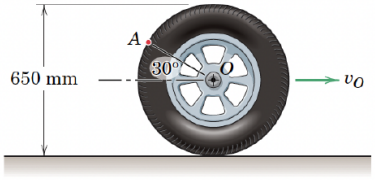
Figure 19.9: MKB 05-143.

https://youtu.be/IG7Qtj2TdFQ

19.19.3 Set 17 – Rolling and Sliding

1. [MKB 05-055] *(ans.* $v_O = 6.93$ *m/s,* $\omega = 21.3$ *rad/s CW)*

5/55 The magnitude of the absolute velocity of point A on the automobile tire is 12 m/s when A is in the position shown. What are the corresponding velocity v_O of the car and the angular velocity ω of the wheel? (The wheel rolls without slipping.)

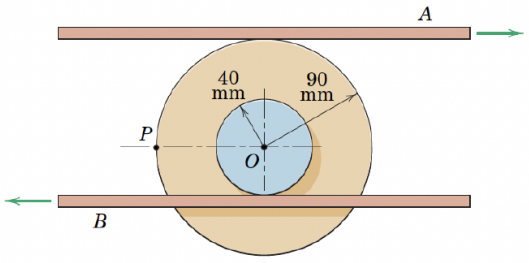


PROBLEM 5/55

Figure 19.10: MKB 05-055.

2. [MKB 05-087] *(ans.* $v_O = 120$ *mm/s,* $v_P = 216$ *mm/s)*

5/87 The attached wheels roll without slipping on the plates A and B, which are moving in opposite directions as shown. If $v_A = 60$ mm/s to the right and $v_B = 200$ mm/s to the left, determine the speeds of the center O and the point P for the position shown.

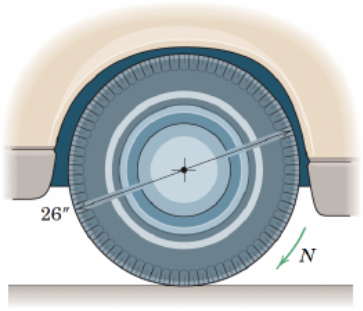


PROBLEM 5/87

Figure 19.11: MKB 05-087.

3. [05-091] *(ans.* $v = 15.71$ *ft/s right,* $v_s = 6.89$ *ft/s left)*

5/91 The rear driving wheel of a car has a diameter of 26 in. and has an angular speed N of 200 rev/min on an icy road. If the instantaneous center of zero velocity is 4 in. above the point of contact of the tire with the road, determine the velocity v of the car and the slipping velocity v_s of the tire on the ice.

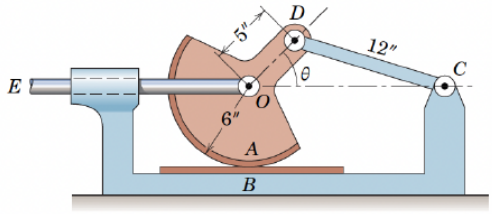


PROBLEM 5/91

Figure 19.12: MKB 05-091.

4. [05-093] *(ans.* $v_A = 9.19$ *ft/sec left)*

5/93 A device which tests the resistance to wear of two materials A and B is shown. If the link EO has a velocity of 4 ft/sec to the right when $\theta = 45°$, determine the rubbing velocity v_A.



PROBLEM 5/93

Figure 19.13: MKB 05-093.

5. [05-100] *(ans.* $\mathbf{v}_A = 5.12\mathbf{E}_x + 2.12\mathbf{E}_y$ *m/s,* $\mathbf{a}_B = -16.25\mathbf{E}_x + 2.5\mathbf{E}_y$ *m/s*^{2}*)*

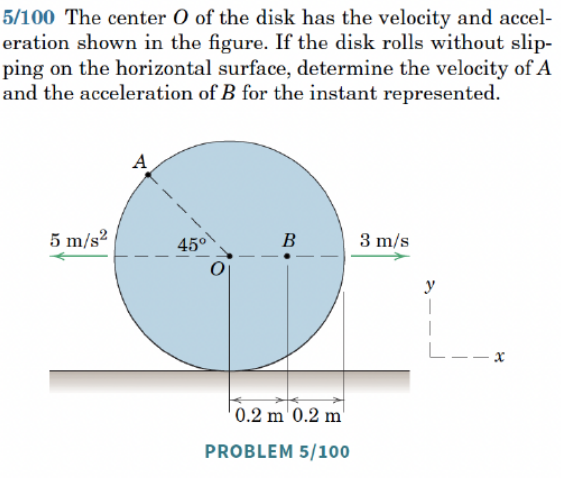


Figure 19.14: MKB 05-100.

19.19.4 Set 18 – Moments of Inertia

1. [MKB B-004] Integration required. *(ans.* $I_{xx} = \frac{3}{10}mr^2$, $I_{yy} = \frac{2}{5}m(r^4 + h^2)/(r^2)$*)*

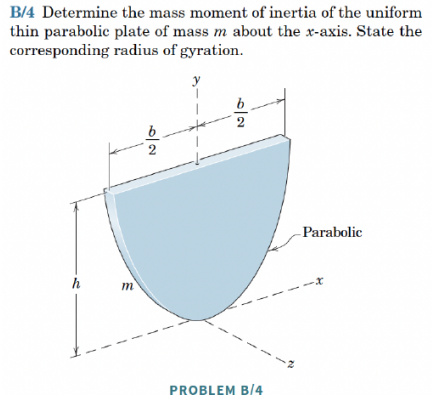
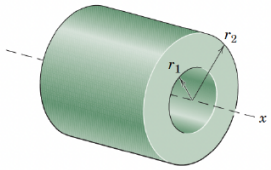


Figure 19.15: MKB B-004.

2. [MKB B-029] Treat the hollow cylinder as a full cylinder of radius r_2 minus a cylinder of radius r_1. *(ans.* $I_{xx} = \frac{1}{2}m(r_1^2 + r_2^2)$*)*

B/29 Determine I_{xx} for the cylinder with a centered circular hole. The mass of the body is m.

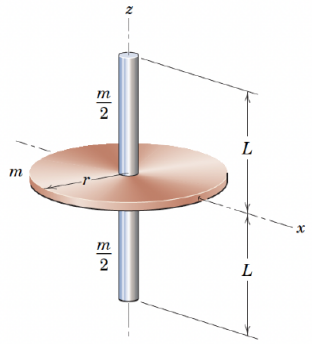


PROBLEM B/29

Figure 19.16: MKB B-029.

3. [B-032] *(ans.* $L = r\sqrt{2\sqrt{3}}$*)*

B/32 Determine the length L of each of the slender rods of mass $m/2$ which must be centrally attached to the faces of the thin homogeneous disk of mass m in order to make the mass moments of inertia of the unit about the x- and z-axes equal.

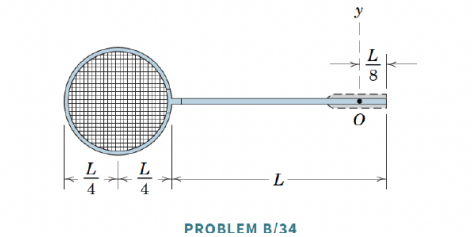


PROBLEM B/32

Figure 19.17: MKB B-032.

4. [B-034] *(ans.* $I_{yy} = \rho L^3 \left(\frac{43}{192} + \frac{83\pi}{128}\right)$*)*

B/34 A badminton racket is constructed of uniform slender rods bent into the shape shown. Neglect the strings and the built-up wooden grip and estimate the mass moment of inertia about the y-axis through O, which is the location of the player's hand. The mass per unit length of the rod material is ρ.



PROBLEM B/34

Figure 19.18: MKB B-034.

20 Kinetics of Rigid Bodies

20.1 Balance Laws for a Rigid Body

The balance laws for a rigid body are **Euler's first law** (Balance of Linear Momentum, BoLM)

$$\mathbf{F} = m\mathbf{a}_C, \tag{20.1}$$

and **Euler's second law** (Balance of Angular Momentum, BoAM), which can be expressed in three equivalent forms:

$$\mathbf{M}^O = \dot{\mathbf{H}}^O \quad \text{about a fixed point } O, \tag{20.2}$$

$$\mathbf{M}^C = \dot{\mathbf{H}}^C \quad \text{about the center of mass } C, \tag{20.3}$$

$$\mathbf{M}^P = \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G} = \dot{\mathbf{H}}^C + (\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{a}_C \quad \text{about any material point } P. \tag{20.4}$$

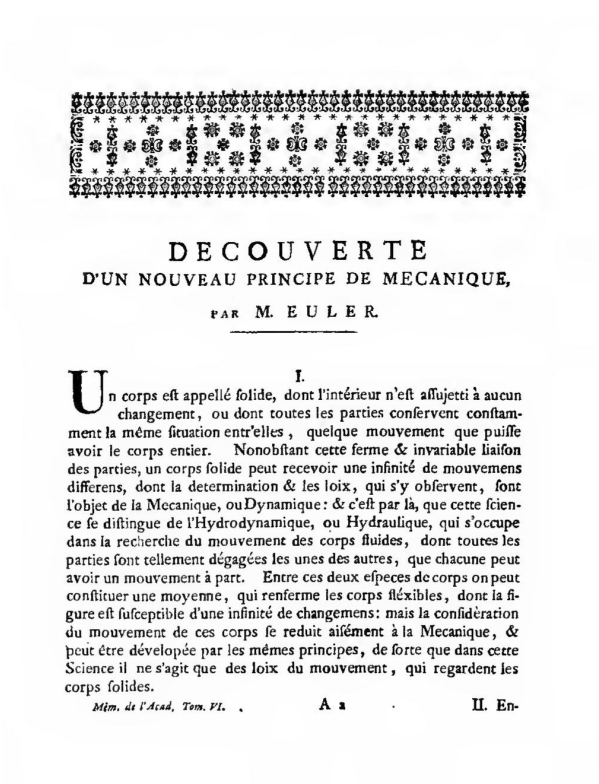
The BoAM is an **independent postulate**, not derivable from the BoLM.

Historical Perspective: Leonhard Euler (1707–1783)

Figure 20.1: Portrait by Emanuel Handmann, 1753 (Wikimedia Commons, public domain).

Swiss mathematician and physicist, one of the most prolific scientists in history. Euler recast Newton's mechanics in the vectorial, differential-equation form used today, and in 1750 he was the first to state the balance of linear and angular momentum as independent postulates for a general body – the two balance laws that bear his name above.

Historical Perspective: Euler's Original Paper (1750)



DECOUVERTE
D'UN NOUVEAU PRINCIPE DE MECANIQUE,

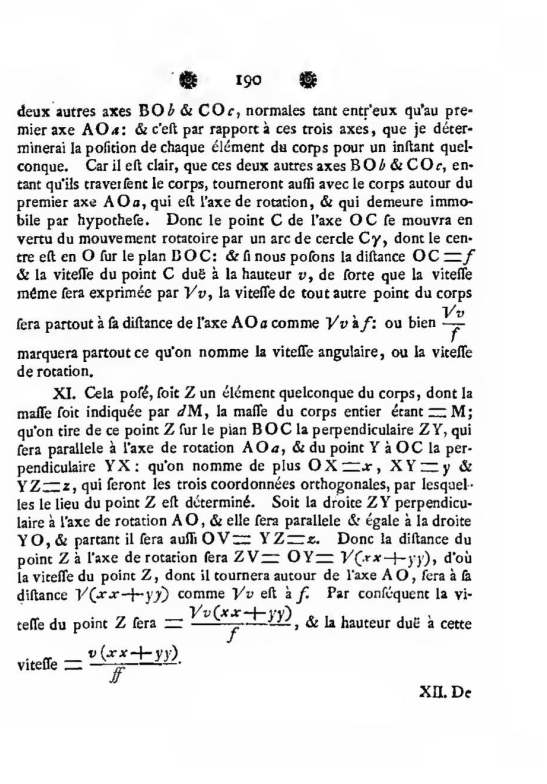
PAR M. EULER.

I.

Un corps eſt appellé ſolide, dont l'intérieur n'eſt aſſujetti à aucun changement, ou dont toutes les parties conſervent conſtamment la même ſituation entr'elles, quelque mouvement que puiſſe avoir le corps entier. Nonobſtant cette ferme & invariable liaiſon des parties, un corps ſolide peut recevoir une infinité de mouvemens differens, dont la determination & les loix, qui s'y obſervent, ſont l'objet de la Mecanique, ou Dynamique: & c'eſt par là, que cette ſcience ſe diſtingue de l'Hydrodynamique, ou Hydraulique, qui s'occupe dans la recherche du mouvement des corps fluides, dont toutes les parties ſont tellement dégagées les unes des autres, que chacune peut avoir un mouvement à part. Entre ces deux eſpeces de corps on peut conſtituer une moyenne, qui renferme les corps fléxibles, dont la figure eſt ſuſceptible d'une infinité de changemens: mais la conſidération du mouvement de ces corps ſe reduit aiſément à la Mecanique, & peut être dévelopée par les mêmes principes, de ſorte que dans cette Science il ne s'agit que des loix du mouvement, qui regardent les corps ſolides.

Mém. de l'Acad. Tom. VI. A a II. En-

(a) Opening page (archive.org, public domain).



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deux autres axes BO*b* & CO*c*, normales tant entr'eux qu'au premier axe AO*a*: & c'eſt par rapport à ces trois axes, que je déterminerai la poſition de chaque élément du corps pour un inſtant quelconque. Car il eſt clair, que ces deux autres axes BO*b* & CO*c*, entant qu'ils traverſent le corps, tourneront auſſi avec le corps autour du premier axe AO*a*, qui eſt l'axe de rotation, & qui demeure immobile par hypotheſe. Donc le point C de l'axe OC ſe mouvra en vertu du mouvement rotatoire par un arc de cercle Cγ, dont le centre eſt en O ſur le plan BOC: & ſi nous poſons la diſtance OC $= f$ & la viteſſe du point C duë à la hauteur v, de ſorte que la viteſſe même ſera exprimée par $\sqrt{v}$, la viteſſe de tout autre point du corps ſera partout à ſa diſtance de l'axe AO*a* comme $\sqrt{v}$ à f: ou bien $\frac{\sqrt{v}}{f}$ marquera partout ce qu'on nomme la viteſſe angulaire, ou la viteſſe de rotation.

XI. Cela poſé, ſoit Z un élément quelconque du corps, dont la maſſe ſoit indiquée par dM, la maſſe du corps entier étant $= M$; qu'on tire de ce point Z ſur le plan BOC la perpendiculaire ZY, qui ſera parallele à l'axe de rotation AO*a*, & du point Y à OC la perpendiculaire YX: qu'on nomme de plus OX $= x$, XY $= y$ & YZ $= z$, qui ſeront les trois coordonnées orthogonales, par leſquelles le lieu du point Z eſt déterminé. Soit la droite ZY perpendiculaire à l'axe de rotation AO, & elle ſera parallele & égale à la droite YO, & partant il ſera auſſi OV $=$ YZ $= z$. Donc la diſtance du point Z à l'axe de rotation ſera ZV $=$ OY $= \sqrt{(xx+yy)}$, d'où la viteſſe du point Z, dont il tournera autour de l'axe AO, ſera à ſa diſtance $\sqrt{(xx+yy)}$ comme $\sqrt{v}$ eſt à f. Par conſéquent la viteſſe du point Z ſera $= \frac{\sqrt{v(xx+yy)}}{f}$, & la hauteur duë à cette viteſſe $= \frac{v(xx+yy)}{ff}$.

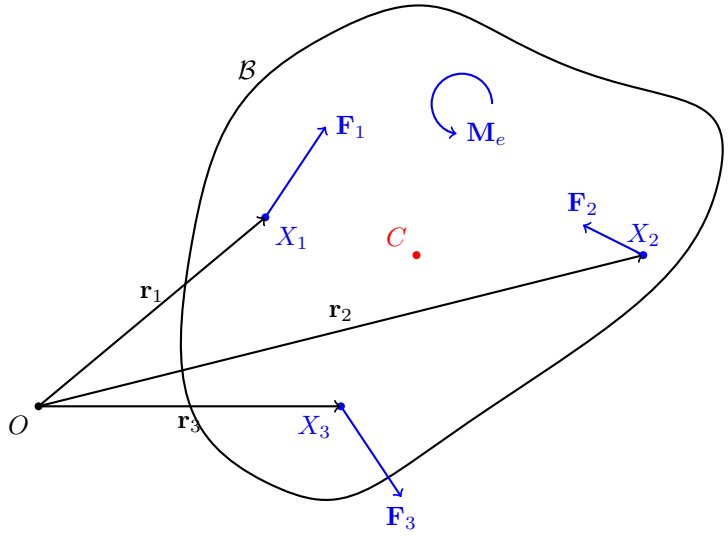
XII. De

(a) The page stating the balance laws (archive.org, public domain).

Euler's "Découverte d'un nouveau principe de mécanique," Mémoires de l'Académie des sciences de Berlin, 1750. In this paper, Euler proposed that $\mathbf{F} = m\mathbf{a}$ (his "first law") should be applied not only to particles, as Newton had done, but independently to every infinitesimal element of a body, and introduced the balance of angular momentum (his "second law") as a separate, independent postulate – the foundation of rigid body dynamics used in this chapter.

20.2 Resultant Forces and Moments

The resultant force $\mathbf{F}$ is the sum of all external forces. The resultant moment about a point includes the moment due to forces plus any applied pure moments $\mathbf{M}_e$ not due to forces.



For K forces $\mathbf{F}_i$ acting at positions $\mathbf{r}_i$ plus a pure moment $\mathbf{M}_e$:

$$\mathbf{F} = \sum_{i=1}^{K} \mathbf{F}_i, \tag{20.5}$$

$$\mathbf{M}^O = \mathbf{M}_e + \sum_{i=1}^{K} \mathbf{r}_i \times \mathbf{F}_i, \tag{20.6}$$

$$\mathbf{M}^C = \mathbf{M}_e + \sum_{i=1}^{K} (\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i. \tag{20.7}$$

Examples of pure moments: reaction moments $\mathbf{M}_R$ at joints; torsional spring moments $-K_T(\theta - \theta_0)\mathbf{E}_z$.

https://www.youtube.com/watch?v=Era7OOKyVEw

20.2.1 Does Weight Result in a Moment About the Center of Mass?

The total weight is $\mathbf{W} = \int_{\mathcal{B}} \mathbf{g}\, dm = \mathbf{g} \int_{\mathcal{B}} dm = m\mathbf{g}$, and its moment about C is:

$$\mathbf{M}^C = \int_{\mathcal{B}} \pi \times \mathbf{g}\, dm = \left(\int_{\mathcal{B}} \pi\, dm \right) \times \mathbf{g} = \mathbf{0}, \tag{20.8}$$

since $\int_{\mathcal{B}} \pi\, dm = \mathbf{0}$ by definition of the center of mass.

20.3 Equivalence of the BoAM Forms

The following devepments are true for rigid bodies where the inertia matrix is constant in the body frame.

20.3.1 BoAM About Any Point P

Starting from $\mathbf{M}^C = \dot{\mathbf{H}}^C$ and replacing $\mathbf{M}^P = \mathbf{M}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{F}$:

$$\mathbf{M}^P = \dot{\mathbf{H}}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{F}. \tag{20.9}$$

Substituting $\dot{\mathbf{H}}^C = \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G} + (\mathbf{r}_P - \mathbf{r}_C) \times m\mathbf{a}_C$ and simplifying:

$$\begin{aligned} \mathbf{M}^P &= \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G} + (\mathbf{r}_P - \mathbf{r}_C) \times m\mathbf{a}_C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{F}, \\ &= \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G}, \end{aligned} \tag{20.10}$$

recovering the BoAM about a general material point P.

20.3.2 BoAM About a Fixed Point O

If P is a fixed point O (so $\mathbf{v}_O = \mathbf{0}$), the above simplifies to $\mathbf{M}^O = \dot{\mathbf{H}}^O$.

Think!

Question: When is the angular momentum of a rigid body about a fixed point O or about its center of mass C conserved? Under what conditions are these anguar momenta conserved in a certain direction (eg. along $\mathbf{E}_z$)?

Answer

https://www.youtube.com/shorts/VmeM0BNnGR0

https://www.youtube.com/watch?v=64t-dVtDwkQ

20.4 Calculating the Derivative of Angular Momentum

Recall that

$$\mathbf{H}^P = \mathbf{I}^P \omega + (\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{v}_P \tag{20.11}$$

if P is taken to be the center of mass C or a fixed point O respectively, we get

$$\begin{aligned}\mathbf{H}^C &= \mathbf{I}^C\omega,\\ \mathbf{H}^P &= \mathbf{I}^P\omega.\end{aligned} \tag{20.12}$$

In component form,

$$\mathbf{H}^C = (I^C_{xx}\omega_x + I^C_{xy}\omega_y + I^C_{xz}\omega_z)\mathbf{e}_x + (I^C_{xy}\omega_x + I^C_{yy}\omega_y + I^C_{yz}\omega_z)\mathbf{e}_y + (I^C_{xz}\omega_x + I^C_{yz}\omega_y + I^C_{zz}\omega_z)\mathbf{e}_z. \tag{20.13}$$

where

$$\omega = \omega_x\mathbf{e}_x + \omega_y\mathbf{e}_y + \omega_z\mathbf{e}_z. \tag{20.14}$$

We need to calculate $\dot{\mathbf{H}}^C$.

$$\dot{\mathbf{H}}^C = \overset{\circ}{\mathbf{H}}{}^C + \omega \times \mathbf{H}^C. \tag{20.15}$$

where $\overset{\circ}{\mathbf{H}}{}^C$ is the corotational rate of $\mathbf{H}$, that is the time derivative of $\mathbf{H}$ that is obtained while keeping $\mathbf{e}_x$, $\mathbf{e}_y$, and $\mathbf{e}_z$ fixed:

$$\overset{\circ}{\mathbf{H}}{}^C = (I^C_{xx}\dot{\omega}_x + I^C_{xy}\dot{\omega}_y + I^C_{xz}\dot{\omega}_z)\mathbf{e}_x + (I^C_{xy}\dot{\omega}_x + I^C_{yy}\dot{\omega}_y + I^C_{yz}\dot{\omega}_z)\mathbf{e}_y + (I^C_{xz}\dot{\omega}_x + I^C_{yz}\dot{\omega}_y + I^C_{zz}\dot{\omega}_z)\mathbf{e}_z. \tag{20.16}$$

For the case where $\omega = \omega\mathbf{E}_z$ considered exclusively in this class, the previous expressions simplify to

$$\begin{aligned}\mathbf{H}^C &= I^C_{xz}\omega_z\mathbf{e}_x + I^C_{yz}\omega_z\mathbf{e}_y + I^C_{zz}\omega_z\mathbf{e}_z,\\ \dot{\mathbf{H}}^C &= (I^C_{xz}\dot{\omega} - I^C_{yz}\omega^2)\mathbf{e}_x + (I^C_{yz}\dot{\omega} + I^C_{xz}\omega^2)\mathbf{e}_y + I^C_{zz}\dot{\omega}\mathbf{E}_z.\end{aligned} \tag{20.17}$$

Analogous developments yield

$$\begin{aligned}\dot{\mathbf{H}}^O &= (I^O_{xz}\dot{\omega} - I^O_{yz}\omega^2)\mathbf{e}_x + (I^O_{yz}\dot{\omega} + I^O_{xz}\omega^2)\mathbf{e}_y + I^O_{zz}\dot{\omega}\mathbf{E}_z,\\ \dot{\mathbf{H}}^C &= (I^C_{xz}\dot{\omega} - I^C_{yz}\omega^2)\mathbf{e}_x + (I^C_{yz}\dot{\omega} + I^C_{xz}\omega^2)\mathbf{e}_y + I^C_{zz}\dot{\omega}\mathbf{E}_z,\\ \dot{\mathbf{H}}^P &= (I^P_{xz}\dot{\omega} - I^P_{yz}\omega^2)\mathbf{e}_x + (I^P_{yz}\dot{\omega} + I^P_{xz}\omega^2)\mathbf{e}_y + I^P_{zz}\dot{\omega}\mathbf{E}_z + \frac{d}{dt}\left((\mathbf{r}_C - \mathbf{r}_P)\times m\mathbf{v}_P\right).\end{aligned} \tag{20.18}$$

For fixed axis rotation, it is generally better to sum the moments about a fixed point passing through the axis of rotation. For general plane motion, there is no fixed point, so the BoAM is either summed about the center of mass or about a general material point P on the body.

! Note!

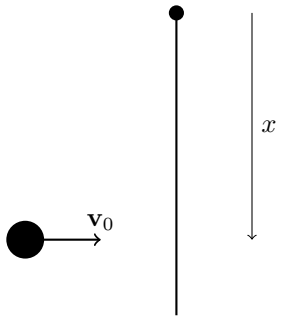
Remember, the BoAM about a general point has additional terms to $\mathbf{I}^P\omega$:

$$\begin{aligned}\mathbf{M}^P &= (I^P_{xz}\dot{\omega} - I^P_{yz}\omega^2)\mathbf{e}_x + (I^P_{yz}\dot{\omega} + I^P_{xz}\omega^2)\mathbf{e}_y + I^P_{zz}\dot{\omega}\mathbf{E}_z + (\mathbf{v}_P - \mathbf{v}_C)\times\mathbf{G},\\ &= (I^C_{xz}\dot{\omega} - I^C_{yz}\omega^2)\mathbf{e}_x + (I^C_{yz}\dot{\omega} + I^C_{xz}\omega^2)\mathbf{e}_y + I^C_{zz}\dot{\omega}\mathbf{E}_z + (\mathbf{r}_C - \mathbf{r}_P)\times m\mathbf{a}_C.\end{aligned} \tag{20.19}$$

20.5 Impact Problems and Center of Percussion

For rigid bodies, impact ideas are the same as in systems of particles.

Example: Consider a projectile fired horizontally at a pendulum (in the horizontal plane, no gravity). Find position x along the pendulum such that the reaction force at the pin is nearly zero during impact. If x is large, the horizontal reaction at O points left; if x is small, it points right. The optimal x that minimizes the reaction is called the **center of percussion**.



20.6 Impulse and Momentum for Rigid Bodies

The **linear impulse - linear momentum equation** is

$$\int_{t_A}^{t^B} \mathbf{F}dt = \mathbf{G}(t_B) - \mathbf{G}(t_A). \tag{20.20}$$

The **angular impulse - angular momentum equations** are equivalently

$$\int_{t_A}^{t_B} \mathbf{M}^C dt = \mathbf{H}^C(t_B) - \mathbf{H}^C(t_A) \quad \text{where } C \text{ is the center of mass,} \tag{20.21}$$

$$\int_{t_A}^{t_B} \mathbf{M}^O dt = \mathbf{H}^O(t_B) - \mathbf{H}^O(t_A) \quad \text{if there is a fixed point } O. \tag{20.22}$$

20.7 Moment-Free Motion of a Rigid Body

The Tennis Racket Problem: A tennis racket is thrown in the air with a certain angular velocity. The racket rotates about its center of mass, and the angular momentum is conserved. The racket has three principal axes of rotation, and the stability of rotation depends on which axis is used. Rotation about the axis with the largest or smallest moment of inertia is stable, while rotation about the intermediate axis is unstable.

This is also known as the Dzhanibekov effect.

You can learn more about this here and here.

https://www.youtube.com/watch?v=1VPfZ_XzisU&t=22s

https://www.youtube.com/watch?v=1VPfZ_XzisU&t=22s

20.8 Summary

For K forces and a moment $\mathbf{M}_e$ acting on a rigid body, the balance laws are:

$$\mathbf{F} = m\frac{d\mathbf{v}_C}{dt}, \qquad \mathbf{M}^O = \dot{\mathbf{H}}^O \quad \text{(or equivalently } \mathbf{M}^C = \dot{\mathbf{H}}^C\text{)}. \tag{20.23}$$

For fixed-axis rotation with $\omega = \omega\mathbf{E}_z$ and $I_{xz} = I_{yz} = 0$:

$$\mathbf{F} = m\dot{\mathbf{v}}_C, \qquad \mathbf{M} = I_{zz}\dot{\omega}\mathbf{E}_z. \tag{20.24}$$

Four classes of applications: (1) purely translational motion ($\omega = \alpha = \mathbf{0}$), (2) rigid body with a fixed point, (3) rolling and sliding rigid bodies, (4) imbalanced rotors.

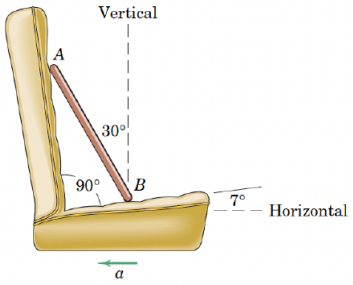
It is important to note that for the second set of applications, the balance law $\mathbf{M}^O = \dot{\mathbf{H}}^O$ is more convenient to use than $\mathbf{M} = \dot{\mathbf{H}}$. The role of $\mathbf{M}_R$ in these problems is to ensure that the axis of rotation remains $\mathbf{E}_z$. Finally, the four steps discussed are used as a guide to solving all of the applications.

20.9 Exercises

20.9.1 Set 19 – Rigid Body Translation

1. [MKB 06-006] *(ans. $a = 5.66\ m/s^2$)*

6/6 A uniform slender rod rests on a car seat as shown. Determine the deceleration a for which the rod will begin to tip forward. Assume that friction at B is sufficient to prevent slipping.



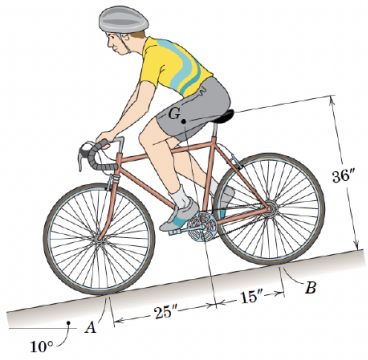
PROBLEM 6/6

Figure 20.4: MKB 06-006.

2. [06-012] *(ans.* $a = 16.43$ *ft/sec*^{2}*)*

$a = 16.43$ ft/sec^2)

6/12 The bicyclist applies the brakes as he descends the 10° incline. What deceleration a would cause the dangerous condition of tipping about the front wheel A? The combined center of mass of the rider and bicycle is at G.

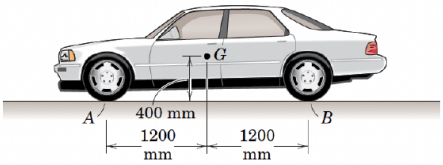


PROBLEM 6/12

Figure 20.5: MKB 06-012.

3. [06-013] *(ans.* $N_A = 6.85$ *kN up,* $N_B = 9.34$ *kN up)*

6/13 **SS** The 1650-kg car has its mass center at G. Calculate the normal forces N_A and N_B between the road and the front and rear pairs of wheels under conditions of maximum acceleration. The mass of the wheels is small compared with the total mass of the car. The coefficient of static friction between the road and the rear driving wheels is 0.80.



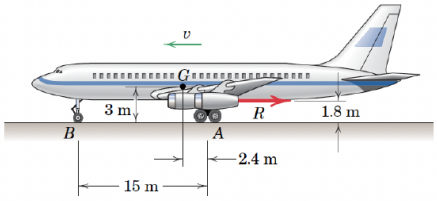
PROBLEM 6/13

Figure 20.6: MKB 06-013.

4. [06-022] *(ans. $N = 257$ kN up)*

$N = 257$ kN up)

6/22 A jet transport with a landing speed of 200 km/h reduces its speed to 60 km/h with a negative thrust R from its jet thrust reversers in a distance of 425 m along the runway with constant deceleration. The total mass of the aircraft is 140 Mg with mass center at G. Compute the reaction N under the nose wheel B toward the end of the braking interval and prior to the application of mechanical braking. At the lower speed, aerodynamic forces on the aircraft are small and may be neglected.



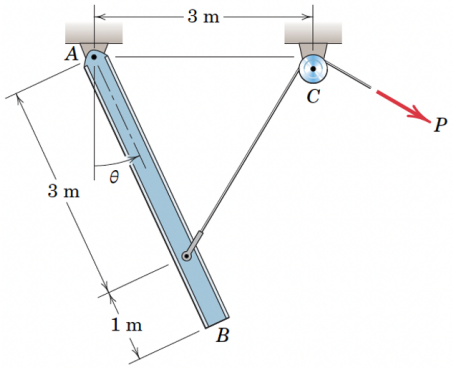
PROBLEM 6/22

Figure 20.7: MKB 06-022.

20.9.2 Set 20 – Fixed Point Rotation

1. [MKB 06-029] *(ans. $\alpha = 1.193$ rad/s^2 CCW, $F_A = 769$ N)*

6/29 The uniform 100-kg beam is freely hinged about its upper end A and is initially at rest in the vertical position with $\theta = 0$. Determine the initial angular acceleration α of the beam and the magnitude F_A of the force supported by the pin at A due to the application of the force $P = 300$ N on the attached cable.

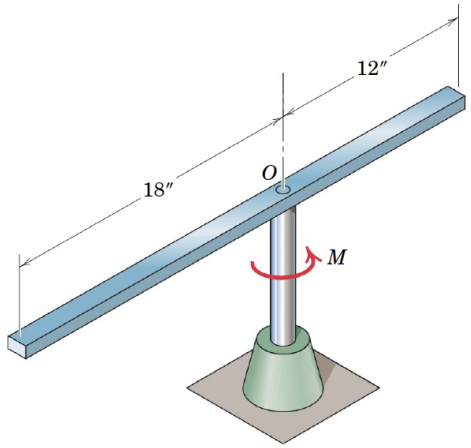


PROBLEM 6/29

Figure 20.8: MKB 06-029.

2. [MKB 06-035] *(ans. $R = 3.57$ lb)*

6/35 The 30-in. slender bar weighs 20 lb and is mounted on a vertical shaft at O. If a torque $M = 100$ lb-in. is applied to the bar through its shaft, calculate the horizontal force R on the bearing as the bar starts to rotate.



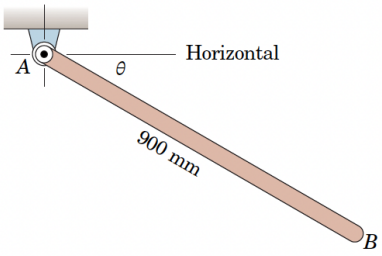
PROBLEM 6/35

Figure 20.9: MKB 06-035.

https://youtu.be/gvBZyEq_bGo

3. [06-037] *(ans. $A = 56.3$ N)*

6/37 The uniform slender bar AB has a mass of 8 kg and swings in a vertical plane about the pivot at A. If $\dot{\theta} = 2$ rad/s when $\theta = 30°$, compute the force supported by the pin at A at that instant.

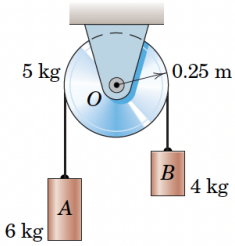


PROBLEM 6/37

Figure 20.10: MKB 06-037.

4. [06-039] *(ans. (a) $\alpha = 7.85$ rad/s^2 CCW, (b) $\alpha = 6.28$ rad/s^2 CCW)*

6/39 Determine the angular acceleration of the uniform disk if (*a*) the rotational inertia of the disk is ignored and (*b*) the inertia of the disk is considered. The system is released from rest, the cord does not slip on the disk, and bearing friction at O may be neglected.

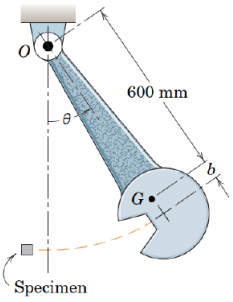


PROBLEM 6/39

Figure 20.11: MKB 06-039.

5. [06-051] *(ans. $b = 40.7$ mm, $R = 167.8$ N)*

6/51 A device for impact testing consists of a 34-kg pendulum with mass center at G and with radius of gyration about O of 620 mm. The distance b for the pendulum is selected so that the force on the bearing at O has the least possible value during impact with the specimen at the bottom of the swing. Determine b and calculate the magnitude of the total force R on the bearing O an instant after release from rest at $\theta = 60°$.



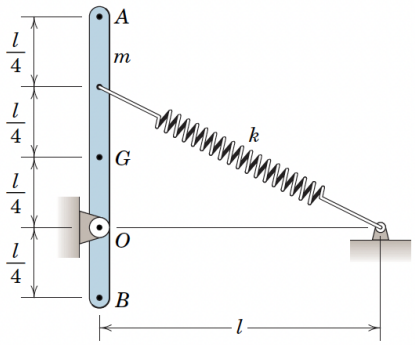
PROBLEM 6/51

Figure 20.12: MKB 06-051.

https://youtu.be/Q7PV2pxBuwc

6. [06-054] *(ans.* $\alpha = \frac{6g}{7l} - \frac{12k}{7m}(\sqrt{5} - \sqrt{3})$*)*

6/54 The spring is uncompressed when the uniform slender bar is in the vertical position shown. Determine the initial angular acceleration α of the bar when it is released from rest in a position where the bar has been rotated 30° clockwise from the position shown. Neglect any sag of the spring, whose mass is negligible.



PROBLEM 6/54

Figure 20.13: MKB 06-054.

7. [MKB 07-066] *Dynamic imbalance of a rotating shaft carrying two offset point masses.*

7/66 The slender shaft carries two offset particles, each of mass m, and rotates about the z-axis with the constant angular rate ω as indicated. Determine the x- and y-components of the bearing reactions at A and B due to the dynamic imbalance of the shaft for the position shown.

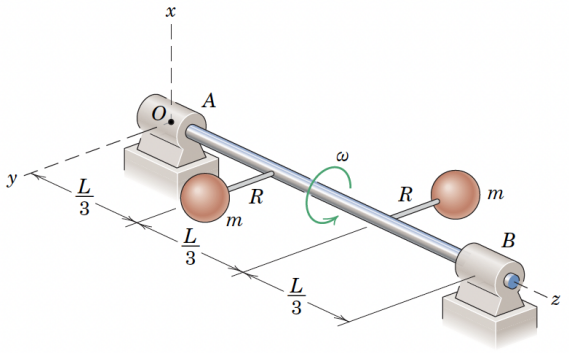


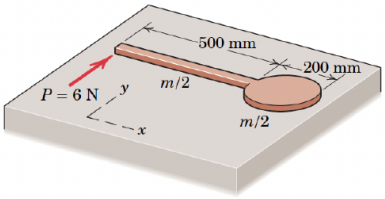
Figure 20.14: MKB 07-066.

https://youtu.be/HGiVnQbItTI

20.9.3 Set 21 – General Plane Motion

1. [MKB 06-061] *(ans.* $\alpha = 48.8\ rad/s^2\ \ CW,\ \bar{a}_x = 0,\ \bar{a}_y = 5\ m/s^2$*)*

6/61 The body consists of a uniform slender bar and a uniform disk, each of mass $m/2$. It rests on a smooth surface. Determine the angular acceleration α and the acceleration of the mass center of the body when the force $P = 6$ N is applied as shown. The value of the mass m of the entire body is 1.2 kg.



PROBLEM 6/61

Figure 20.15: MKB 06-061.

2. [MKB 06-062] *(ans. A:* $\alpha_A = \frac{g}{r}\sin\theta,\ \mu_s = 0$*; B:* $\alpha_B = \frac{g}{2r}\sin\theta,\ \mu_s = \frac{1}{2}\tan\theta$*)*

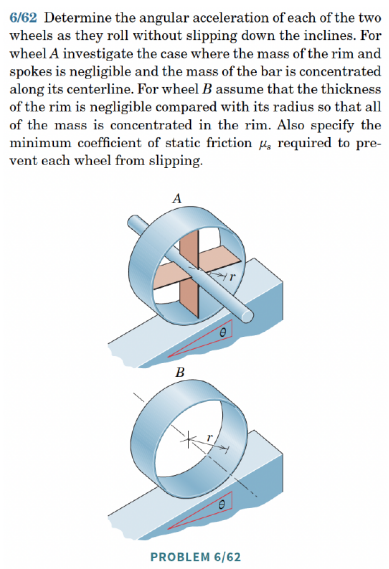


Figure 20.16: MKB 06-062.

3. [06-063] *(ans.* $\bar{a} = 13.80$ *ft/sec*^{2} *down incline,* $F = 1.714$ *lb up incline)*

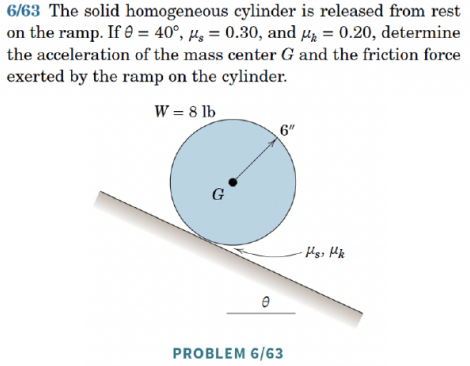
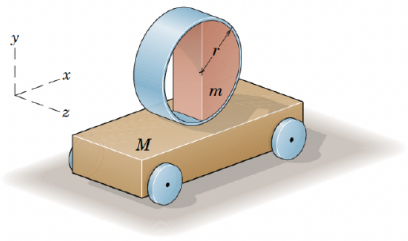


Figure 20.17: MKB 06-063.

4. [06-070] *(ans.* $a = \frac{8(m+M)g}{3\pi(m+3M)}$ *left,* $\alpha = \frac{8mg}{3\pi r(m+3M)}$ *CW)*

6/70 The system of Prob. 6/18 is repeated here. If the hoop- and semicylinder-assembly is centered on the top of the stationary cart and the system is released from rest, determine the initial acceleration a of the cart and the angular acceleration α of the hoop and semicylinder. Friction between the hoop and cart is sufficient to prevent slip. Motion takes place in the x-y plane. Neglect the mass of the cart wheels and any friction in the wheel bearings.



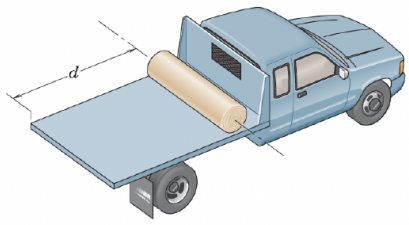
PROBLEM 6/70

Figure 20.18: MKB 06-070.

https://youtu.be/3UvEbjlaBwc

5. [06-076] *(ans.* $s = \frac{3d}{2}$*)*

6/76 The truck, initially at rest with a solid cylindrical roll of paper in the position shown, moves forward with a constant acceleration a. Find the distance s which the truck goes before the paper rolls off the edge of its horizontal bed. Friction is sufficient to prevent slipping.

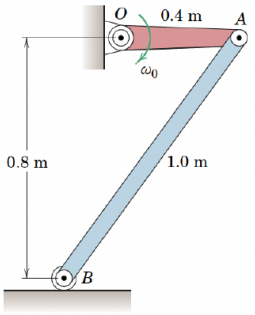


PROBLEM 6/76

Figure 20.19: MKB 06-076.

6. [06-077] *(ans.* $N_B = 36.4\ N\ up$*)*

6/77 The crank OA rotates in the vertical plane with a constant clockwise angular velocity ω_0 of 4.5 rad/s. For the position where OA is horizontal, calculate the force under the light roller B of the 10-kg slender bar AB.



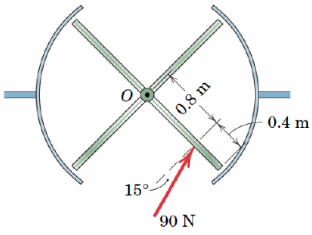
PROBLEM 6/77

Figure 20.20: MKB 06-077.

20.9.4 Set 22 – Impulse and Momentum for Rigid Bodies

1. [MKB 06-136] *(ans.* $\omega = 1.811$ *rad/s CCW)*

6/136 A person who walks through the revolving door exerts a 90-N horizontal force on one of the four door panels and keeps the 15° angle constant relative to a line which is normal to the panel. If each panel is modeled by a 60-kg uniform rectangular plate which is 1.2 m in length as viewed from above, determine the final angular velocity ω of the door if the person exerts the force for 3 seconds. The door is initially at rest and friction may be neglected.

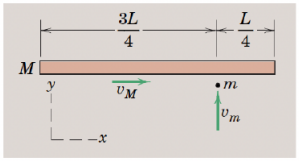


PROBLEM 6/136

Figure 20.21: MKB 06-136.

2. [MKB 06-142] *(ans.* $\mathbf{v} = \frac{Mu_M\mathbf{E}_x + m\mathbf{v}_m}{M+m}$, $\omega = \frac{12vm}{L(4M+7m)}$ *CCW)*

6/142 A uniform slender bar of mass M and length L is translating on the smooth horizontal x-y plane with a velocity v_M when a particle of mass m traveling with a velocity v_m as shown strikes and becomes embedded in the bar. Determine the final linear and angular velocities of the bar with its embedded particle.

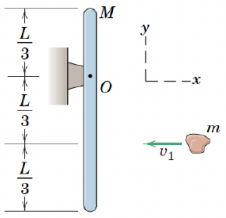


PROBLEM 6/142

Figure 20.22: MKB 06-142.

3. [06-145] *(ans.* $\omega = \frac{3mv_1}{(M+m)L}$ *CW)*

6/145 The wad of clay of mass m is initially moving with a horizontal velocity v_1 when it strikes and sticks to the initially stationary uniform slender bar of mass M and length L. Determine the final angular velocity of the combined body and the x-component of the linear impulse applied to the body by the pivot O during the impact.



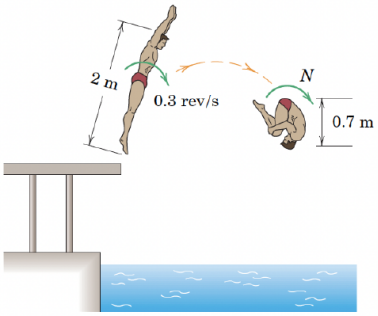
PROBLEM 6/145

Figure 20.23: MKB 06-145.

4. [06-146] *(ans.* $N = 2.04$ *rev/s)*

$N = 2.04$ rev/s)

6/146 Just after leaving the platform, the diver's fully extended 80-kg body has a rotational speed of 0.3 rev/s about an axis normal to the plane of the trajectory. Estimate the angular velocity N later in the dive when the diver has assumed the tuck position. Make reasonable assumptions concerning the mass moment of inertia of the body in each configuration.

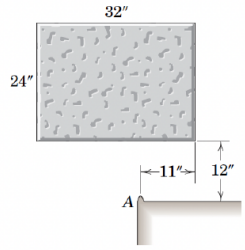


PROBLEM 6/146

Figure 20.24: MKB 06-146.

5. [06-148] *(ans.* $\omega = 1.593$ *rad/s CCW,* $n = 91.7\%$*)*

6/148 The uniform concrete block, which weighs 171 lb and falls from rest in the horizontal position shown, strikes the fixed corner A, and pivots around it with no rebound. Calculate the angular velocity ω of the block immediately after it hits the corner and the percentage loss n of energy due to the impact.

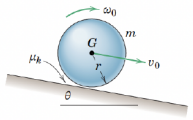


PROBLEM 6/148

Figure 20.25: MKB 06-148.

6. [06-155] *(ans.* $t = \frac{2v_0}{g(7\mu_k \cos\theta - 2\sin\theta)}$*)*

6/155 The homogeneous sphere of mass m and radius r is projected along the incline of angle θ with an initial speed v_0 and no angular velocity ($\omega_0 = 0$). If the coefficient of kinetic friction is μ_k, determine the time duration t of the period of slipping. In addition, state the velocity v of the mass center G and the angular velocity ω at the end of the period of slipping.



PROBLEM 6/155

Figure 20.26: MKB 06-155.

https://youtu.be/0N8tDgnrXNA

21 The Work-Energy Theorem for a Rigid Body

The **Koenig decomposition** for the kinetic energy of a rigid body is:

$$T = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \mathbf{H}^C \cdot \omega. \tag{21.1}$$

In this class, $\omega = \omega \mathbf{E}_z$, so $\frac{1}{2}\mathbf{H}^C \cdot \omega = \frac{1}{2} I_{zz} \omega^2$.

For fixed point rotation about O, the kinetic energy simplifies to:

$$T = \frac{1}{2} \mathbf{H}^O \cdot \omega \tag{21.2}$$

which further simplifies to $T = \frac{1}{2} I_{zz}^O \omega^2$ if $\omega = \omega \mathbf{E}_z$.

The **work-energy theorem** for a rigid body has two equivalent forms:

$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega = \sum_{i=1}^{N} \mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_e \cdot \omega. \tag{21.3}$$

Integrating from t_A to t_B:

$$T_B - T_A = W_{\mathbf{F},AB} + W_{\mathbf{M},AB} = \sum_{i=1}^{N} W_{\mathbf{F}_i,AB} + W_{\mathbf{M}_e,AB}, \tag{21.4}$$

where

$$W_{\mathbf{F},AB} = \int_{t_A}^{t_B} \mathbf{F} \cdot \mathbf{v}_C \, dt, \qquad W_{\mathbf{M},AB} = \int_{t_A}^{t_B} \mathbf{M}^C \cdot \omega \, dt, \tag{21.5}$$

$$W_{\mathbf{F}_i,AB} = \int_{t_A}^{t_B} \mathbf{F}_i \cdot \mathbf{v}_i \, dt, \qquad W_{\mathbf{M}_e,AB} = \int_{t_A}^{t_B} \mathbf{M}_e \cdot \omega \, dt. \tag{21.6}$$

Notice: the work of a force uses the velocity of its **point of application**.

https://youtu.be/acmYFwl8rRo

21.1 Koenig's Decomposition

By definition, the kinetic energy T of a rigid body is the (continuous) sum of the kinetic energies of the differential masses composing it:

$$T = \frac{1}{2}\int_{\mathcal{B}} \mathbf{v}\cdot\mathbf{v}\,\rho\,dv. \tag{21.7}$$

We can introduce the $\mathbf{v}_C$ term through

$$\begin{aligned}
&\mathbf{v} = \mathbf{v}_C + \omega\times\pi,\\
&\pi = \mathbf{r} - \mathbf{r}_C,\\
&\omega = \omega_x\mathbf{e}_x + \omega_y\mathbf{e}_y + \omega_z\mathbf{e}_z.
\end{aligned} \tag{21.8}$$

Since

$$\mathbf{v}\cdot\mathbf{v} = (\mathbf{v}_C+\omega\times\pi)\cdot(\mathbf{v}_C+\omega\times\pi) = \mathbf{v}_C\cdot\mathbf{v}_C + \mathbf{v}_C\cdot(\omega\times\pi) + (\omega\times\pi)\cdot\mathbf{v}_C + (\omega\times\pi)\cdot(\omega\times\pi), \tag{21.9}$$

and the two middle terms are equal by commutativity of the dot product, substituting the above expressions in the kinetic energy integral, we get

$$T = \frac{1}{2}\int_{\mathcal{B}} \left(\mathbf{v}_C\cdot\mathbf{v}_C + 2\mathbf{v}_C\cdot(\omega\times\pi) + (\omega\times\pi)\cdot(\omega\times\pi)\right) dm. \tag{21.10}$$

For the first two terms in the above integral,

$$\begin{aligned}
&\frac{1}{2}\int_{\mathcal{B}} \mathbf{v}_C\cdot\mathbf{v}_C\,dm = \frac{\mathbf{v}_C\cdot\mathbf{v}_C}{2}\int dm = \frac{1}{2}m\mathbf{v}_C\cdot\mathbf{v}_C,\\
&\int_{\mathcal{B}} \mathbf{v}_C\cdot(\omega\times\pi)\,dm = \mathbf{v}_C\cdot\left(\omega\times\int\pi\,dm\right) = 0.
\end{aligned} \tag{21.11}$$

Thus,

$$T = \frac{1}{2}m\mathbf{v}_C\cdot\mathbf{v}_C + \frac{1}{2}\int_{\mathcal{B}}(\omega\times\pi)\cdot(\omega\times\pi)\,dm. \tag{21.12}$$

We can simplify $(\omega\times\pi)\cdot(\omega\times\pi)$ using the scalar triple product identity $(\mathbf{A}\times\mathbf{B})\cdot\mathbf{C} = \mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})$ with $\mathbf{A}=\omega$, $\mathbf{B}=\pi$, $\mathbf{C}=\omega\times\pi$:

$$(\omega\times\pi)\cdot(\omega\times\pi) = \omega\cdot(\pi\times(\omega\times\pi)). \tag{21.13}$$

The inner cross product expands via the BAC-CAB identity, $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$, with $\mathbf{a}=\pi, \mathbf{b}=\omega, \mathbf{c}=\pi$:

$$\pi\times(\omega\times\pi) = (\pi\cdot\pi)\omega - (\pi\cdot\omega)\pi. \tag{21.14}$$

Recall that

$$\mathbf{H}^C = \int_{\mathcal{B}} \pi \times (\omega \times \pi)\, dm = \int_{\mathcal{B}} ((\pi \cdot \pi)\omega - (\pi \cdot \omega)\pi)\, dm. \tag{21.15}$$

Putting the last two results together, and pulling ω out of the integral since it is the same for every material point of the rigid body:

$$\frac{1}{2}\int_{\mathcal{B}} (\omega \times \pi) \cdot (\omega \times \pi)\, dm = \frac{1}{2}\omega \cdot \int_{\mathcal{B}} \pi \times (\omega \times \pi)\, dm = \frac{1}{2}\omega \cdot \mathbf{H}^C = \frac{1}{2}\mathbf{H}^C \cdot \omega. \tag{21.16}$$

Hence, we obtain the Koenig decomposition:

$$T = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2}\mathbf{H}^C \cdot \omega. \tag{21.17}$$

https://youtu.be/3HRYCW7wiRs

21.1.1 Fixed Point Rotation

In the case of fixed point rotation, $\mathbf{v}_C = \omega \times (\mathbf{r}_C - \mathbf{r}_O)$ and $\mathbf{v}_O = \mathbf{0}$. Since $\mathbf{G} = m\mathbf{v}_C$ is the linear momentum,

$$\frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C = \frac{1}{2}(m\mathbf{v}_C) \cdot \mathbf{v}_C = \frac{1}{2}\mathbf{G} \cdot \mathbf{v}_C = \frac{1}{2}\mathbf{G} \cdot (\omega \times \mathbf{r}_{C/O}), \tag{21.18}$$

so, using the cyclical property of the scalar triple product, $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}) = \mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})$:

$$\begin{aligned} T &= \frac{1}{2}\mathbf{G} \cdot (\omega \times \mathbf{r}_{C/O}) + \frac{1}{2}\mathbf{H}^C \cdot \omega \\ &= \frac{1}{2}\omega \cdot (\mathbf{r}_{C/O} \times \mathbf{G}) + \frac{1}{2}\mathbf{H}^C \cdot \omega \\ &= \frac{1}{2}(\mathbf{r}_{C/O} \times \mathbf{G} + \mathbf{H}^C) \cdot \omega \\ &= \frac{1}{2}\mathbf{H}^O \cdot \omega, \end{aligned} \tag{21.19}$$

where the second line follows from the cyclical identity, the third line factors ω out of the sum of two dot products, and the last line uses the identity

$$\mathbf{H}^O = \mathbf{H}^C + \mathbf{r}_{C/O} \times \mathbf{G}, \tag{21.20}$$

obtained by writing

$$\mathbf{H}^O = \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_O) \times \mathbf{v}\, dm = \int_{\mathcal{B}} (\pi + \mathbf{r}_{C/O}) \times \mathbf{v}\, dm = \int_{\mathcal{B}} \pi \times \mathbf{v}\, dm + \mathbf{r}_{C/O} \times \int_{\mathcal{B}} \mathbf{v}\, dm, \tag{21.21}$$

and noting that $\int_{\mathcal{B}} \pi \times \mathbf{v}\, dm = \mathbf{H}^C$ (by the same steps used above, since $\int \pi\, dm = \mathbf{0}$) while $\int_{\mathcal{B}} \mathbf{v}\, dm = m\mathbf{v}_C = \mathbf{G}$.

https://youtu.be/3bu7kY8JLpM

21.2 Derivation of the Work-Energy Theorem

Taking the time derivative of $T = \frac{1}{2}m\mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2}\mathbf{H}^C \cdot \omega$:

$$\dot{T} = \frac{1}{2}m\dot{\mathbf{v}}_C \cdot \mathbf{v}_C + \frac{1}{2}m\mathbf{v}_C \cdot \dot{\mathbf{v}}_C + \frac{1}{2}\dot{\mathbf{H}}^C \cdot \omega + \frac{1}{2}\mathbf{H}^C \cdot \dot{\omega}. \tag{21.22}$$

We need to show that $\dot{\mathbf{H}}^C \cdot \omega = \mathbf{H}^C \cdot \dot{\omega}$.

$$\begin{aligned} \alpha = \dot{\omega} &= \frac{d}{dt}(\omega_x \mathbf{e}_x + \omega_y \mathbf{e}_y + \omega_z \mathbf{e}_z) \\ &= \dot{\omega}_x \mathbf{e}_x + \dot{\omega}_y \mathbf{e}_y + \dot{\omega}_z \mathbf{e}_z + \omega_x \dot{\mathbf{e}}_x + \omega_y \dot{\mathbf{e}}_y + \omega_z \dot{\mathbf{e}}_z \\ &= \dot{\omega}_x \mathbf{e}_x + \dot{\omega}_y \mathbf{e}_y + \dot{\omega}_z \mathbf{e}_z + \omega \times (\omega_x \mathbf{e}_x + \omega_y \mathbf{e}_y + \omega_z \mathbf{e}_z) \\ &= \dot{\omega}_x \mathbf{e}_x + \dot{\omega}_y \mathbf{e}_y + \dot{\omega}_z \mathbf{e}_z. \end{aligned} \tag{21.23}$$

A direct calculation using this expression for α shows that

$$\mathbf{H}^C \cdot \dot{\omega} = (I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z)\dot{\omega}_x + (I_{xy}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z)\dot{\omega}_y + (I_{xz}\omega_x + I_{yz}\omega_y + I_{zz}\omega_z)\dot{\omega}_z. \tag{21.24}$$

To compute the corresponding expression for $\dot{\mathbf{H}}^C \cdot \omega$, recall $\dot{\mathbf{H}}^C = \overset{\circ}{\mathbf{H}}{}^C + \omega \times \mathbf{H}^C$, so

$$\dot{\mathbf{H}}^C \cdot \omega = \overset{\circ}{\mathbf{H}}{}^C \cdot \omega + (\omega \times \mathbf{H}^C) \cdot \omega = \overset{\circ}{\mathbf{H}}{}^C \cdot \omega, \tag{21.25}$$

since $(\omega \times \mathbf{H}^C) \cdot \omega = 0$ (the vector $\omega \times \mathbf{H}^C$ is perpendicular to ω). Using the component expression for $\overset{\circ}{\mathbf{H}}{}^C$ and dotting with ω:

$$\overset{\circ}{\mathbf{H}}{}^C \cdot \omega = (I_{xx}\dot{\omega}_x + I_{xy}\dot{\omega}_y + I_{xz}\dot{\omega}_z)\omega_x + (I_{xy}\dot{\omega}_x + I_{yy}\dot{\omega}_y + I_{yz}\dot{\omega}_z)\omega_y + (I_{xz}\dot{\omega}_x + I_{yz}\dot{\omega}_y + I_{zz}\dot{\omega}_z)\omega_z. \tag{21.26}$$

Comparing this term by term with the expression for $\mathbf{H}^C \cdot \dot{\omega}$ above, and using the symmetry of the inertia tensor ($I_{xy} = I_{yx}$, $I_{xz} = I_{zx}$, $I_{yz} = I_{zy}$), every term matches (e.g. $I_{xy}\dot{\omega}_x\omega_y = I_{xy}\omega_x\dot{\omega}_y$). Hence $\dot{\mathbf{H}}^C \cdot \omega = \mathbf{H}^C \cdot \dot{\omega}$. Consequently,

$$\dot{T} = \frac{1}{2}m\dot{\mathbf{v}}_C \cdot \mathbf{v}_C + \frac{1}{2}m\mathbf{v}_C \cdot \dot{\mathbf{v}}_C + \frac{1}{2}\dot{\mathbf{H}}^C \cdot \omega + \frac{1}{2}\dot{\mathbf{H}}^C \cdot \omega, \tag{21.27}$$

which implies that

$$\dot{T} = m\dot{\mathbf{v}}_C \cdot \mathbf{v}_C + \dot{\mathbf{H}}^C \cdot \omega. \tag{21.28}$$

Invoking the balance of linear momentum and the balance of angular momentum, we obtain the work-energy theorem:

$$\dot{T} = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega. \tag{21.29}$$

This is a natural extension of the work-energy theorem for a single particle.

https://youtu.be/vCROTYHJRow

21.3 Alternative Form

Recall,

$$\begin{aligned}\mathbf{F} &= \sum_{i=1}^{K} \mathbf{F}_i \\ \mathbf{M}^C &= \mathbf{M}_e + \sum_{i=1}^{K} (\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i\end{aligned} \tag{21.30}$$

Hence, the mechanical power of the resultant forces and moments can be written as

$$\mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega = \left(\sum_{i=1}^{K} \mathbf{F}_i \right) \cdot \mathbf{v}_C + \mathbf{M}_e \cdot \omega + \left(\sum_{i=1}^{K} (\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i \right) \cdot \omega. \tag{21.31}$$

With $\mathbf{a} = \omega$, $\mathbf{b} = \mathbf{r}_i - \mathbf{r}_C$, $\mathbf{c} = \mathbf{F}_i$, the identity $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$ gives

$$((\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i) \cdot \omega = \omega \cdot ((\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i) = \mathbf{F}_i \cdot (\omega \times (\mathbf{r}_i - \mathbf{r}_C)) \,. \tag{21.32}$$

Substituting this into the expression for $\mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega$ above:

$$\mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega = \sum_{i=1}^{K} \mathbf{F}_i \cdot \mathbf{v}_C + \mathbf{M}_e \cdot \omega + \sum_{i=1}^{K} \mathbf{F}_i \cdot (\omega \times (\mathbf{r}_i - \mathbf{r}_C)) = \sum_{i=1}^{K} \mathbf{F}_i \cdot (\mathbf{v}_C + \omega \times (\mathbf{r}_i - \mathbf{r}_C)) + \mathbf{M}_e \cdot \omega. \tag{21.33}$$

Noting that $\mathbf{v}_i = \mathbf{v}_C + \omega \times (\mathbf{r}_i - \mathbf{r}_C)$, we find that

$$\mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega = \sum_{i=1}^{K} \mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_e \cdot \omega. \tag{21.34}$$

In conclusion, we have an alternative form of the work-energy theorem that proves to be far easier to use in applications:

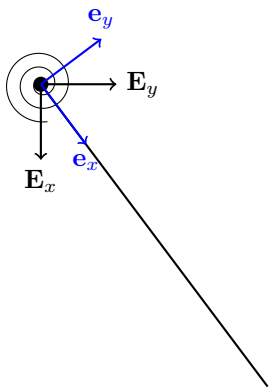
$$\dot{T} = \sum_{i=1}^{K} \mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_e \cdot \omega. \tag{21.35}$$

Each force is dotted with the velocity of its point of application.

21.4 Conservative Moments

21.4.1 Torsional Spring

A torsional spring is just a model for some kind of elasticity at the joint.



A torsional spring with stiffness K creates a couple $\mathbf{M}_S = -K\theta\mathbf{E}_z$. Its power is:

$$\mathbf{M}_S \cdot \omega = -K\theta\dot{\theta} = -\frac{d}{dt}\left(\frac{K\theta^2}{2}\right). \tag{21.36}$$

Hence $U = \frac{1}{2}K\theta^2$ for the torsional spring.

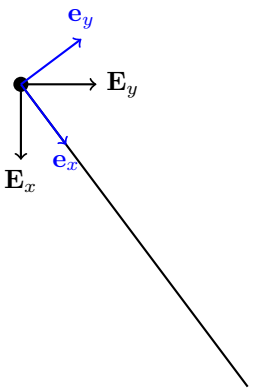
21.4.2 Constant Couple

A constant couple $\mathbf{M}_e = M_e\mathbf{E}_z$ has power $\mathbf{M}_e \cdot \omega = M_e\dot{\theta} = -\frac{d}{dt}(-M_e\theta)$. Hence $U = -M_e\theta$.

> **!** Note!
>
> These results apply only for $\omega = \omega\mathbf{E}_z$. Constant couples are not conservative in general.

21.5 Examples

21.5.1 Bar Pendulum Fixed at Its End

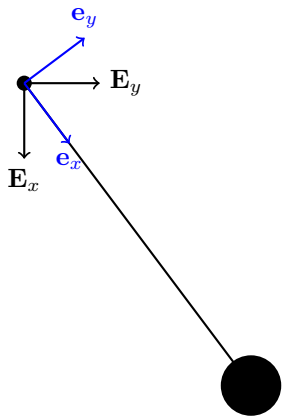


A bar of length ℓ and mass m pinned at its end. The energy is conserved:

$$U = -\frac{mg\ell}{2}\cos\theta, \tag{21.37}$$

$$E = \frac{m\ell^2}{6}\dot{\theta}^2 - \frac{mg\ell}{2}\cos\theta = \text{const}. \tag{21.38}$$

21.5.2 Bar Pendulum With End Mass

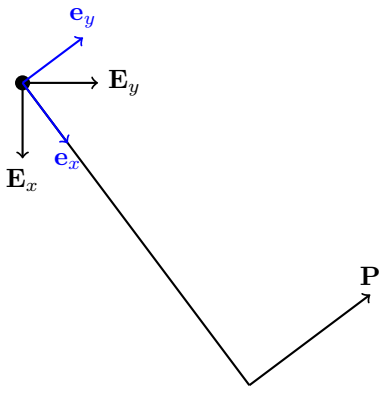


Adding a dead mass m_e at the end:

$$U = -\frac{mg\ell}{2}\cos\theta - m_e g\ell\cos\theta. \tag{21.39}$$

21.5.3 Follower Force

What happens if there is a follower force applied at the end of the rod always perpendicular to it?



If instead a follower force $\mathbf{P}$ is applied perpendicular to the bar at its end, this force is **nonconservative** (its direction actively changes with configuration). Energy is not conserved.

21.6 Energy Conservation

As with particles and systems of particles, this theorem can be used to establish conservation of the total mechanical energy of a rigid body, $E = T + U$, when all work-doing forces and moments are conservative.

21.7 Summary

Koenig decomposition of kinetic energy:

$$T = \tfrac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \tfrac{1}{2} \mathbf{H}^C \cdot \omega. \tag{21.40}$$

If a point O has zero velocity: $T = \tfrac{1}{2} \mathbf{H}^O \cdot \omega$.

Work–energy theorem for a rigid body:

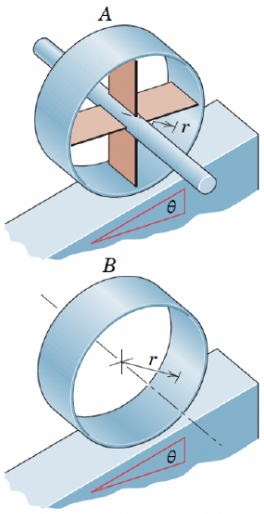
$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M} \cdot \omega = \sum_i \mathbf{F}_i \cdot \mathbf{v}_i + M_e \omega. \tag{21.41}$$

21.8 Exercises

The following problems are from Set 23 – Work–Energy Theorem for Rigid Bodies.

1. [MKB 06-096] *(ans.* $v_A = \sqrt{2gx \sin\theta}$, $v_B = \sqrt{gx \sin\theta}$*)*

6/96 The two wheels of Prob. 6/62, shown again here, represent two extreme conditions of distribution of mass. For case A all of the mass m is assumed to be concentrated in the center of the hoop in the axial bar of negligible diameter. For case B all of the mass m is assumed to be concentrated in the rim. Determine the speed of the center of each hoop after it has traveled a distance x down the incline from rest. The hoops roll without slipping.

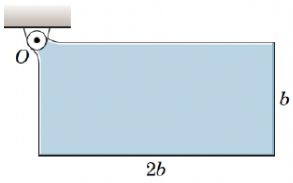


PROBLEM 6/96

Figure 21.1: MKB 06-096.

2. [MKB 06-099] *(ans.* $\omega_{\max} = 0.861\sqrt{g/b}$*)*

6/99 The uniform rectangular plate is released from rest in the position shown. Determine the maximum angular velocity ω during the ensuing motion. Friction at the pivot is negligible.

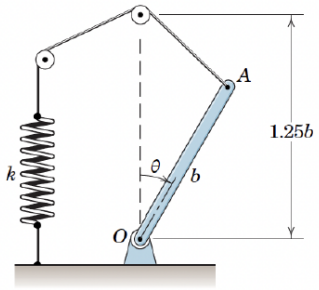


PROBLEM 6/99

Figure 21.2: MKB 06-099.

3. [06-108] *(ans. (a)* $k = 93.3$ *N/m; (b)* $\omega = 1.484$ *rad/s CW)*

6/108 The system is at rest with the spring unstretched when $\theta = 0$. The 5-kg uniform slender bar is then given a slight clockwise nudge. The value of b is 0.4 m. (a) If the bar comes to momentary rest when $\theta = 40°$, determine the spring constant k. (b) For the value $k = 90$ N/m, find the angular velocity of the bar when $\theta = 25°$.

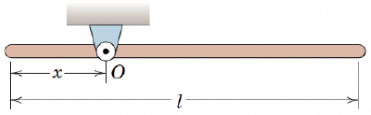


PROBLEM 6/108

Figure 21.3: 06-108.

4. [06-112] *(ans.* $x = 0.211l$, $\omega_{\max} = 1.861\sqrt{g/l}$ *CW)*

6/112 For the pivoted slender rod of length l, determine the distance x for which the angular velocity will be a maximum as the bar passes the vertical position after being released in the horizontal position shown. State the corresponding angular velocity.



PROBLEM 6/112

Figure 21.4: 06-112.

5. [06-118] *(ans.* $v_A = \sqrt{3}\sqrt{\frac{M\theta}{m} - gb(1 - \cos\theta)}$ *right)*

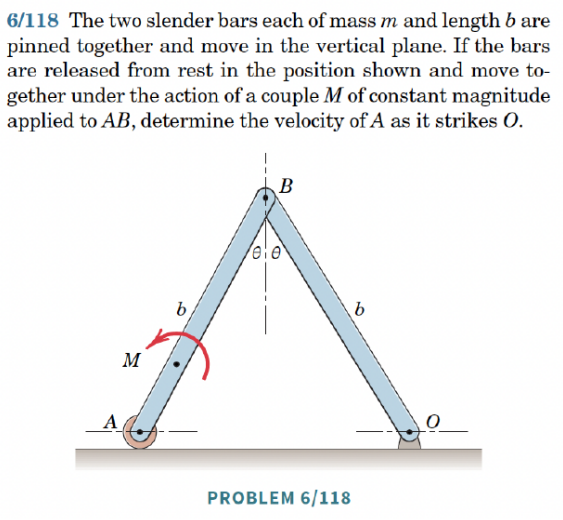


Figure 21.5: 06-118.

https://youtu.be/-jdslhyHAIY

22 Systems of Particles and Rigid Bodies

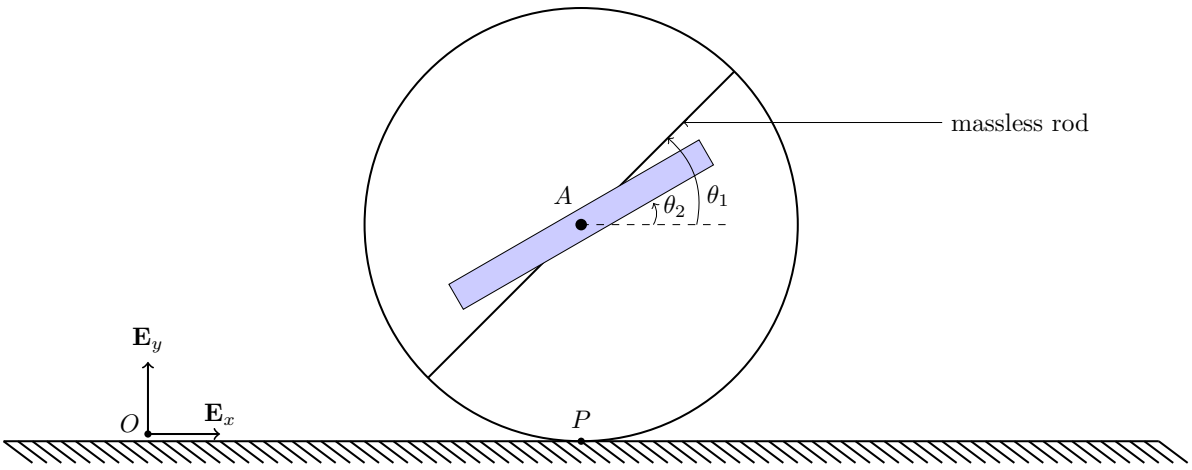
The balance laws developed separately for particles, systems of particles, and rigid bodies apply just as well to *combinations* of these — assemblies of rigid bodies and particles connected by pins, rods, or motors. The following example illustrates how to set up such a problem: identifying the kinematic constraints between the bodies, drawing free-body diagrams of the combined system and of its parts, and applying the balance laws to each.

22.1 Example: BB8

Consider a spherical robot composed of a spherical shell of mass m and radius R and a bar of mass $2m$ and length $\ell < 2R$ pinned at point A, its center, to the center of the shell as shown in the figure below. A geared motor drive at A provides an internal moment which rotates the bar. The friction between the robot and the ground is sufficient to prevent slipping.

Throughout the motion, the position vector from the origin to the center of the robot is $\mathbf{r}_A = x\mathbf{E}_x + R\mathbf{E}_y$.

The angular velocity of the spherical shell is $\omega_1 = \dot{\theta}_1 \mathbf{E}_z$ and the angular velocity of the bar of length ℓ is $\omega_2 = \dot{\theta}_2 \mathbf{E}_z$.

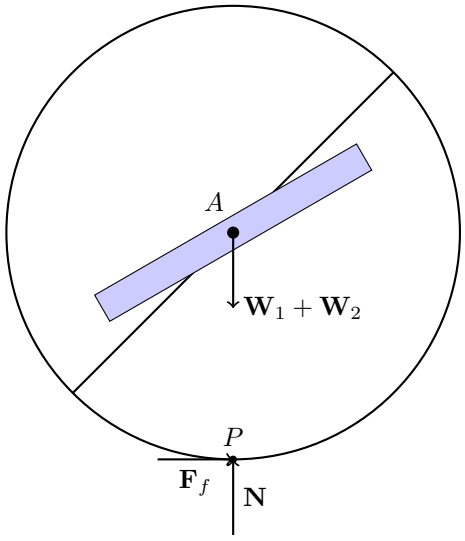


(a) Draw the free-body diagram of (1) the combined system of the spherical shell and the bar of length ℓ, (2) the spherical shell with its massless diameter alone, and (3) the bar of length ℓ alone.

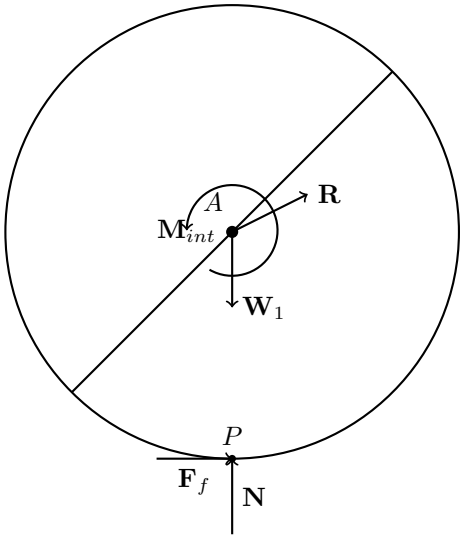
(b) We wish to determine the motion of the robot, i.e., we want to find $x(t)$, $\theta_1(t)$, $\theta_2(t)$, resulting from the moments applied by the motor. How do we proceed?

22.1.1 Free-Body Diagrams

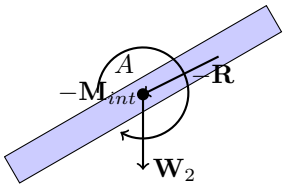
Combined system (shell + bar):



Spherical shell alone:



Bar alone:



where

$$\mathbf{F}_f = F_f \mathbf{E}_x. \tag{22.1}$$

22.1.2 Kinematics

By differentiating $\mathbf{r}_A$, we get

$$\begin{aligned} \mathbf{v}_A &= \dot{x}\mathbf{E}_x, \\ \mathbf{a}_A &= \ddot{x}\mathbf{E}_x. \end{aligned} \tag{22.2}$$

The roll-without-slip condition between the sphere and the ground, $\mathbf{v}_P = \mathbf{0}$, yields a relationship between x and θ_1:

$$\begin{aligned} \mathbf{v}_C &= \omega_1 \times R\mathbf{E}_y, \\ \dot{x} &= -R\dot{\theta}_1. \end{aligned} \tag{22.3}$$

22.1.3 Balance of Linear Momentum (Combined System)

$$\mathbf{F} = (m_1 + m_2)\mathbf{a}_A. \tag{22.4}$$

> **Note**
>
> This example is left as a set-up exercise: complete the balance of linear momentum in components, then use the balance of angular momentum (about A or about P) for the combined system, the shell alone, and the bar alone, together with the kinematic constraint above, to solve for $x(t)$, $\theta_1(t)$, $\theta_2(t)$.

Part V

Lagrangian Mechanics

23 Lagrange's Equations of Motion for a System of Particles

23.1 Constraints

Put in simple terms, a holonomic constraint is a constraint which can be written in terms of position coordinates, e.g. $x = 0$. A nonholonomic constraint is one which cannot be written in terms of position coordinates, e.g. $\mathbf{v} \cdot \mathbf{e}_\theta = 0$.

Think!

Question: Consider a particle constrained to move on a horizontal surface $z = 0$. Is this constraint holonomic?

Answer

...

Think!

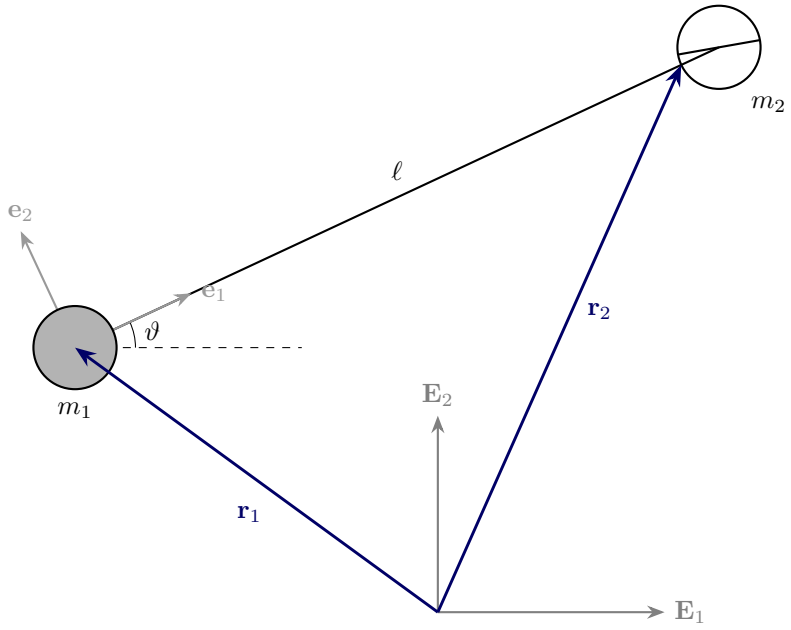
Question: Consider two particles connected by a rigid massless rod of length ℓ. What constraint is this system of two particles subject to? Is this constraint holonomic or nonholonomic?

Answer

...

Think!

Question: Consider the system in the previous question where m_2 is subject to a knife-edge constraint $\mathbf{v}_2 \cdot \mathbf{e}_2 = 0$. Is this constraint holonomic or nonholonomic.



Answer

...

Frobenius' integrability criteria may be used to determine definitely if a constraint or a system of constraints is holonomic or not. The criterion is beyond the scope of the class. Here, we restrict our attention to holonomic constraints.

23.2 Lagrange's Prescription on Constraints

Consider a constraint of the form

$$\mathbf{f}_1 \cdot \mathbf{v}_1 + ... + \mathbf{f}_N \cdot \mathbf{v}_N + e = 0,$$

then the associated Lagrange prescription of the constraint force is $\mu \mathbf{f}_K$ applied at particle K, $K = 1, ..., n$.

Think!

Question: Prescribe the constraint force for a particle constrained on a plane $z = 0$.

Answer

This constraint may be written as $\mathbf{v} \cdot \mathbf{E}_z = 0$ and the constraint force acting on the particle is $\mathbf{F} = \mu \mathbf{E}_z$.

Think!

Question: Prescribe the constraint force acting on a simple pendulum associated with the constraint that the length of the pendulum is constant.

Answer

A simple pendulum has the constraint $\mathbf{r} = 0$ which may be written as $\mathbf{v} \cdot \mathbf{e}_r = 0$, then the constraint force acting on the particle is $\mathbf{F} = \mu \mathbf{e}_r$.

23.3 Unconstrained Equations

Consider a system of n particles with $3n$ coordinates. The system is subject to m holonomic constraints, so the number of degrees of freedom of the system is $3n - m$. The generalized coordinates of the system may be listed as

$$q = \begin{bmatrix} x^1 & y^1 & z^1 & \cdots & x^n & y^n & z^n \end{bmatrix}^T \tag{23.1}$$

and the corresponding generalized velocities are

$$u = \begin{bmatrix} v^{x1} & v^{y1} & v^{z1} & \cdots & v^{xn} & v^{yn} & v^{zn} \end{bmatrix}^T \tag{23.2}$$

We define u so that later we set $u = \dot{q}$.

We arrange the generalized coordinates such that the first $3n - m$ coordinates are the unconstrained degrees of freedom and the last m coordinates are equal to the holonomic constraints, and can be later set to zero.

Think!

Question: Generalized coordinates for simple harmonic oscillator.

Answer

$q = [x \ y \ z]$, $u = [v_x \ v_y \ v_z]$.

Think!

Question: Generalized coordinates for simple pendulum.

Answer

$q = [\theta \ \beta \ z]$, $u = [u_\theta \ u_\beta \ u_z]$ where $\beta = r - \ell$.

The kinetic energy of the system is T, then Lagrange's equations of motion of the system are

$$\frac{d}{dt}\left(\frac{\partial T}{\partial u^i}\right) - \frac{\partial T}{\partial q^i} = \sum_{K=1}^{n} \mathbf{F}^K \cdot \frac{\partial \mathbf{r}^K}{\partial q^i} \tag{23.3}$$

Equivalently, using the Lagrangian $L = T - U$, where U is the potential energy, the equations of motion of the system may be written as

$$\frac{d}{dt}\left(\frac{\partial L}{\partial u^i}\right) - \frac{\partial L}{\partial q^i} = \sum_{K=1}^{n} \mathbf{F}^K_{nonc} \cdot \frac{\partial \mathbf{r}^K}{\partial q^i} \tag{23.4}$$

where $\mathbf{F}^K$ is the sum of external forces acting on particle K while $\mathbf{F}^K_{nonc}$ is the sum of nonconservative forces acting on K.

23.3.1 Advantage of Lagrange's Equations

If the system is subject to m holonomic constraints, if you choose the last m generalized coordinates to be equal to the constraints, then the first $m - n$ Lagrange's equations of motion will be uncoupled from the holonomic constraint forces, these are the equations of motion. The last $n - m$ equations can be solved for the holonomic constraint forces. If one does not wish to solve for these forces, one can just use the constrained kinetic energy.

23.4 Constrained Equations

Consider a system of n particles with $3n$ degrees of freedom and m holonomic constraints prescribed using Lagrange's equations of motion. The unconstrained degrees of freedom of the system are arranged in an array of generalized coordinates q and generalized velocities u, and the constrained kinetic energy of the system is $\tilde{T}$, then Lagrange's equations of motion of the system are

$$\frac{d}{dt}\left(\frac{\partial \tilde{T}}{\partial u^i}\right) - \frac{\partial \tilde{T}}{\partial q^i} = \sum_{K=1}^{n} \mathbf{F}^K \cdot \frac{\partial \mathbf{r}^K}{\partial q^i} \tag{23.5}$$

Equivalently, using the constrained Lagrangian $\tilde{L} = \tilde{T} - \tilde{U}$, where $\tilde{U}$ is the constrained potential energy, the equations of motion of the system may be written as

$$\frac{d}{dt}\left(\frac{\partial \tilde{L}}{\partial u^i}\right) - \frac{\partial \tilde{L}}{\partial q^i} = \sum_{K=1}^{n} \mathbf{F}^K_{nonc} \cdot \frac{\partial \mathbf{r}^K}{\partial q^i} \tag{23.6}$$

If the forces are prescribed with Lagrange's prescription, then $\sum_{K=1}^{n} \mathbf{F}^K \cdot \frac{\partial \mathbf{r}^K}{\partial q^i}$ and $\sum_{K=1}^{n} \mathbf{F}^K_{nonc} \cdot \frac{\partial \mathbf{r}^K}{\partial q^i}$ reduce to $\sum_{K=1}^{n} \mathbf{F}_{appliedc} \cdot \frac{\partial \mathbf{r}^K}{\partial q^i}$ and $\sum_{K=1}^{n} \mathbf{F}_{applied,nconc} \cdot \frac{\partial \mathbf{r}^K}{\partial q^i}$ respectively, that is the applied forces and the nonconservative applied forces.

In many cases, the nonconservative applied forces are zero, hence the commonly known homogeneous form of Lagrange's equation

$$\frac{d}{dt}\left(\frac{\partial \tilde{L}}{\partial u^i}\right) - \frac{\partial \tilde{L}}{\partial q^i} = 0. \tag{23.7}$$

23.5 Example: Simple Harmonic Oscillator

Consider a simple harmonic oscillator of mass m connected to a linear spring of unstretched length ℓ_0 and stiffness k. The origin is taken at the location of m when the spring is unstretched so that the position vector of m is $\mathbf{r} = x\mathbf{E}_x$ and x denotes the stretch of the spring. The velocity of m is $\mathbf{v} = v\mathbf{E}_x$. Find the equation of motion of the mass m.

Using unconstrained Lagrangian:

$$\begin{aligned} q^1 &= x, \qquad q^2 = y, \qquad q^3 = z, \\ u^1 &= v_x, \qquad u^2 = v_y, \qquad u^3 = v_z, \end{aligned} \tag{23.8}$$

Lagrangian

$$L = T - U = \frac{m}{2}\left(v_x^2 + v_y^2 + v_z^2\right) - mgz \tag{23.9}$$

Constraint forces $\mu_1\mathbf{E}_y$, $\mu_2\mathbf{E}_z$.

Equation along z,

$$\dot{\mathbf{v}}_z + mg = \mu_2. \tag{23.10}$$

Using constrained Lagrangian:

The kinetic energy of the spring is $\tilde{T} = \frac{1}{2}v^2$ and potential energy is $\tilde{U} = \frac{1}{2}kx^2$, thus the Lagrangian is $\tilde{L} = \frac{1}{2}v^2 - \frac{1}{2}kx^2$. Here, m has one degree of freedom, so $q = [x]$ and $u = [v]$, so the equation of motion of the system is obtained …

23.6 Example: Simple Pendulum

$$\begin{aligned} \mathbf{r} &= \ell\mathbf{e}_x, \\ \mathbf{v} &= \ell\dot{\theta}\mathbf{e}_y \end{aligned} \tag{23.11}$$

Gravitational potential energy

23.7 Example: Double Pendulum

$$\begin{aligned}
\mathbf{r}_1 &= \ell_{1\,1}\mathbf{e}_x, \\
\mathbf{v}_1 &= \ell_1\dot{\theta}_{1\,1}\mathbf{e}_y, \\
\mathbf{r}_2 &= \mathbf{r}_1 + \ell_{2\,2}\mathbf{e}_x, \\
\mathbf{v}_2 &= \mathbf{v}_1 + \ell_2\dot{\theta}_{2\,2}\mathbf{e}_y
\end{aligned} \tag{23.12}$$

Gravitational potential energy

23.8 Alternative Principles of Dynamics

Source: O'Reilly's textbook on Intermediate Dynamics, second edition, Section 4.12

- Jean Bernoulli's principle of virtual work (1717) - for statics
- D'Alembert's principle (1743) - extension of the principle of virtual work for dynamics
- Gauss' principle of least constraint (1829)
- Hamilton's principle (1835)

23.8.1 Principle of Virtual Work and D'Alembert's Principle

Consider the principle of virtual work and D'Alembert's Principle applied to a system of particles. We assume that the particles are subject to a single constraint:

$$\mathbf{f}_1 \cdot \mathbf{v}_1 + ... + \mathbf{f}_N \cdot \mathbf{v}_N + e = 0.$$

The principle of virtual work and D'Alembert's principle collectively state that the motion of the system of particles is such that the following equation is satisfied:

$$(\mathbf{F}_{a1} - m_1\ddot{\mathbf{r}}_1) \cdot \mathbf{d}_1 + ... + (\mathbf{F}_{aN} - m_N\ddot{\mathbf{r}}_N) \cdot \mathbf{d}_N = 0, \tag{23.13}$$

for all possible choices of the vectors $\mathbf{d}_1, ..., \mathbf{d}_N$ that satisfy the condition

$$\mathbf{f}_1 \cdot \mathbf{d}_1 + ... + \mathbf{f}_N \cdot \mathbf{d}_N = 0. \tag{23.14}$$

The vectors $\mathbf{d}_K$ are known as the virtual displacements and are usually denoted by $\delta\mathbf{r}_K$. The virtual work performed by the applied force $\mathbf{F}_{ak}$ is defined as $\mathbf{F}_{ak} \cdot \mathbf{d}_K$; thus the equation above states that the combined virtual work of the applied forces $\mathbf{F}_{aK}$ and the inertial forces $-m_K\ddot{\mathbf{r}}_K$ is zero.

How to obtain the equations of motion of the system of particles from the equation above?

Introduce a Lagrange multiplier, which we denote by the scalar function $\mu = \mu(t)$, to accommodate the single constraint on the vectors $\mathbf{d}_K$:

$$\sum_{K=1}^{N} (\mathbf{F}_{aK} - m\ddot{\mathbf{r}}_N) \cdot \mathbf{d}_K + \mu \sum_{K=1}^{N} \mathbf{f}_K \cdot \mathbf{d}_K = 0. \tag{23.15}$$

As a consequence of the Lagrange multiplier μ, the vectors $\mathbf{d}_K$ can be varied independently. For the previous equation to hold for all such displacements, it is necessary and sufficient that

$$m_K \ddot{\mathbf{r}}_K = \mathbf{F}_{aK} + \mu \mathbf{f}_K \qquad (K = 1, ..., N). \tag{23.16}$$

These equations are none other than the balances of linear momenta for a system of particles subject to the above constraint, for which the constraint forces are prescribed by use of Lagrange's prescription:

$$\mathbf{F}_{cK} = \mu \mathbf{f}_K \qquad (K = 1, ..., N). \tag{23.17}$$

23.8.2 Gauss' Principle of Least Action

Suppose we have a system of N particles that are subject to the constraint

$$\mathbf{f}_1 \cdot \mathbf{v}_1 + ... + \mathbf{f}_N \cdot \mathbf{v}_N + e = 0.$$

and suppose that the constraint forces are prescribed by using Lagrange's prescription $\mathbf{F}_{cK} = \mu \mathbf{f}_K$, $(K = 1, ..., N)$. Then, in any motion of the system that satisfies the constraint, the constraint forces $\mathbf{F}_{cK} = \mu \mathbf{f}_K$ are the least needed to ensure that the constraint is satisfied. That is, Lagrange's prescription is in a sense optimal!

23.8.3 Hamilton's Principle

For ease of exposition, we restrict our discussion to a single particle of mass m and suppose that the coordinates $q = (q^1, q^2, q^2)$ are used to parametrize the position vector. Hamilton's principle states that the motion of the system between a given initial configuration $q(t_0)$ and a given final configuration $q(t_1)$ is such that it extremizes the integral:

$$I = \int_{t_0}^{t_1} L dt \tag{23.18}$$

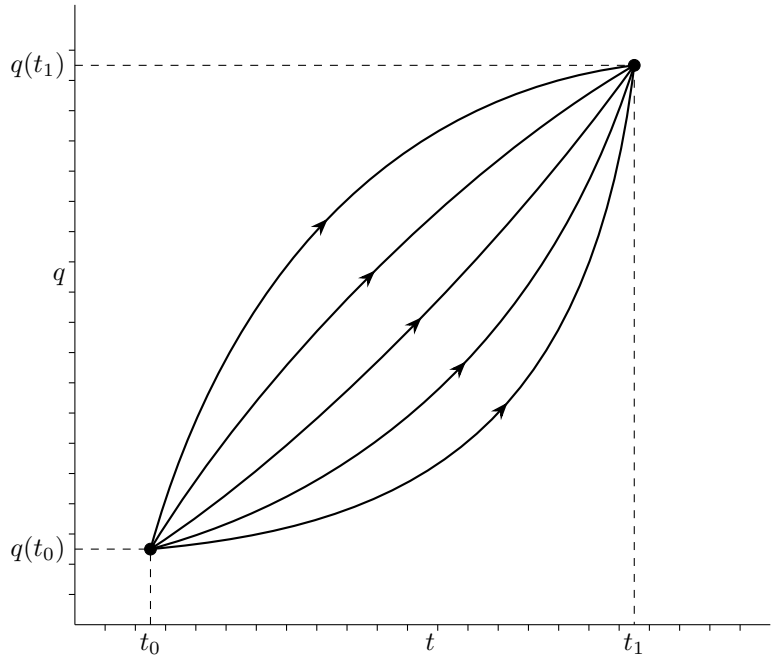
where the Lagrangian $L = T(q^i, u^i) - U(q^i)$ and t_0 and t_1 are fixed instances of time.

There are an infinite number of paths $q(t)$ that can connect two possible configurations, and so finding one that extremizes I appears to be a daunting task.

The necessary conditions for the components of $q(t)$ to extremize I are that $q(t)$ satisfy the following differential equations:

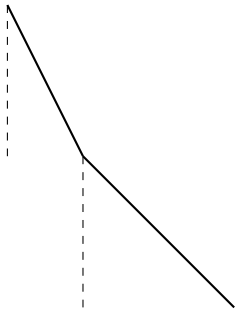
$$\frac{d}{dt}\left(\frac{\partial L}{\partial u^i}\right) - \frac{\partial L}{\partial q^i} = 0 \qquad (k = 1, 2, 3). \tag{23.19}$$

In the context of extremizing I (calculus of variations), these are known as the Euler-Lagrange equations.



23.9 Exercises

Double pendulum



24 Assignments

This chapter collects the computational assignments for the course. Each section is a self-contained assignment: follow the steps, then do the exercises at the end.

The assignments use Python. You can run the code on your own computer (see What you need) or in Google Colab, which needs only a browser.

24.1 Solving ODEs and Animating a Simple Pendulum

Most equations of motion in this course cannot be solved by hand. In this assignment you will solve the equation of motion of a simple pendulum numerically in Python, plot the result, and turn it into an animation like the one below.

../figures/assignments/pendulum_animation.mp4

By the end you should be able to:

1. Rewrite a second-order equation of motion as a system of first-order ODEs.
2. Solve it with `solve_ivp` from SciPy and plot the solution.
3. Convert the solution into positions and animate it with Matplotlib.
4. Judge whether a numerical solution can be trusted.

> **Code you can run on this page**
>
> Cells with a **Run Code** button run Python directly in your browser, with nothing to install. Edit the code and run it as often as you like. The first run downloads Python and its packages, which can take up to a minute; after that, runs are quick.

24.1.1 What you need

- **Python 3** with three packages: `numpy` (arrays and math), `scipy` (the ODE solver) and `matplotlib` (plots and animations). Install them with

```
pip install numpy scipy matplotlib
```

- **ffmpeg**, the program Matplotlib uses to save the animation as an `.mp4` video. On macOS install it with `brew install ffmpeg`; on Windows, download it from ffmpeg.org and add it to your PATH. With Anaconda, `conda install -c conda-forge ffmpeg` works on any system.
- An editor such as VS Code, or a Jupyter notebook.

> **No installation? Use Google Colab**
>
> Google Colab already has `numpy`, `scipy`, `matplotlib` and `ffmpeg`, and the saved `.mp4` appears in the Files panel on the left. In a notebook, an animation does not play with `plt.show()`. Display it with these two lines instead:
>
> ```
> from IPython.display import HTML
> HTML(ani.to_jshtml())
> ```

24.1.2 Step 1: The equation of motion

In Example: Nonlinear Pendulum we found the equation of motion of a particle of mass m on a massless string of length ℓ:

$$\ddot{\theta} + \frac{g}{\ell}\sin(\theta) = 0, \tag{24.1}$$

where θ is measured from the downward vertical.

This equation is **nonlinear** because of $\sin(\theta)$, and it has no solution in terms of elementary functions. We need its solution $\theta(t)$ for given initial conditions $\theta(0) = \theta_0$ and $\dot{\theta}(0) = \dot{\theta}_0$. This is an *initial value problem*.

24.1.3 Step 2: Rewrite it as a first-order system

Numerical ODE solvers work with systems of **first-order** equations, $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$. To bring our equation to this form, define the state

$$x_1 = \theta, \qquad x_2 = \dot{\theta}. \tag{24.2}$$

Differentiating each and using the equation of motion gives

$$\dot{x}_1 = x_2, \tag{24.3}$$

$$\dot{x}_2 = -\frac{g}{\ell}\sin(x_1), \tag{24.4}$$

with initial conditions $x_1(0) = \theta_0$ and $x_2(0) = \dot{\theta}_0$.

Think!

Question: How many first-order equations would you need for a double pendulum, which has two degrees of freedom?

Answer

Four. Each degree of freedom contributes a coordinate and its rate: $x_1 = \theta_1$, $x_2 = \theta_2$, $x_3 = \dot{\theta}_1$, $x_4 = \dot{\theta}_2$. In general, n second-order equations become $2n$ first-order equations.

This exercise is interactive: complete it in the online version of these notes.

24.1.4 Step 3: Solve with `solve_ivp`

First, import the packages and set the parameters.

```
import numpy as np
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt
import matplotlib.animation as animation

g = 9.81    # Gravitational acceleration in SI units
l = 1       # Length of pendulum chord in meters
```

Next, add the right-hand side $\mathbf{f}(t, \mathbf{x})$, the function `dxdt` you wrote above.

```
# Define the ODE function
def dxdt(t, x):
    xdot1 = x[1]
    xdot2 = -g/l*np.sin(x[0])
    return [xdot1, xdot2]
```

Now choose the time interval and the initial conditions, and call the solver. Here the pendulum is released from rest at $\theta_0 = \pi/4$ (45°).

```
# Define the time span and initial condition
t_span = (0, 5)          # Time interval: from 0 to 5 s
x0 = [np.pi/4, 0]        # theta(0) = pi/4, thetadot(0) = 0

# Solve the ODE
solution = solve_ivp(dxdt, t_span, x0, t_eval=np.linspace(0, 5, 100))
```

```
theta = solution.y[0]
thetadot = solution.y[1]
```

The arguments of `solve_ivp` are:

Argument	Meaning
`dxdt`	The function $\mathbf{f}(t, \mathbf{x})$
`t_span`	Start and end times
`x0`	Initial state $[\theta_0, \dot{\theta}_0]$
`t_eval`	The times at which you want the solution stored (here 100 equally spaced times)

The result `solution` holds the times in `solution.t` and the states in `solution.y`: row 0 is θ, row 1 is $\dot{\theta}$. By default the solver uses `RK45`, a Runge–Kutta method that adjusts its own step size.

Plot $\theta(t)$ to look at the solution:

```
plt.plot(solution.t, theta)
plt.xlabel("t (s)")
plt.ylabel("theta (rad)")
plt.show()
```

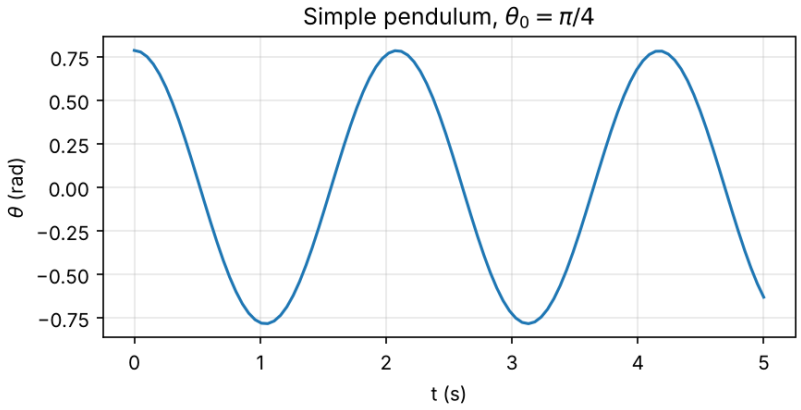


Figure 24.1: $\theta(t)$ for $\theta_0 = \pi/4$, $\ell = 1$ m.

Think!

Question: For small angles, $\sin(\theta) \approx \theta$ and the period is $T_0 = 2\pi\sqrt{\ell/g} \approx 2.01$ s. Read the period off the plot above. Is it the same?

Answer

The first maximum after release is at about $t = 2.09$ s, so the period is about 2.09 s, roughly 4% longer than T_0. The period of the nonlinear pendulum grows with the amplitude. At 45°, $\sin(\theta)$ is already noticeably smaller than θ, so the restoring torque is weaker than the linear model predicts.

An interactive version of this code is available in the online version of these notes.

24.1.5 Step 4: From angle to position

To draw the pendulum we need the position of the mass. With the pivot at the origin, x to the right and y up:

$$x = \ell \sin(\theta), \qquad y = -\ell \cos(\theta). \tag{24.5}$$

```
x = l*np.sin(theta)
y = -l*np.cos(theta)
```

`theta` is an array, so `x` and `y` are arrays too: one position for each of the 100 times.

24.1.6 Step 5: Animate the pendulum

An animation is a figure whose contents are updated frame by frame. First, set up the figure with the pendulum in its initial position: a line for the string and a circle for the mass.

```
# Set up the figure and axis
fig, ax = plt.subplots()          # creating figure and axes objects
ax.set_xlim([-1.2*l, 1.2*l])      # limits of x-axis
ax.set_ylim([-1.2*l, 0.2*l])      # limits of y-axis
ax.set_aspect('equal')

# Set axis labels and title
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Animating motion of pendulum")
```

```
# Drawing a line from the origin to the mass m.
# This line represents the chord of the pendulum.
line, = ax.plot([0, x[0]], [0, y[0]], color='black')
circle = plt.Circle((x[0], y[0]), 0.1, color='blue', fill=True)
ax.add_patch(circle)
```

> **Why `set_aspect('equal')`?**
>
> Without it, one unit on the x-axis and one unit on the y-axis can have different lengths on screen. The circle would look like an ellipse and the string would appear to stretch as it swings.

Next, write an `update` function. Matplotlib calls it once per frame with the frame number, and it moves the line and the circle to that frame's position.

```
# Update function for animation
def update(frame):
    line.set_xdata([0, x[frame]])
    line.set_ydata([0, y[frame]])
    circle.center = (x[frame], y[frame])
    return line, circle,
```

Finally, create the animation, save it as an `.mp4` video and show it.

```
# Create animation
ani = animation.FuncAnimation(fig, update, frames=np.size(x), interval=50, blit=True)
ani.save('pendulum_animation.mp4', writer='ffmpeg', fps=20)

plt.show()
```

- `frames=np.size(x)` makes one frame per stored time (100 frames).
- `interval=50` shows each frame for 50 ms. Because the 100 stored times are 0.05 s apart, the animation plays in real time.
- `fps=20` sets the frame rate of the saved file, again 1/0.05 s.

> **Error when saving?**
>
> If `ani.save` fails with a message like `MovieWriter ffmpeg unavailable`, Python cannot find ffmpeg. Install it (see What you need) and restart your editor or terminal.

24.1.7 The complete script

Full code: animation_simple_pendulum.py

```
# xdot1 = x2
# xdot2 = -g/l*sin(x1)

import numpy as np
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt
import matplotlib.animation as animation

g = 9.81    # Gravitational acceleration in SI units
l = 1       # Length of pendulum chord in meters

# Define the ODE function
def dxdt(t, x):
    xdot1 = x[1]
    xdot2 = -g/l*np.sin(x[0])
    return [xdot1, xdot2]

# Define the time span and initial condition
t_span = (0, 5)          # Time interval: from 0 to 5 s
x0 = [np.pi/4, 0]        # theta(0) = pi/4, thetadot(0) = 0

# Solve the ODE
solution = solve_ivp(dxdt, t_span, x0, t_eval=np.linspace(0, 5, 100))

theta = solution.y[0]
thetadot = solution.y[1]

x = l*np.sin(theta)
y = -l*np.cos(theta)

# Set up the figure and axis
fig, ax = plt.subplots()          # creating figure and axes objects
ax.set_xlim([-1.2*l, 1.2*l])      # limits of x-axis
ax.set_ylim([-1.2*l, 0.2*l])      # limits of y-axis
ax.set_aspect('equal')

# Set axis labels
ax.set_xlabel("x")
ax.set_ylabel("y")

# Set plot title
ax.set_title("Animating motion of pendulum")

# Drawing a line from the origin to the mass m.
# This line represents the chord of the pendulum.
line, = ax.plot([0, x[0]], [0, y[0]], color='black')
circle = plt.Circle((x[0], y[0]), 0.1, color='blue', fill=True)
ax.add_patch(circle)

# Update function for animation
def update(frame):
    line.set_xdata([0, x[frame]])
    line.set_ydata([0, y[frame]])
```

24.1.8 Video walkthrough

The two videos below go through the same procedure. Watch them if you want to see the code written and run step by step.

https://www.youtube.com/watch?v=RMkMK32vqyM

https://www.youtube.com/watch?v=pj7kitk0VcI

24.1.9 Exercises

1. Large swings. Release the pendulum from rest at $\theta_0 = 170^\circ$ (remember to convert to radians with `np.radians(170)`) and solve over 10 s. Plot $\theta(t)$ and animate it. Does the motion make physical sense?

> **Hint**
>
> A pendulum released from rest cannot rise higher than where it started, so θ should stay between -170° and 170°. If your plot leaves that range, the problem is not the physics. Look at the optional arguments `rtol` and `atol` of `solve_ivp`, and see Exercise 2.

2. Can you trust the solver? For the undamped pendulum, the energy per unit mass

$$e = \frac{1}{2}\ell^2\dot{\theta}^2 + g\ell\,(1 - \cos(\theta)) \tag{24.6}$$

should stay constant. Solve for $\theta_0 = \pi/4$ over 60 s, compute e at every stored time, and plot it. Then solve again with `rtol=1e-8, atol=1e-10` and compare. What does this tell you about the default settings?

3. Linear versus nonlinear. On the same axes, plot $\theta(t)$ from the nonlinear equation and from the linearized equation $\ddot{\theta} + \frac{g}{\ell}\theta = 0$, first for $\theta_0 = 5^\circ$ and then for $\theta_0 = 60^\circ$. When is the linear model good enough?

An interactive starter cell for this exercise is available in the online version of these notes.

4. Add damping. Air resistance adds a term to the equation of motion:

$$\ddot{\theta} + \frac{c}{m}\dot{\theta} + \frac{g}{\ell}\sin(\theta) = 0. \tag{24.7}$$

Rewrite it as a first-order system, change `dxdt` accordingly, and animate the motion for $c/m = 0.5\ \mathrm{s}^{-1}$. Plot the energy e from Exercise 2. How does it change over time, and why?

References

Lubliner, Jacob, and Panayiotis Papadopoulos. 2017. *Introduction to Solid Mechanics: An Integrated Approach*. Springer. https://doi.org/10.1007/978-1-4614-6768-7.

Part VI

Appendices

25 Mathematical Background for Conservative Forces

25.1 Gradient

In coordinate-free terms, the gradient of a function $f(\mathbf{r})$ is defined by

$$df = \nabla f \cdot d\mathbf{r}, \tag{25.1}$$

where df is the total infinitesimal change in f for an infinitesimal displacement $d\mathbf{r}$.

25.1.1 Cartesian Coordinates

For $f = f(x, y, z)$ with $d\mathbf{r} = dx\mathbf{E}_x + dy\mathbf{E}_y + dz\mathbf{E}_z$:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz. \tag{25.2}$$

Since this must hold for all independent variations dx, dy, dz:

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{E}_x + \frac{\partial f}{\partial y}\mathbf{E}_y + \frac{\partial f}{\partial z}\mathbf{E}_z. \tag{25.3}$$

25.1.2 Cylindrical-Polar Coordinates

For $f = f(r, \theta, z)$ with $d\mathbf{r} = dr\mathbf{e}_r + r\,d\theta\mathbf{e}_\theta + dz\mathbf{E}_z$:

$$df = \frac{\partial f}{\partial r}dr + \frac{\partial f}{\partial \theta}d\theta + \frac{\partial f}{\partial z}dz = \nabla f \cdot (dr\mathbf{e}_r + r\,d\theta\mathbf{e}_\theta + dz\mathbf{E}_z). \tag{25.4}$$

Matching coefficients:

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{E}_z. \tag{25.5}$$

25.1.3 Curvilinear Coordinates

> **Note**
>
> This section is beyond the scope of this course. See O. M. O'Reilly's *Intermediate Dynamics* Section 1.6 for more information.

Consider a curvilinear coordinate system $\{q^1, q^2, q^3\}$ defined by $q^i = \hat{q}^i(x_1, x_2, x_3)$, assumed locally invertible. The **covariant basis vectors** are:

$$\mathbf{a}_i = \frac{\partial \mathbf{r}}{\partial q^i} = \sum_{k=1}^{3} \frac{\partial \hat{x}^k}{\partial q^i} \mathbf{E}_k. \tag{25.6}$$

The **contravariant basis vectors** are the gradients of the curvilinear coordinates:

$$\mathbf{a}^k = \nabla q^k = \sum_{i=1}^{3} \frac{\partial \hat{q}^k}{\partial x_i} \mathbf{E}_i. \tag{25.7}$$

25.2 Condition for a Vector Field to Be Conservative

A force $\mathbf{F} = F_x \mathbf{E}_x + F_y \mathbf{E}_y + F_z \mathbf{E}_z$ is derivable from a potential if $F_x = \partial f/\partial x$, $F_y = \partial f/\partial y$, $F_z = \partial f/\partial z$. This requires:

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}, \qquad \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}, \qquad \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}, \tag{25.8}$$

summarized by $\operatorname{curl} \mathbf{F} = \mathbf{0}$.

This condition is both necessary and sufficient for $\mathbf{F}$ to be derivable from a potential (see this proof).

25.3 Conservative Forces

A conservative force $\mathbf{F}_c$ is derivable from a potential U:

$$\mathbf{F}_c = -\nabla U. \tag{25.9}$$

The work of a conservative force depends only on the endpoints:

$$W_{\mathbf{F}_c, AB} = \int_{\mathbf{r}(t_A)}^{\mathbf{r}(t_B)} \mathbf{F}_c \cdot d\mathbf{r} = -(U_B - U_A) = U_A - U_B. \tag{25.10}$$

25.3.1 Constant Force

For a constant force $\mathbf{C}$, the potential is $U = -\mathbf{C} \cdot \mathbf{r}$. In Cartesians:

$$\nabla(\mathbf{C} \cdot \mathbf{r}) = \frac{\partial \mathbf{C} \cdot \mathbf{r}}{\partial x}\mathbf{E}_x + \frac{\partial \mathbf{C} \cdot \mathbf{r}}{\partial y}\mathbf{E}_y + \frac{\partial \mathbf{C} \cdot \mathbf{r}}{\partial z}\mathbf{E}_z = \mathbf{C}. \tag{25.11}$$

Hence $\mathbf{C} = -\nabla(-\mathbf{C} \cdot \mathbf{r})$.

25.3.2 Spring Force

For the spring force $\mathbf{F}_s = -K(r - \ell_0)\mathbf{e}_r$ (taking the anchor at the origin), the potential is $U = \frac{1}{2}K(r - \ell_0)^2$. Using the polar gradient:

$$\nabla\left(\frac{1}{2}K(r - \ell_0)^2\right) = K(r - \ell_0)\mathbf{e}_r, \tag{25.12}$$

so $\mathbf{F}_s = -\nabla U$.

25.3.3 Gravitational Force

For Newton's law of gravitation, the force on particle 1 due to particle 2 is

$$\mathbf{F}_1 = G\frac{m_1 m_2}{\left\|\mathbf{r}_{2/1}\right\|^3}\mathbf{r}_{2/1}. \tag{25.13}$$

Using the polar gradient with $\mathbf{r}_{2/1} = r\mathbf{e}_r$, one can show that $U = -Gm_1m_2/r$ is the corresponding potential.

26 Sets and Linear Function Spaces

> **Note**
>
> This appendix is taken from Professor Panayiotis Papadopoulos's notes for UC Berkeley's ME280A course, Section 2.1.

26.1 Sets

A *set* $\mathcal{U}$ is a collection of elements. Elements of the set are denoted $u \in \mathcal{U}$. A set can be finite or infinite, and may have algebraic structure imposed on it.

26.2 Linear (Vector) Spaces

A set $\mathcal{V}$ is a **linear space** (or vector space) over the real numbers $\mathbb{R}$ if it is equipped with two operations:

1. **Addition:** $\mathbf{u} + \mathbf{v} \in \mathcal{V}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$,
2. **Scalar multiplication:** $\alpha\mathbf{u} \in \mathcal{V}$ for all $\alpha \in \mathbb{R}$, $\mathbf{u} \in \mathcal{V}$,

satisfying the standard axioms (commutativity, associativity, existence of zero element, existence of additive inverse, distributivity).

Examples of linear spaces: $\mathbb{R}^n$; the space of continuous functions on an interval; the space of $n \times n$ matrices.

26.3 Linear Transformations

A map $\mathbf{T} : \mathcal{V} \to \mathcal{W}$ between two linear spaces is a **linear transformation** if:

$$\mathbf{T}(\mathbf{u} + \mathbf{v}) = \mathbf{T}\mathbf{u} + \mathbf{T}\mathbf{v} \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathcal{V}, \tag{26.1}$$

$$\mathbf{T}(\alpha\mathbf{u}) = \alpha\mathbf{T}\mathbf{u} \quad \text{for all } \alpha \in \mathbb{R},\ \mathbf{u} \in \mathcal{V}. \tag{26.2}$$

When $\mathcal{V} = \mathcal{W}$ is the space of 3D vectors $\mathbb{E}^3$, a linear transformation from $\mathbb{E}^3$ to $\mathbb{E}^3$ is called a **tensor**.

> **!** Note!
>
> The rotation $\mathbf{Q}$ is a linear transformation from $\mathbb{E}^3$ to $\mathbb{E}^3$, so it is a tensor. This justifies calling it the **rotation tensor** throughout the kinematics of rigid bodies chapter.

26.4 Basis and Dimension

A set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a **basis** for $\mathcal{V}$ if: - The vectors are **linearly independent**: $\sum_i \alpha_i \mathbf{e}_i = \mathbf{0}$ implies $\alpha_i = 0$ for all i. - They **span** $\mathcal{V}$: every $\mathbf{v} \in \mathcal{V}$ can be written as $\mathbf{v} = \sum_i v_i \mathbf{e}_i$.

The number of basis vectors is the **dimension** of the space.

For 3D Euclidean space $\mathbb{E}^3$, any orthonormal set $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ with $\mathbf{E}_i \cdot \mathbf{E}_j = \delta_{ij}$ is a basis.